Company / customer	Spacemaker Inc Limited
Drawing number	C40572
Drawing Revision	C
Project description	SpaceX

ELECTRICAL DRAWINGS FOR VTD OUTBOUND PANELS 6-10

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: TITLE PAGE / COVER SHEET

REV	DESCRIPTION	DATE

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07663

JOB # C40572

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55.0	SPARE
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56.1	BOM
56.2	BOM

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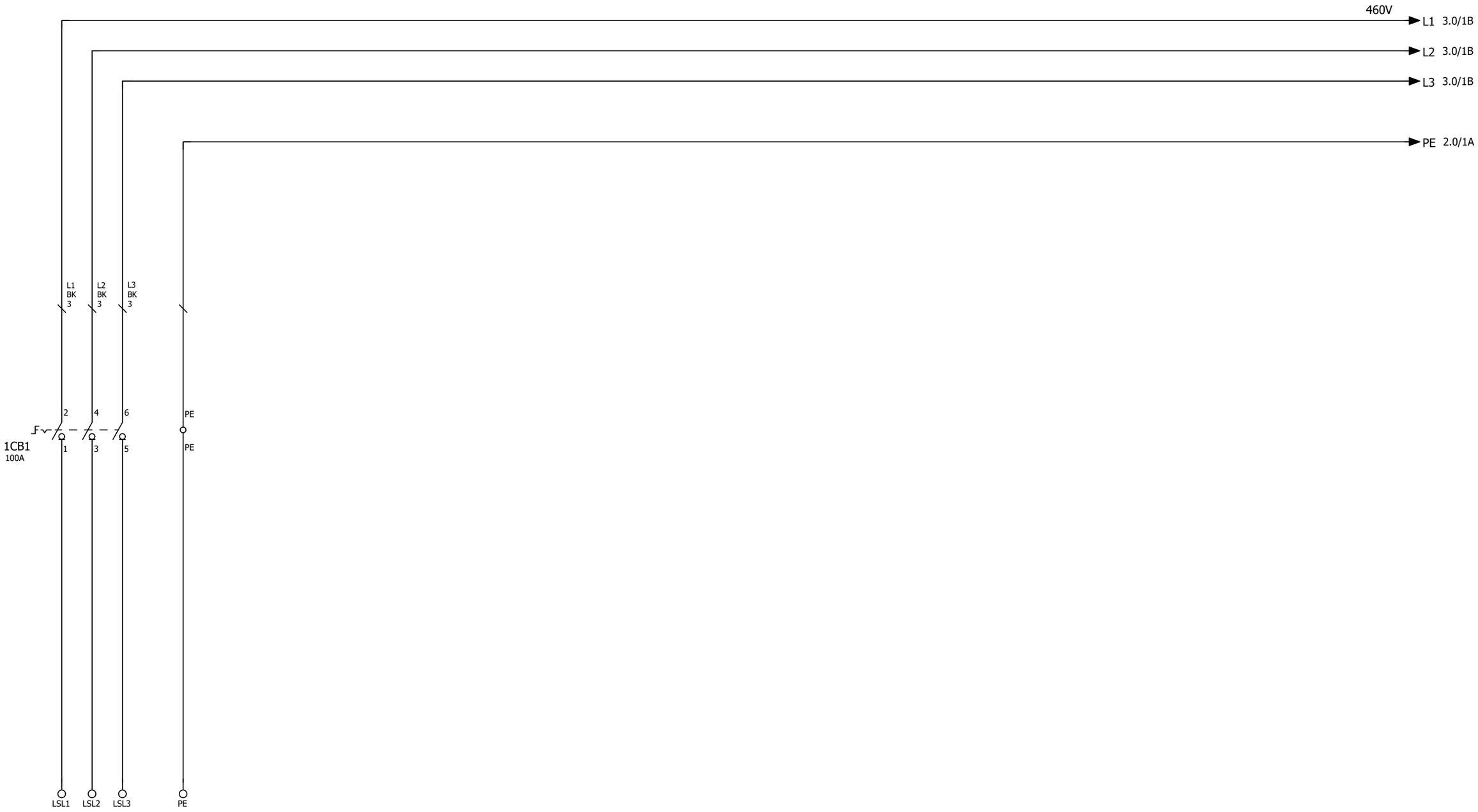

LINE ENTRY
460V/3PH/60HZ
AVAILABLE FAULT LEVEL
MUST NOT EXCEED 10kA r.m.s.
SCCR 10kA r.m.s.

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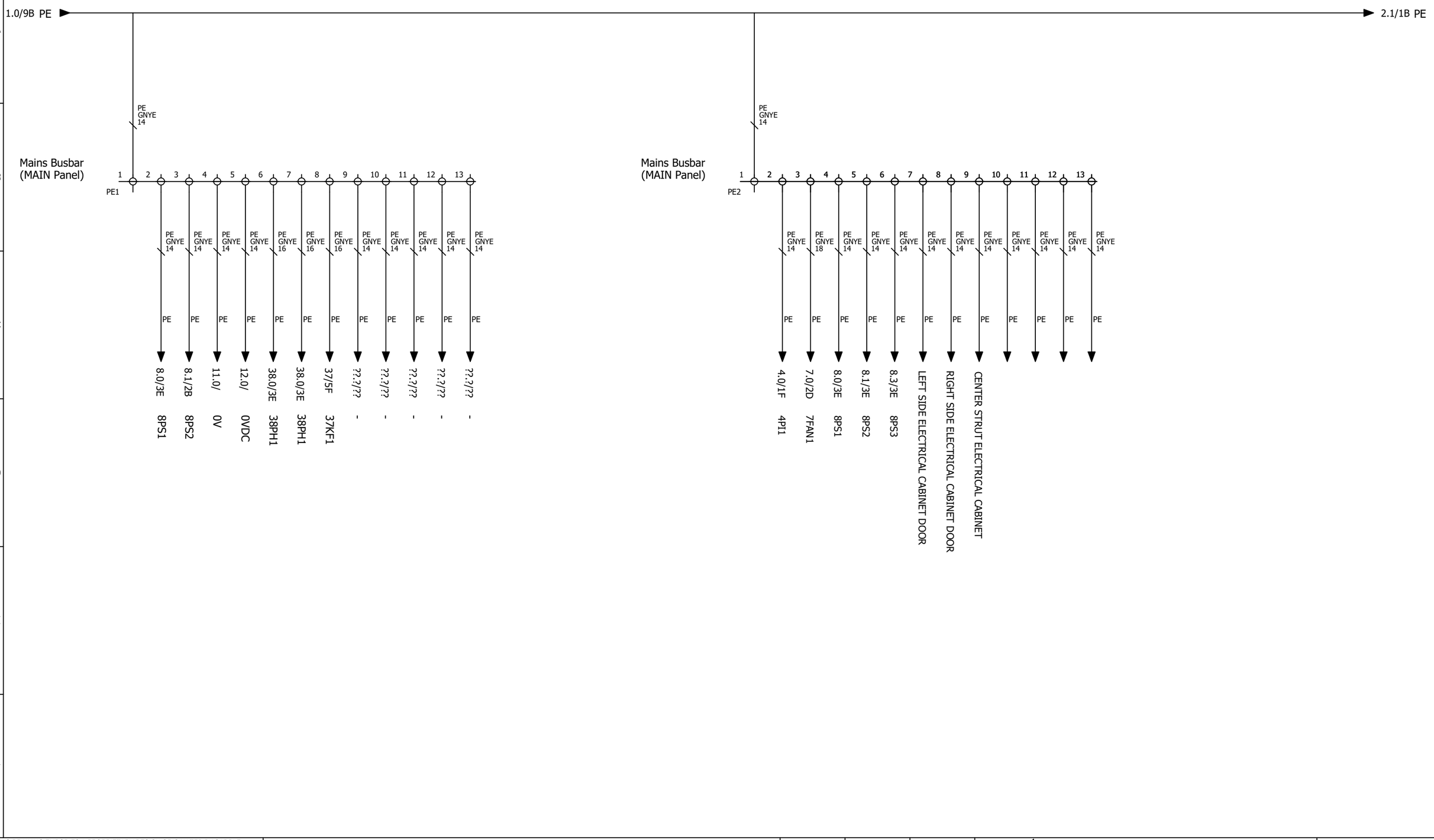

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PAGE DESCRIPTION: MAIN EARTH BARS

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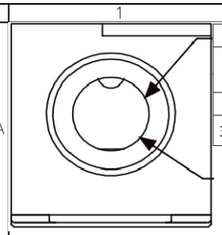
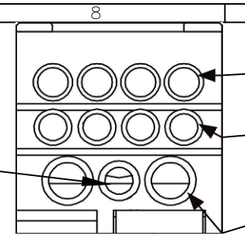


DB250		
Size	Torque	Strip lenght
35...120 mm²	19.0 Nm	27 mm

250A
DISTRIBUTION
BLOCK

Size	Torque	Strip lenght
2,5...16 mm²	2.7 Nm	16 mm



Size (4x)	Torque	Strip lenght
2,5...16 mm²	2.7 Nm	11 mm
Size (4x)	Torque	Strip lenght
2,5...10 mm²	2.7 Nm	10 mm
Size (2x)	Torque	Strip lenght
6...35 mm²	4.4 Nm	16 mm

460V

1.0/9A L1 → L1 3.1/1A

1.0/9A L2 → L2 3.1/1A

1.0/9A L3 → L3 3.1/1A

460V

PI1
PRE MADE CABLE

L1 BK 14 | L2 BK 14 | L3 BK 14

L1 BK 10 | L2 BK 10 | L3 BK 10

4.0/1A L1 | 4.0/1A L2 | 4.0/1A L3 → 4PI1

7/1A L1 | 7/1A L2 | 7/1A L3 → 7CB1

8/1A L1 | 8/1A L2 | 8/1A L3 → 8CB1

8.1/1A L1 | 8.1/1A L2 | 8.1/1A L3 → 8CB3

8.3/1A L1 | 8.3/1A L2 | 8.3/1A L3 → 8CB5

15.1/4E L1 | 15.1/4E L2 | 15.1/5E L3 → 15CB1

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18.1/1E L1 | 18.1/1E L2 | 18.1/1E L3 → 18CB1

L1 | L2 | L3 → SPARE

20.1/1E L1 | 20.1/1E L2 | 20.1/1E L3 → 20CB1

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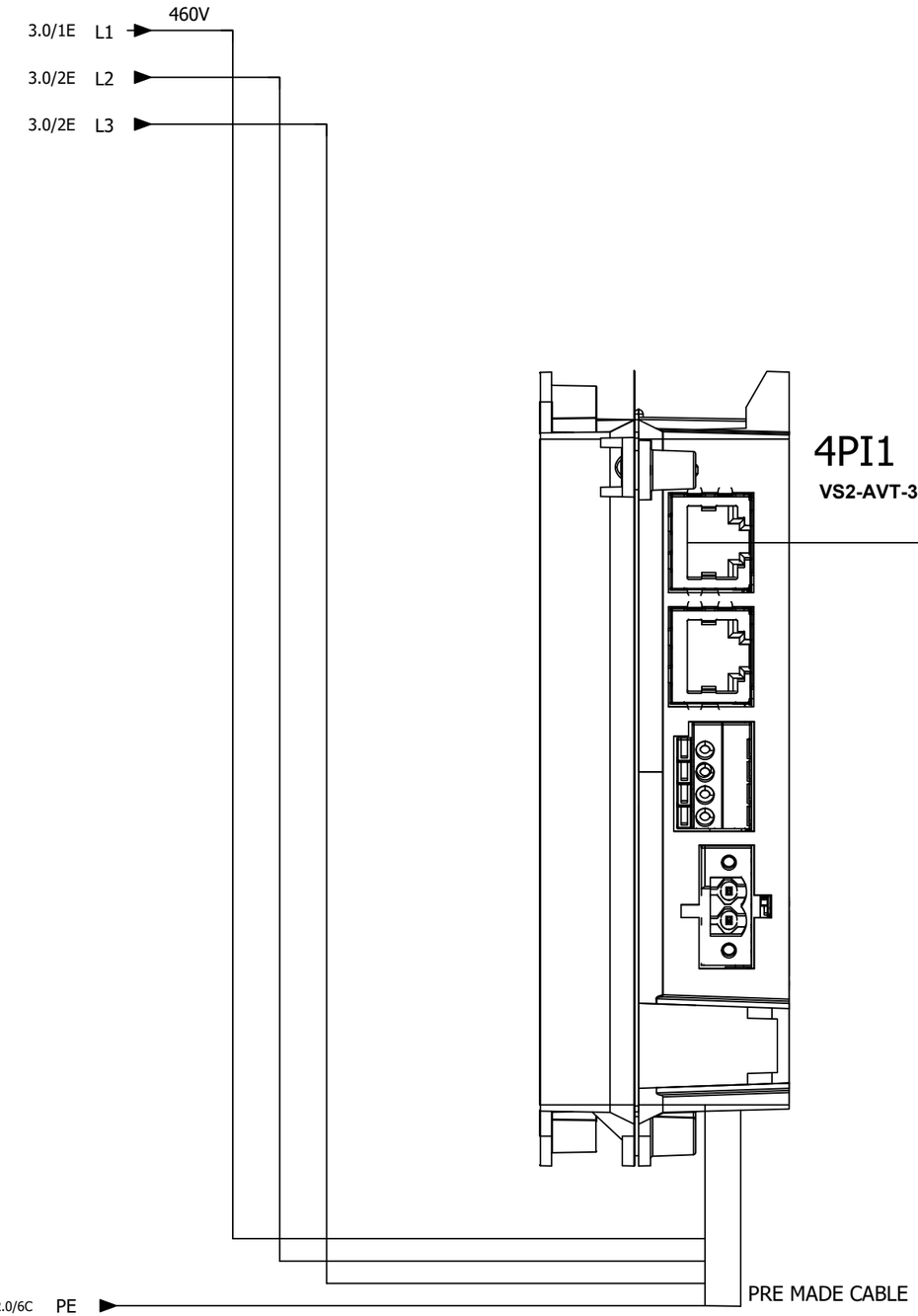

4PI1
VS2-AVT-3PB-16

PANDUIT

VOLTAGE
INDICATOR AC

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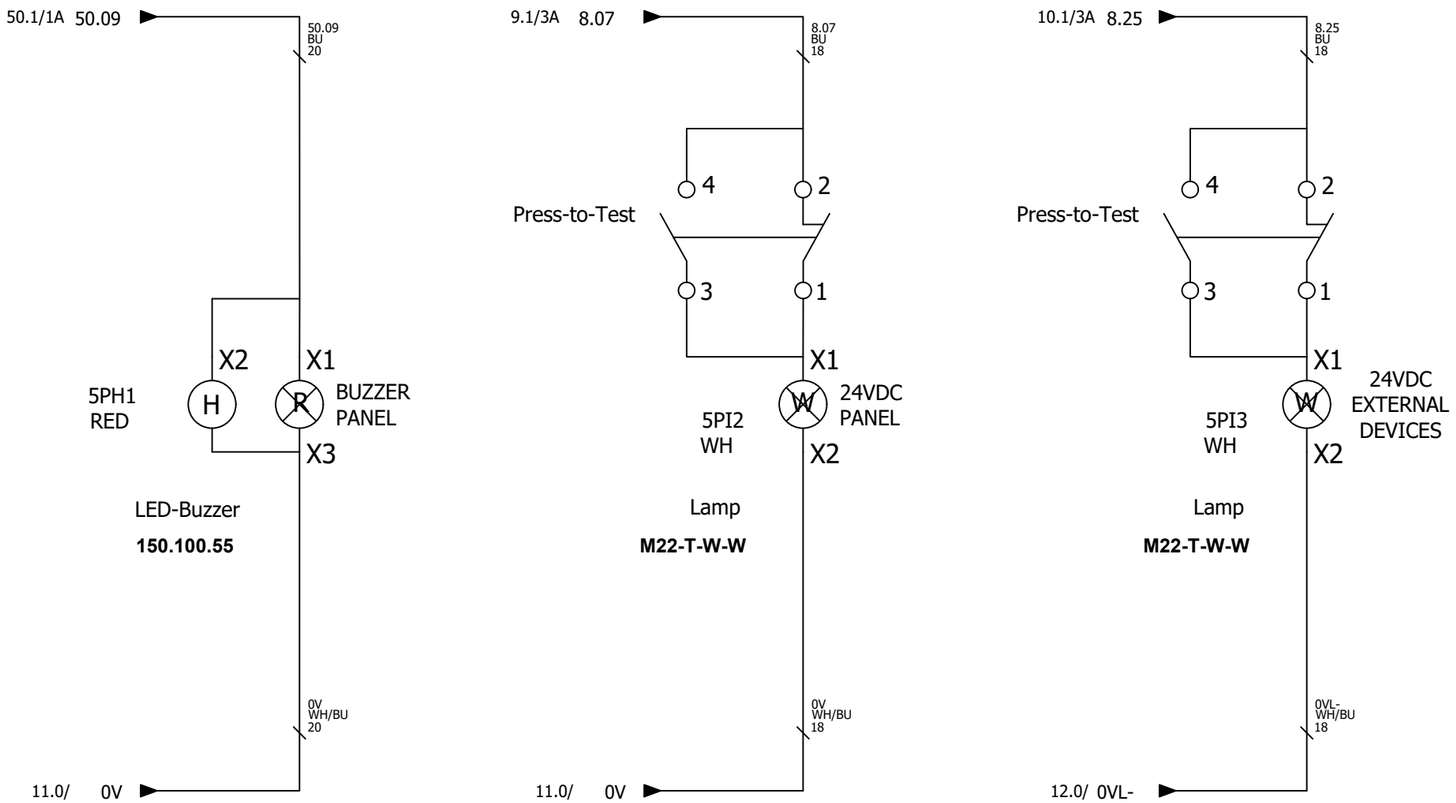

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 24V DC POWER INDICATORS

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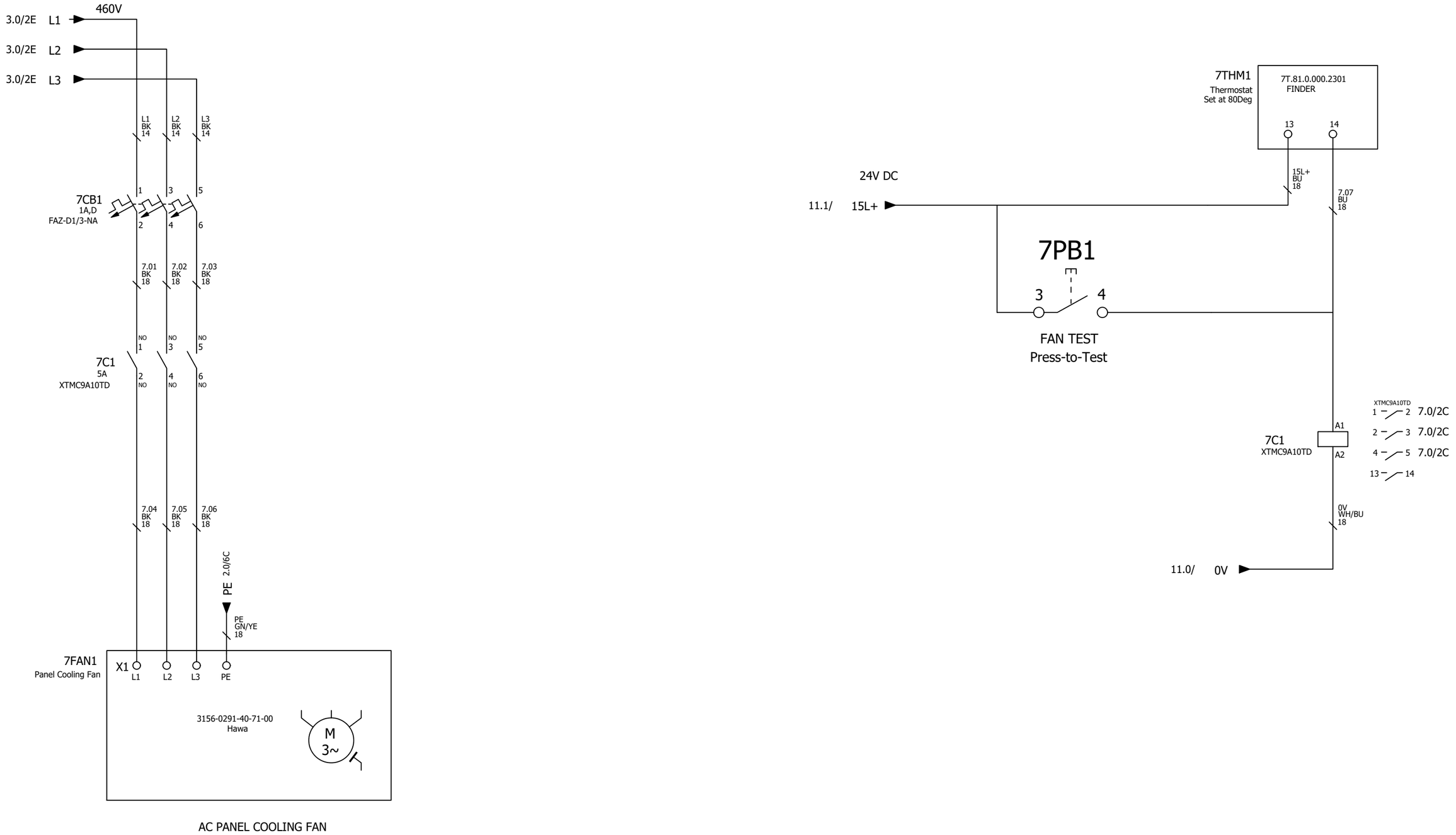

AC PANEL COOLING FAN

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: PANEL COOLING

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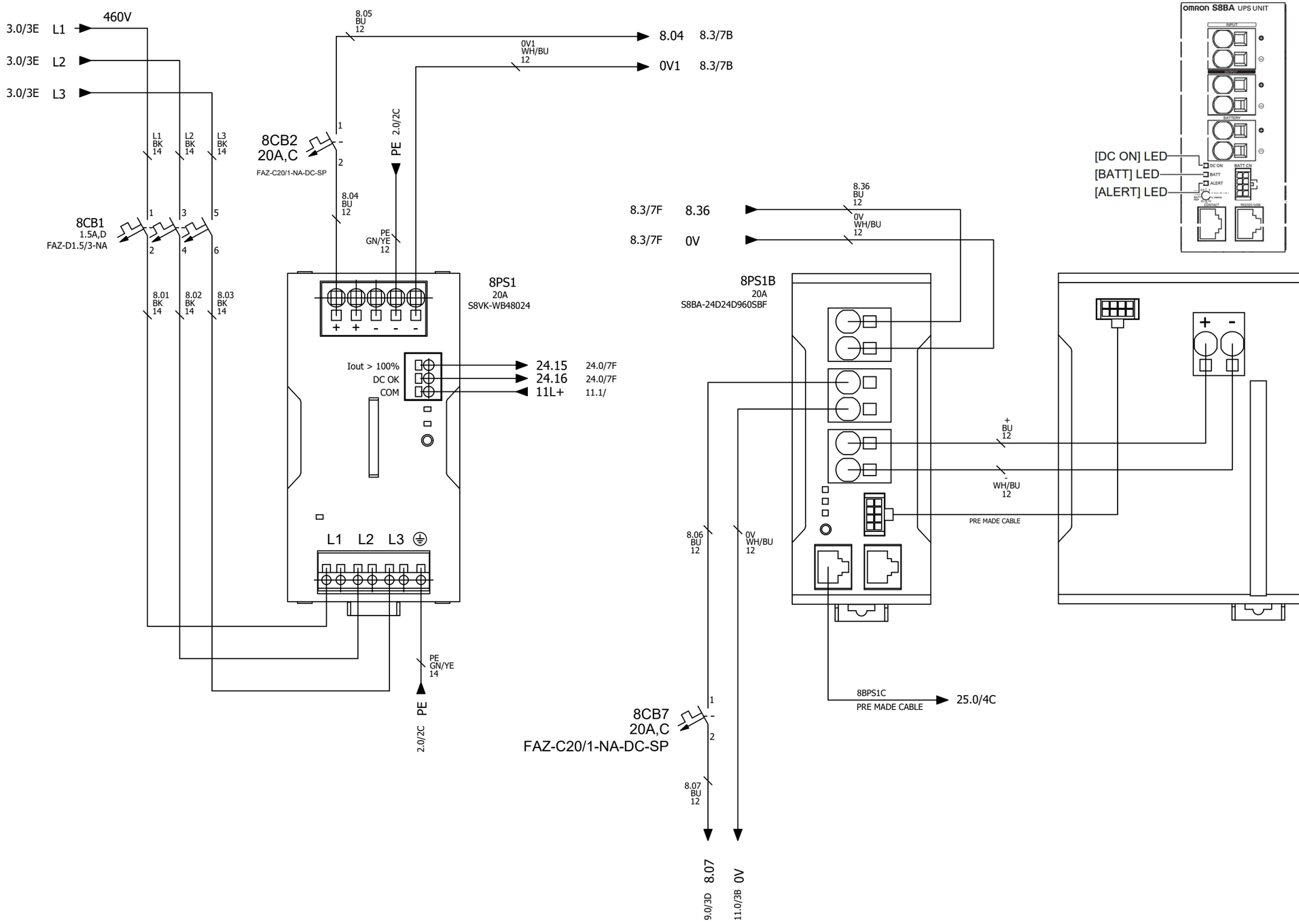

Pin number	Cable color	Item
1	White/orange	Backup signal output (BU)
2	Orange	Remote ON/OFF input (-)
3	White/green	Trouble signal output (TR)
4	Blue	COMMON (COM)
5	White/blue	Battery LOW signal output (BL)
6	Green	Backup stop signal input (BS)
7	White/brown	Battery Replacement Signal output (WB)
8	Brown	Remote ON/OFF input (+)

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CONTROL POWER SUPPLIES MAIN PANEL

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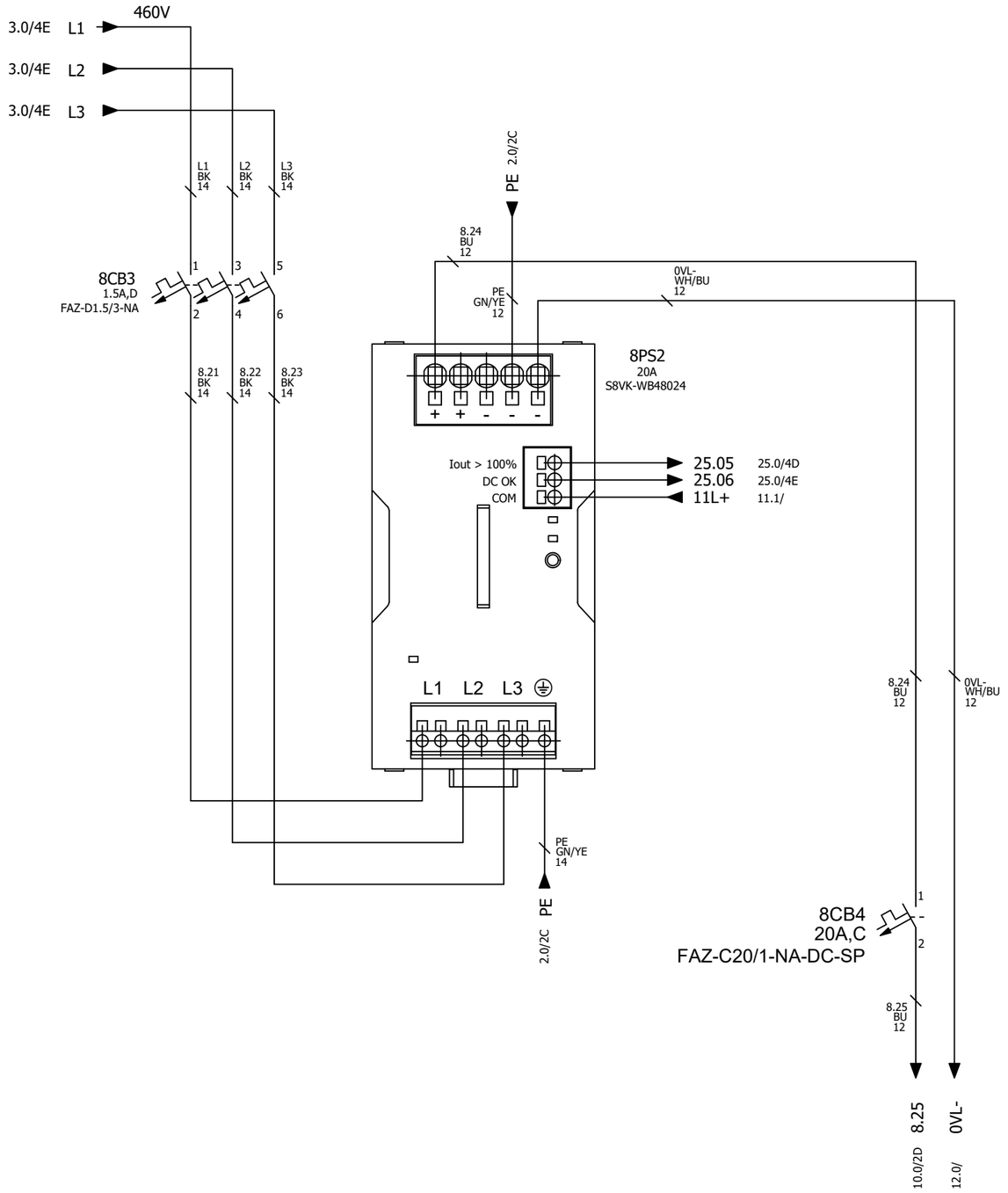

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CONTROL POWER SUPPLIES MAIN PANEL FOR REMOTE COMPONENTS

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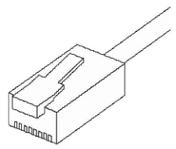
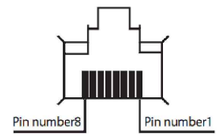
I/O signal port (RJ45 connector)

Outlook of the connector	Pin number	Cable color	Item
	1	White/orange	Backup signal output (BU)
	2	Orange	Remote ON/OFF input (-)
	3	White/green	Trouble signal output (TR)
	4	Blue	COMMON (COM)
	5	White/blue	Battery LOW signal output (BL)
	6	Green	Backup stop signal input (BS)
	7	White/brown	Battery Replacement Signal output (WB)
	8	Brown	Remote ON/OFF input (+)



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8BPS1

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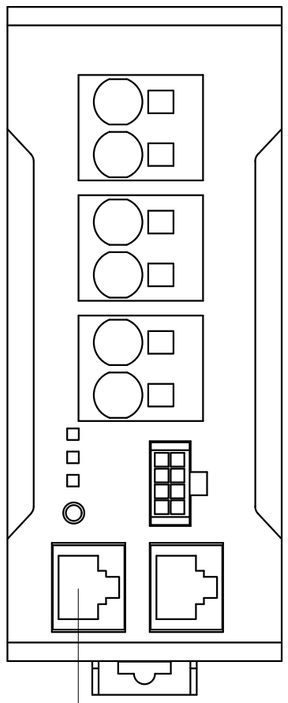

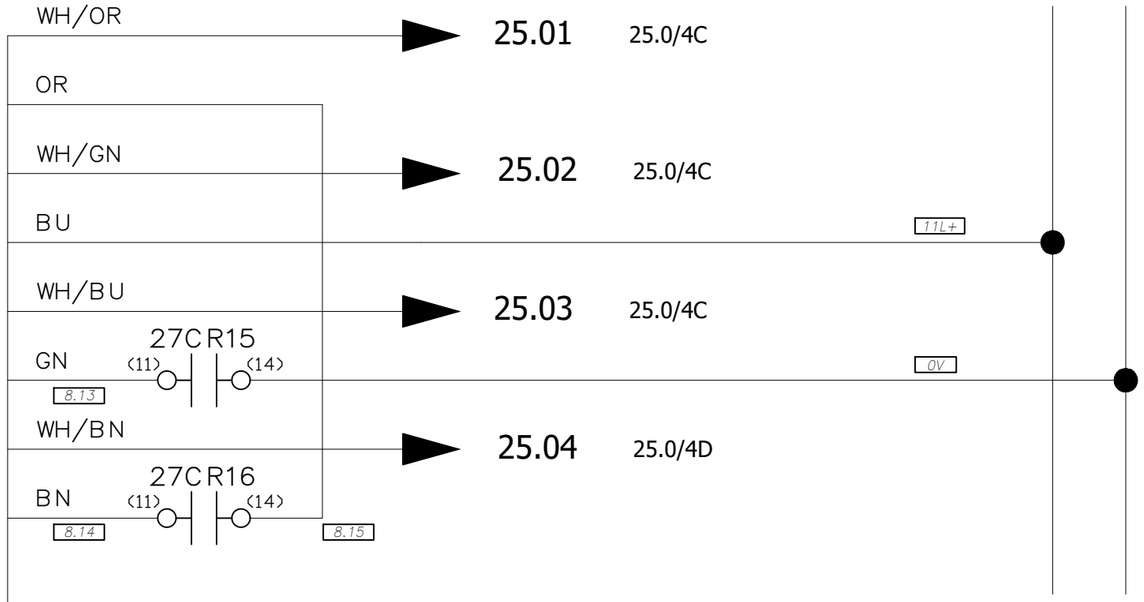

8BPS1C

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PAGE DESCRIPTION: CONTROL POWER SUPPLIES MAIN PANEL UPS I/O

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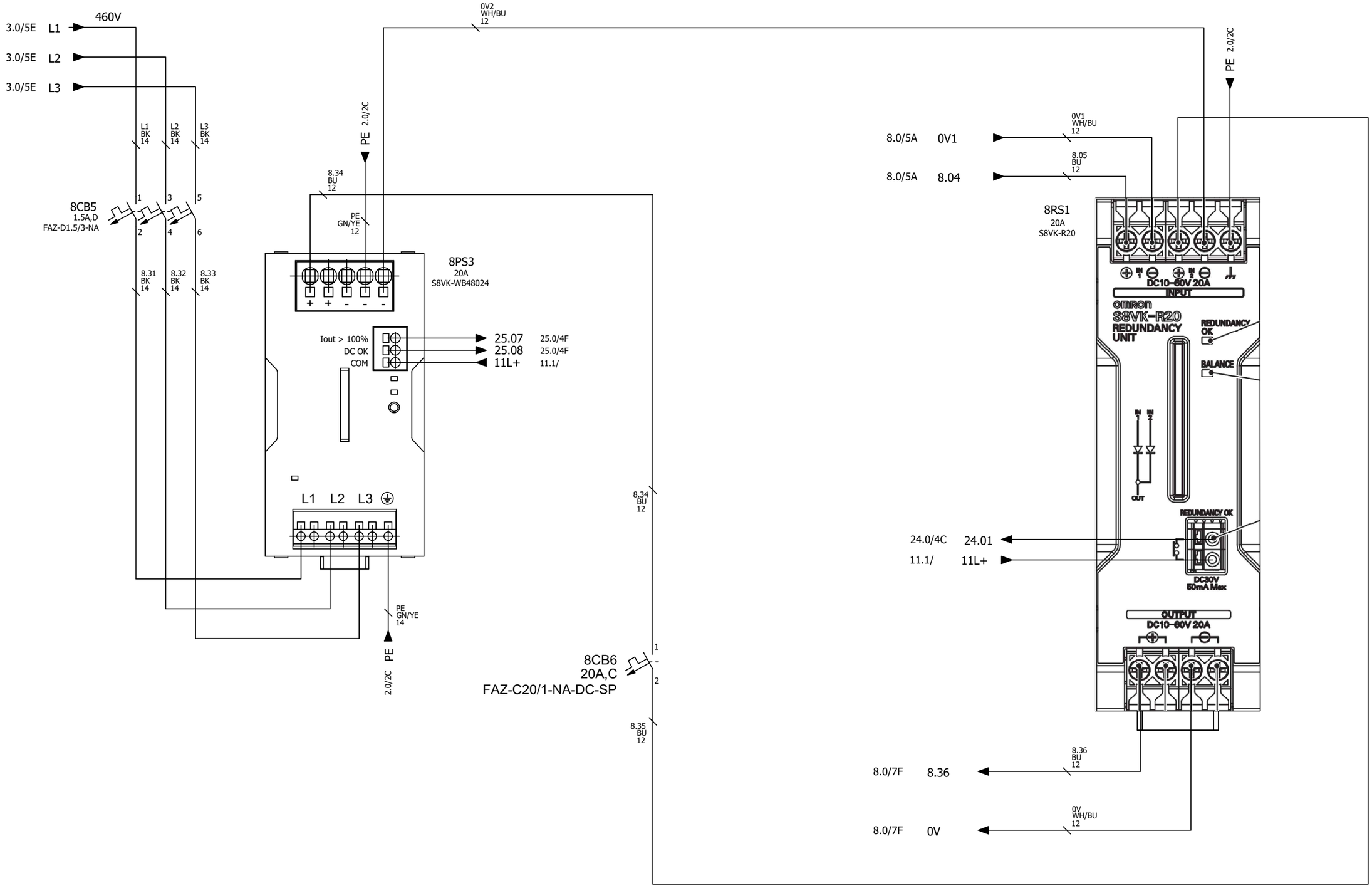

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PAGE DESCRIPTION: CONTROL POWER SUPPLIES MAIN PANEL REDUNDANT 24 VDC

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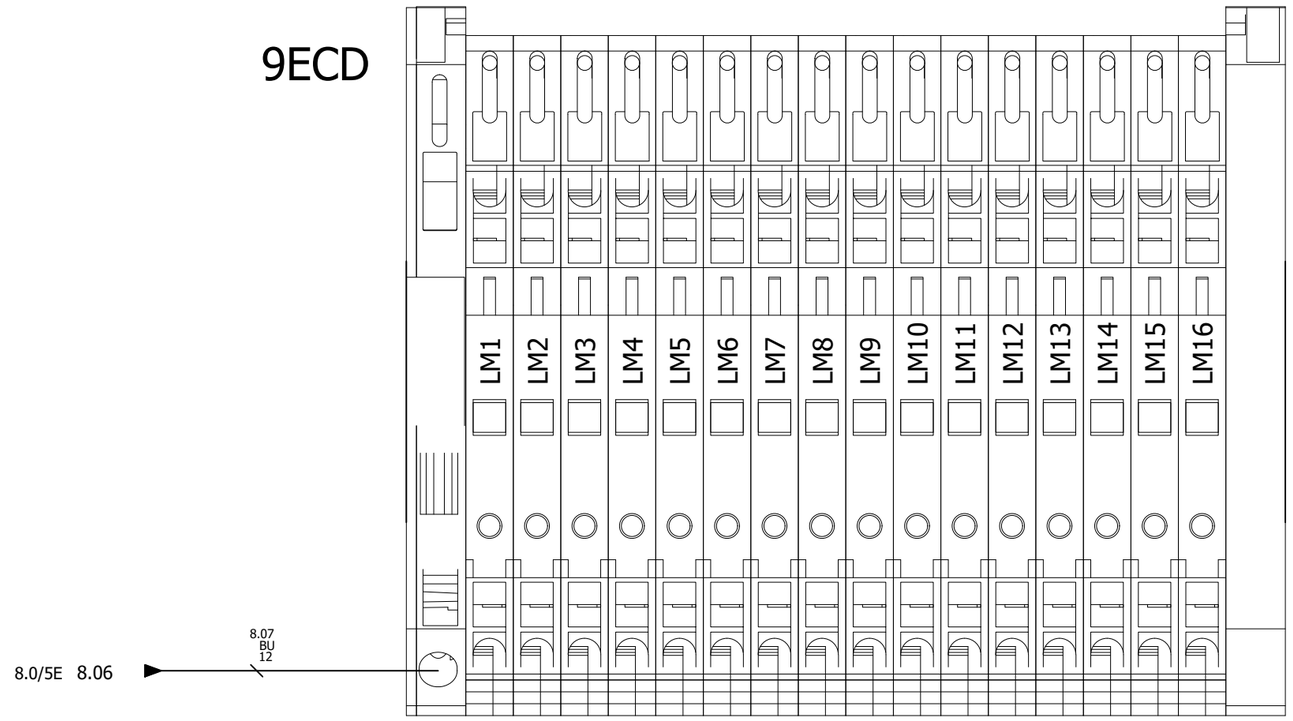


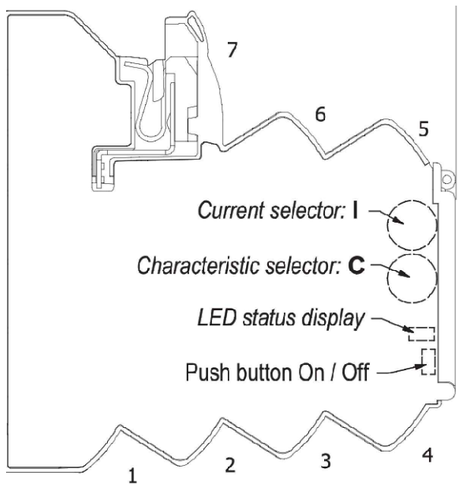
LOAD MONITORING(LM) PIN ASSIGNMENT

PIN 1–OUT+(LOAD)
PIN 2–CTRL INPUT (SET/RESET)
PIN 3–STATUS OUTPUT
PIN 4–NC
PIN 5–0V
PIN 6–SUPPLY (ALTERNATIVE)
PIN 7–SUPPLY

TRIP CURVES:

1. CHARACTERISTIC FAST
2. CHARACTERISTIC MEDIUM
3. CHARACTERISTIC SLOW–1
4. CHARACTERISTIC SLOW–2
5. CHARACTERISTIC SLOW–3

ELECTRONIC CIRCUIT BREAKER (LM)



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PAGE DESCRIPTION: 24V ELECTRONIC LOAD MONITORING MAIN PANEL

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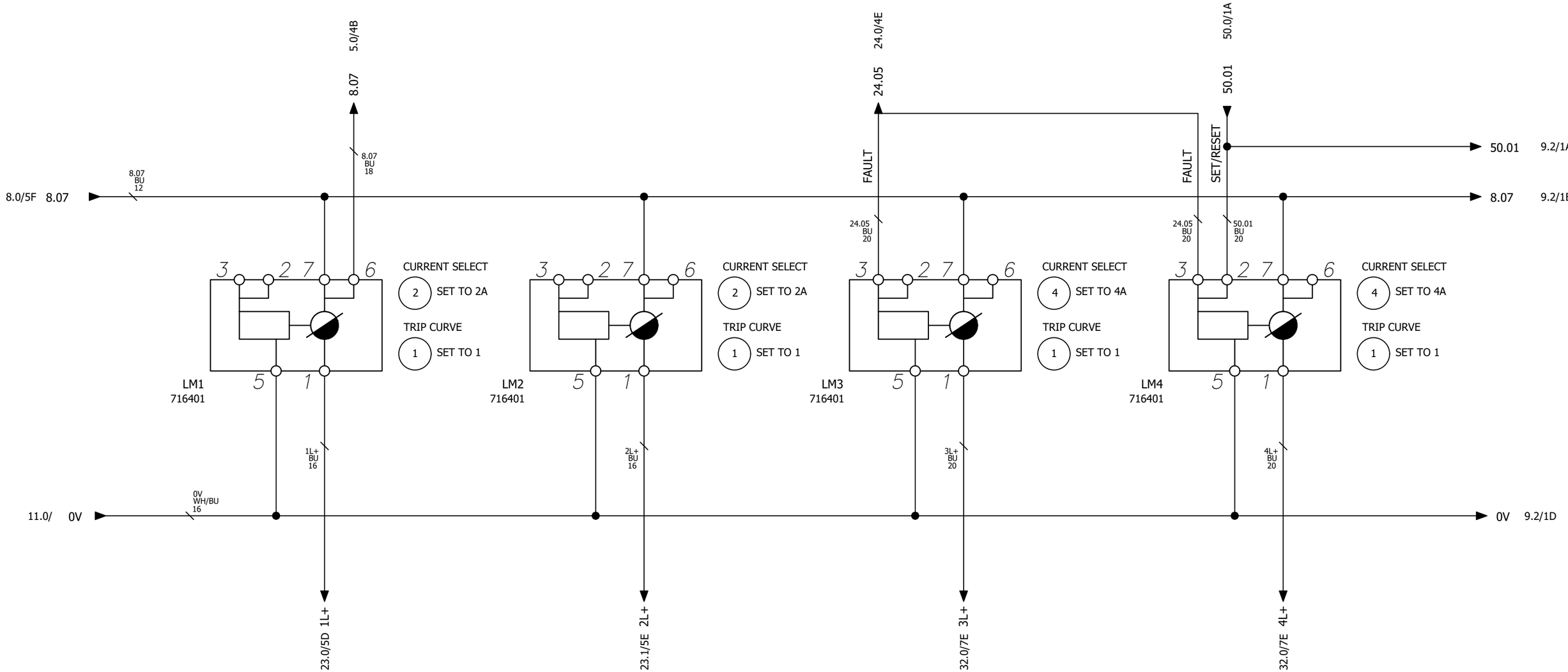

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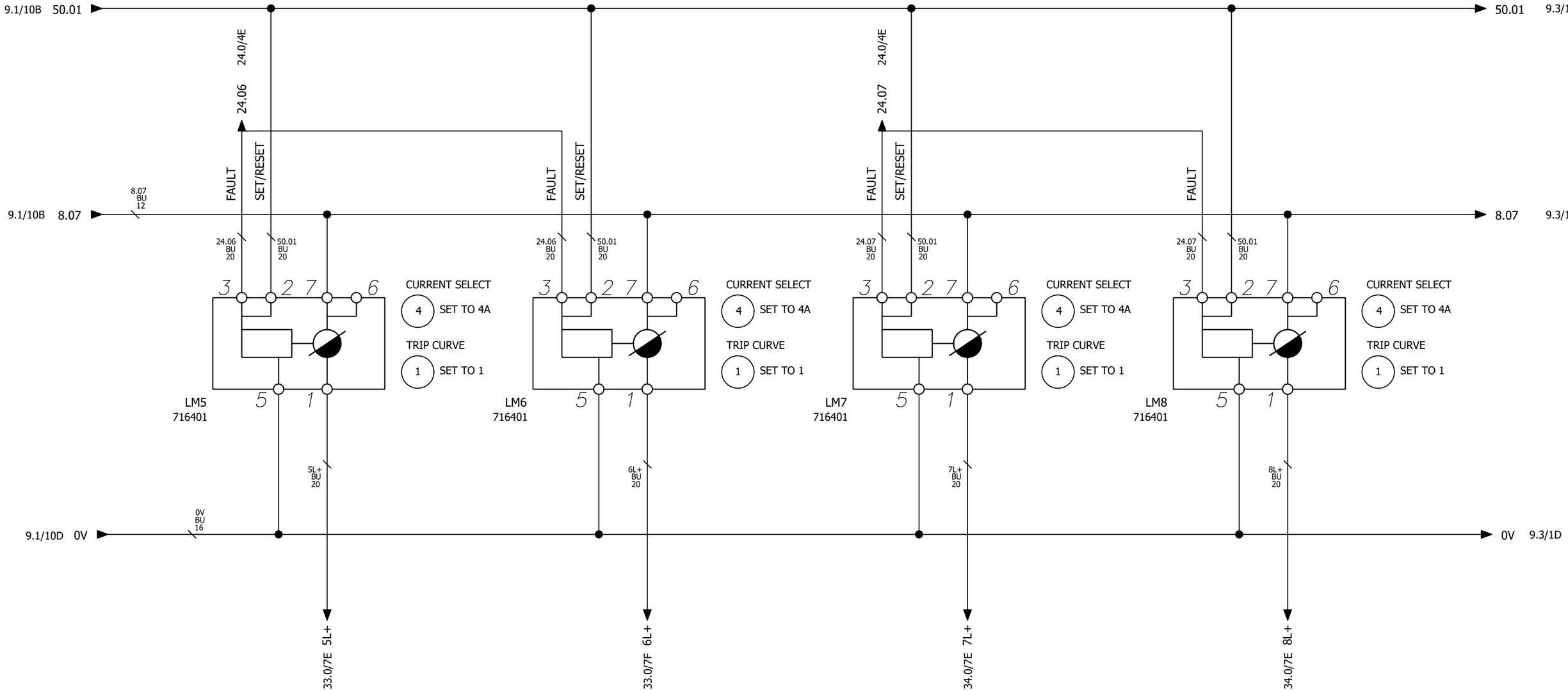

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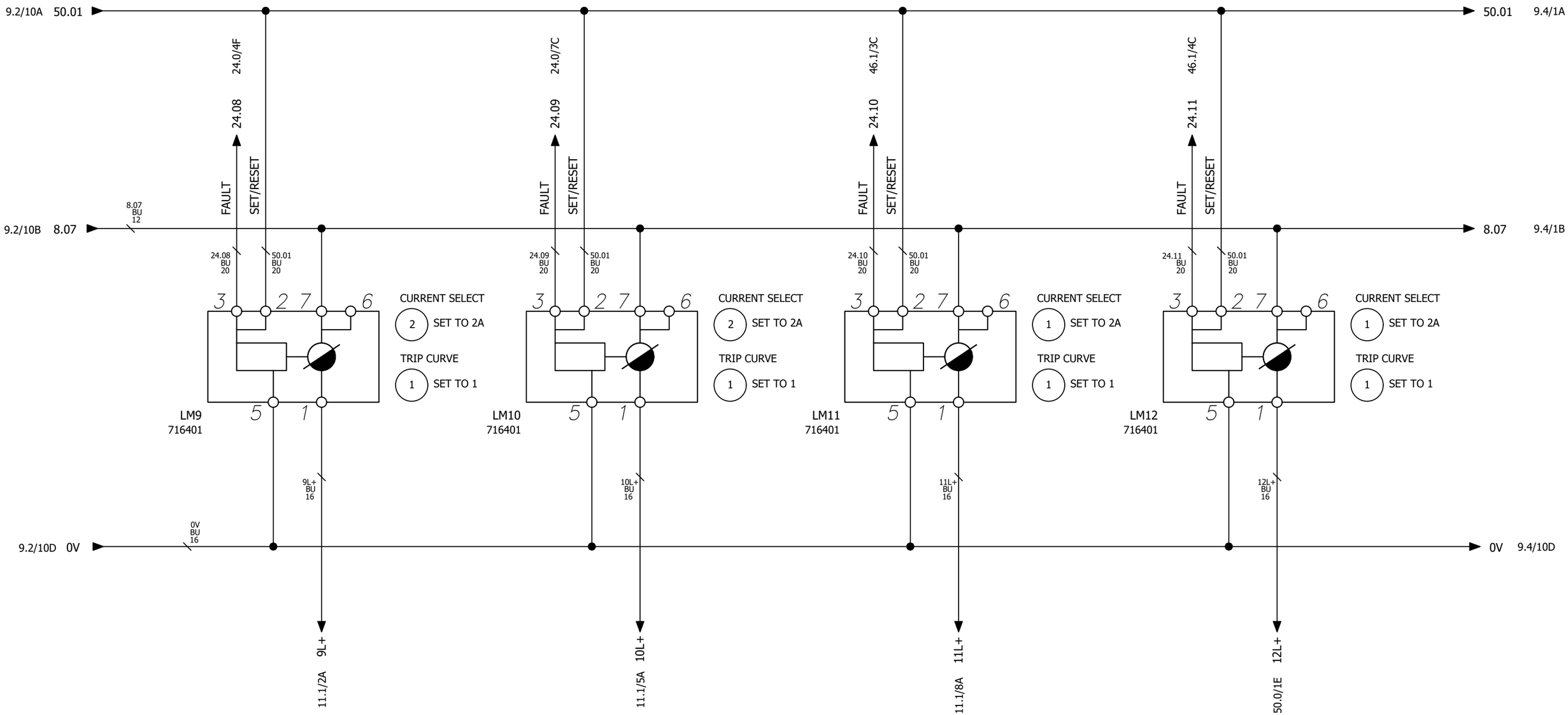

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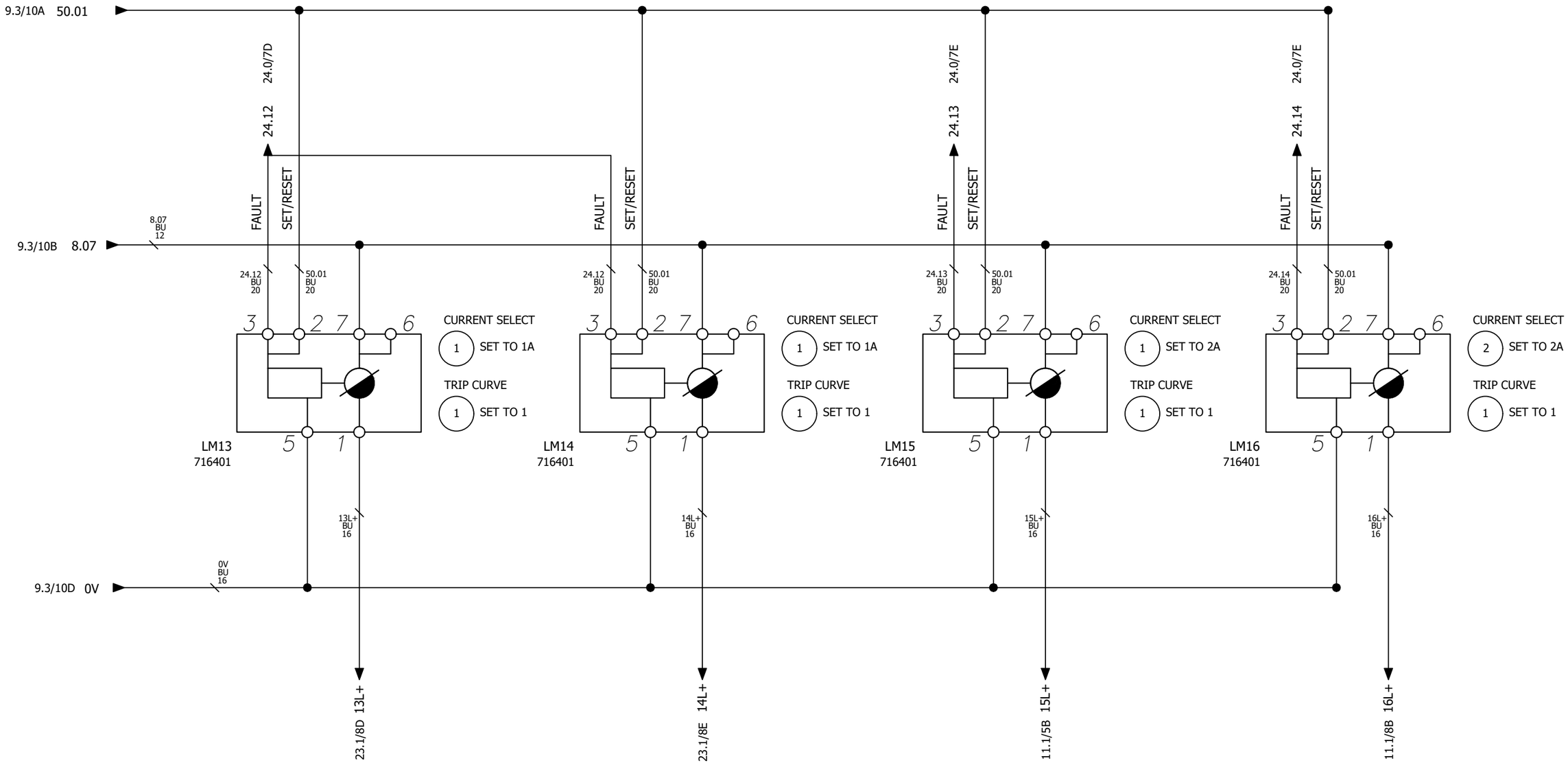

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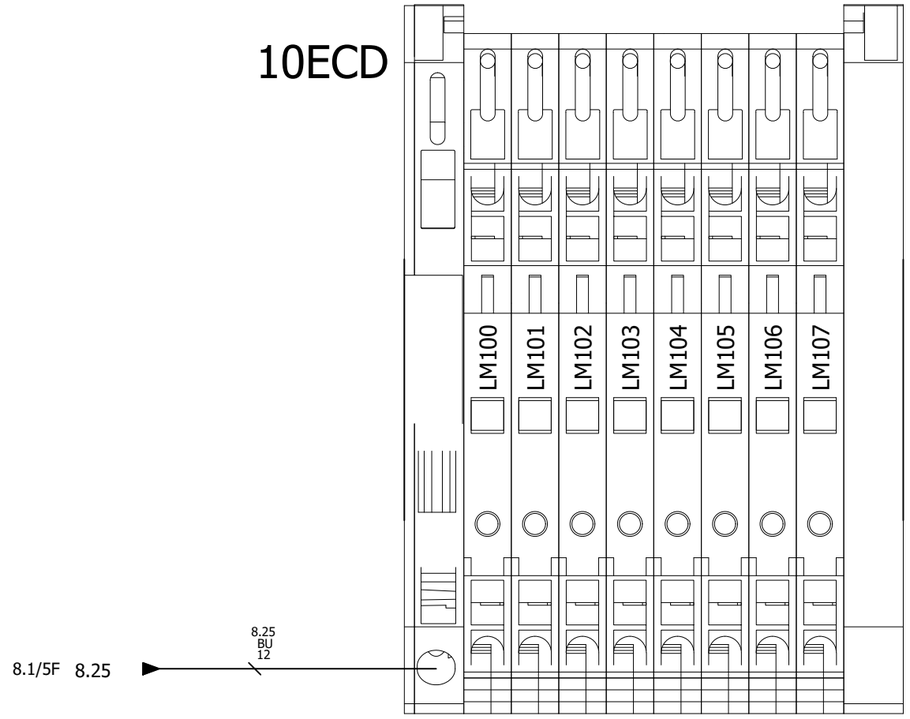

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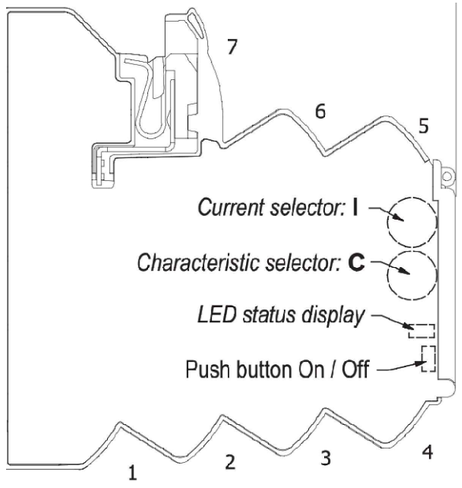

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PAGE DESCRIPTION: 24V ELECTRONIC LOAD MONITORING MAIN PANEL FOR REMOTE COMPONENTS

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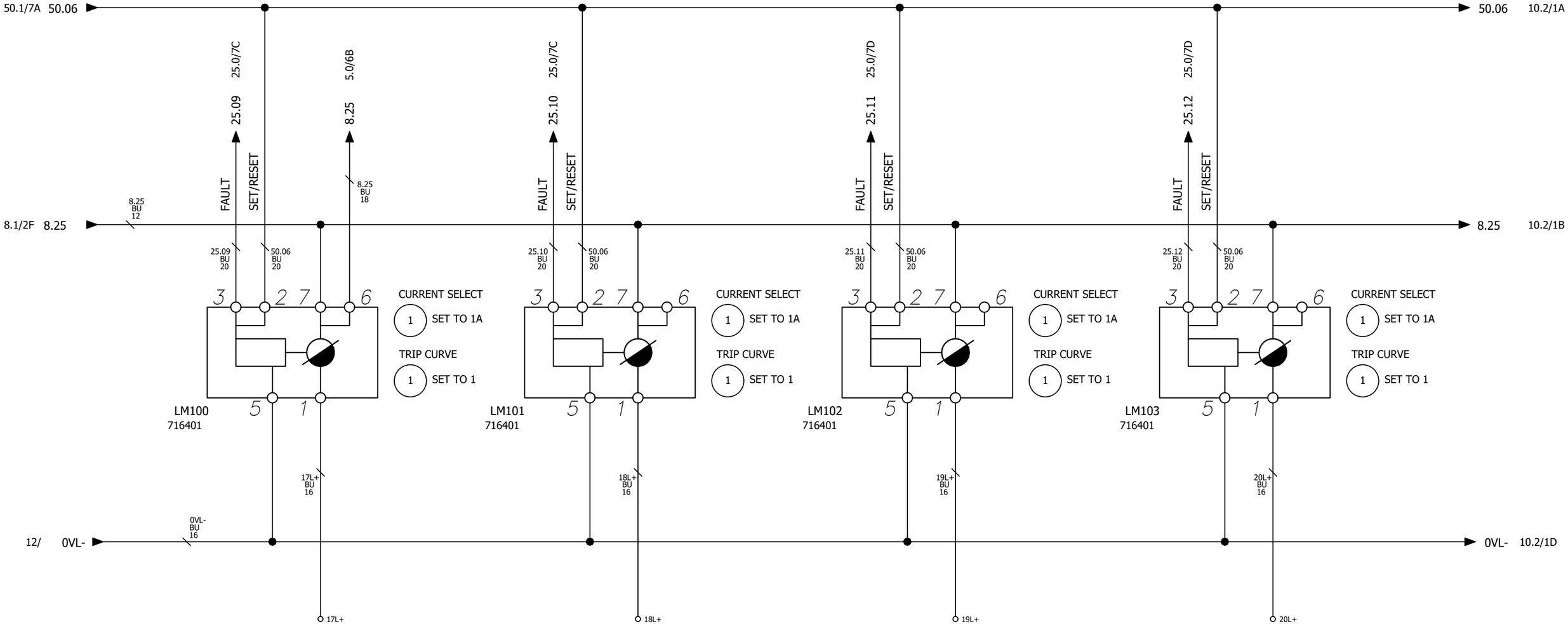

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 24V DISTRIBUTION MAIN PANEL FOR REMOTE COMPONENTS

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TOTAL PAGES: 171

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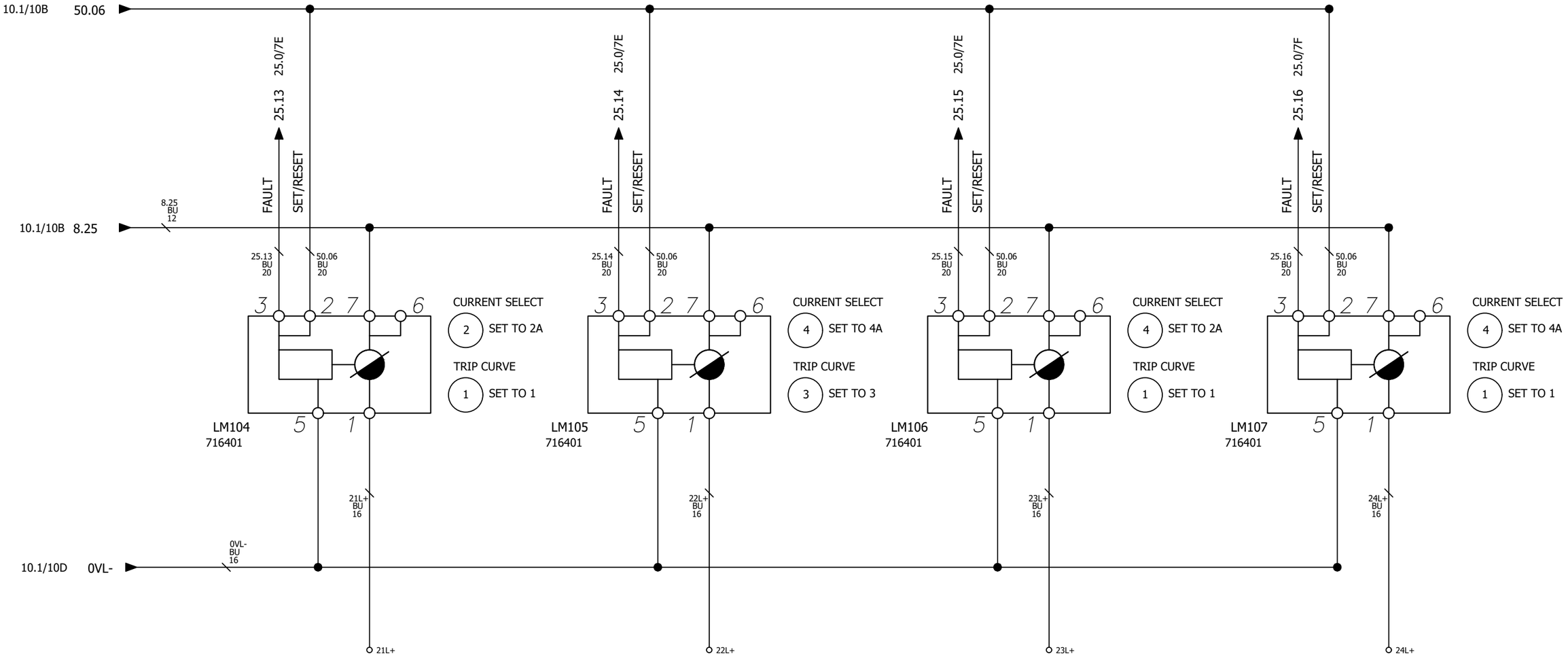

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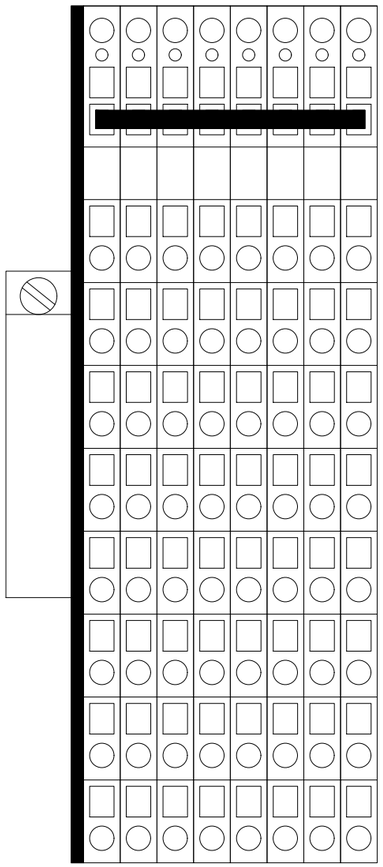

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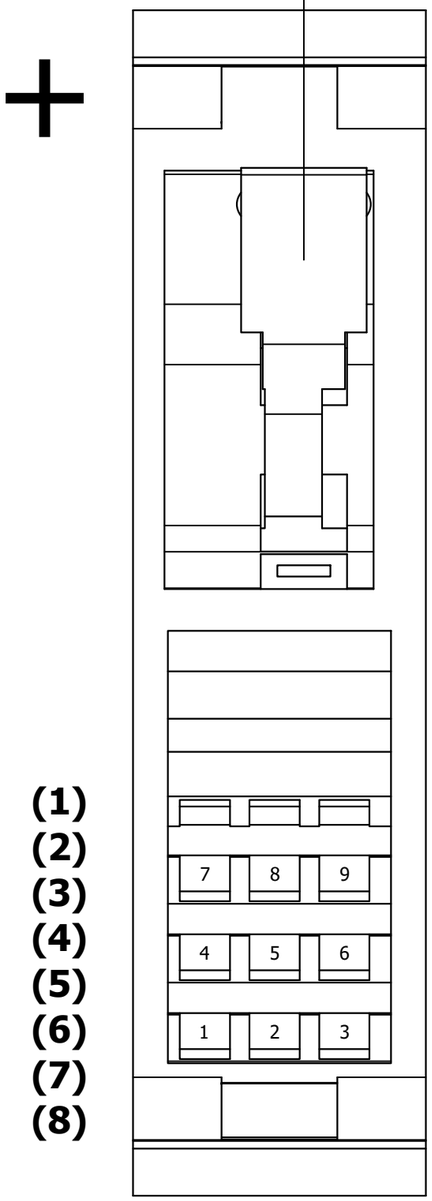
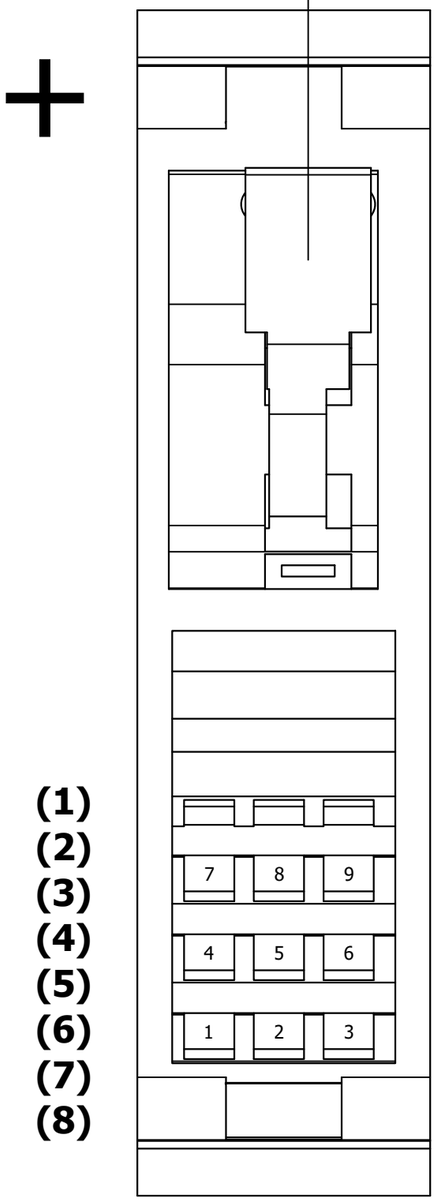
9.3/3E 9L+

9L+



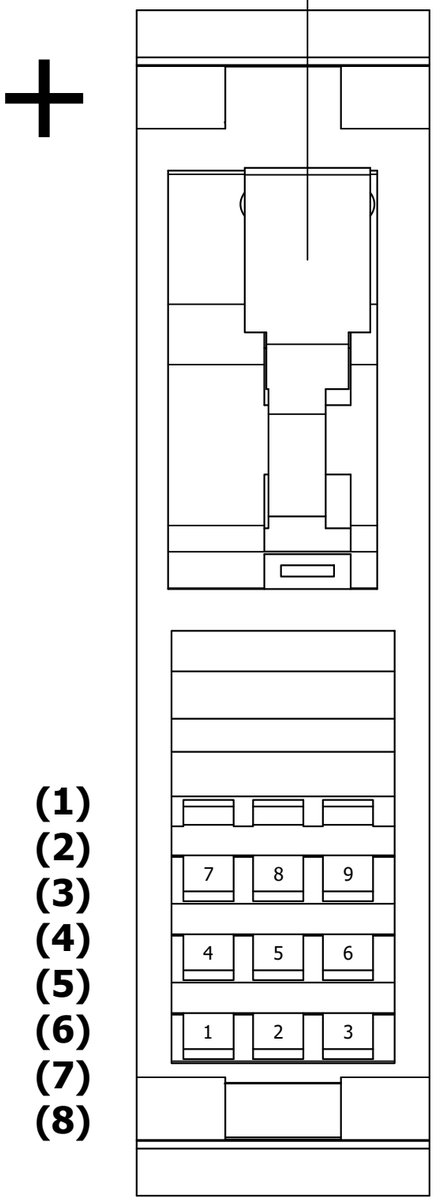
9.3/5E 10L+

10L+



9.3/7E 11L+

11L+



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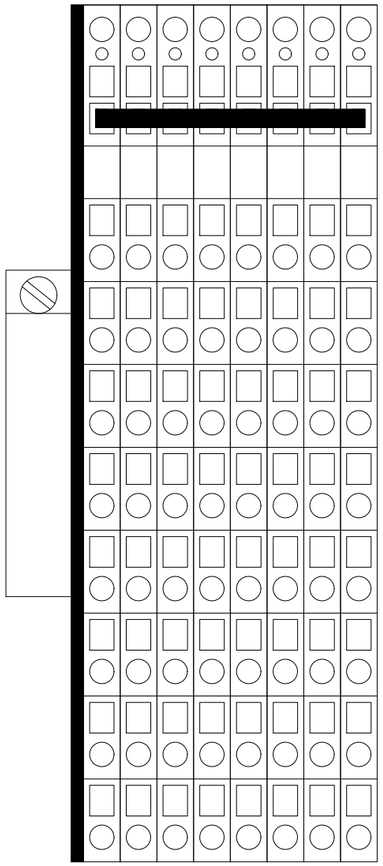

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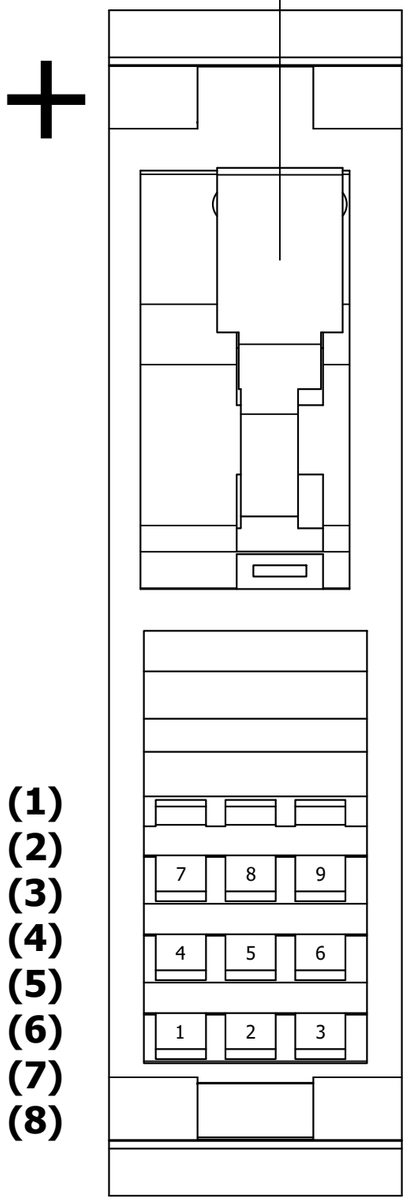
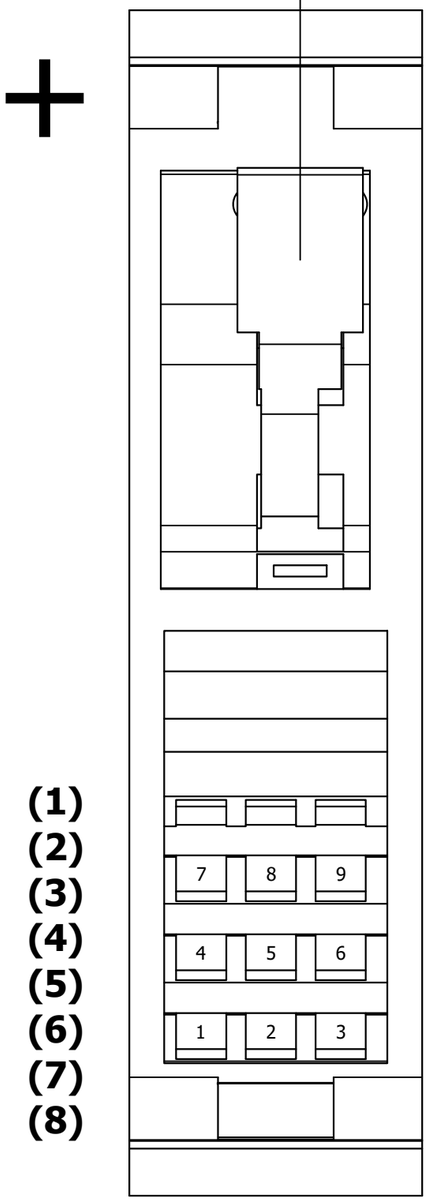
10.1/7E 19L+

19L+



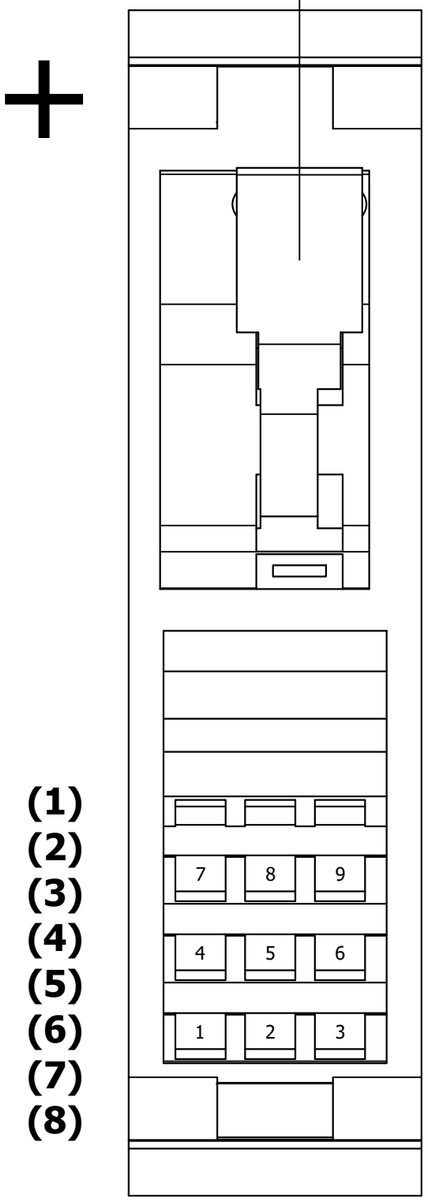
10.1/9E 20L+

20L+



10.2/3E 21L+

21L+



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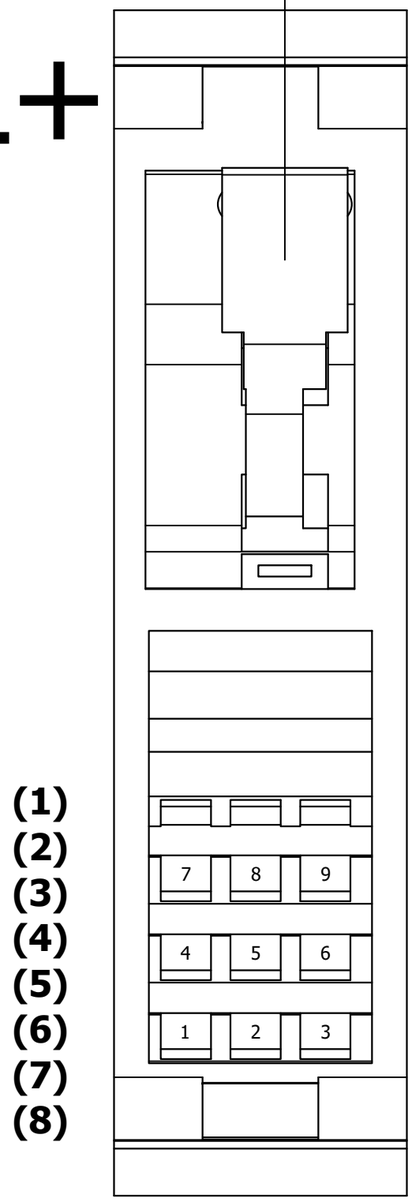
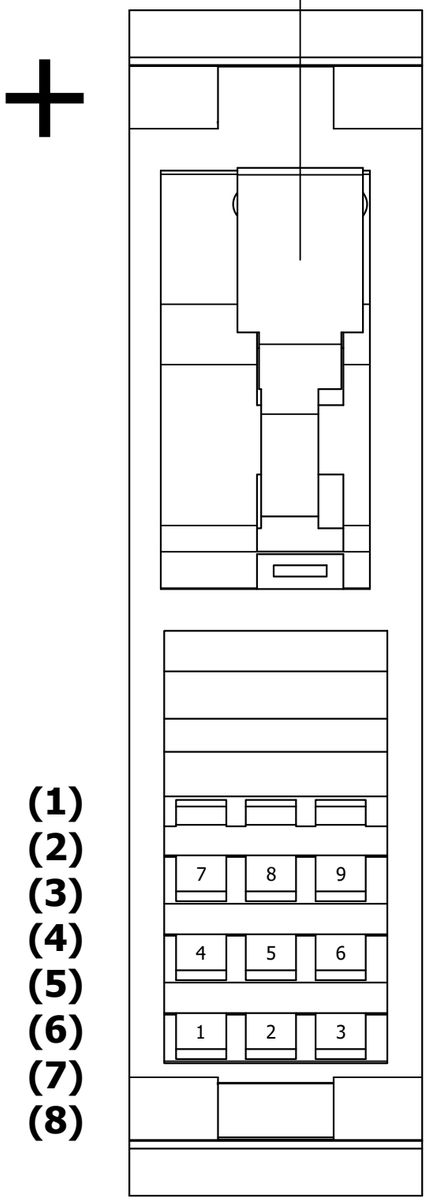
10.2/5E 22L+

22L+



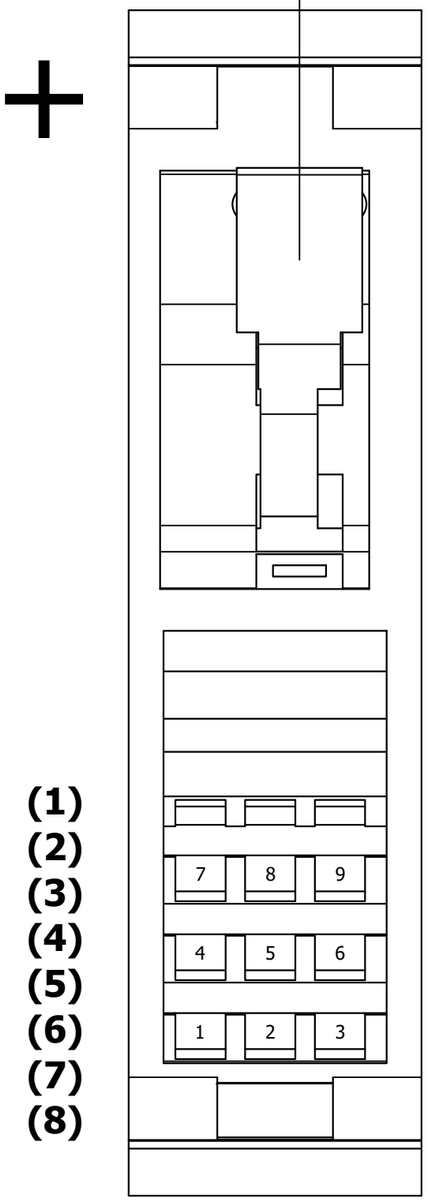
10.2/7E 23L+

23L+



10.2/9E 24L+

24L+



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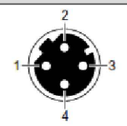
15VFD1

X4233: Ethernet fieldbus

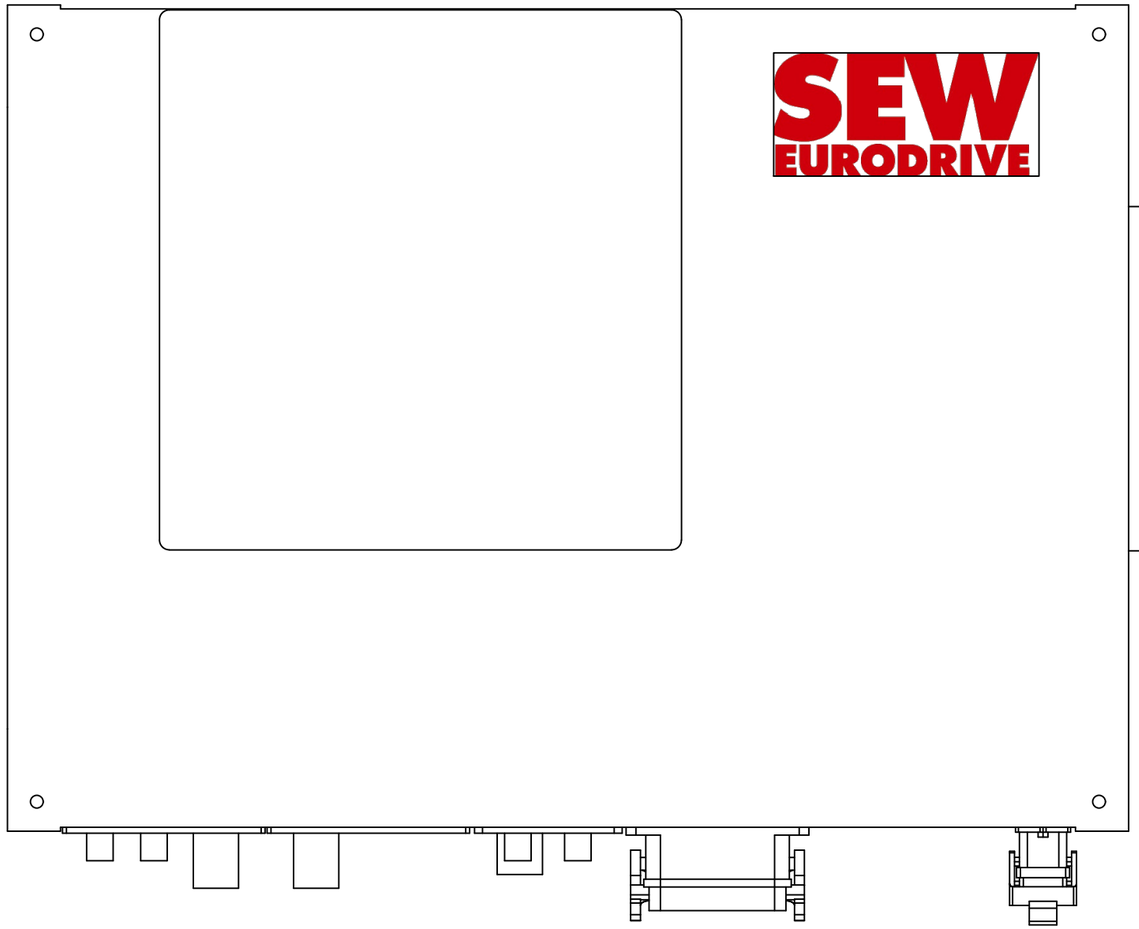
Function
Ethernet fieldbus interface, 4-pin

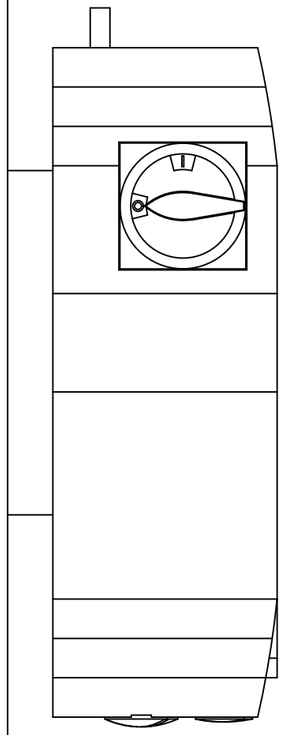
Connection type
M12, 4-pin, female, D-coded

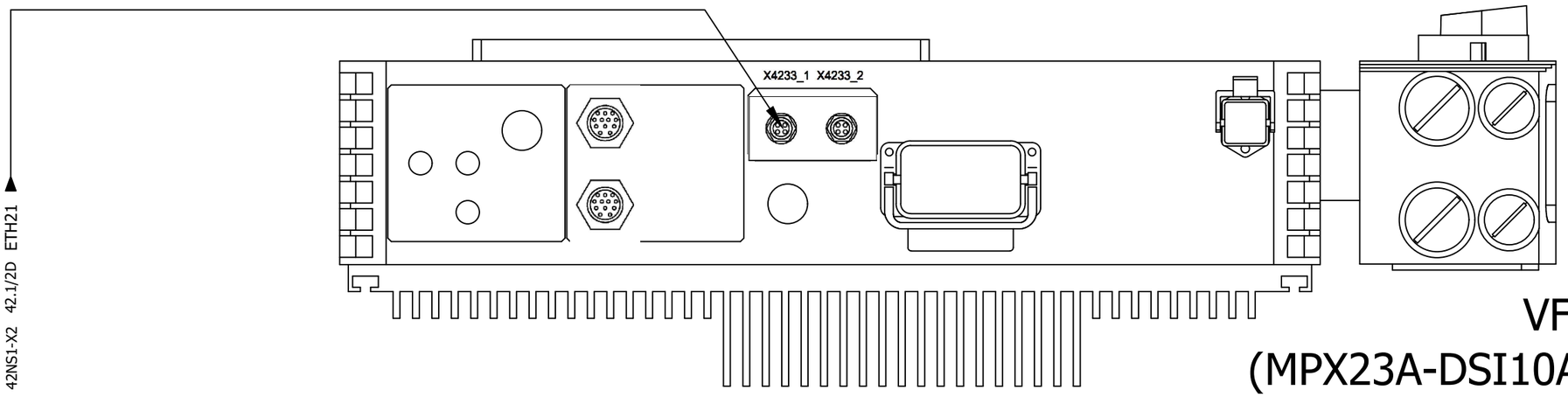
Wiring diagram



No.	Name	Function
1	TX+	Sending cable (+)
2	RX+	Receiving cable (+)
3	TX-	Sending cable (-)
4	RX-	Receiving cable (-)





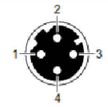


X4224: Ethernet engineering

Function
Ethernet engineering interface, 4-pin

Connection type
M12, 4-pin, female, D-coded

Connection diagram



No.	Name	Function
1	TX+	Sending cable (+)
2	RX+	Receiving cable (+)
3	TX-	Sending cable (-)
4	RX-	Receiving cable (-)

VFD 15kW
(MPX23A-DSI10A-MPI-0460-5E3-CA-C)

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PAGE DESCRIPTION: VTD STATIC AREA DRIVE 15VFD1

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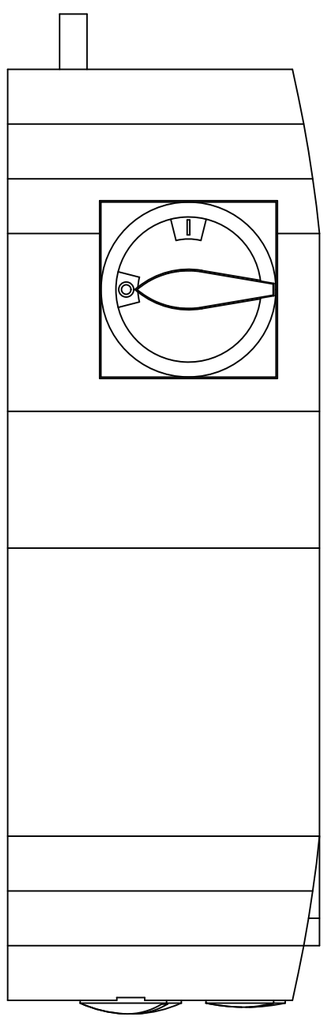
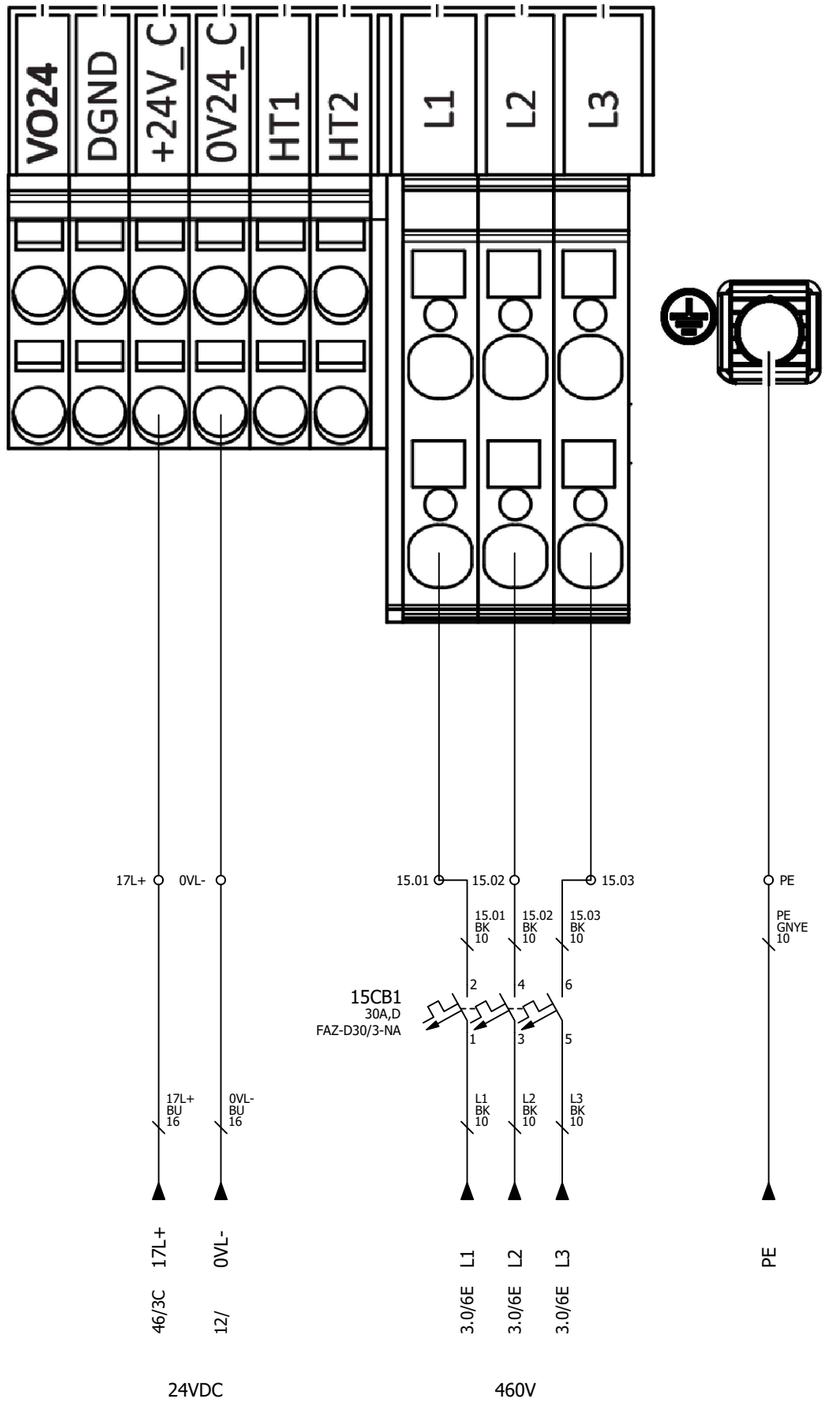
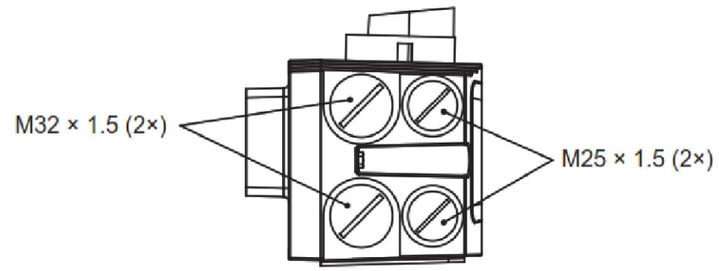




Interface box	PZM2xA..
Line terminals L1 – L3	57 A at 16 mm²
24 V terminals	41 A at 6 mm²



Name	Function
VO24	DC 24 V output
DGND	Not assigned
+24V_C	DC 24 V input
0V24_C	0V24 reference potential – input
HT1	Auxiliary terminal for additional voltage levels (without internal function)
HT2	Auxiliary terminal for additional voltage levels (without internal function)
L1	Line connection, phase 1
L2	Line connection, phase 2
L3	Line connection, phase 3
PE	Protective earth connection

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PAGE DESCRIPTION: VTD STATIC AREA DRIVE POWER FEED

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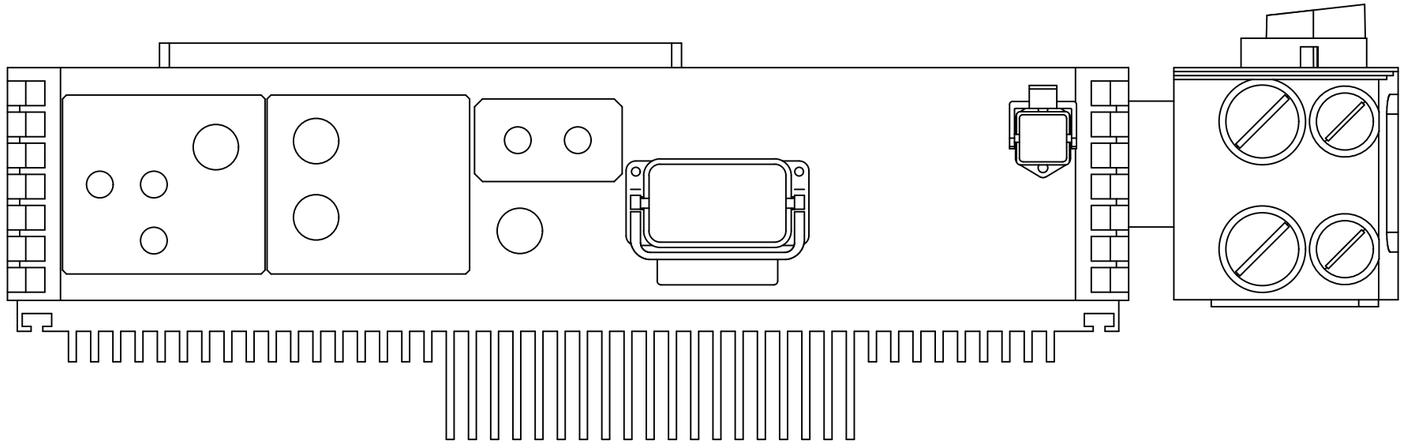
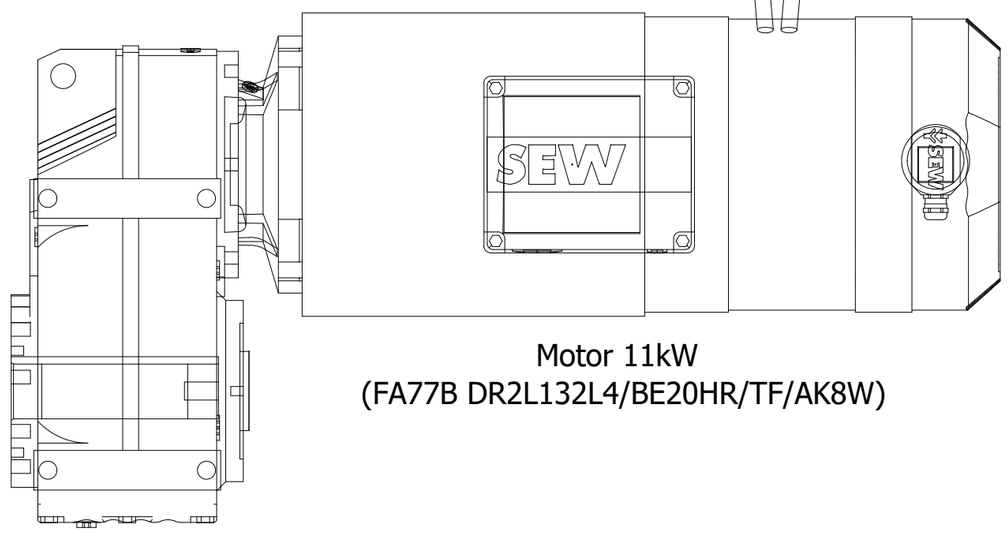
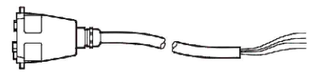
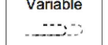


M23, coding ring: brown, male — Open, core end sleeves	CE/UL: 28126009	LEONI LEHC® 005272	Variable	1.5 mm² / AC 500 V



Motor 11kW
(FA77B DR2L132L4/BE20HR/TF/AK8W)

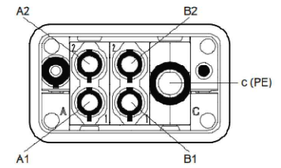
Cable cross section 16 mm²

Connection cable	Conformity/ part number	Cable type	Length/installation type	Cable cross section/operating voltage
 Han® 10 B, male — open	CE/UL: 28178599	HELUKABEL TOPSERV 109 Li9YC11Y-HF	Variable 	16 mm² / AC 500 V

Function
Connection of motor phases

Connection type
Han-Modular® 10 B, female

Connection diagram



[a] Han® C module, male		
Contact	**Function**	
1	U	Motor connection, phase U
2	V	Motor connection, phase V

[b] Han® EE module, male		
Contact	**Function**	
1	W	Motor connection, phase W
2	Res.	Reserved

[c] Articulated frame		
Contact	**Function**	
–	PE	Protective earth connection

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD STATIC AREA DRIVE

REV	DESCRIPTION	DATE

DRAWN BY:	APPROVED FOR BUILD BY:	DRAWING NUMBER:
PAUL	Carlos A	C40572
DATE 9/16/26	DATE 9/15/26	CURRENT REVISION: C

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TOTAL PAGES: 171

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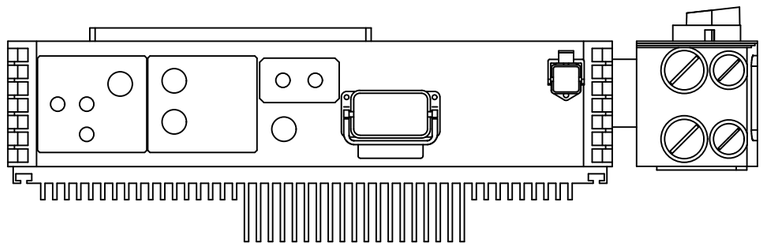
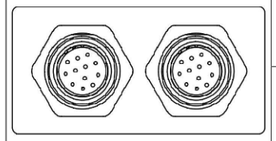


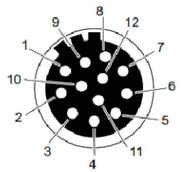
Connections			Encoder type
[1] [2] 	[1]	X3222	Multi distance encoder: HIPERFACE®, SSI, Sin/ Cos, HTL
	[2]	X3011	Motor encoder: HIPERFACE®, sin/cos, TTL, HTL, RS422

X3222: Multi-distance encoder

Function
Connection for HIPERFACE®, SSI, sin/cos, TTL, HTL and RS422 encoders

Connection type
M23, insulator, P part, 12-pin, female, +20°-coded

Wiring diagram



No.	Name	Function
1	CLK (C)	Clock line (Signal track C (K0))
2	/CLK (/C)	Negated clock line (Negated signal track C (/K0))
3	A	Signal track A (K1)
4	/A	Negated signal track A (/K1)
5	B	Signal track B (K2)
6	/B	Negated signal track B (/K2)
7	Data-	Data line (-)
8	Data+	Data line (+)
9	GND	Reference potential
10	+24V	DC 24 V output Total current load of DC 24 V encoder supply ≤ 400 mA
11	GND	Reference potential
12	+12V	DC 12 V output Total current load of DC 12 V encoder supply ≤ 650 mA

Part number: 18121942 Cable design: (6X2X0.25)	Variable length	HIPERFACE®/SSI encoder (24 V)
 M23, 12-pin, 20°-coded ↔ Open with conductor end sleeves		

Part number	Signal name	Color coding
18121942	CLK	Brown
	/CLK	White
	A	Red
	/A	Blue
	B	Yellow
	/B	Green
	Data-	Purple
	Data+	Black
	GND	Gray/Pink + Pink
	+24V	Red/Blue + Gray

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD STATIC AREA DRIVE SSI CONNECTION

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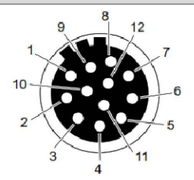
X3222: Multi-distance encoder

Function
Connection for HIPERFACE®, SSI, sin/cos, TTL, HTL and RS422 encoders

Connection type
M23, insulator, P part, 12-pin, female, +20°-coded

Wiring diagram



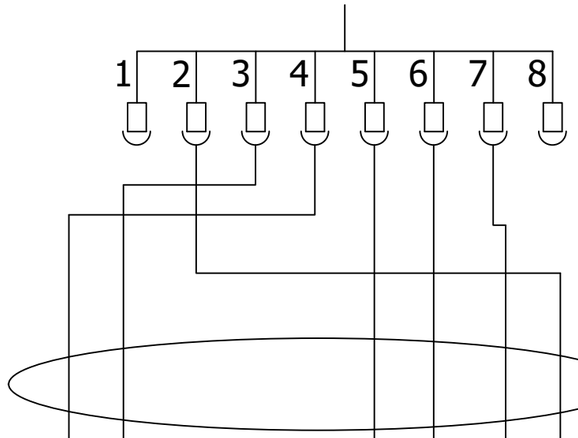
No.	Name	Function
1	CLK (C)	Clock line (Signal track C (K0))
2	/CLK (/C)	Negated clock line (Negated signal track C (/K0))
3	A	Signal track A (K1)
4	/A	Negated signal track A (/K1)
5	B	Signal track B (K2)
6	/B	Negated signal track B (/K2)
7	Data-	Data line (-)
8	Data+	Data line (+)
9	GND	Reference potential
10	+24V	DC 24 V output Total current load of DC 24 V encoder supply ≤ 400 mA
11	GND	Reference potential
12	+12V	DC 12 V output Total current load of DC 12 V encoder supply ≤ 650 mA

PORT # X3222 (male connector
M23, 12-pin, +20°-coded)

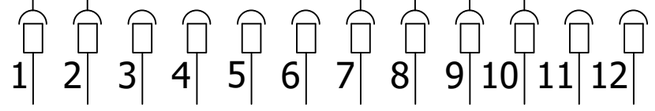
SICK DL50-P2228	WIRE COLOR	SIGNAL	SEW X3222	SEW SIGNAL
PIN 4	YELLOW	RX+ / CLK+	PIN 1	CLK
PIN 3	GREEN	RX− / CLK−	PIN 2	/CLK
PIN 6	PINK	TX+ / DATA+	PIN 8	DATA+
PIN 5	GRAY	TX− / DATA−	PIN 7	DATA−
PIN 2	BROWN	L+	PIN 10	+24 VDC
PIN 7	BLUE	M / 0V	PIN 9	GND
PIN 1	WHITE	Q1	NOT REQUIRED FOR SSI	
PIN 8	RED	Q2/MF	NOT REQUIRED FOR SSI	

SSI ENCODER
(male connector M12,
8-pin, A-coded)

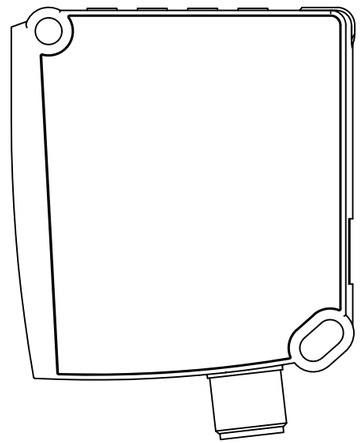
15DE1



15VFD1

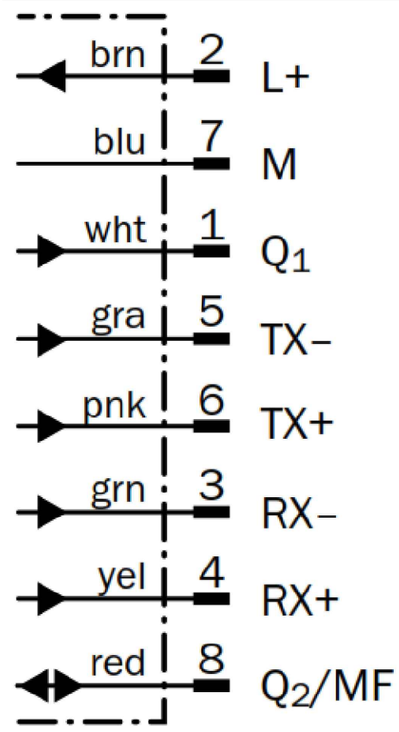


PORT # X3222 (male connector
M23, 12-pin, +20°-coded)



DL50-P2228

(Male connector M12, 8-pin, A-coded)



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD STATIC AREA DRIVE SSI CONNECTION

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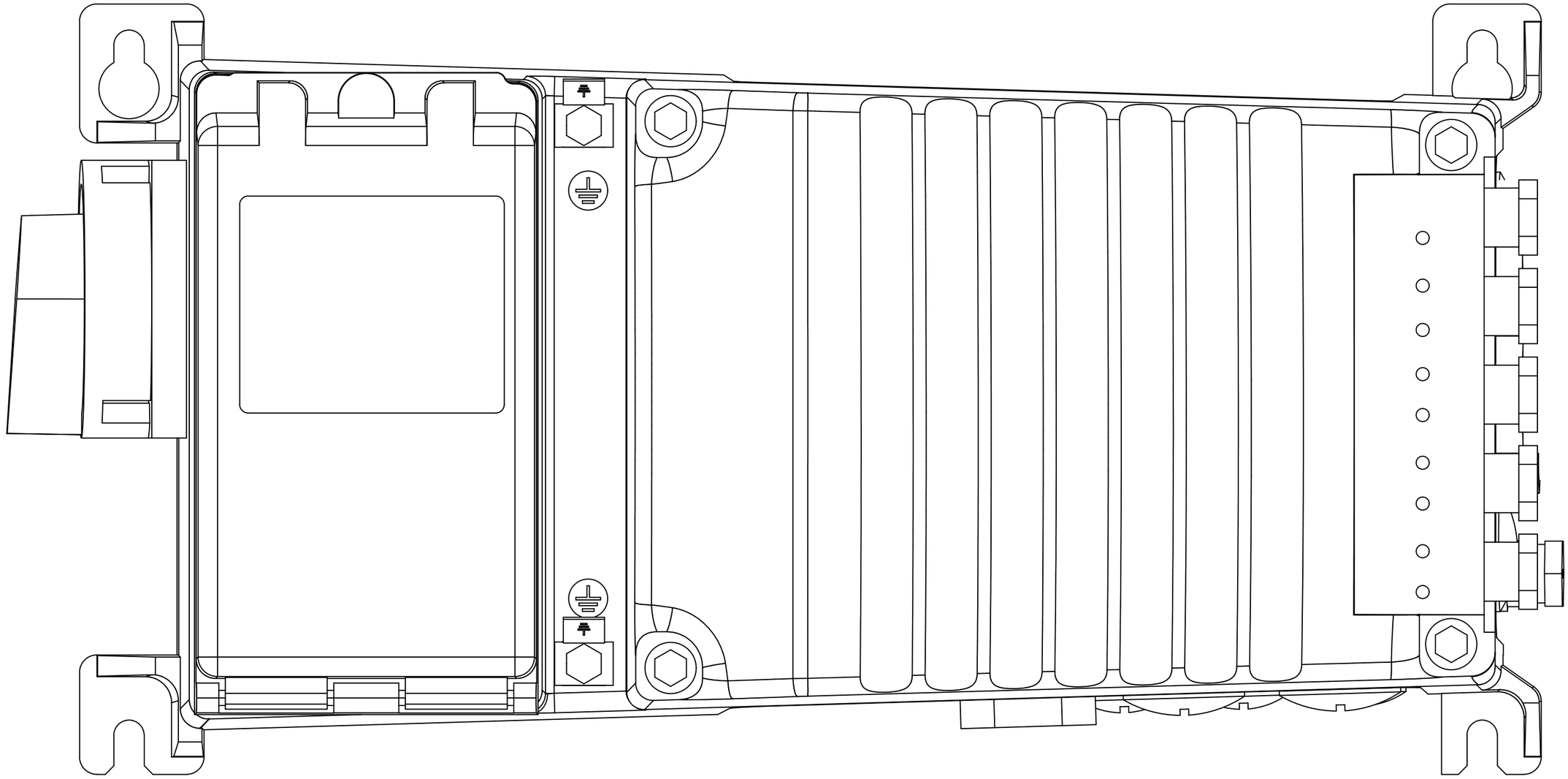

16VFD1

VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

VTD MOBILE EXTRACTOR CONVEYOR AREA VFD

ALL .75 VFD REFERENCES FOUND IN THIS SECTION

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PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA DRIVE 16VFD1

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TOTAL PAGES: 171

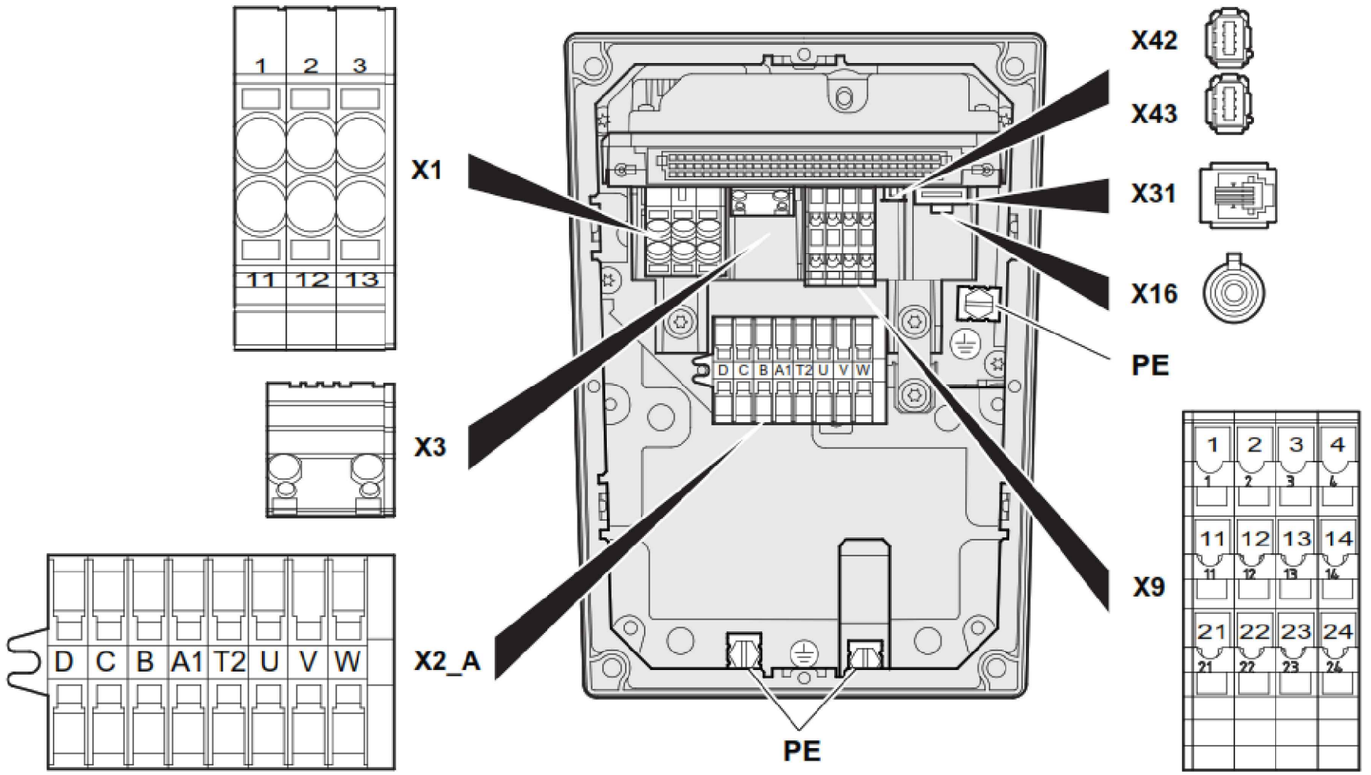
SHEET 16.0 OF 56

X2_A terminals for motor and brake	Brake control /B for AC brakes 100 – 525 V.
D	**Connection of brake 14** • Brake coil: White • Hybrid cable: Violet (D)
C	**Connection of brake 13** • Brake coil: Red • Hybrid cable: Pink (C)
B	**Connection of brake 15** • Brake coil: Blue • Hybrid cable: Orange (B)
A	Reserved • Hybrid cable: Yellow (A)
U	**Motor connection, phase U**
V	**Motor connection, phase V**
W	**Motor connection, phase W**

Assignment				
Terminal	**No.**	**Name**	**Marking**	**Function**
X1 **line terminals**	1	L1	Brown	Line connection, phase L1 – IN
	2	L2	Black	Line connection, phase L2 – IN
	3	L3	Gray	Line connection, phase L3 – IN
	11	L1	Brown	Line connection, phase L1 – OUT
	12	L2	Black	Line connection, phase L2 – OUT
	13	L3	Gray	Line connection, phase L3 – OUT
⏚	–	PE	–	PE connection
X3 **braking resistor terminals**	1	BW	–	Braking resistor connection
	2	BW	–	Braking resistor connection

Assignment				
Terminal	**No.**	**Name**	**Marking**	**Function**
X9 **control terminals**	1	F_STO_P1	Yellow	Input STO+
	2	F_STO_P1	Yellow	Input STO+ (to loop through)
	3	0V24_out	–	0V24 reference potential for DC 24 V auxiliary output
	4	24V_out	–	DC 24 V auxiliary output (**only for internal voltage supply of for non-safety-related enable signal via STO input**)
	11	F_STO_M	Yellow	Input STO_ground
	12	F_STO_M	Yellow	Input STO_ground (to loop through)
	13	24V_in	–	DC 24 V input
	14	24V_in	–	DC 24 V input (to loop through)
	21	F_STO_P2	Yellow	Input STO+
	22	F_STO_P2	Yellow	Input STO+ (to loop through)
	23	0V24_in	–	0V24 reference potential
	24	0V24_in	–	0V24 reference potential (to loop through)
X42 **EtherCAT®/SBus^{PLUS} IN**	1	TX+	–	Transmit line +
	2	TX-	–	Transmit line -
	3	RX+	–	Receive line +
	6	RX-	–	Receive line -
X43 **EtherCAT®/SBus^{PLUS} OUT**	1	TX+	–	Transmit line +
	2	TX-	–	Transmit line -
	3	RX+	–	Receive line +
	6	RX-	–	Receive line -
X31 **engineering interface**	–	–	–	Engineering interface with DC 24 V supply voltage for use in conjunction with suitable SEW engineering options
X16 **MOVILINK® DDI interface**	1	DDI	–	MOVILINK® DDI supply/communication
	2	DDI_GND	–	MOVILINK® DDI reference potential

5.5.1 Design MMF1., MMF31



27021626766467339

16VFD1

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA DRIVE

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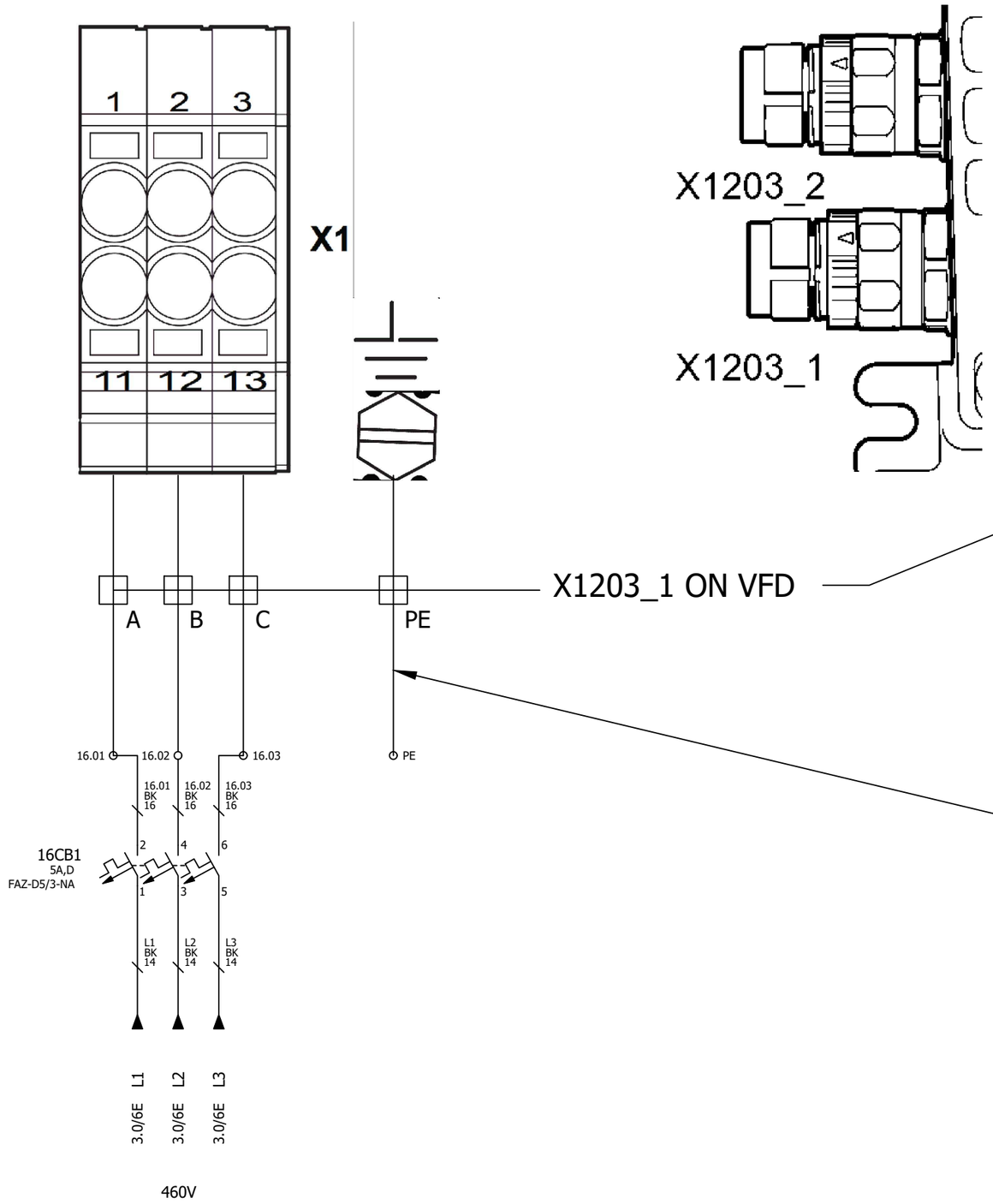
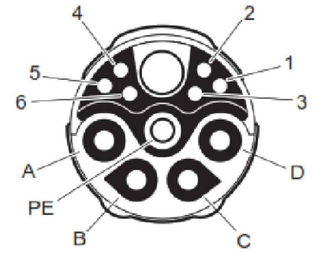


Connection diagram



Assignment

Contact	Function	
A	L1	Line connection, phase L1
B	L2	Line connection, phase L2
C	L3	Line connection, phase L3
D	Res.	Reserved
PE	PE	Protective earth connection
1	Res.	Reserved
2	Res.	Reserved
3	Res.	Reserved
4	Res.	Reserved
5	Res.	Reserved
6	Res.	Reserved

Open — M23, coding ring: black, male	UL: 18153283	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

16VFD1

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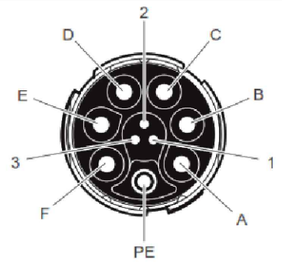
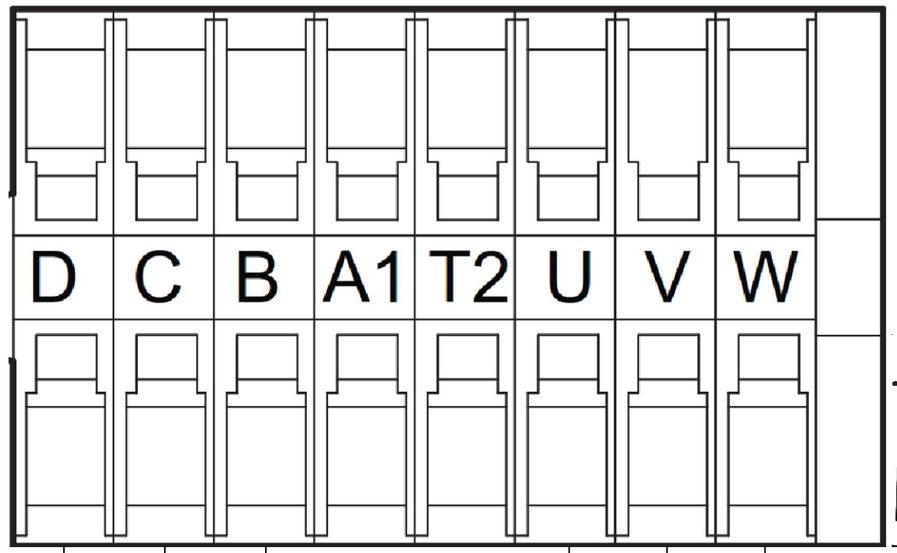
SHEET 16.2 OF 56

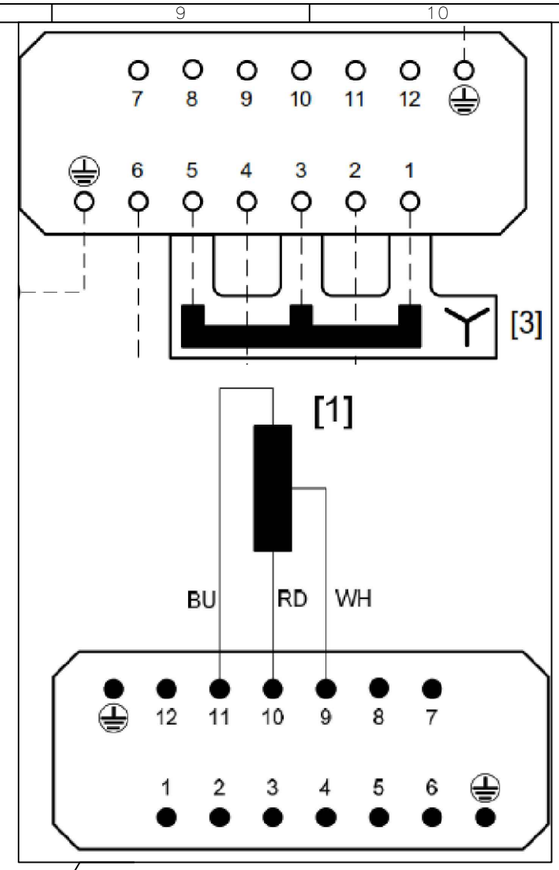
Connection diagram



Assignment

Contact	Function	
A	U	Motor connection, phase U
B	V	Motor connection, phase V
C	W	Motor connection, phase W
D	13	Brake connection 13
E	14	Brake connection 14
F	15	Brake connection 15
PE	PE	Protective earth connection
1	Temp+	Temperature sensor+ connection[1)]
2	res.	Reserved
3	Temp -	Connection [1)]

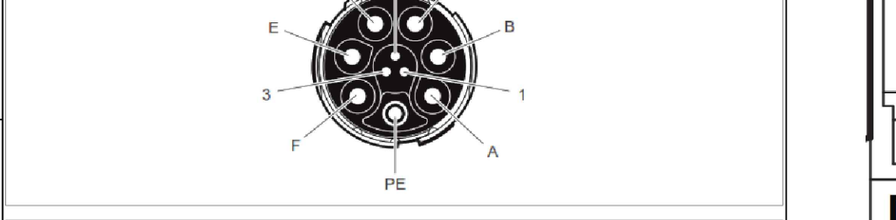




ON MOTOR

X2041 ON VFD

E D F A B C



9 10 11 2 4 6

WH RD BU U1 V1 W1 PE

1HP/.75kW
460V
60HZ
1.56A

M
3 ~

16M1

16VFD1

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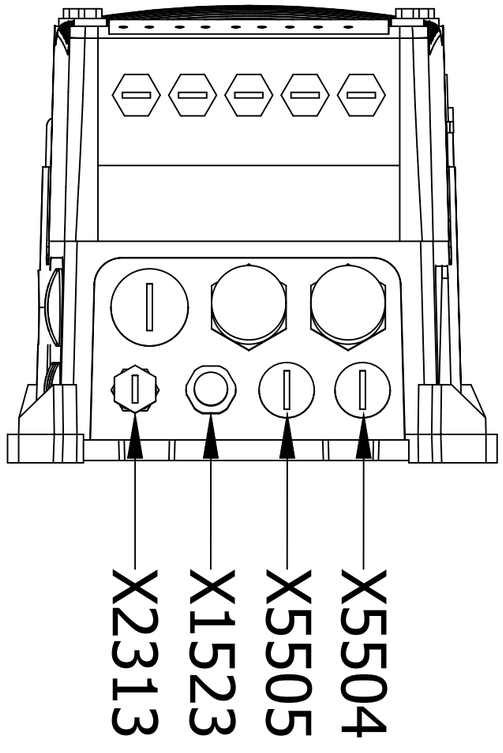
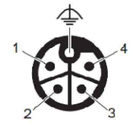


X1523

Function
Input DC 24 V backup voltage

Connection type
M12, 5-pin, male, L-coded, color: light gray

Connection diagram

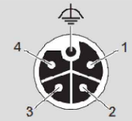


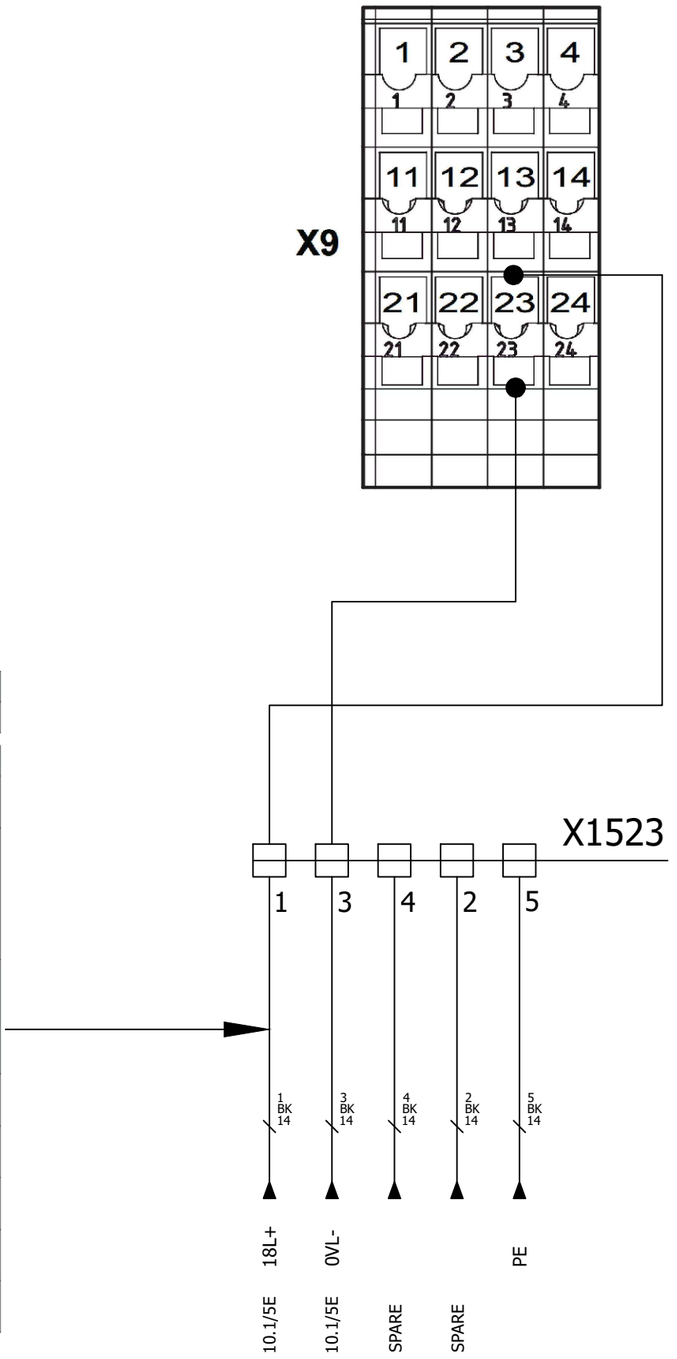
Assignment

Contact	Function	
1	+24V/L1	DC 24 V input/L1 (for backup mode)
2	0V24/N2	0V24 reference potential/N2 (for DC 24 V /BES brake rectifier)
3	0V24/N1	0V24 reference potential/N1 (for backup mode)
4	+24V/L2	DC 24 V connection/L2 (for DC 24 V /BES brake rectifier)
⏚	FE	Functional earth

Part numbers
28117786

Assembly

Open cable end			Description	Prefabricated connector	
					
Core color/ core cross section	Identification	Assembly		Signal	Contact
Black 2.5 mm²	1	Not prefabricated	DC 24 V output/L1 (for backup voltage/supply)	+24V/L1	1
Black 2.5 mm²	2	Not prefabricated	0V24 reference potential/N2 (for DC 24 V /BES brake rectifier)	0V24/N2	2
Black 2.5 mm²	3	Not prefabricated	0V24 reference potential/N1 (for backup voltage/supply)	0V24/N1	3
Black 2.5 mm²	4	Not prefabricated	DC 24 V output/L2 (for DC 24 V /BES brake rectifier)	+24V/L2	4
Black 2.5 mm²	5	Not prefabricated	Functional earth	FE	⏚



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD 24V DC POWER

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DATE	DATE	CURRENT REVISION:
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JOB # C40572

TOTAL PAGES: 171

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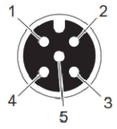
X5504: STO (3 cores)

Function
Connection for safe torque off (STO, 3 cores)

Connection type
M12, 5-pin, female, A-coded, color: yellow

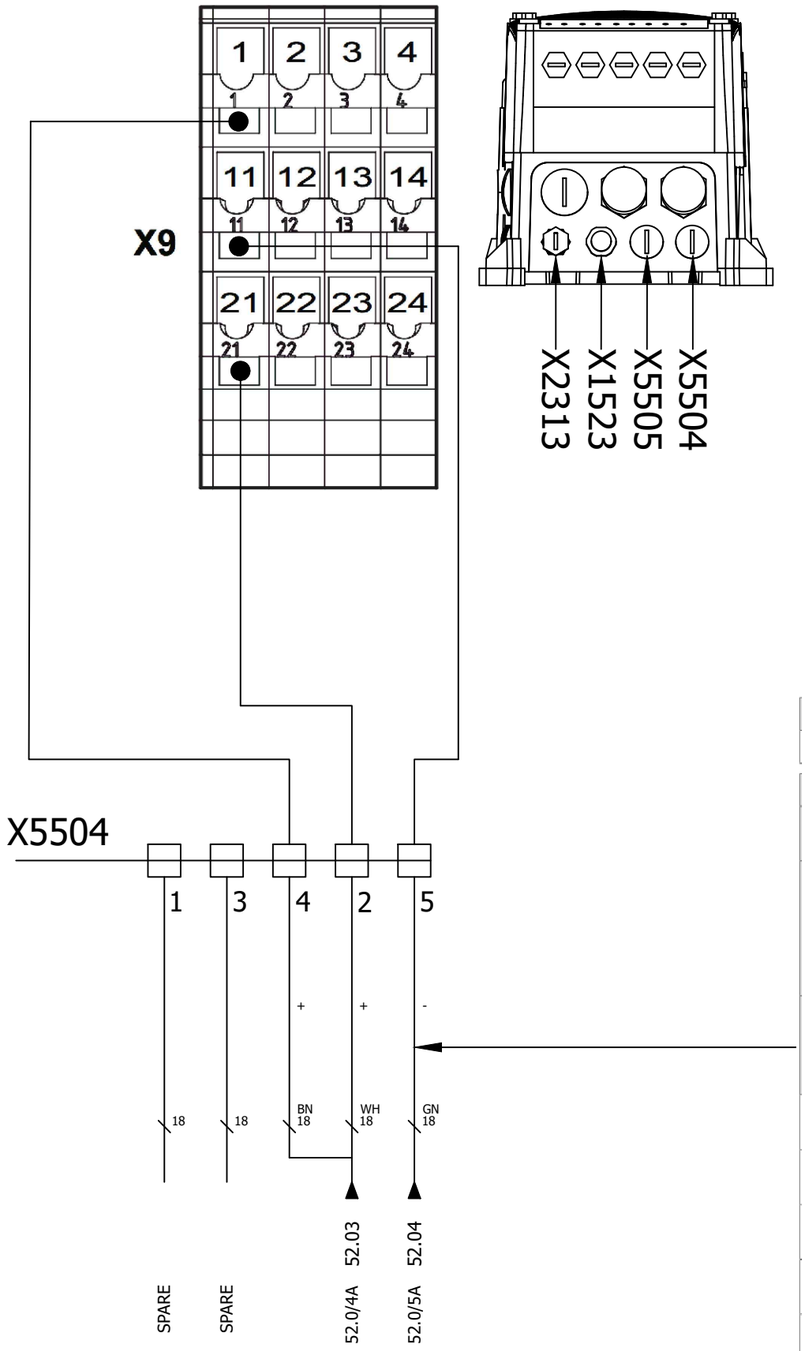
Connection diagram



Assignment

Contact	Signal	Description
1	24V_OUT	DC 24 V auxiliary output (permitted only for operation with STO jumper plug)
2	F_STO_P2	F_STO_P2 connection
3	0V24_OUT	0V24 reference potential for DC 24 V auxiliary output (permitted only for operation with STO jumper plug)
4	F_STO_P1	F_STO_P1 connection
5	F_STO_M	F_STO_M connection

Do not use the 24 V output (pins 1 and 3) for safety-related applications.



Plug connectors

Designation	Coding ring/ color	Function
X5504	Yellow	STO (3-core connection)[1]
X5505	Yellow	STO (3-core connection)[1]
X1523	Light gray	DC 24 V backup voltage – input[2]
X2313	Light gray	DC 24 V backup voltage – output
X5004_1	Black	Digital inputs[3]
X5004_2	Black	Digital inputs[3]
X5004_3	Black	Digital inputs[3]

Open / M12, 5-pin, A-coded, male	CE/UL: 28110943	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V

Part numbers

28110978, 28110943

Assembly

Open cable end			Description	Prefabricated plug connectors	
Core color/ core cross section	Identi-fication	Assembly		Signal	Contact
[1]	–	Not pre-fabricated	DC 24 V auxiliary output	24V_OUT	1
White 0.75 mm²	–	Not pre-fabricated	F_STO_P2 connection	F_STO_P2	2
[1]	–	Not pre-fabricated	0V24 reference potential for DC 24 V auxiliary output	0V24_OUT	3
Brown 0.75 mm²	–	Not pre-fabricated	F_STO_P1 connection	F_STO_P1	4
Green 0.75 mm²	–	Not pre-fabricated	F_STO_M connection	F_STO_M	5

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PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD STO

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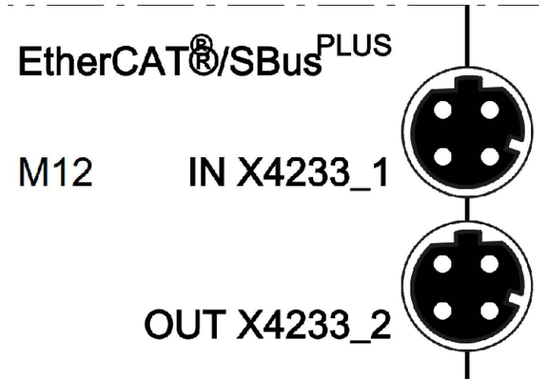


X4233_1: EtherCAT®/SBus^PLUS interface, IN

The following table shows information about this connection:

Function
EtherCAT®/SBus^PLUS interface, IN

Connection type
M12, 4-pin, female, D-coded, color: black

Connection diagram

Assignment		
Contact	**Signal**	**Description**
1	TX+	Transmit line (+)
2	RX+	Receive line (+)
3	TX-	Transmit line (-)
4	RX-	Receive line (-)

X4233_2: EtherCAT®/SBus^PLUS interface, OUT

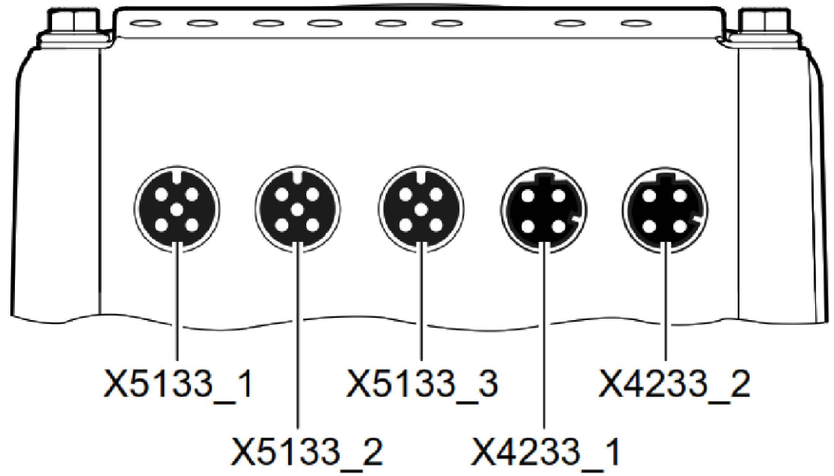
The following table shows information about this connection:

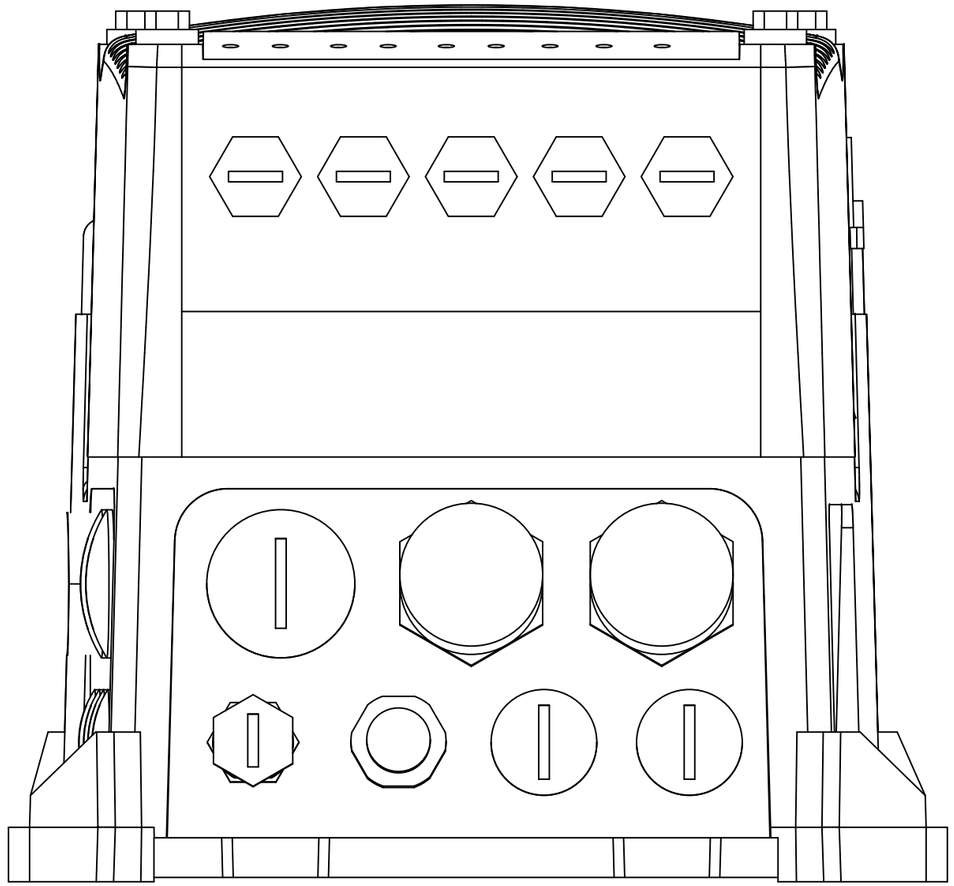
Function
EtherCAT®/SBus^PLUS interface, OUT

Connection type
M12, 4-pin, female, D-coded, color: black

Connection diagram

Assignment		
Contact	**Signal**	**Description**
1	TX+	Transmit line (+)
2	RX+	Receive line (+)
3	TX-	Transmit line (-)
4	RX-	Receive line (-)





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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD EtherCAT INTERFACE

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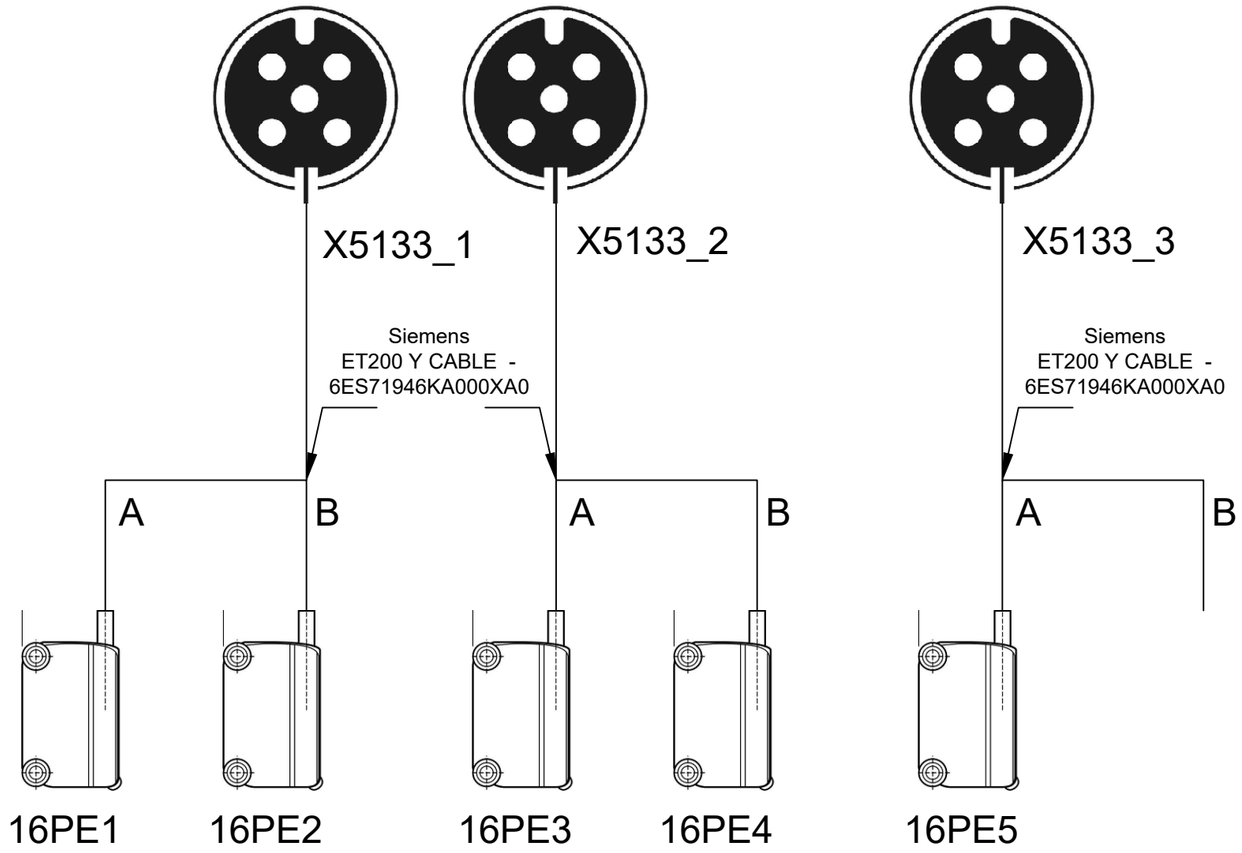
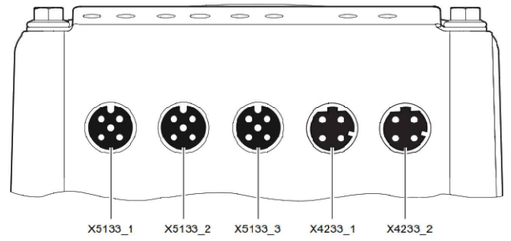


Plug connector positions at the DSI.. electronics cover

The following figure shows an example of the plug connector positions:



18014420217585035

Plug connector	Function
X5133_1[1)]	Digital inputs DI01 and DI02
X5133_2[1)]	Digital inputs DI03 and DI04
X5133_3[1)]	Digital inputs/outputs DIO01 and DIO02

X5133_1: Digital inputs

The following table shows information about this connection:

Function
Digital inputs

Connection type
M12, 5-pin, female, A-coded, color: black

Connection diagram

Assignment

Contact	Signal	Description
1	+24V	DC 24 V sensor supply
2	DI02	Sensor input DI02
3	0V24	0V24 reference potential for sensors
4	DI01	Sensor input DI01
5	FE	Functional earth

X5133_2: Digital inputs

The following table shows information about this connection:

Function
Digital inputs

Connection type
M12, 5-pin, female, A-coded, color: black

Connection diagram

Assignment

Contact	Signal	Description
1	+24V	DC 24 V sensor supply
2	DI04	Sensor input DI04
3	0V24	0V24 reference potential for sensors
4	DI03	Sensor input DI03
5	FE	Functional earth

X5133_3: Digital inputs/outputs

The following table shows information about this connection:

Function
Digital inputs/outputs

Connection type
M12, 5-pin, female, A-coded, color: black

Connection diagram

Assignment

Contact	Signal	Description
1	+24V	DC 24 V sensor/actuator supply
2	DIO02	Sensor input DIO02/actuator output DIO02
3	0V24	0V24 reference potential for sensors/actuators
4	DIO01	Sensor input DIO01/actuator output DIO01
5	FE	Functional earth

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

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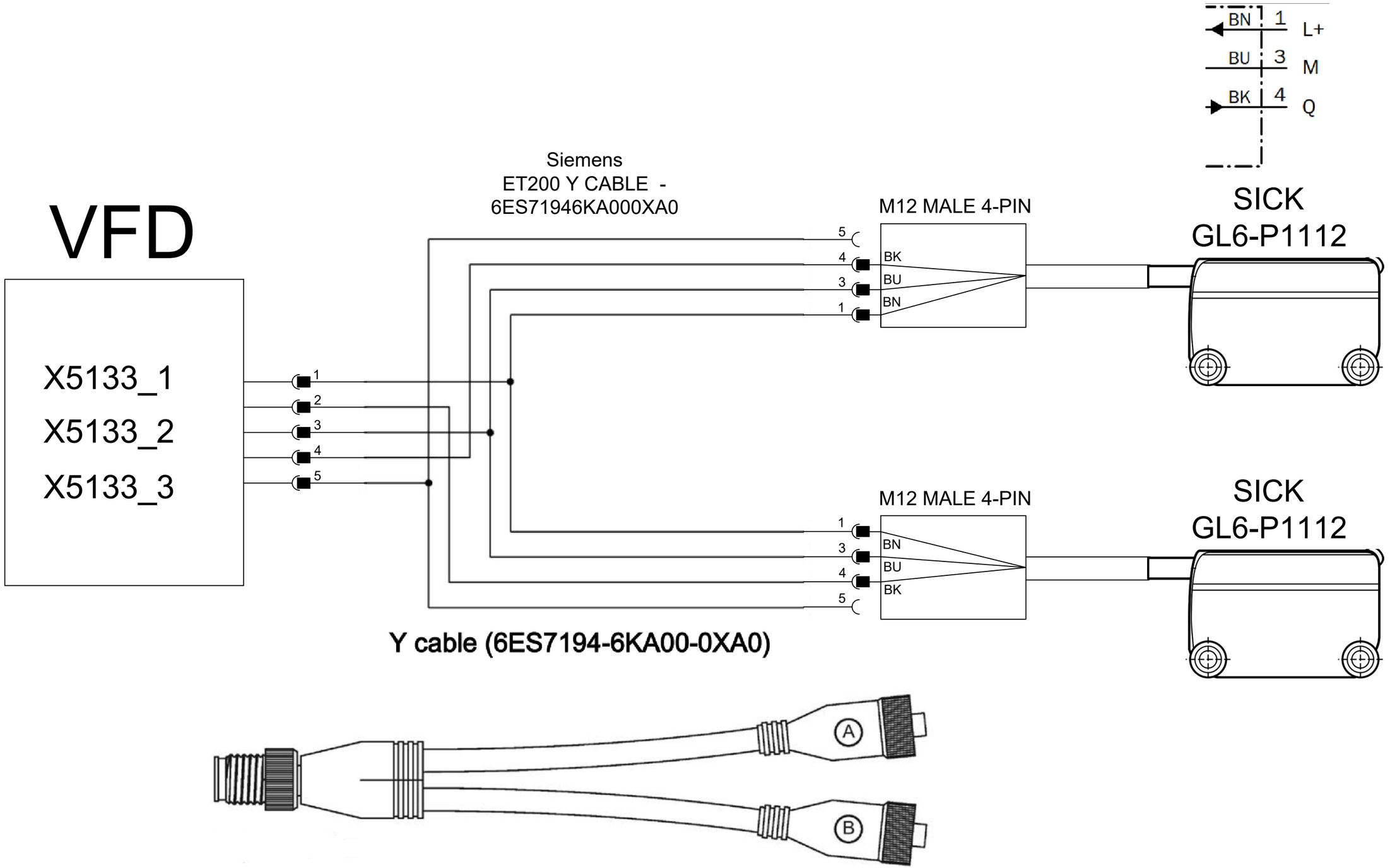

Y cable (6ES7194-6KA00-0XA0)

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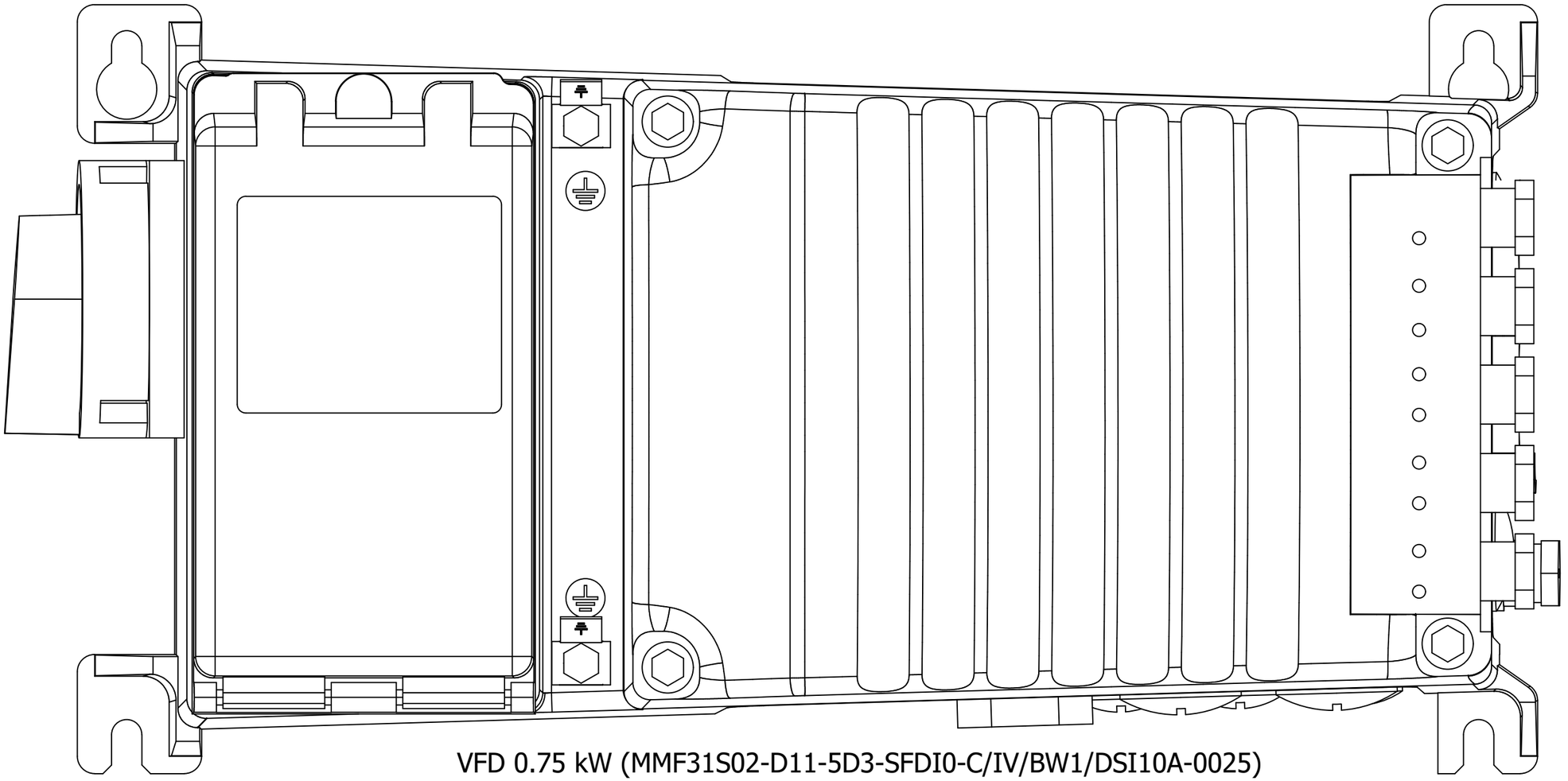

VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

BUFFER CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 4 VFD's

17VFD1	17VFD2	17VFD3	17VFD4
CONVEYOR WITH STOPPER GROUND LEVEL	CONVEYOR WITH STOPPER LEVEL 1	CONVEYOR WITH STOPPER LEVEL 2	CONVEYOR WITH STOPPER LEVEL 3

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1

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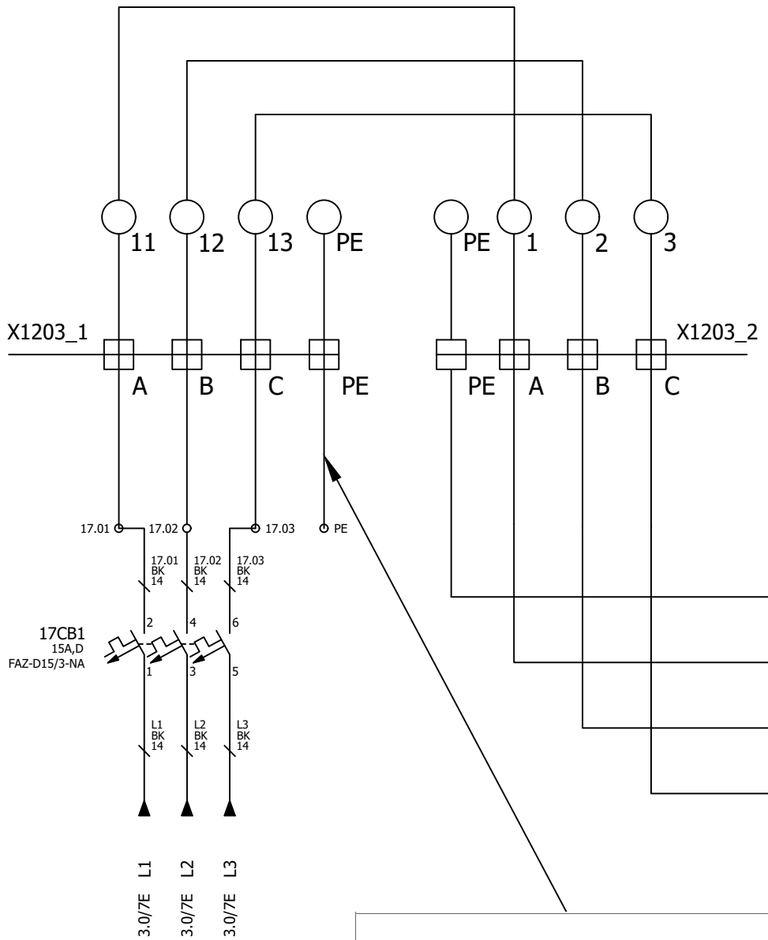
CONVEYOR WITH STOPPER GROUND LEVEL

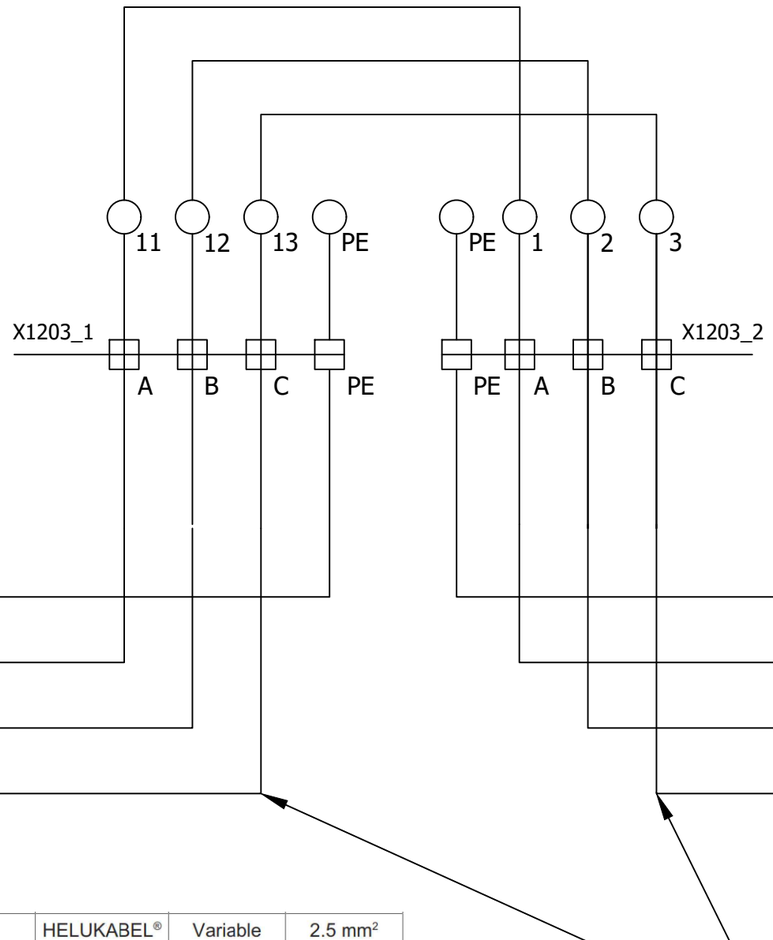
17VFD1



Open — M23, coding ring: black, male	UL: 18153283	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

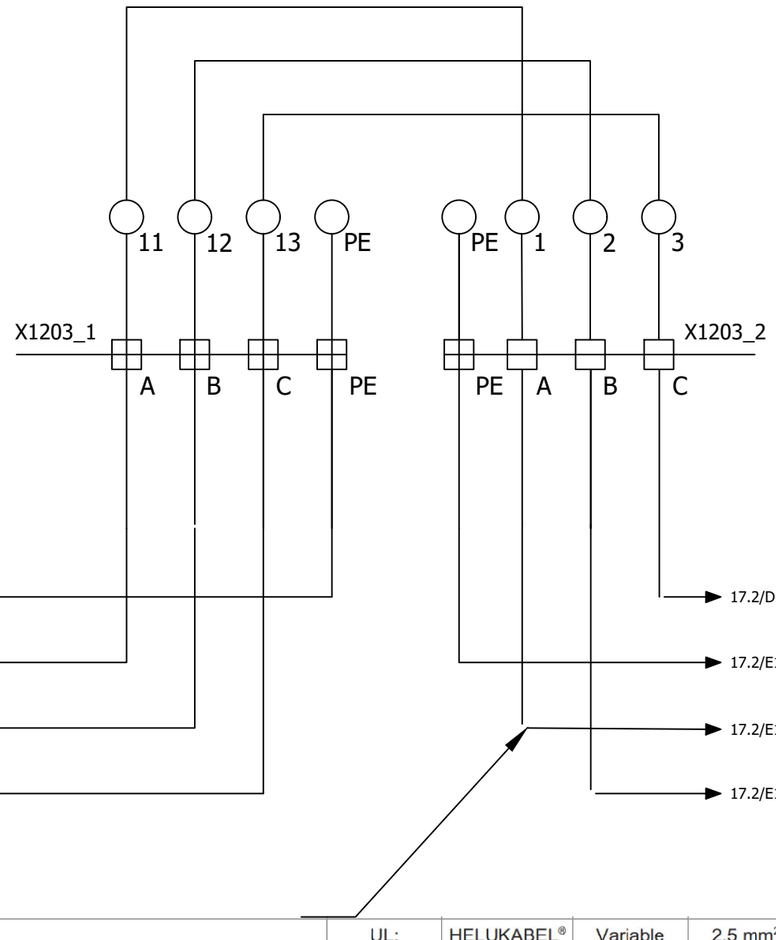
CONVEYOR WITH STOPPER LEVEL 1

17VFD2



CONVEYOR WITH STOPPER LEVEL 2

17VFD3



M23, coding ring: black, male — M23, coding ring: black, male	UL: 18153267	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 POWER

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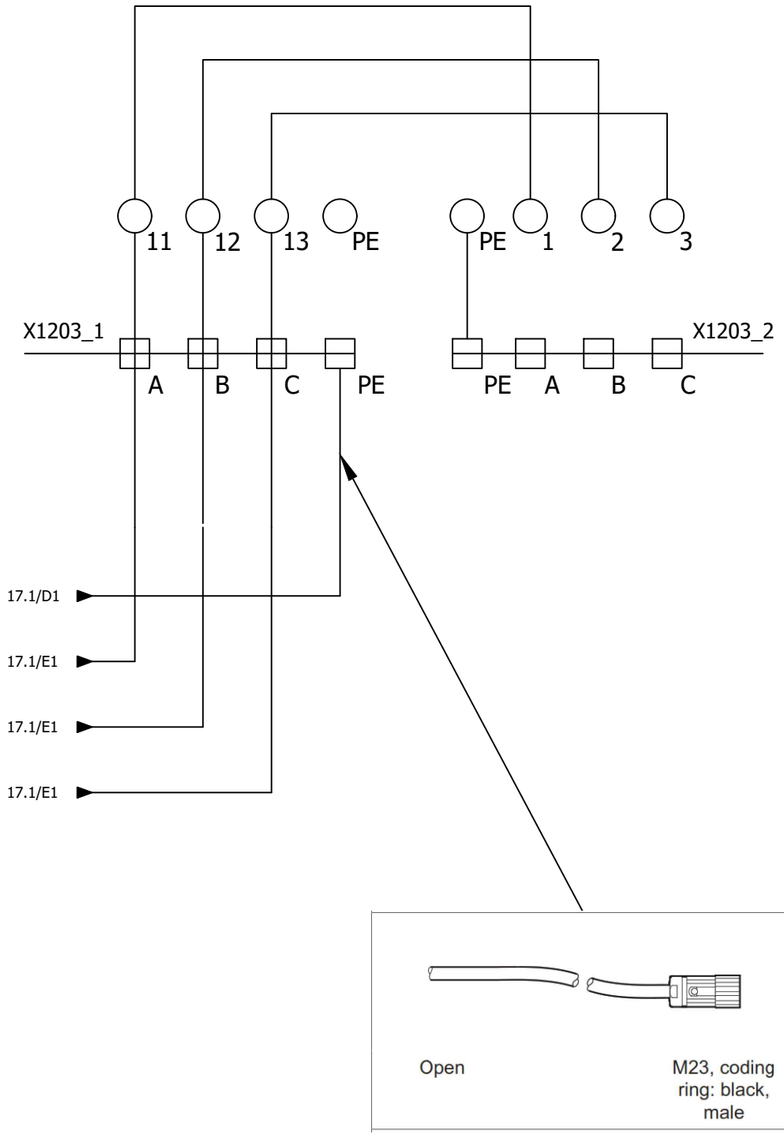
CONVEYOR
WITH STOPPER
LEVEL 3

17VFD4



Open M23, coding ring: black, male	UL: 18153283	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 POWER

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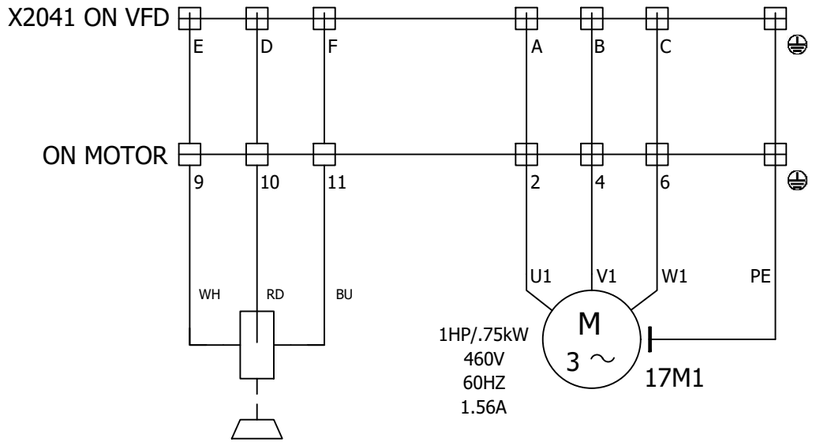
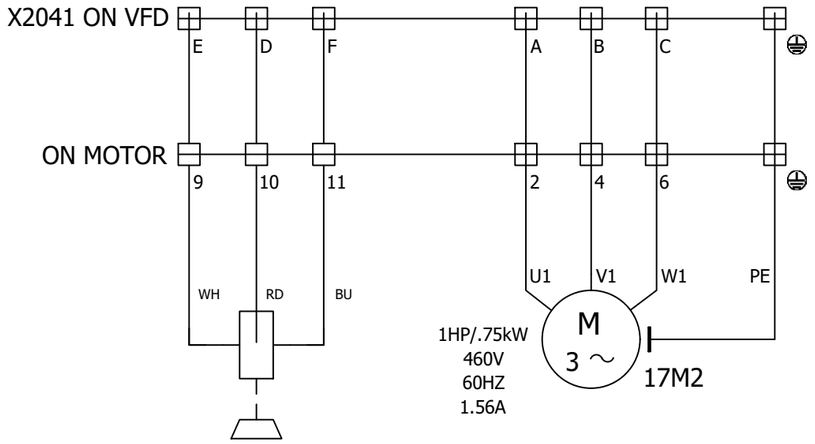
CONVEYOR WITH STOPPER GROUND LEVEL

17VFD1



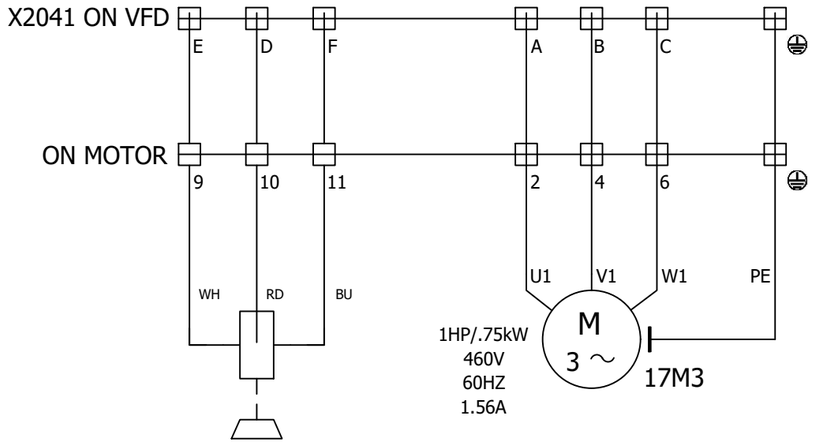
CONVEYOR WITH STOPPER LEVEL 2

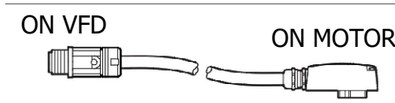
17VFD2



CONVEYOR WITH STOPPER LEVEL 3

17VFD3





M23, coding ring: brown, male	IS2, female, ⅄	CE/UL: 28125940	LEONI LEHC® 000749	Variable	1.5 mm² / AC 500 V

Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 MOTOR POWER

REV	DESCRIPTION	DATE

DRAWN BY:	APPROVED FOR BUILD BY:	DRAWING NUMBER:
PAUL	Carlos A	C40572
DATE	DATE	CURRENT REVISION:
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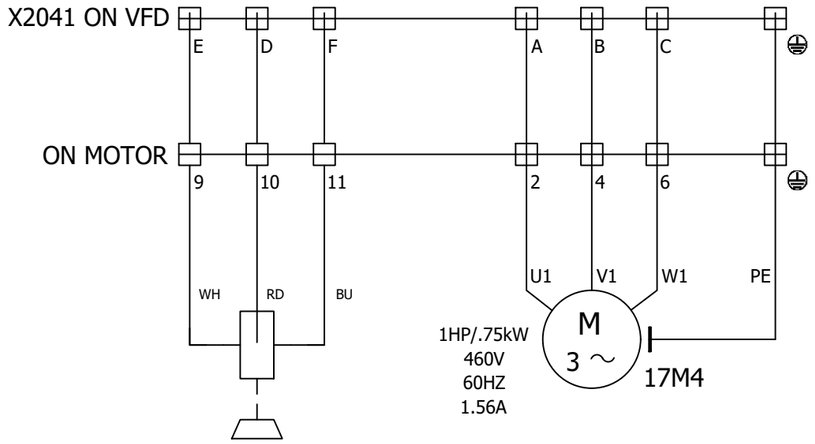
JOB # C40572

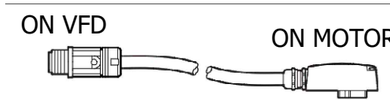
TOTAL PAGES: 171

SHEET 17.3 OF 56

CONVEYOR
WITH STOPPER
LEVEL 3

17VFD4





M23, coding ring: brown, male	IS2, female, ⅄	CE/UL: 28125940	LEONI LEHC® 000749	Variable	1.5 mm² / AC 500 V

Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 MOTOR POWER

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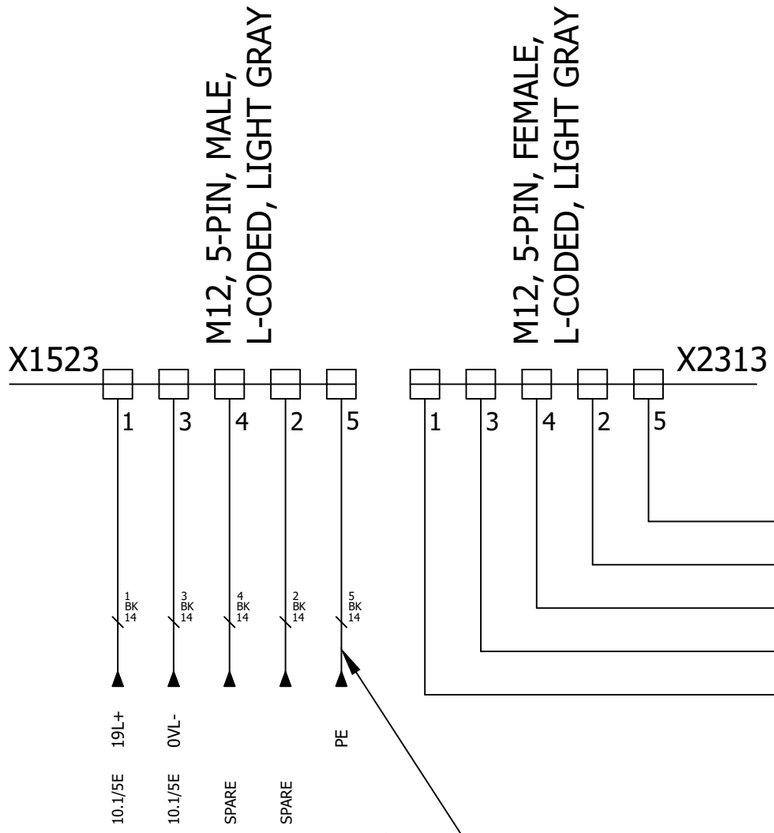
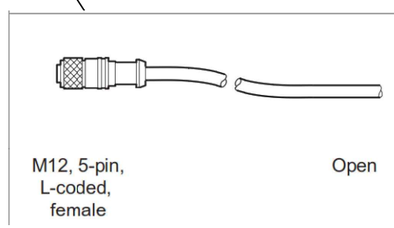
TOTAL PAGES: 171

SHEET 17.4 OF 56

CONVEYOR WITH STOPPER GROUND LEVEL

17VFD1



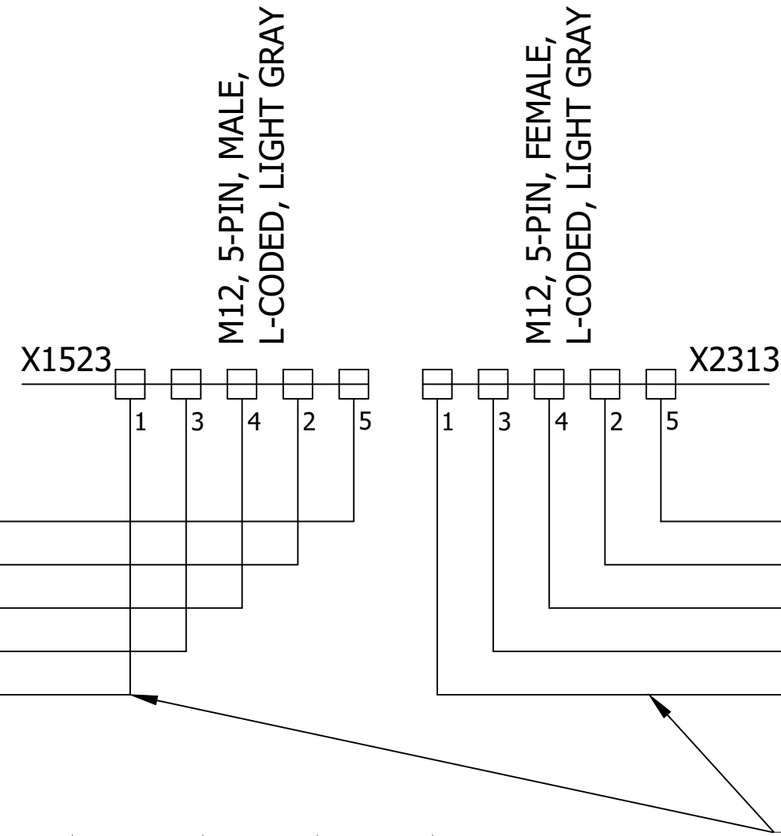


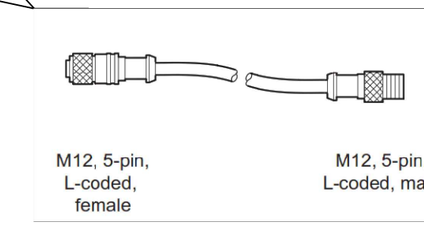
M12, 5-pin, L-coded, female — Open

CE/UL:	HELUKABEL® JZ-500	Variable	5 × 2.5 mm² / DC 60 V
28117786			

CONVEYOR WITH STOPPER LEVEL 1

17VFD2



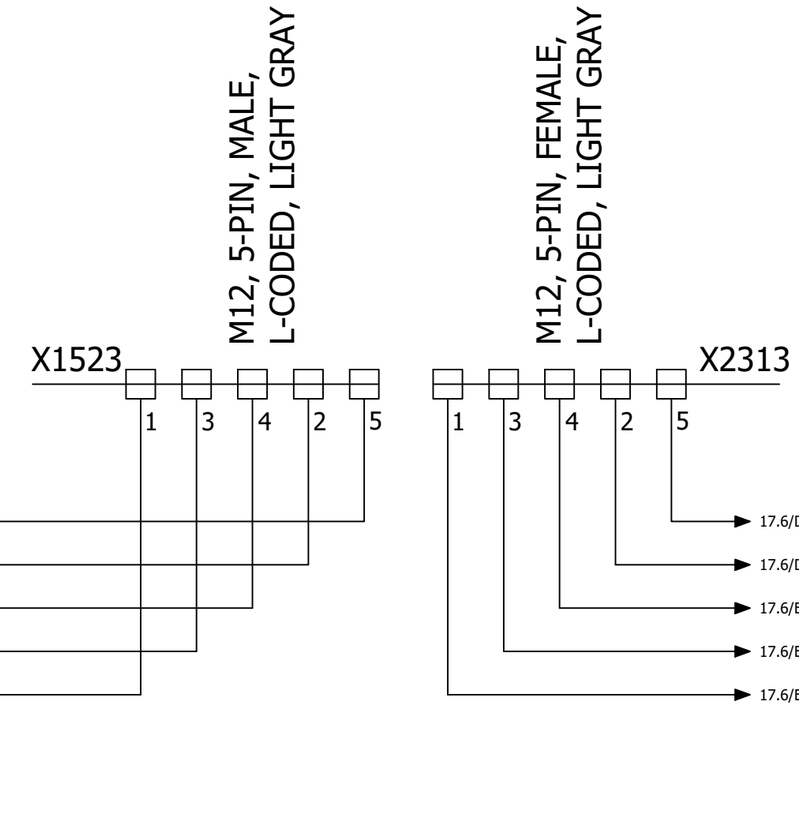


M12, 5-pin, L-coded, female — M12, 5-pin, L-coded, male

CE/UL:	HELUKABEL® JZ-500	Variable	5 × 2.5 mm² / DC 60 V
28114345			

CONVEYOR WITH STOPPER LEVEL 2

17VFD3



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER

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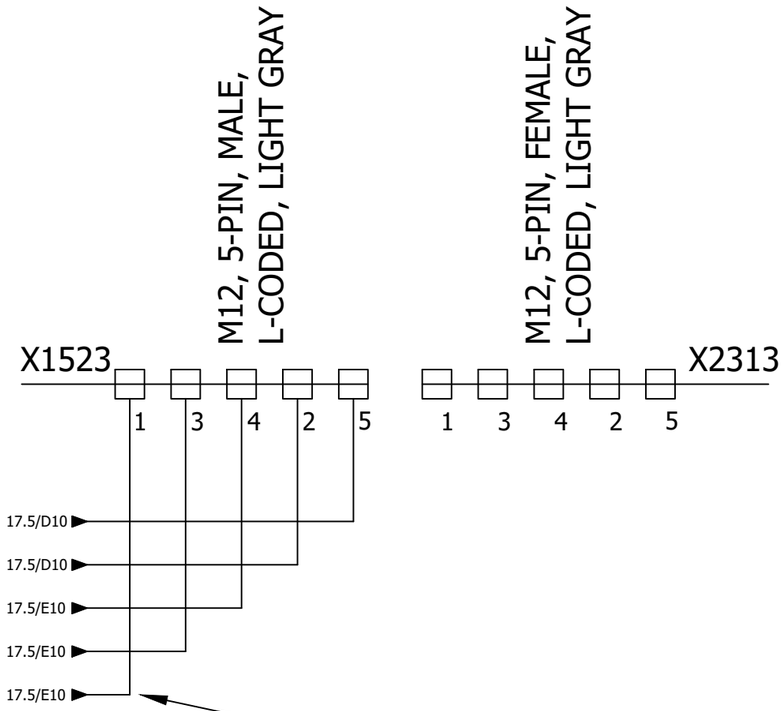
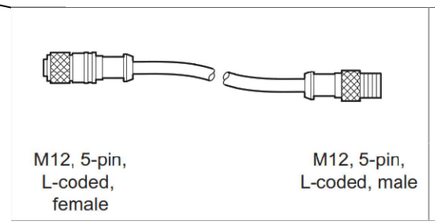
SHEET 17.5 OF 56

CONVEYOR
WITH STOPPER
LEVEL 3

17VFD4





CE/UL: 28114345	HELUKABEL® JZ-500	Variable	5 × 2.5 mm² / DC 60 V	

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

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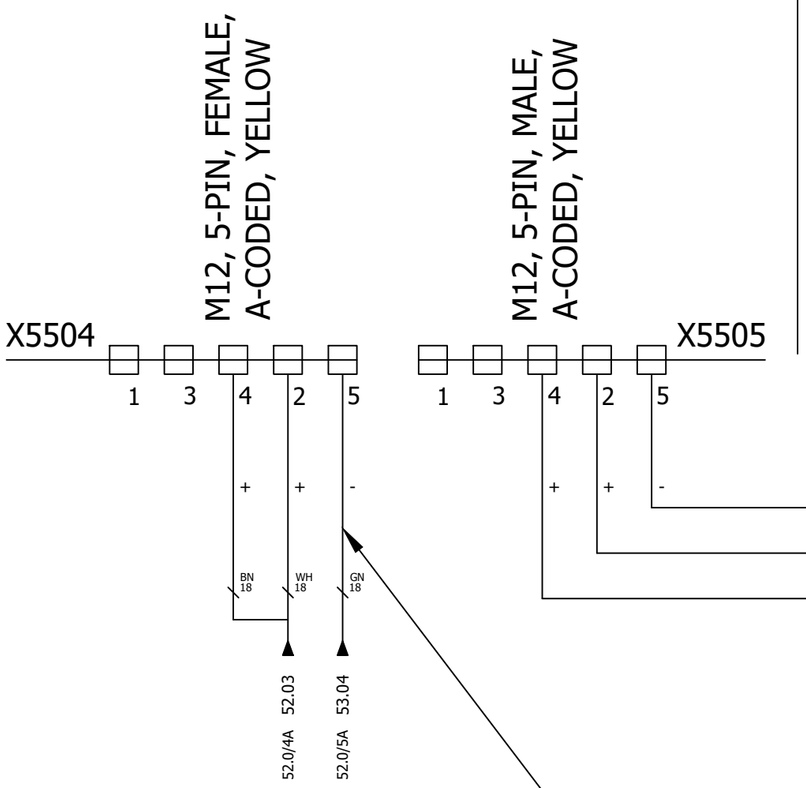
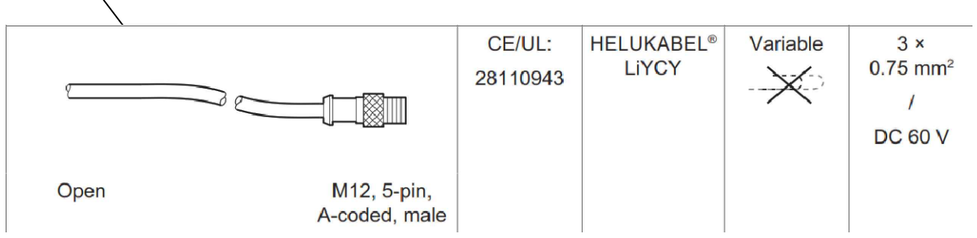
SHEET 17.6 OF 56

CONVEYOR WITH STOPPER GROUND LEVEL

17VFD1

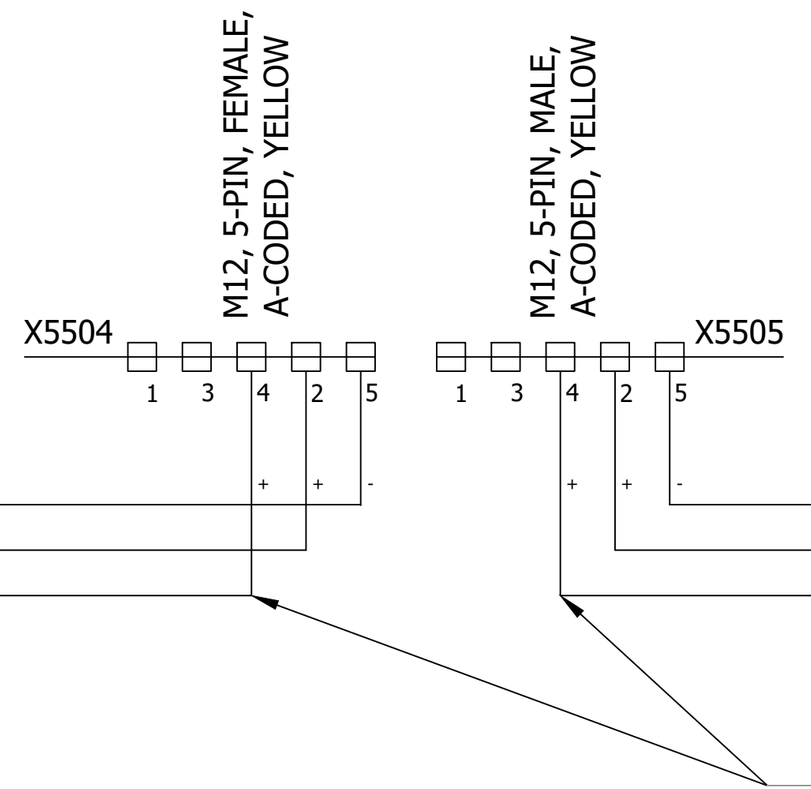




Open — M12, 5-pin, A-coded, male	CE/UL: 28110943	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V

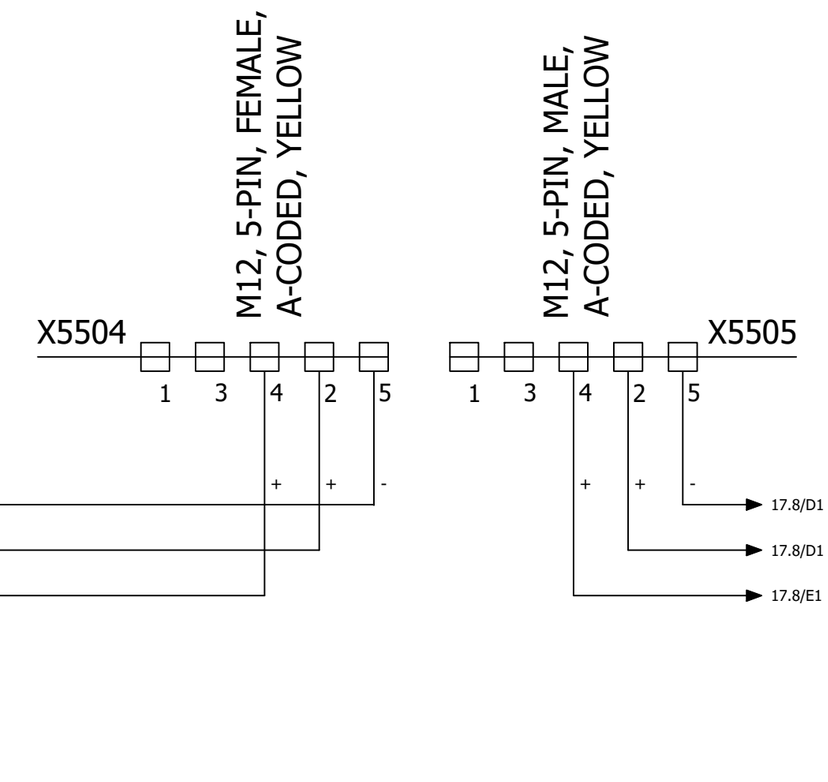
CONVEYOR WITH STOPPER LEVEL 1

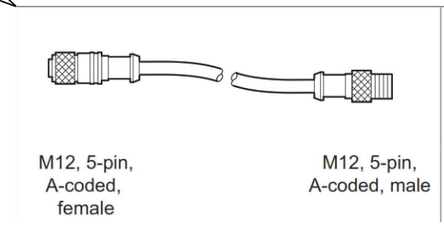
17VFD2



CONVEYOR WITH STOPPER LEVEL 2

17VFD3





M12, 5-pin, A-coded, female — M12, 5-pin, A-coded, male	CE/UL: 28110935	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 VFD STO

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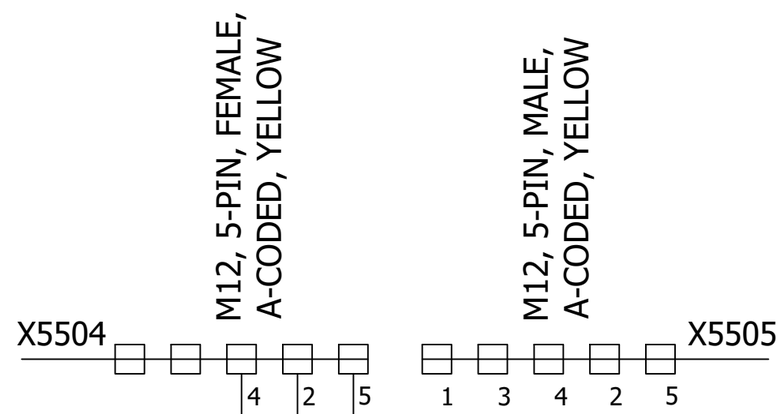
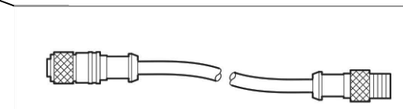
TOTAL PAGES: 171

SHEET 17.7 OF 56

CONVEYOR
WITH STOPPER
LEVEL 3

17VFD4





M12, 5-pin, A-coded, female

M12, 5-pin, A-coded, male

CE/UL: 28110935	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V	

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 VFD STO

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TOTAL PAGES: 171

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Assignment		
Contact	**Signal**	**Description**
1	TX+	Transmit line (+)
2	RX+	Receive line (+)
3	TX-	Transmit line (-)
4	RX-	Receive line (-)

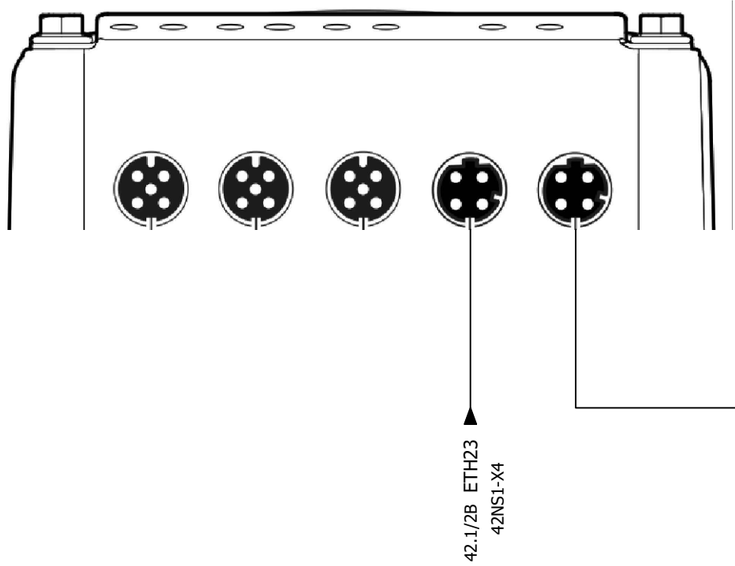
17VFD1

X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK

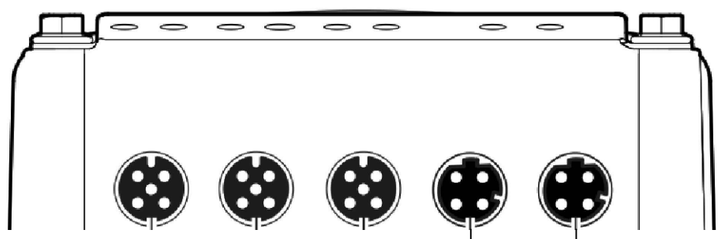
EtherCAT®/SBus^{PLUS}

M12 IN X4233_1

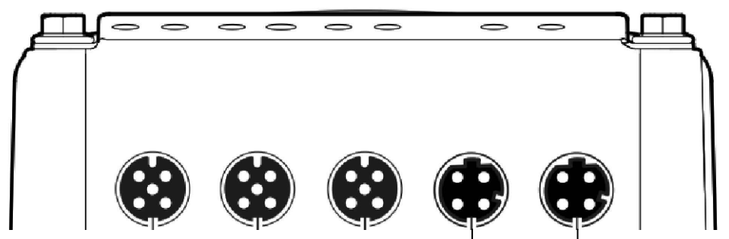
OUT X4233_2



17VFD2



17VFD3



CONVEYOR
WITH STOPPER
GROUND LEVEL

CONVEYOR
WITH STOPPER
LEVEL 1

CONVEYOR
WITH STOPPER
LEVEL 2

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE

REV	DESCRIPTION	DATE

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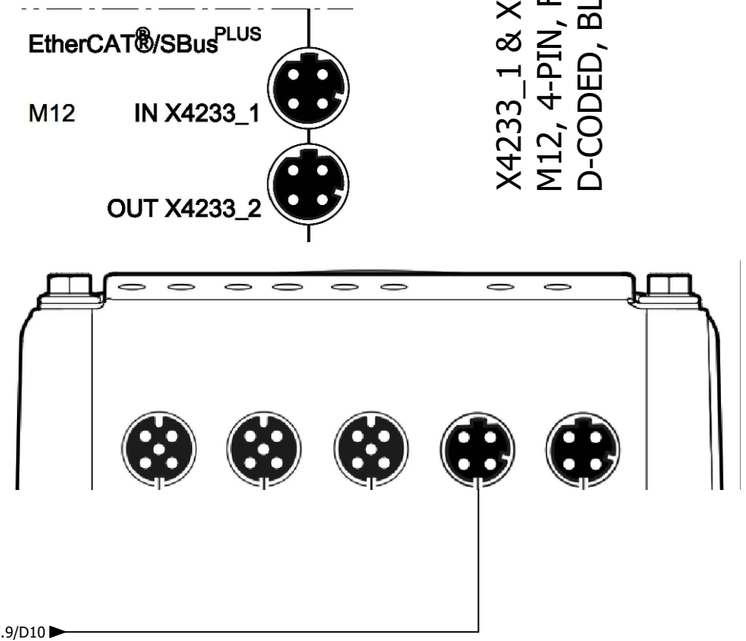
JOB # C40572

TOTAL PAGES: 171

SHEET 17.9 OF 56

Assignment		
Contact	**Signal**	**Description**
1	TX+	Transmit line (+)
2	RX+	Receive line (+)
3	TX-	Transmit line (-)
4	RX-	Receive line (-)

17VFD4



CONVEYOR WITH STOPPER LEVEL3

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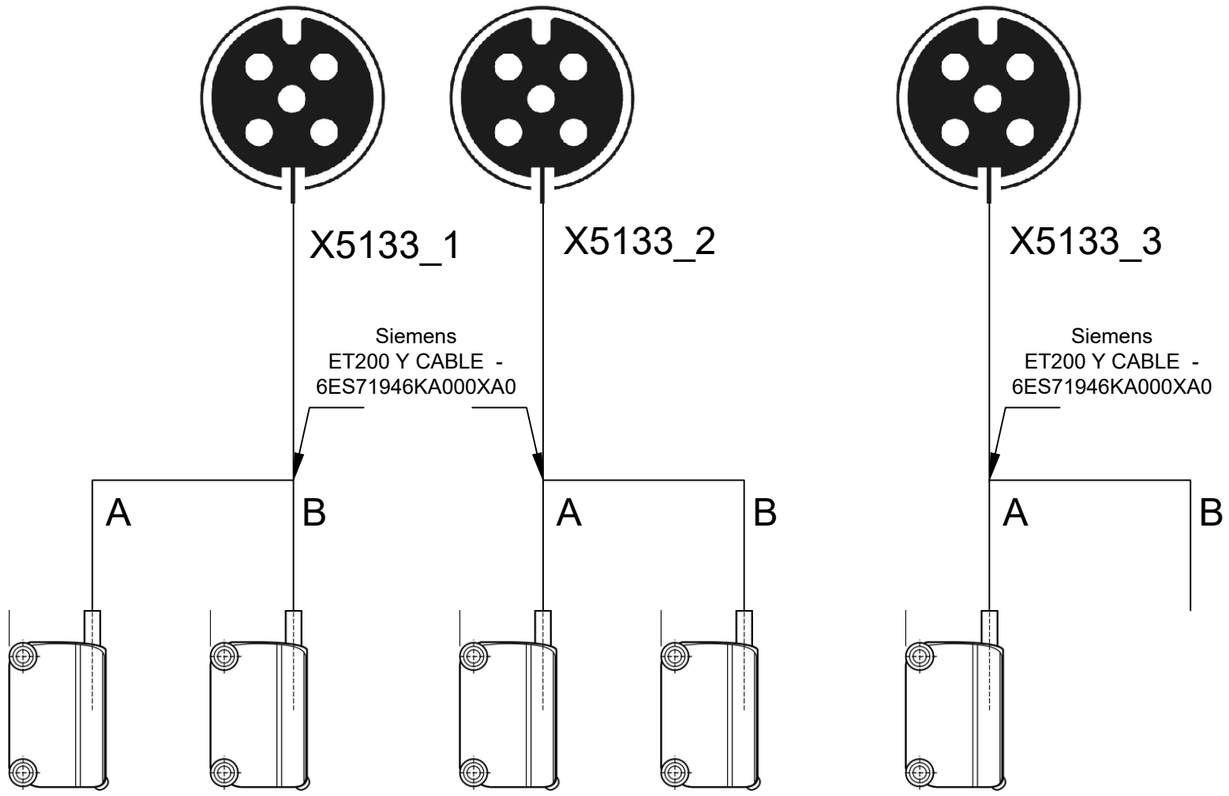
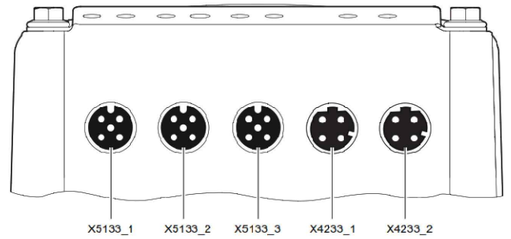


17PE1	17PE2	17PE3	17PE4	17PE5	TO 17VFD1
17PE7	17PE8	17PE9	17PE10	17PE11	TO 17VFD2
17PE13	17PE14	17PE15	17PE16	17PE17	TO 17VFD3
17PE19	17PE20	17PE21	17PE22	17PE23	TO 17VFD4

Plug connector positions at the DSI.. electronics cover

The following figure shows an example of the plug connector positions:



18014420217585035

Plug connector	Function
X5133_1[1)]	Digital inputs DI01 and DI02
X5133_2[1)]	Digital inputs DI03 and DI04
X5133_3[1)]	Digital inputs/outputs DIO01 and DIO02

X5133_1: Digital inputs

The following table shows information about this connection:

Function
Digital inputs

Connection type
M12, 5-pin, female, A-coded, color: black

Connection diagram

Assignment

Contact	Signal	Description
1	+24V	DC 24 V sensor supply
2	DI02	Sensor input DI02
3	0V24	0V24 reference potential for sensors
4	DI01	Sensor input DI01
5	FE	Functional earth

X5133_2: Digital inputs

The following table shows information about this connection:

Function
Digital inputs

Connection type
M12, 5-pin, female, A-coded, color: black

Connection diagram

Assignment

Contact	Signal	Description
1	+24V	DC 24 V sensor supply
2	DI04	Sensor input DI04
3	0V24	0V24 reference potential for sensors
4	DI03	Sensor input DI03
5	FE	Functional earth

X5133_3: Digital inputs/outputs

The following table shows information about this connection:

Function
Digital inputs/outputs

Connection type
M12, 5-pin, female, A-coded, color: black

Connection diagram

Assignment

Contact	Signal	Description
1	+24V	DC 24 V sensor/actuator supply
2	DIO02	Sensor input DIO02/actuator output DIO02
3	0V24	0V24 reference potential for sensors/actuators
4	DIO01	Sensor input DIO01/actuator output DIO01
5	FE	Functional earth

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

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TOTAL PAGES: 171

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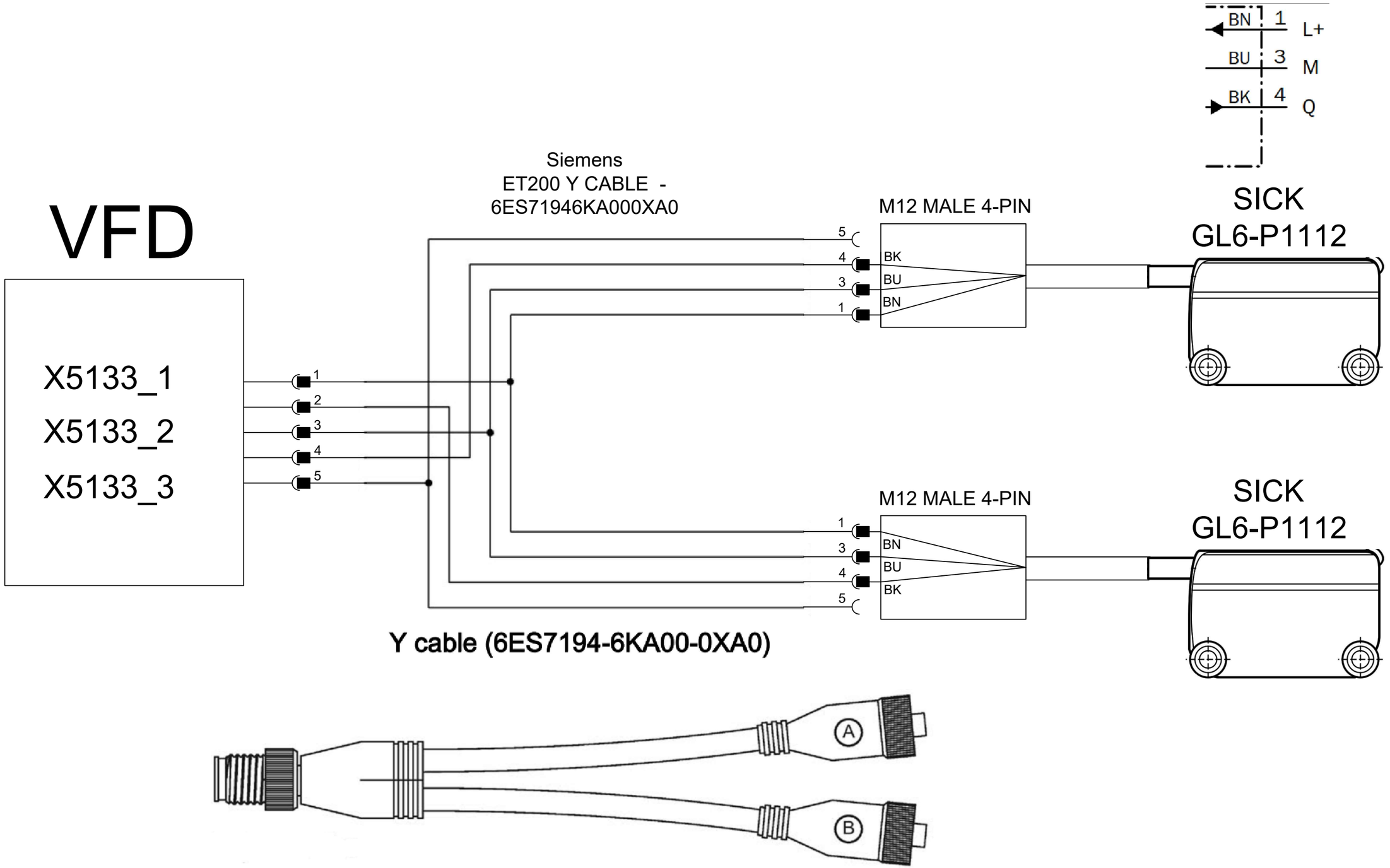

Y cable (6ES7194-6KA00-0XA0)

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

REV	DESCRIPTION	DATE

DRAWN BY:	APPROVED FOR BUILD BY:	DRAWING NUMBER:
PAUL	Carlos A	C40572
DATE	DATE	CURRENT REVISION:
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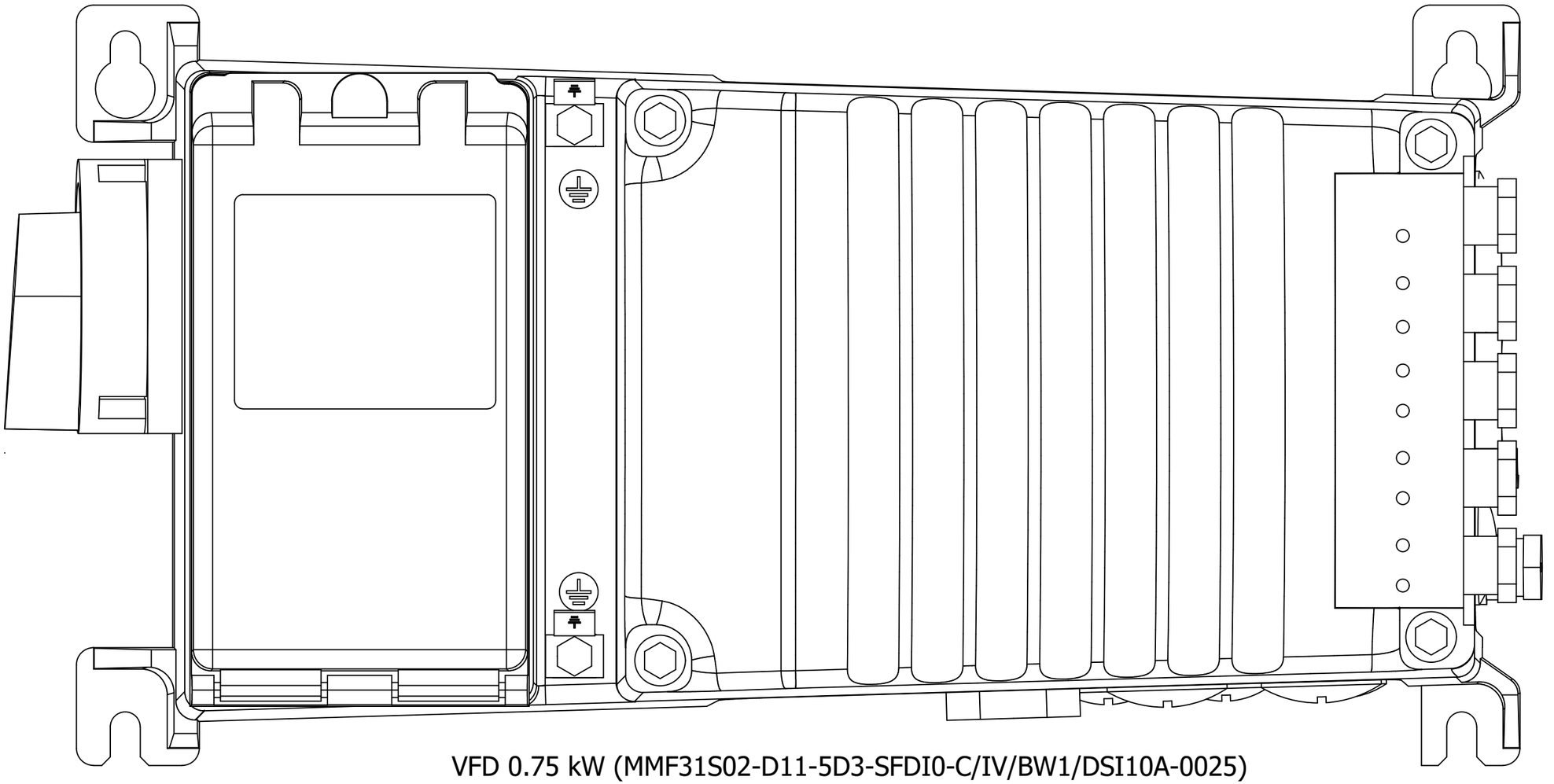
JOB # C40572

TOTAL PAGES: 171

SHEET 17.12 OF 56



VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

BUFFER CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

18VFD1	18VFD2	18VFD3
CONVEYOR WITH STOPPER LEVEL 4	CONVEYOR WITH STOPPER LEVEL 5	CONVEYOR WITH STOPPER LEVEL 6

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2

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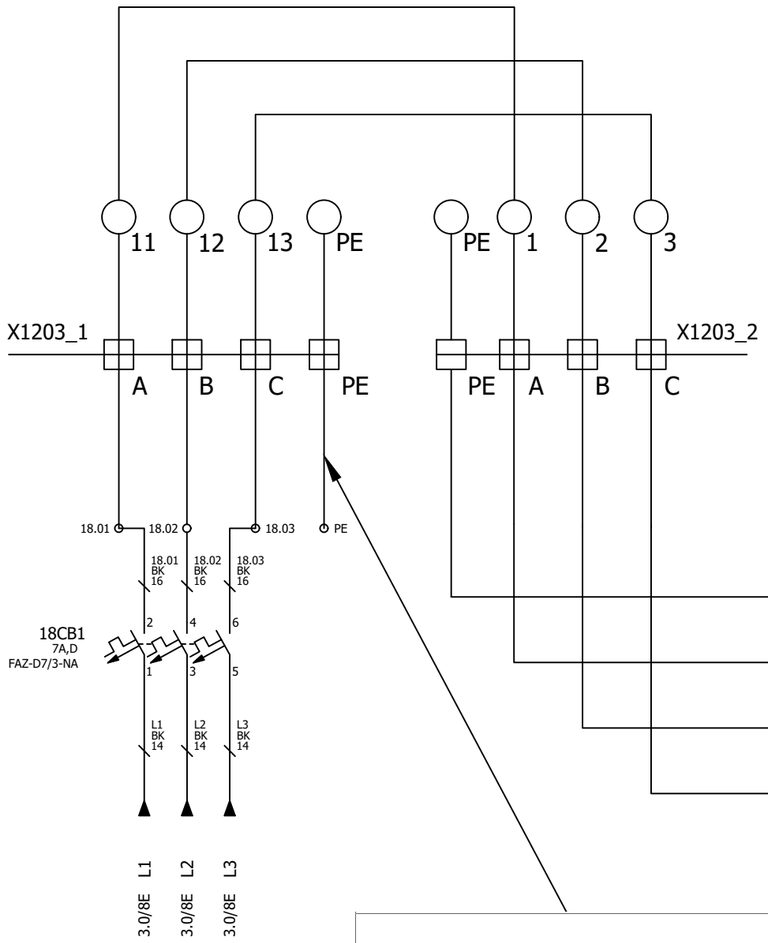
TOTAL PAGES: 171

SHEET 18.0 OF 56

CONVEYOR WITH STOPPER LEVEL 4

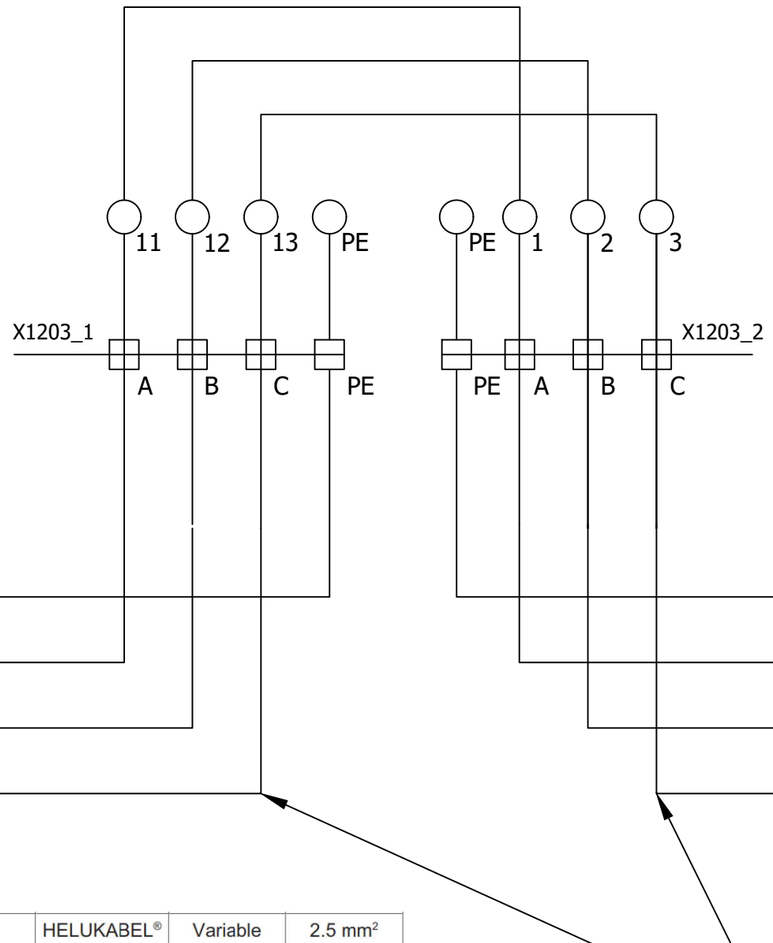
18VFD1



Open — M23, coding ring: black, male	UL: 18153283	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

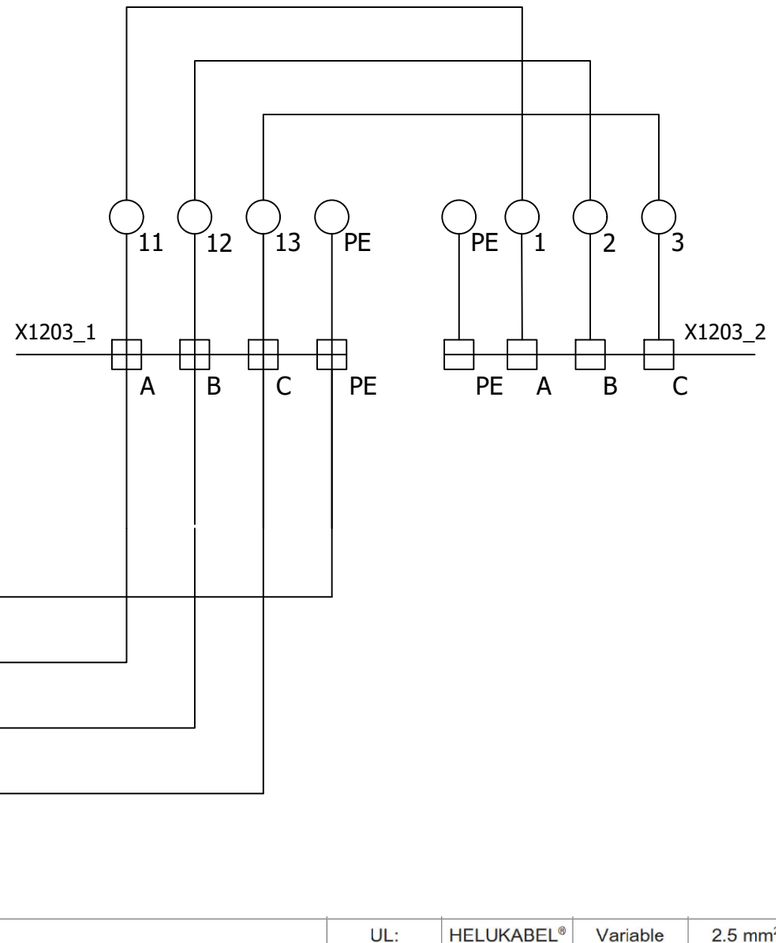
CONVEYOR WITH STOPPER LEVEL 5

18VFD2



CONVEYOR WITH STOPPER LEVEL 6

18VFD3



M23, coding ring: black, male — M23, coding ring: black, male	UL: 18153267	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 POWER

REV	DESCRIPTION	DATE

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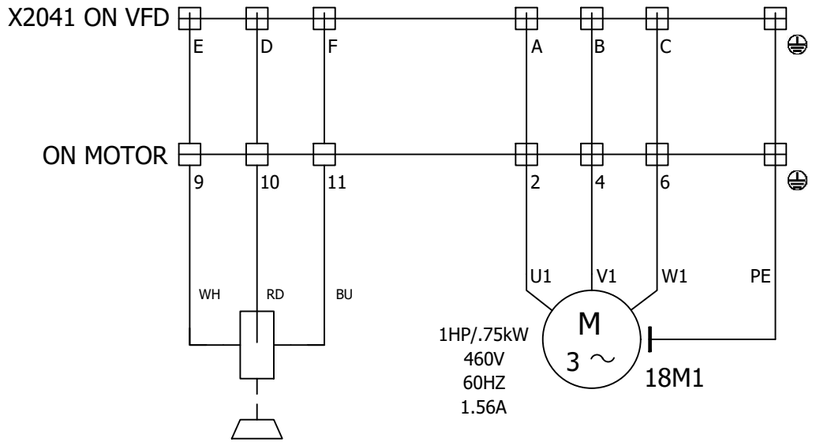
TOTAL PAGES: 171

SHEET 18.1 OF 56

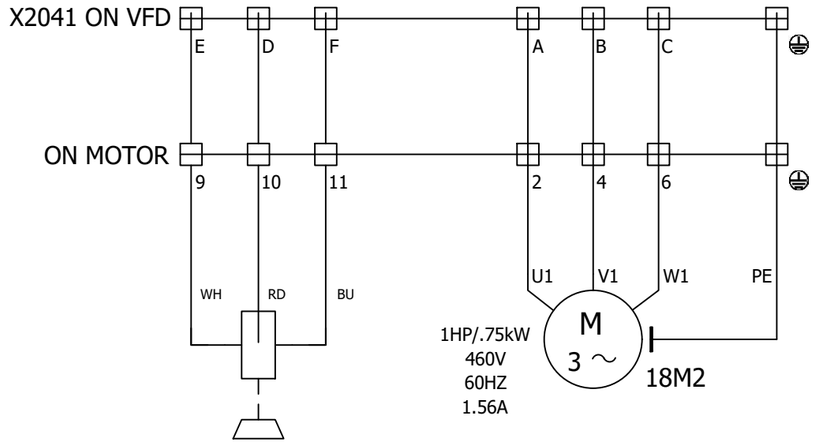
CONVEYOR WITH STOPPER LEVEL 4

18VFD1



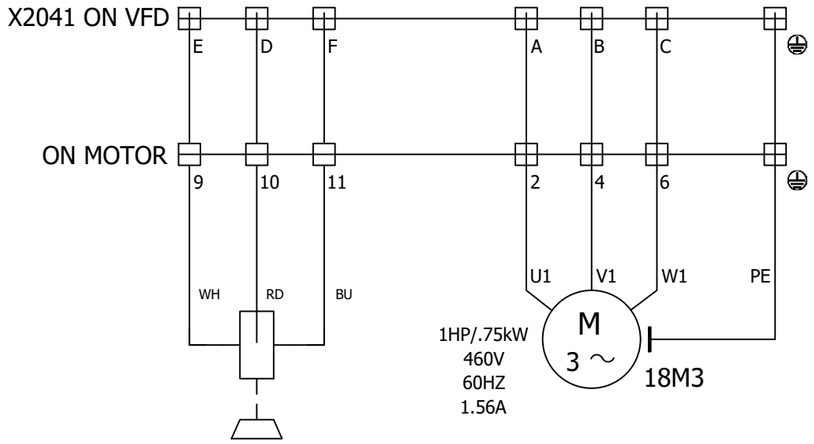
CONVEYOR WITH STOPPER LEVEL 5

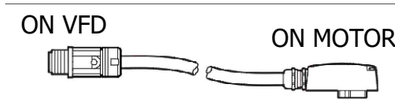
18VFD2



CONVEYOR WITH STOPPER LEVEL 6

18VFD3





M23, coding ring: brown, male	IS2, female, ⅄	CE/UL: 28125940	LEONI LEHC® 000749	Variable	1.5 mm² / AC 500 V

Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 MOTOR POWER

REV	DESCRIPTION	DATE

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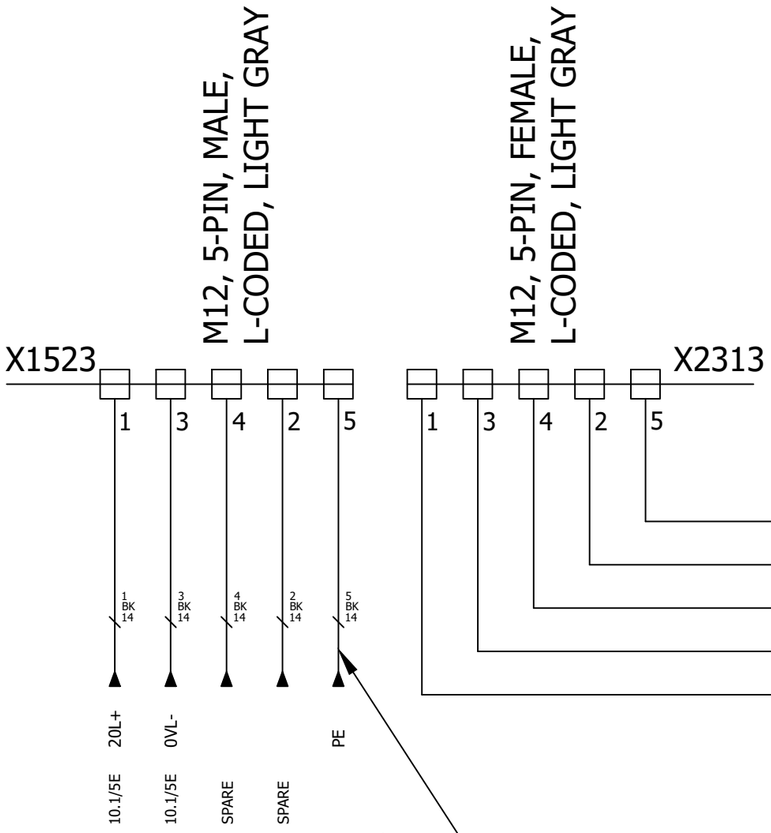
JOB # C40572

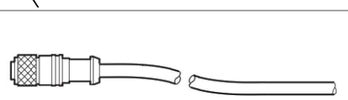
TOTAL PAGES: 171

SHEET 18.2 OF 56

CONVEYOR WITH STOPPER LEVEL 4

18VFD1



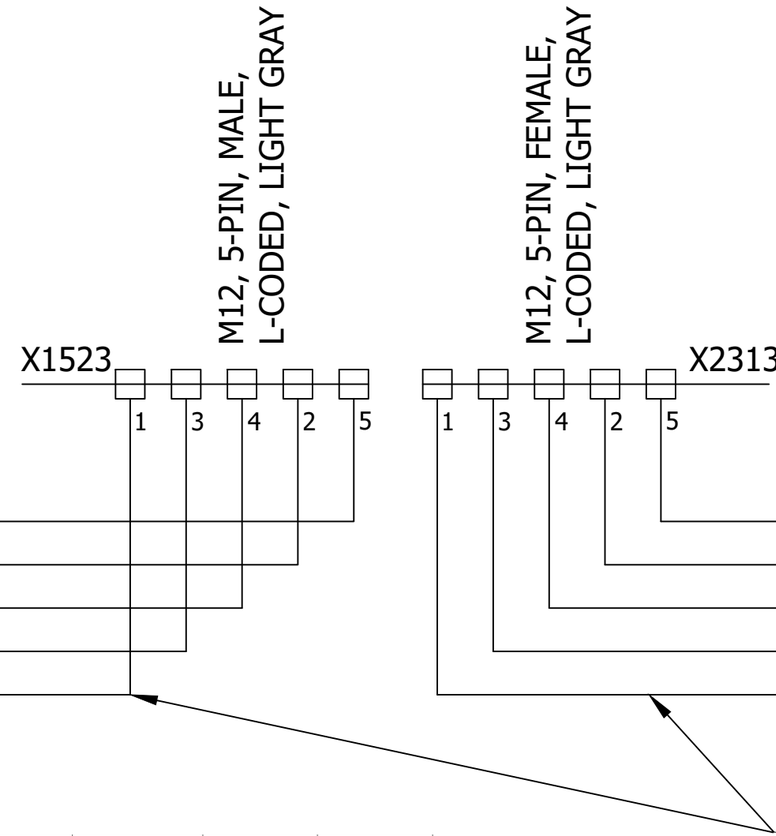


M12, 5-pin, L-coded, female — Open

CE/UL:	HELUKABEL®	Variable	5 × 2.5 mm² / DC 60 V
28117786	JZ-500	✕	

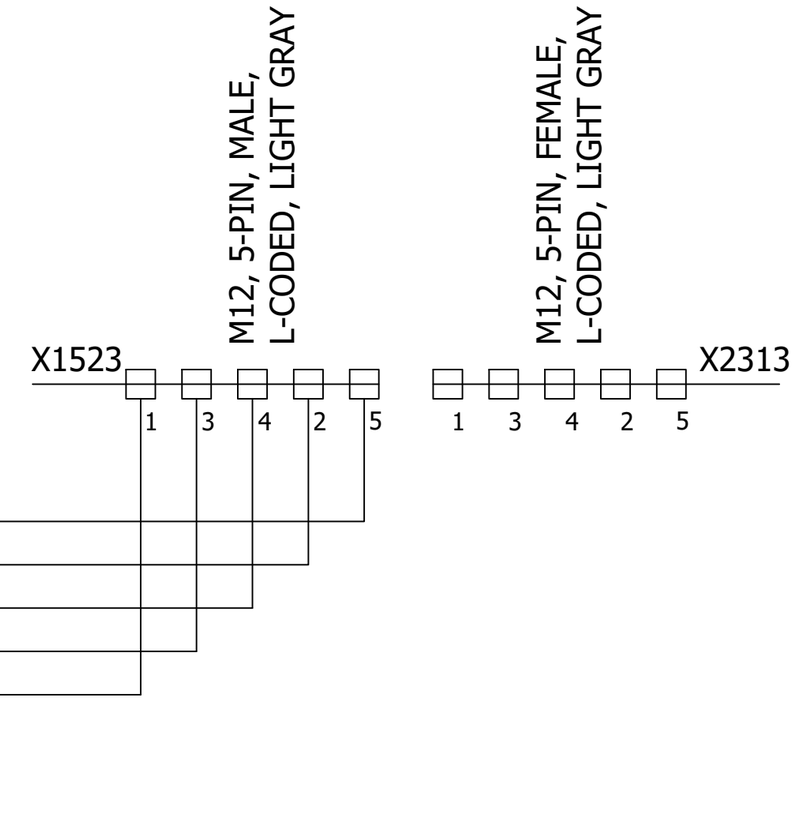
CONVEYOR WITH STOPPER LEVEL 5

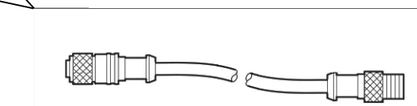
18VFD2



CONVEYOR WITH STOPPER LEVEL 6

18VFD3





M12, 5-pin, L-coded, female — M12, 5-pin, L-coded, male

CE/UL:	HELUKABEL®	Variable	5 × 2.5 mm² / DC 60 V
28114345	JZ-500	✕	

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 VFD 24VDC POWER

REV	DESCRIPTION	DATE

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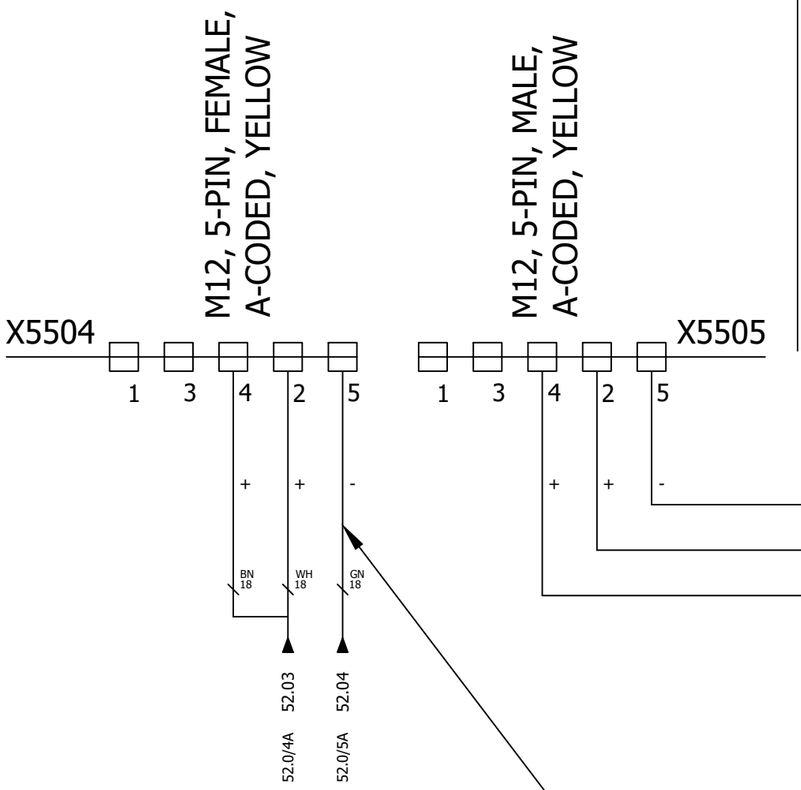
JOB # C40572

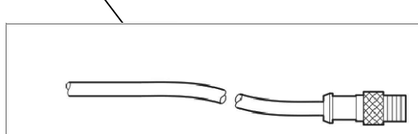
TOTAL PAGES: 171

SHEET 18.3 OF 56

CONVEYOR WITH STOPPER LEVEL 4

18VFD1

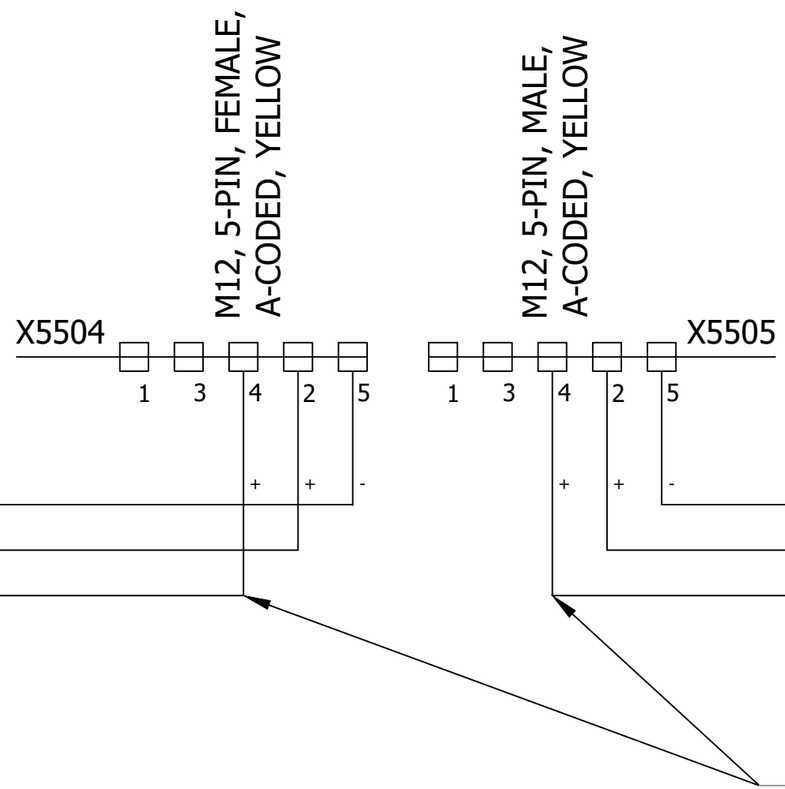




Open — M12, 5-pin, A-coded, male	CE/UL: 28110943	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V

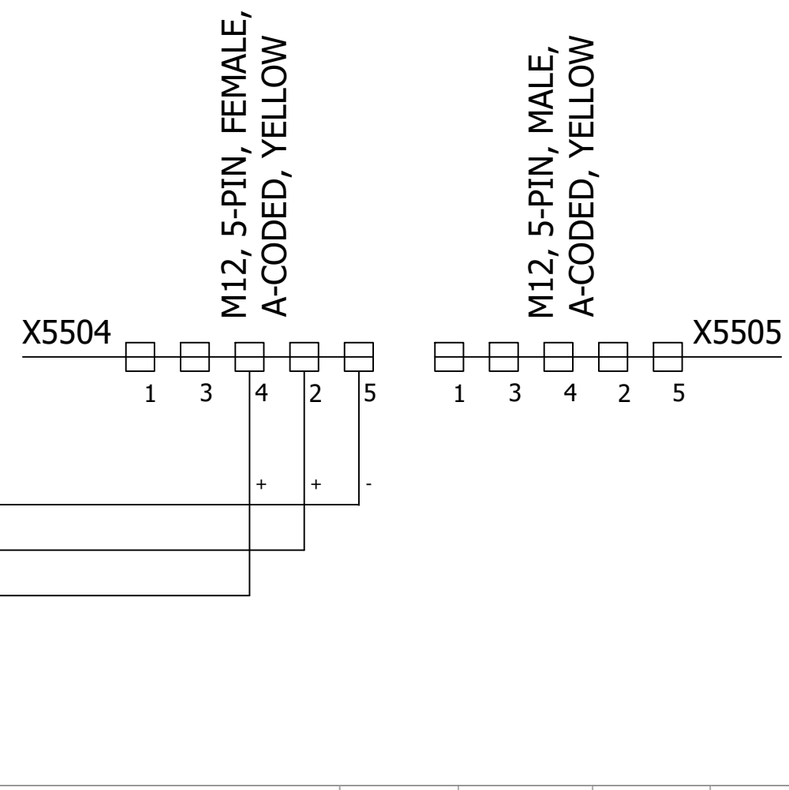
CONVEYOR WITH STOPPER LEVEL 5

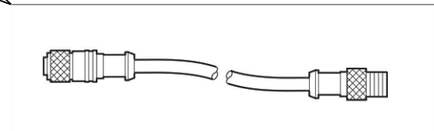
18VFD2



CONVEYOR WITH STOPPER LEVEL 6

18VFD3





M12, 5-pin, A-coded, female — M12, 5-pin, A-coded, male	CE/UL: 28110935	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 VFD STO

REV	DESCRIPTION	DATE

DRAWN BY:	APPROVED FOR BUILD BY:	DRAWING NUMBER:
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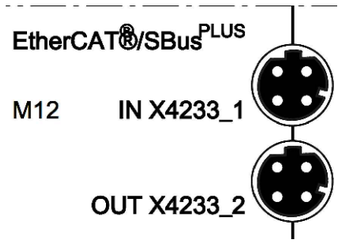
TOTAL PAGES: 171

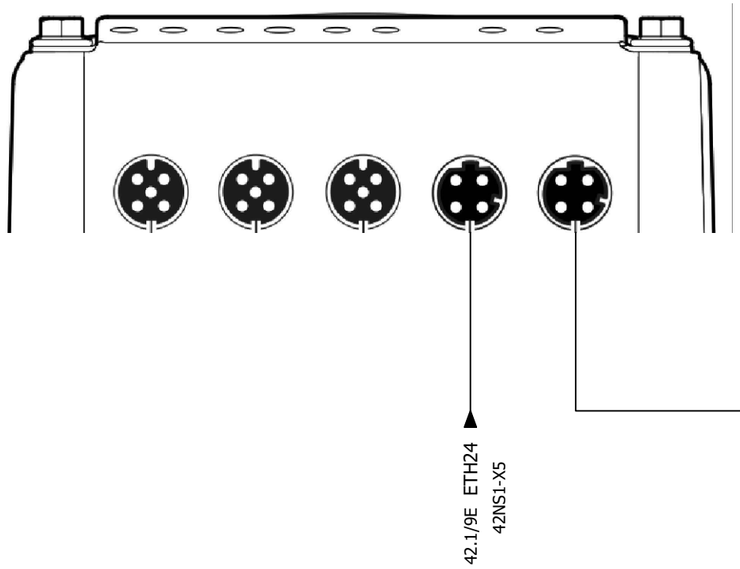
SHEET 18.4 OF 56

Assignment		
Contact	**Signal**	**Description**
1	TX+	Transmit line (+)
2	RX+	Receive line (+)
3	TX-	Transmit line (-)
4	RX-	Receive line (-)

18VFD1

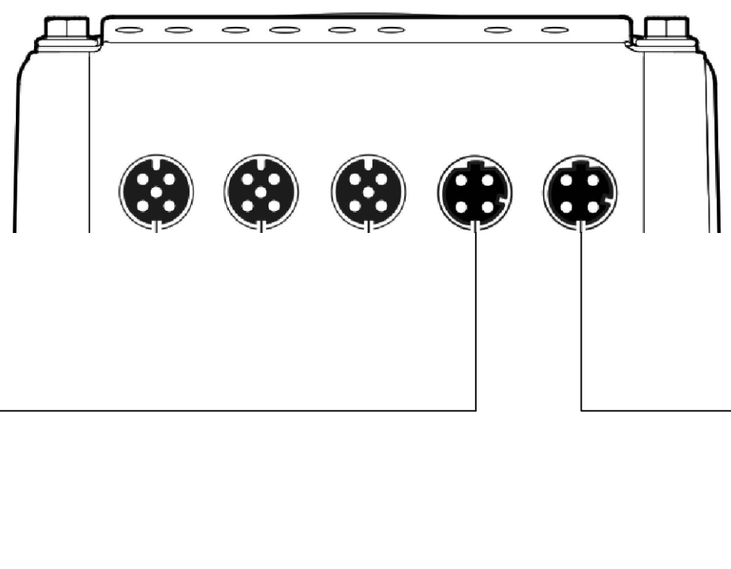


X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK



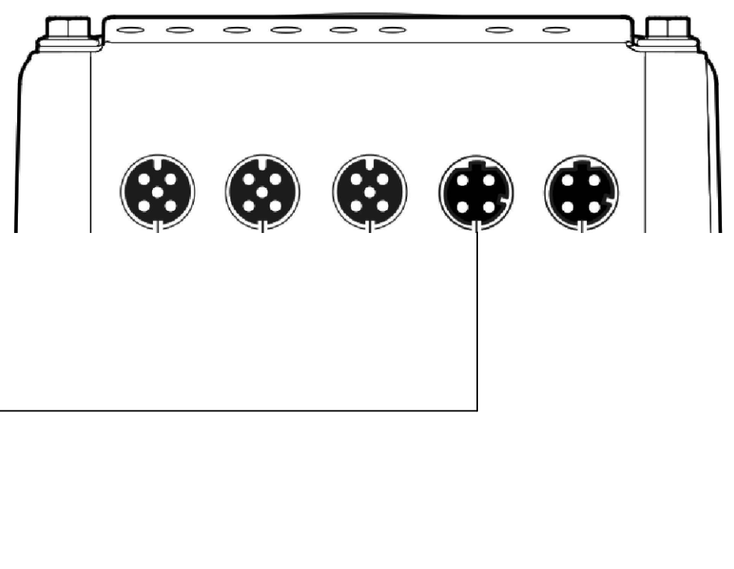
CONVEYOR
WITH STOPPER
LEVEL 4

18VFD2



CONVEYOR
WITH STOPPER
LEVEL 5

18VFD3



CONVEYOR
WITH STOPPER
LEVEL 6

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 VFD EtherCAT INTERFACE

REV	DESCRIPTION	DATE

DRAWN BY:	APPROVED FOR BUILD BY:	DRAWING NUMBER:
PAUL	Carlos A	C40572
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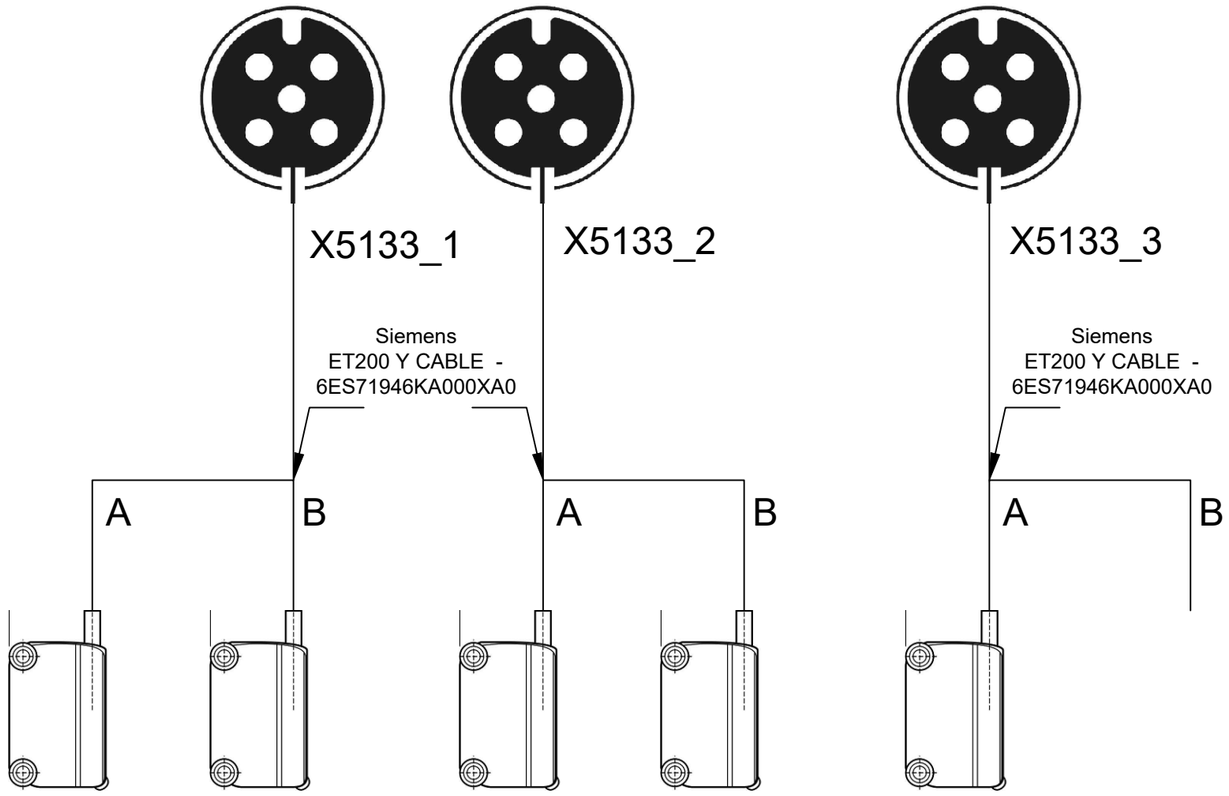
TOTAL PAGES: 171

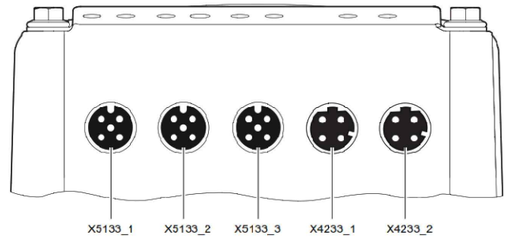
SHEET 18.5 OF 56



18PE1	18PE2	18PE3	18PE4	18PE5	TO 18VFD1
18PE7	18PE8	18PE9	18PE10	18PE11	TO 18VFD2
18PE13	18PE14	18PE15	18PE16	18PE17	TO 18VFD3

Plug connector positions at the DSI.. electronics cover

The following figure shows an example of the plug connector positions:



18014420217585035

Plug connector	Function
X5133_1[1)]	Digital inputs DI01 and DI02
X5133_2[1)]	Digital inputs DI03 and DI04
X5133_3[1)]	Digital inputs/outputs DIO01 and DIO02

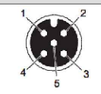
X5133_1: Digital inputs

The following table shows information about this connection:

Function
Digital inputs

Connection type
M12, 5-pin, female, A-coded, color: black

Connection diagram



Assignment

Contact	Signal	Description
1	+24V	DC 24 V sensor supply
2	DI02	Sensor input DI02
3	0V24	0V24 reference potential for sensors
4	DI01	Sensor input DI01
5	FE	Functional earth

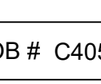
X5133_2: Digital inputs

The following table shows information about this connection:

Function
Digital inputs

Connection type
M12, 5-pin, female, A-coded, color: black

Connection diagram



Assignment

Contact	Signal	Description
1	+24V	DC 24 V sensor supply
2	DI04	Sensor input DI04
3	0V24	0V24 reference potential for sensors
4	DI03	Sensor input DI03
5	FE	Functional earth

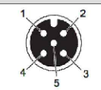
X5133_3: Digital inputs/outputs

The following table shows information about this connection:

Function
Digital inputs/outputs

Connection type
M12, 5-pin, female, A-coded, color: black

Connection diagram



Assignment

Contact	Signal	Description
1	+24V	DC 24 V sensor/actuator supply
2	DIO02	Sensor input DIO02/actuator output DIO02
3	0V24	0V24 reference potential for sensors/actuators
4	DIO01	Sensor input DIO01/actuator output DIO01
5	FE	Functional earth

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

REV	DESCRIPTION	DATE

DRAWN BY:	APPROVED FOR BUILD BY:	DRAWING NUMBER:
PAUL	Carlos A	C40572
DATE 9/16/26	DATE 9/15/26	CURRENT REVISION: C

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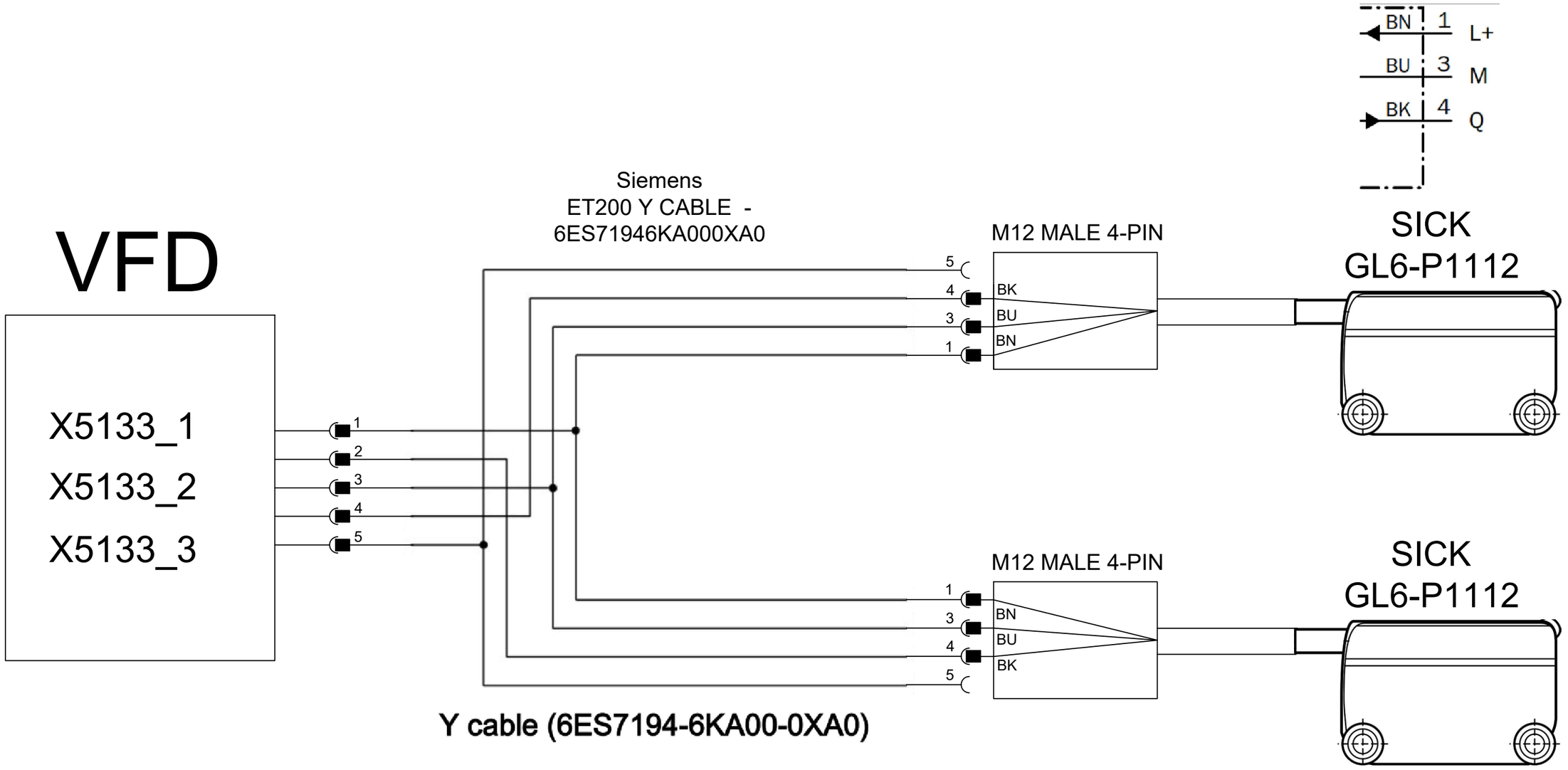
JOB # C40572

TOTAL PAGES: 171

SHEET 18.6 OF 56



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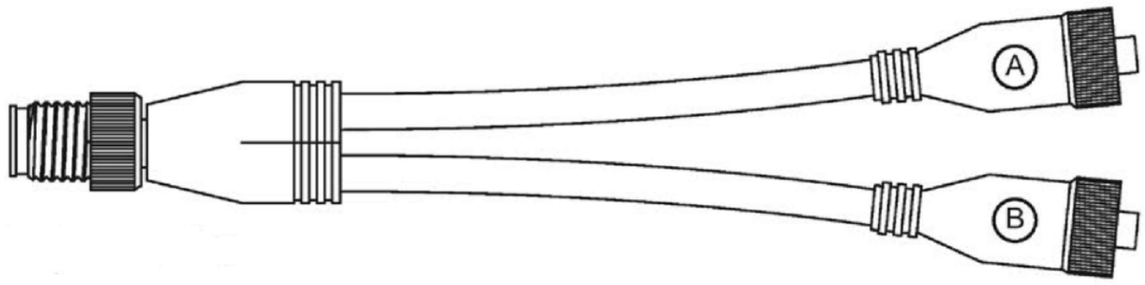

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: SPARE

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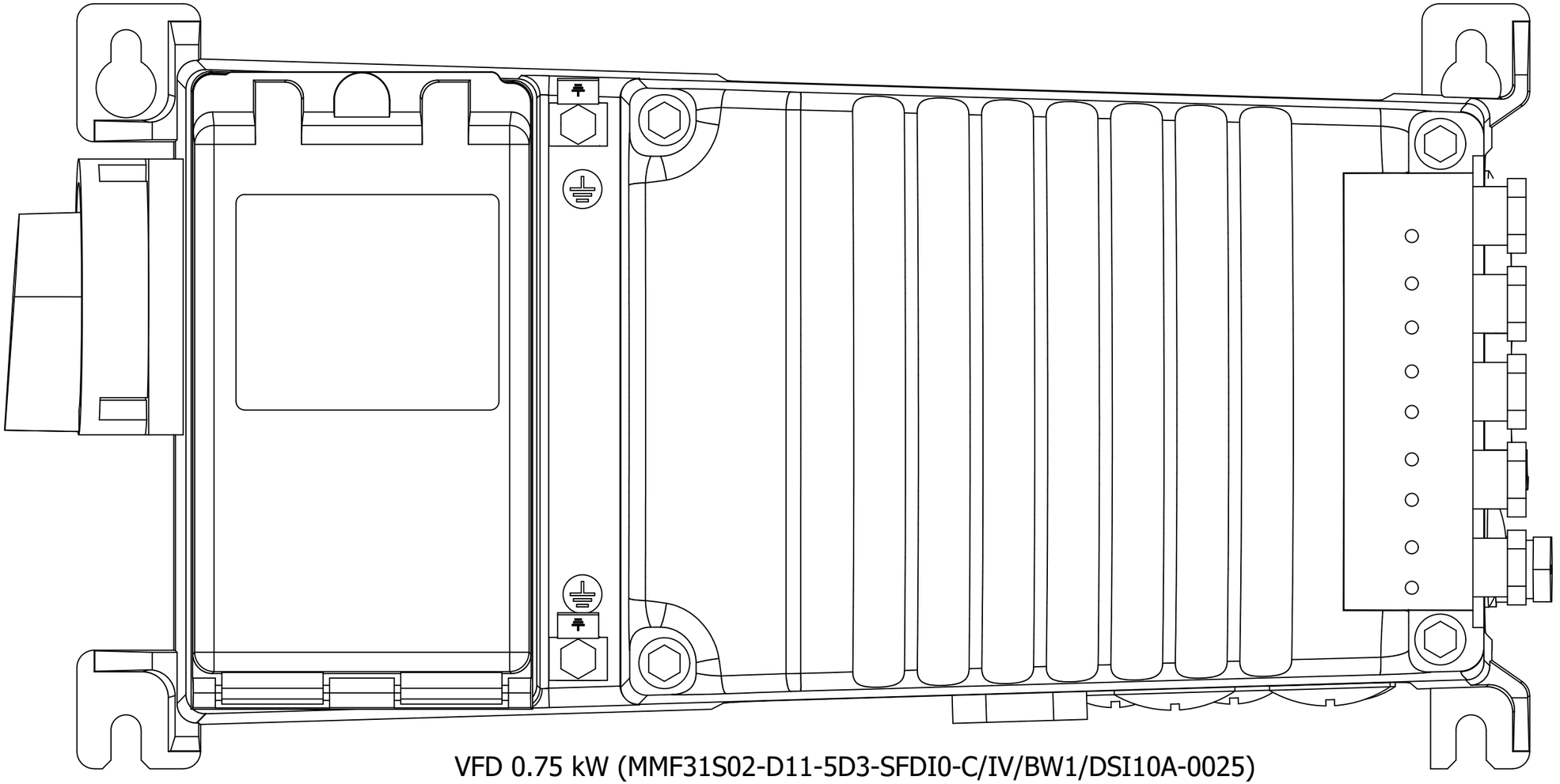
TOTAL PAGES: 171

SHEET 19.0 OF 56



VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

CAROUSEL CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 5 VFD's

20VFD1	20VFD2	20VFD3	20VFD4	20VFD5
ROLLER CONVEYOR 1	ROLLER CONVEYOR 2	ROLLER CONVEYOR 3	ROLLER CONVEYOR 4	ROLLER CONVEYOR 5

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1

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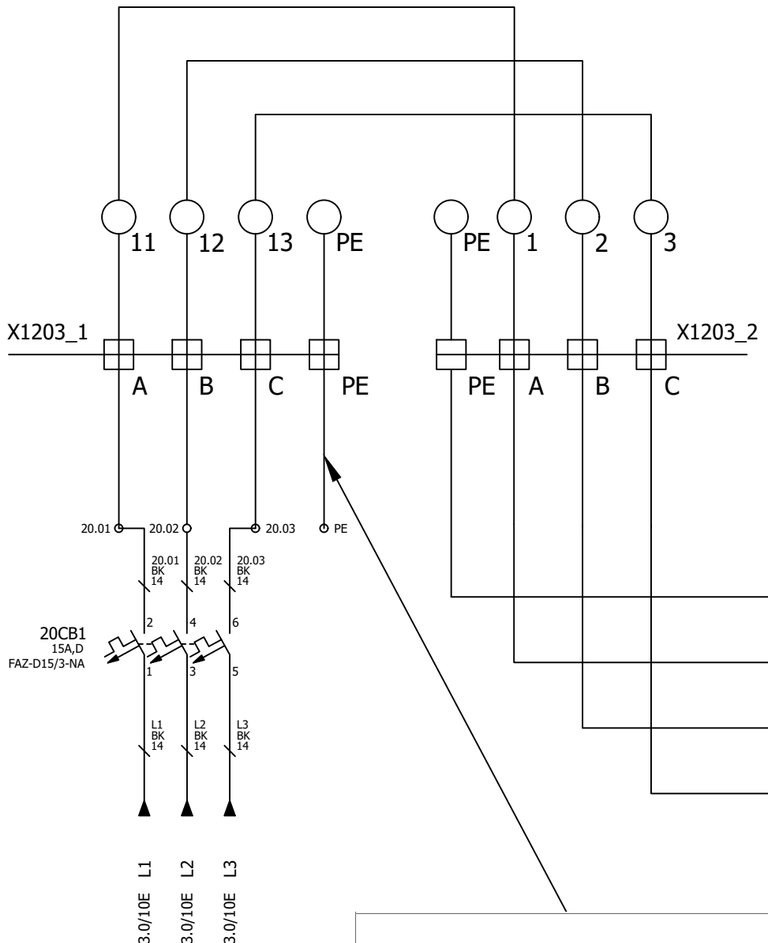
TOTAL PAGES: 171

SHEET 20.0 OF 56

ROLLER CONVEYOR 1

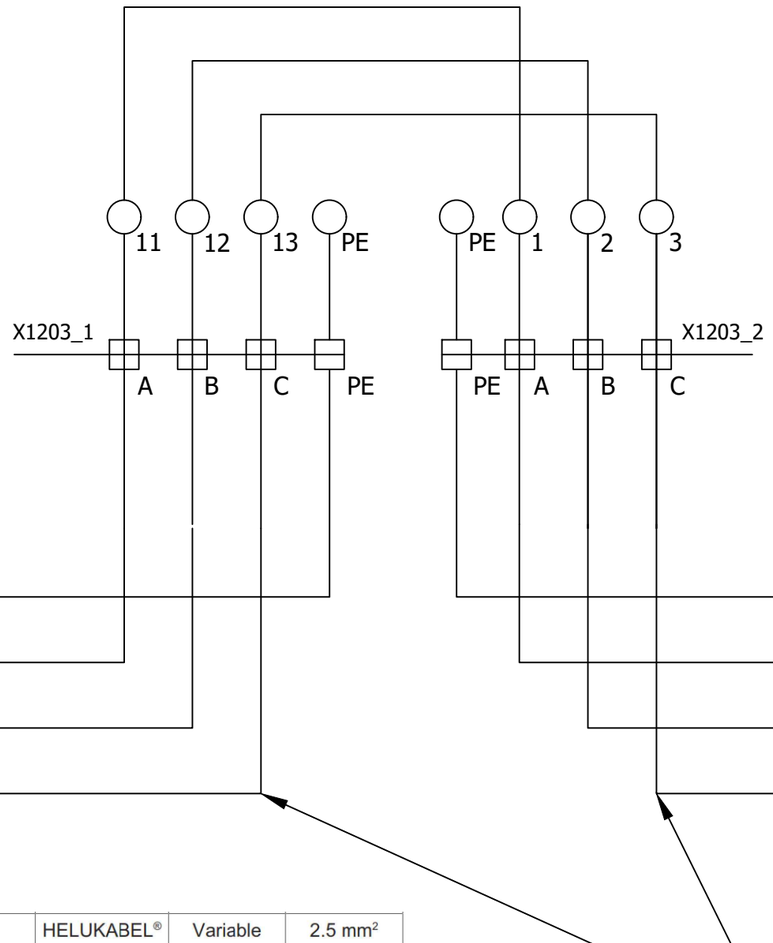
20VFD1



Open — M23, coding ring: black, male	UL: 18153283	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

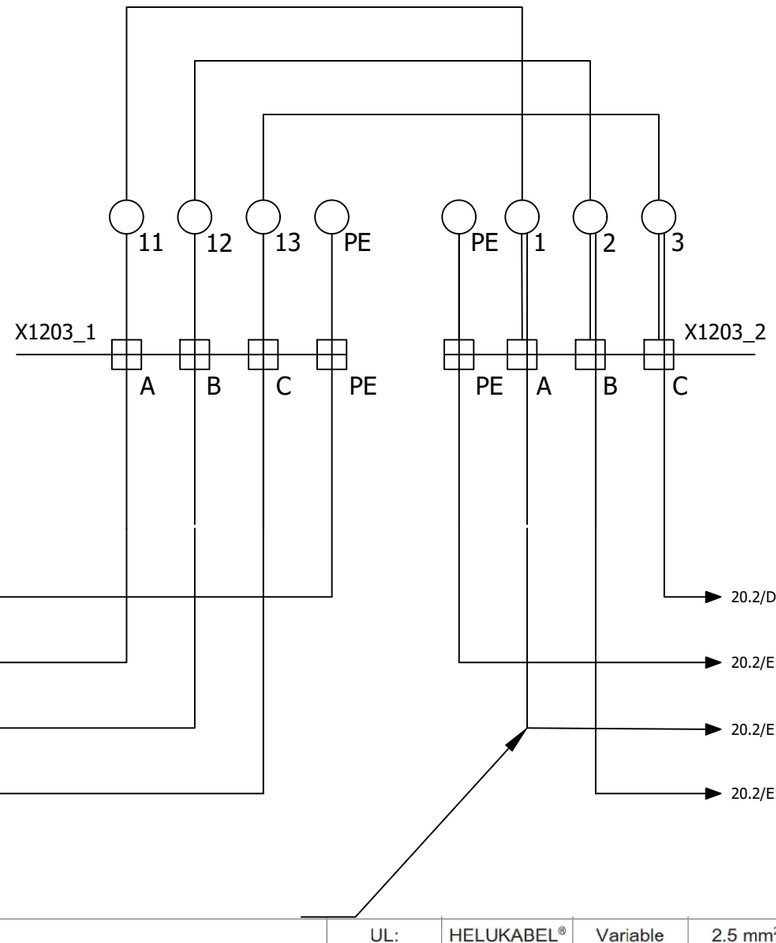
ROLLER CONVEYOR 2

20VFD2



ROLLER CONVEYOR 3

20VFD3



M23, coding ring: black, male — M23, coding ring: black, male	UL: 18153267	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 POWER

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TOTAL PAGES: 171

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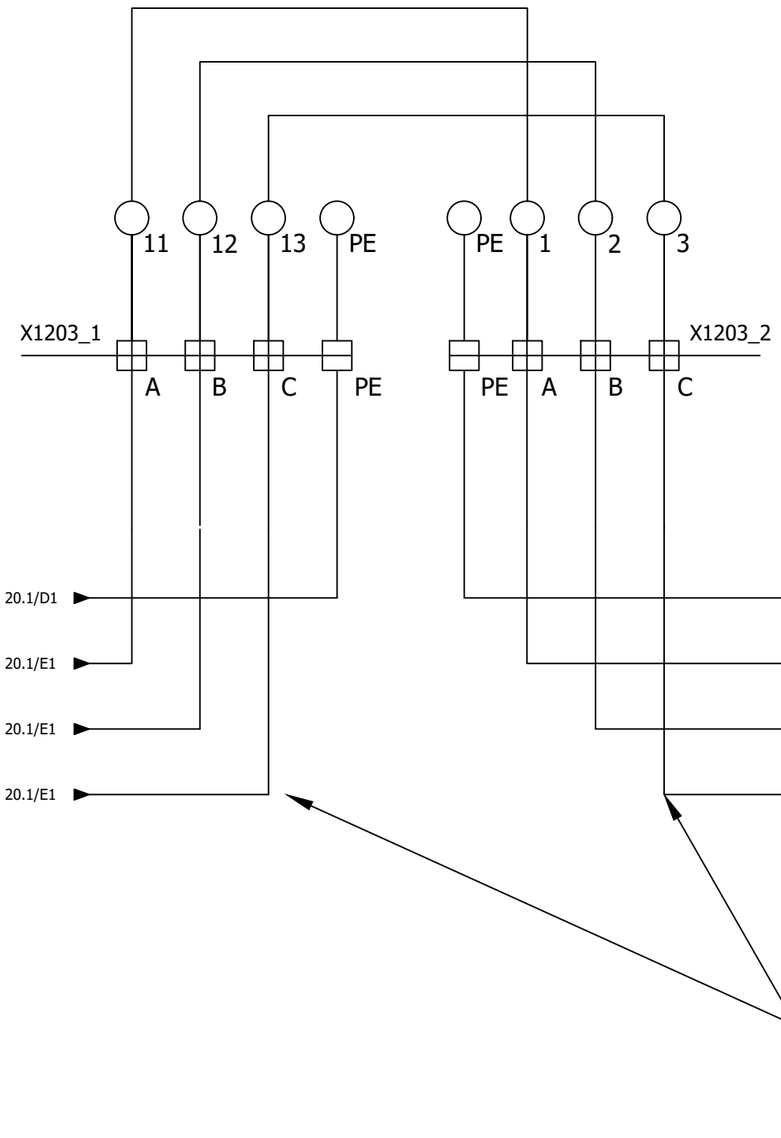
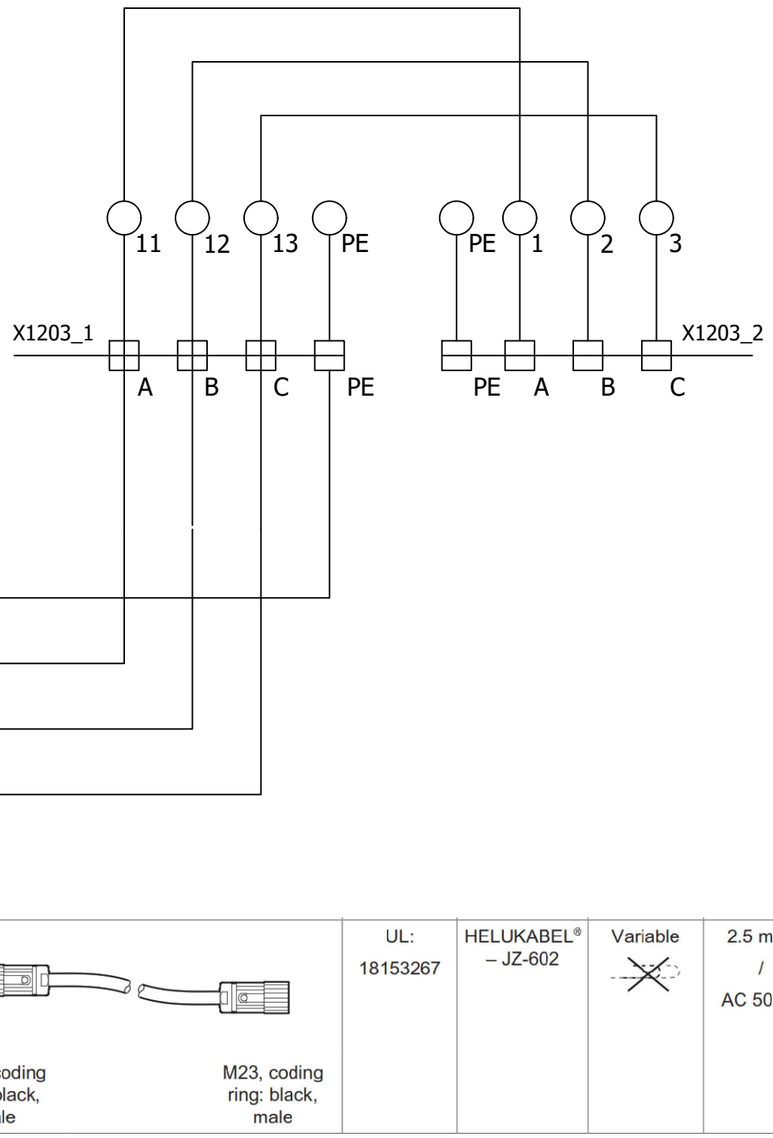
ROLLER CONVEYOR 4

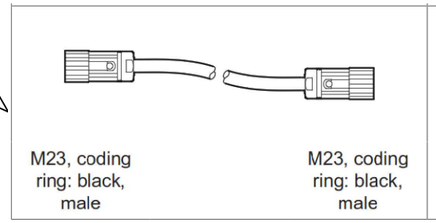
20VFD4



ROLLER CONVEYOR 5

20VFD5





UL:	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V
18153267			

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

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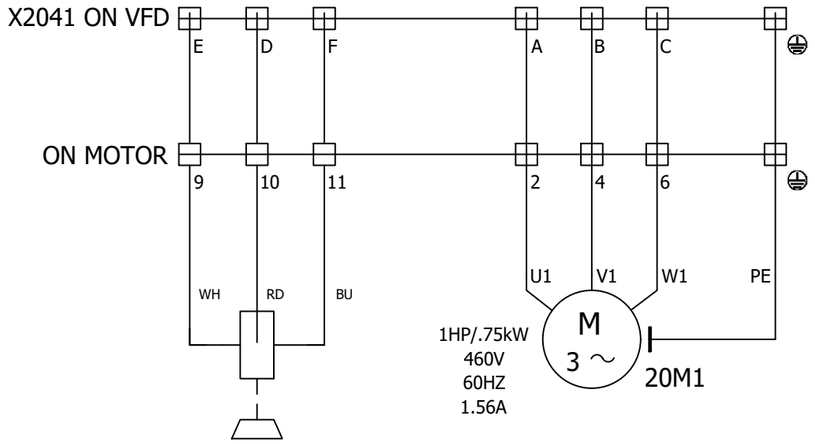
TOTAL PAGES: 171

SHEET 20.2 OF 56

ROLLER CONVEYOR 1

20VFD1



ROLLER CONVEYOR 2

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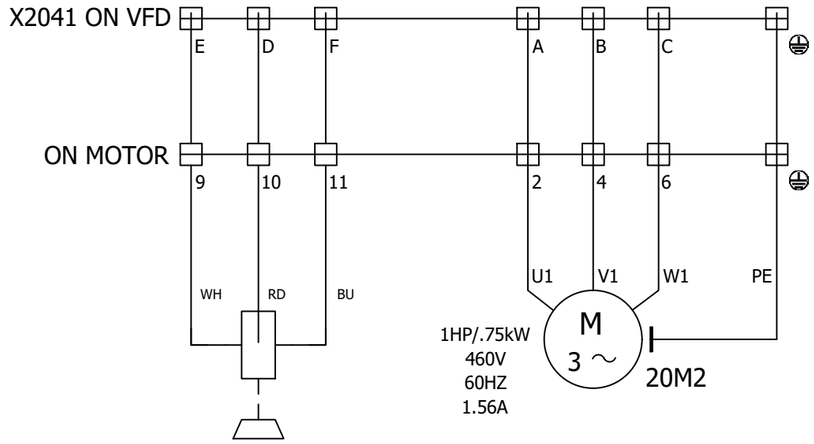
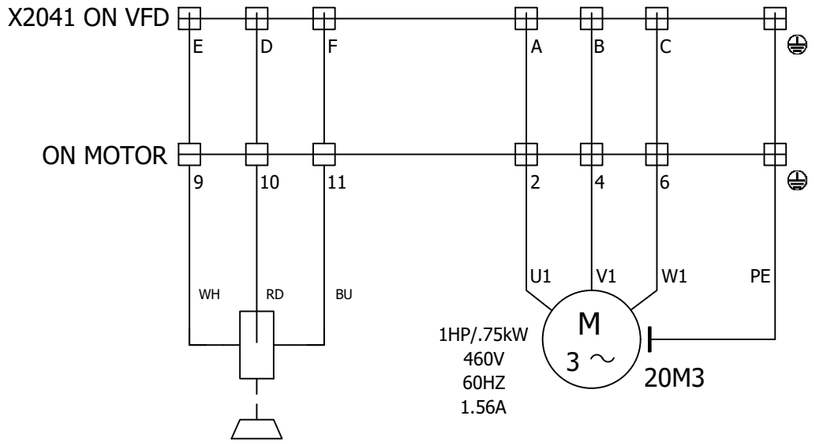
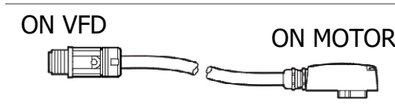


ROLLER CONVEYOR 3

20VFD3





M23, coding ring: brown, male	IS2, female, ⅄	CE/UL: 28125940	LEONI LEHC® 000749	Variable	1.5 mm² / AC 500 V

Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 MOTOR POWER

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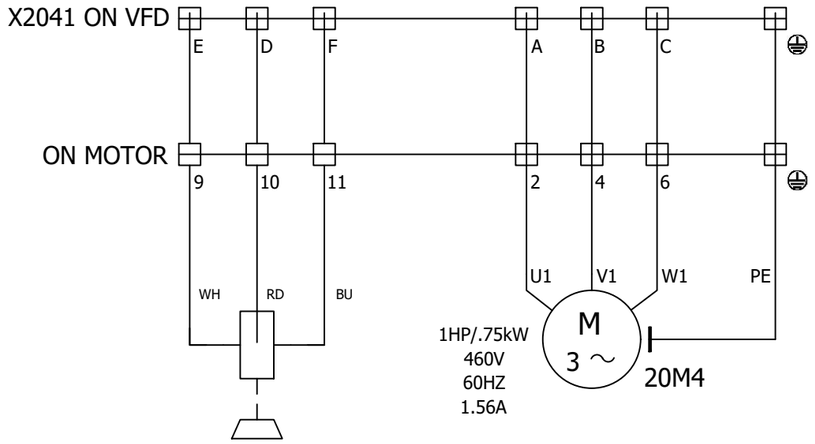
TOTAL PAGES: 171

SHEET 20.3 OF 56

ROLLER CONVEYOR 4

20VFD4



ROLLER CONVEYOR 5

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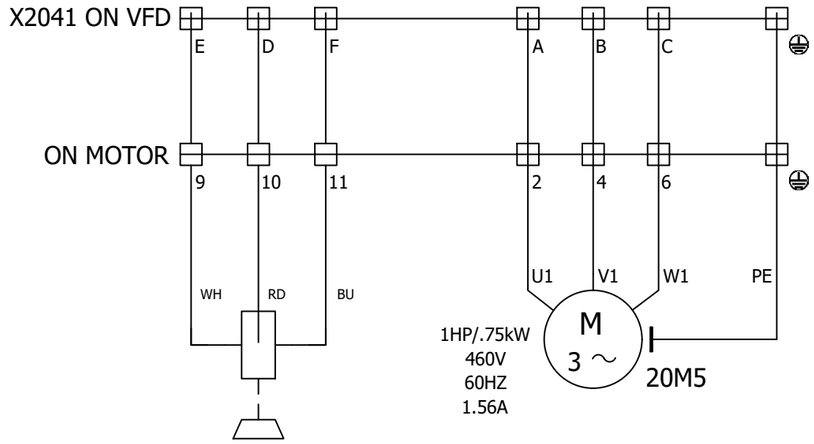

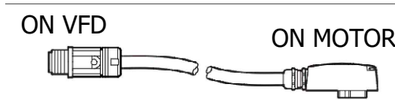

M23, coding ring: brown, male	IS2, female, ⅄	CE/UL: 28125940	LEONI LEHC® 000749	Variable	1.5 mm² / AC 500 V

Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 MOTOR POWER

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TOTAL PAGES: 171

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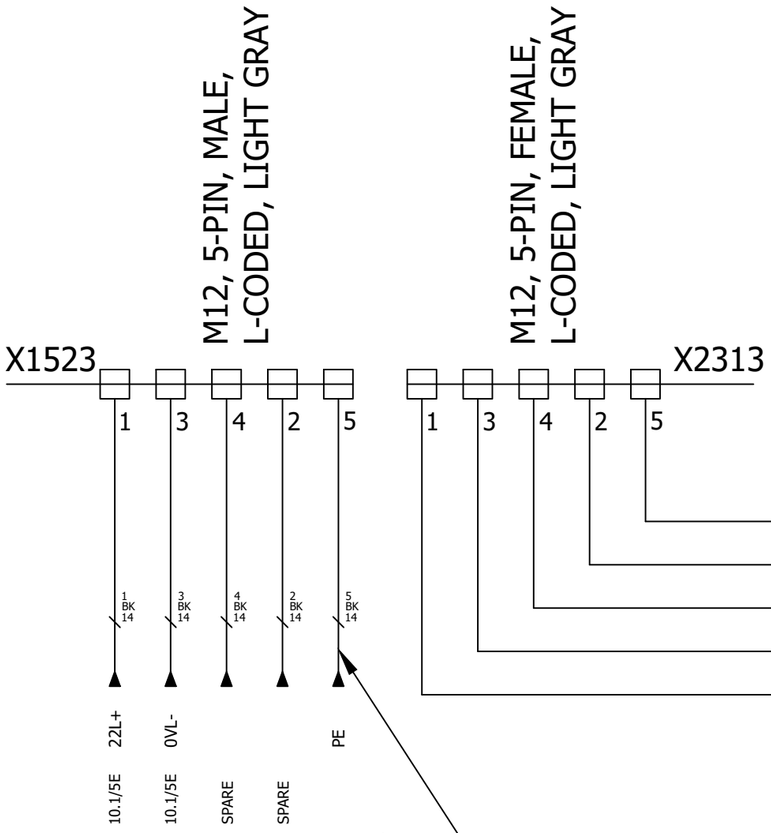
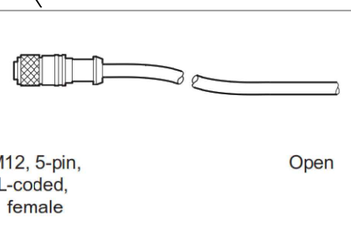
ROLLER CONVEYOR 1

20VFD1



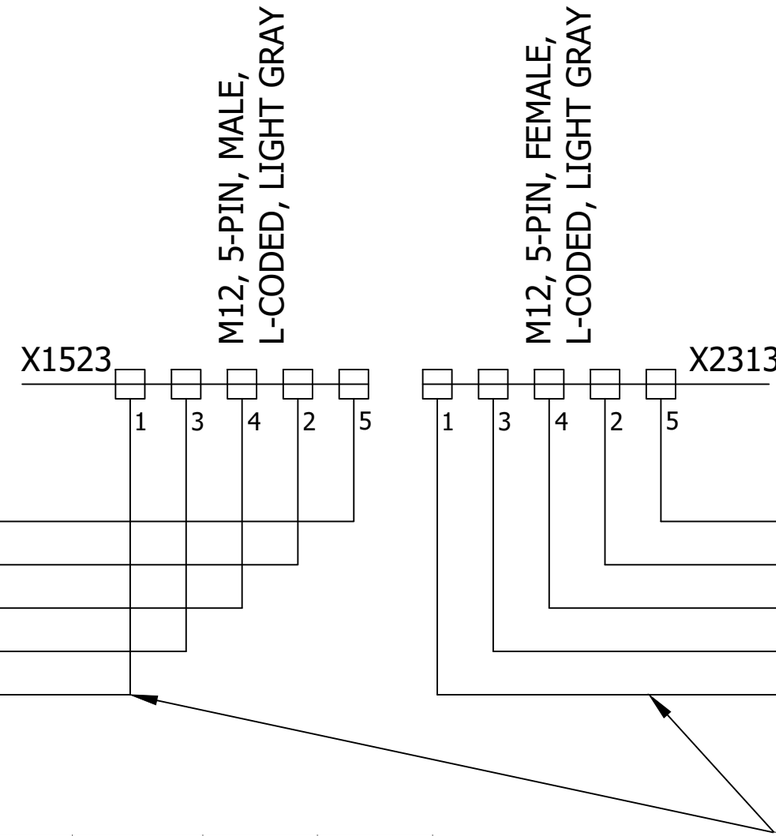


M12, 5-pin, L-coded, female — Open

CE/UL:	HELUKABEL® JZ-500	Variable	5 × 2.5 mm² / DC 60 V
28117786			

ROLLER CONVEYOR 2

20VFD2



ROLLER CONVEYOR 3

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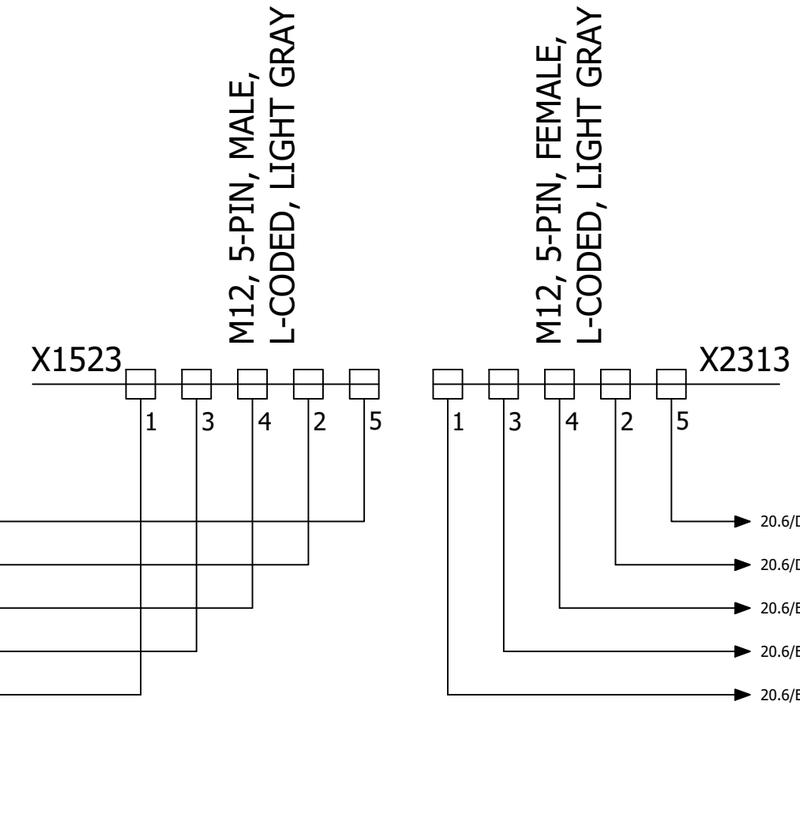

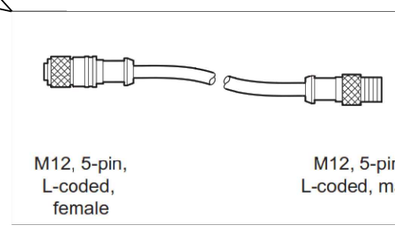

M12, 5-pin, L-coded, female — M12, 5-pin, L-coded, male

CE/UL:	HELUKABEL® JZ-500	Variable	5 × 2.5 mm² / DC 60 V
28114345			

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER

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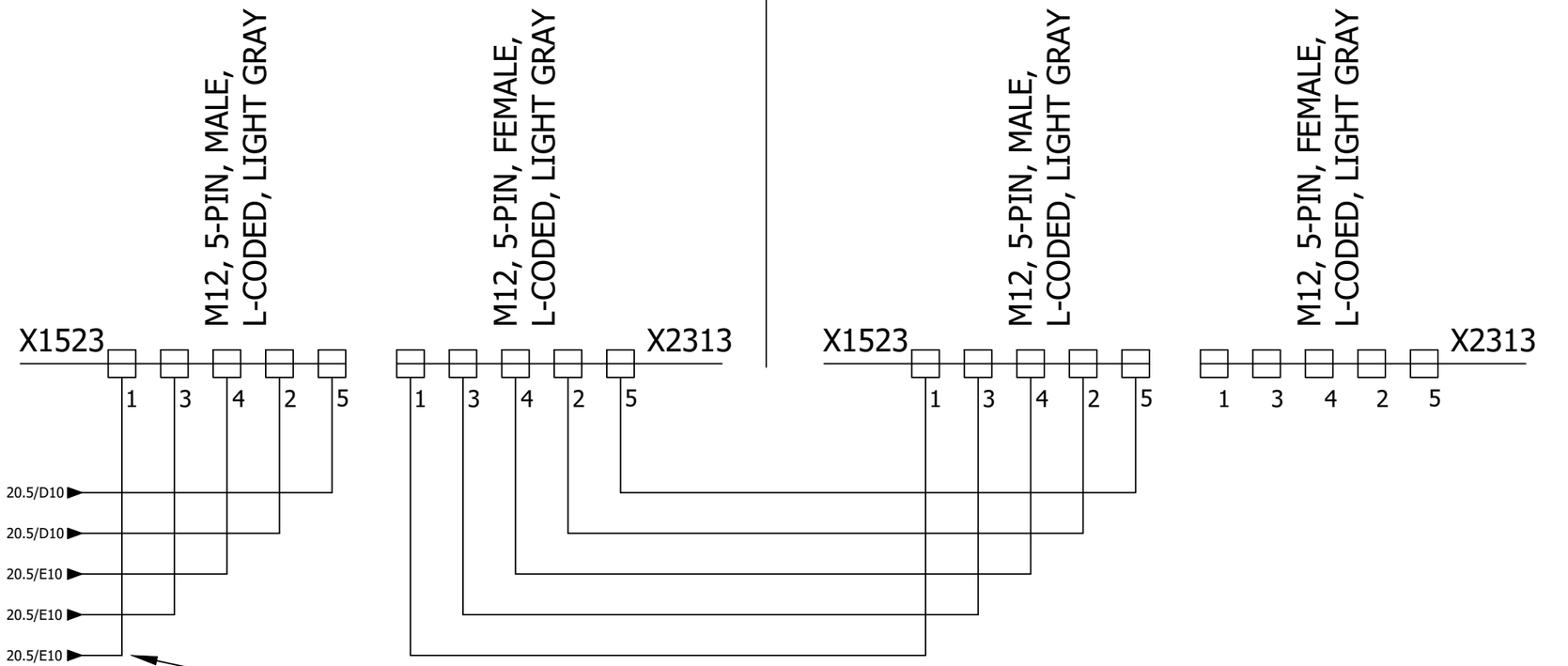
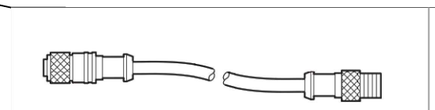
CAROUSEL TURNTABLE CONVEYOR

20VFD4



ROLLER CONVEYOR 5

20VFD5



M12, 5-pin, L-coded, female

M12, 5-pin, L-coded, male

CE/UL:	HELUKABEL®	Variable	5 × 2.5 mm² / DC 60 V
28114345	JZ-500		

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PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER

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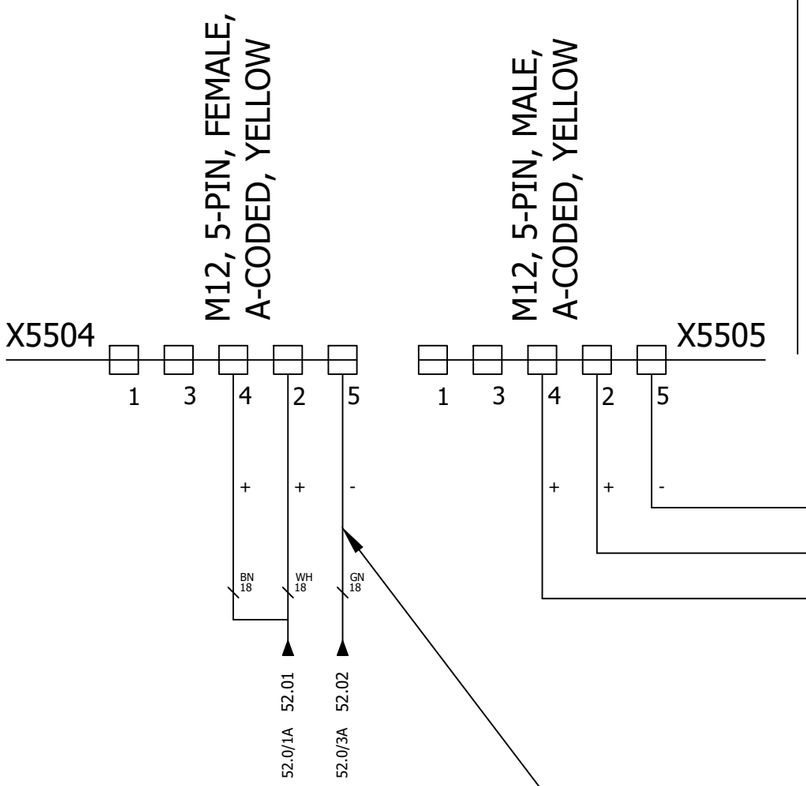
ROLLER CONVEYOR 1

20VFD1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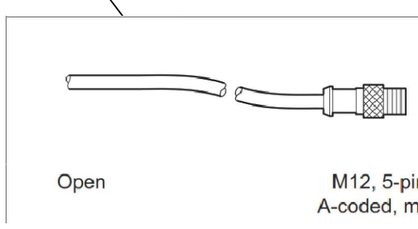
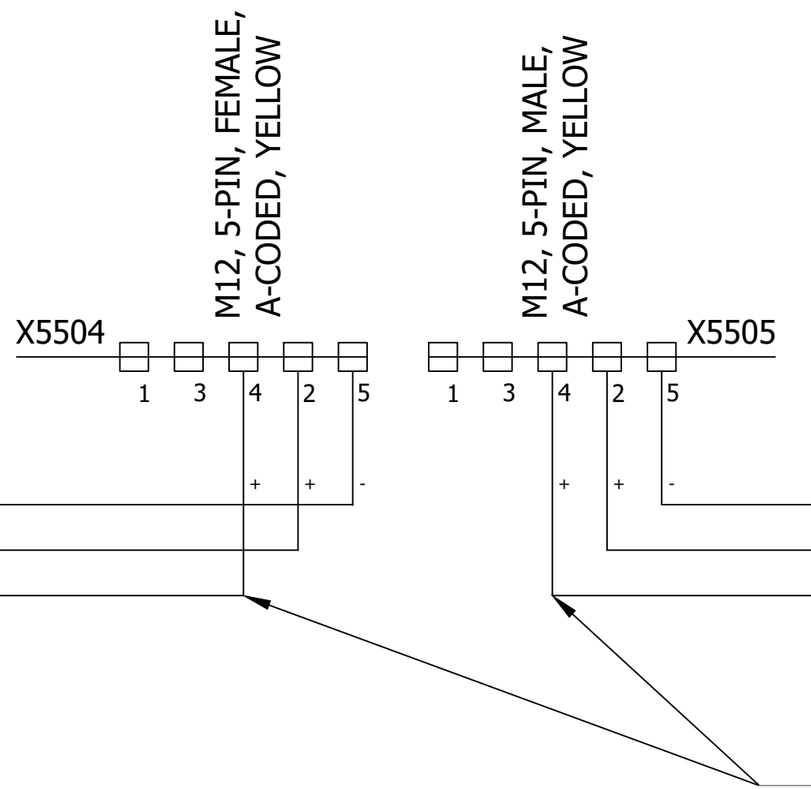
M12, 5-PIN, MALE, A-CODED, YELLOW
X5504
X5505
1 3 4 2 5
+ + -
BN 18
WH 18
GN 18
52.01
52.0/1A
52.02
52.0/3A



Open — M12, 5-pin, A-coded, male	CE/UL: 28110943	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V

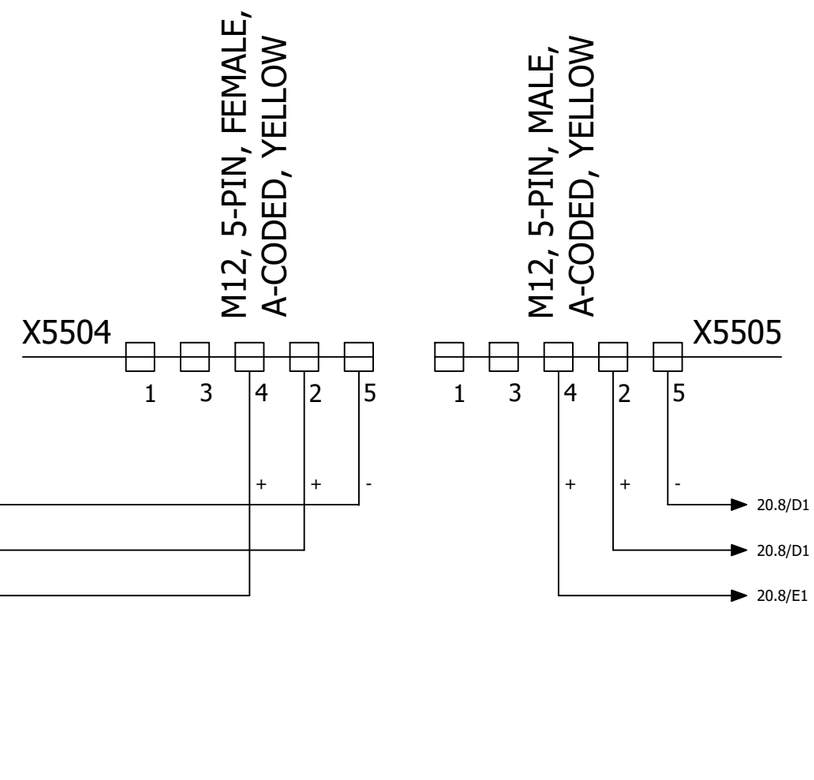
ROLLER CONVEYOR 2

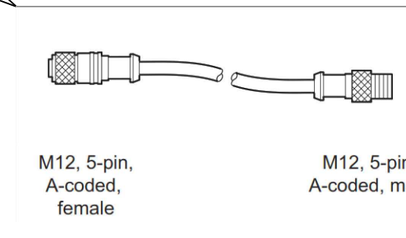
20VFD2



ROLLER CONVEYOR 3

20VFD3





M12, 5-pin, A-coded, female — M12, 5-pin, A-coded, male	CE/UL: 28110935	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD STO

REV	DESCRIPTION	DATE

DRAWN BY:	APPROVED FOR BUILD BY:	DRAWING NUMBER:
PAUL	Carlos A	C40572
DATE 9/16/26	DATE 9/15/26	CURRENT REVISION: C

Saddle Brook Controls
280 N Midland Ave.
Saddle Brook, NJ
07663

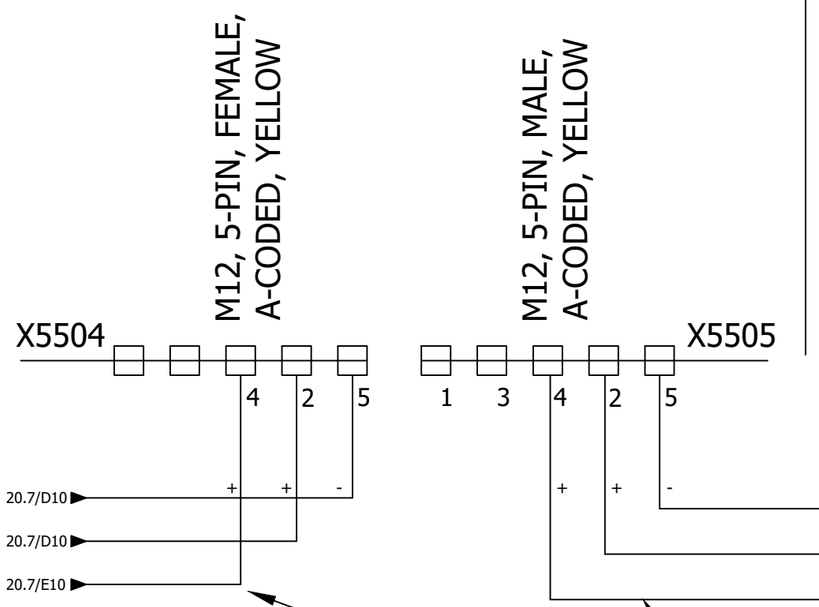
JOB # C40572

TOTAL PAGES: 171

SHEET 20.7 OF 56

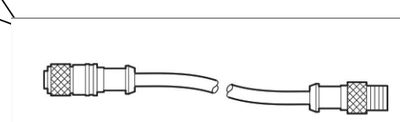
ROLLER CONVEYOR 1

20VFD4



ROLLER CONVEYOR 5

20VFD5



M12, 5-pin, A-coded, female

M12, 5-pin, A-coded, male

CE/UL:	HELUKABEL®	Variable	3 × 0.75 mm² / DC 60 V
28110935	LiYCY		

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD STO

REV	DESCRIPTION	DATE

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JOB # C40572

TOTAL PAGES: 171

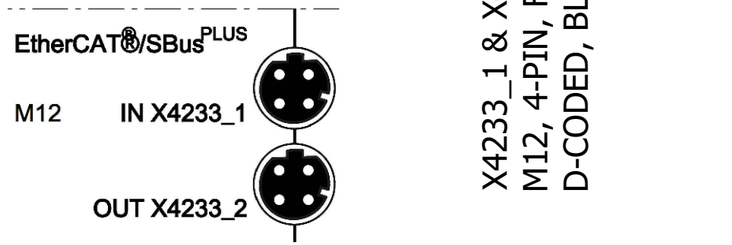
SHEET 20.8 OF 56

Assignment		
Contact	**Signal**	**Description**
1	TX+	Transmit line (+)
2	RX+	Receive line (+)
3	TX-	Transmit line (-)
4	RX-	Receive line (-)

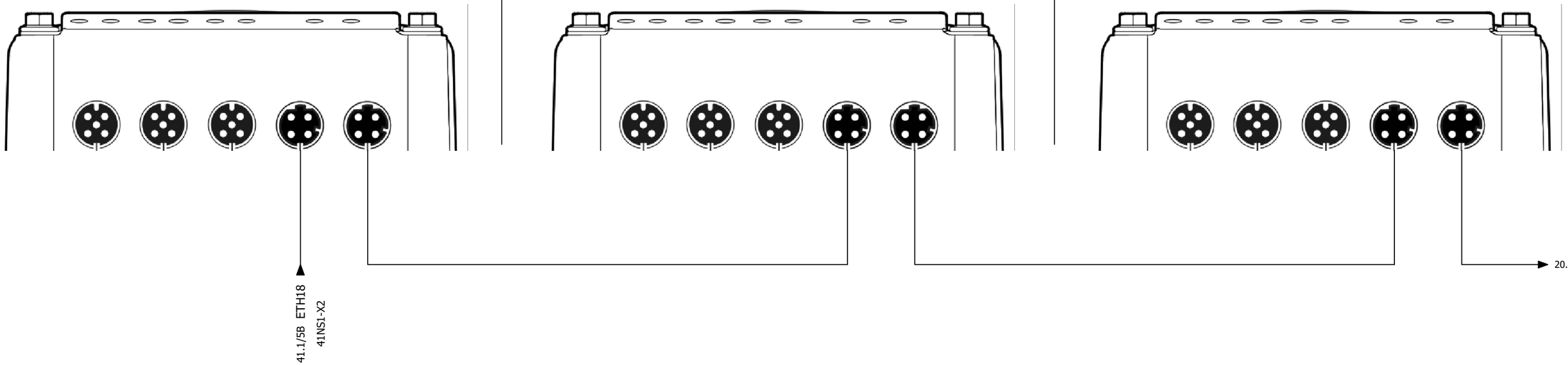
20VFD1

20VFD2

20VFD3



X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK



ROLLER CONVEYOR 1

ROLLER CONVEYOR 2

ROLLER CONVEYOR 3

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE

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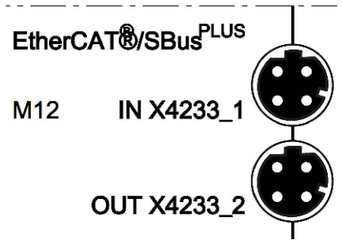
JOB # C40572

TOTAL PAGES: 171

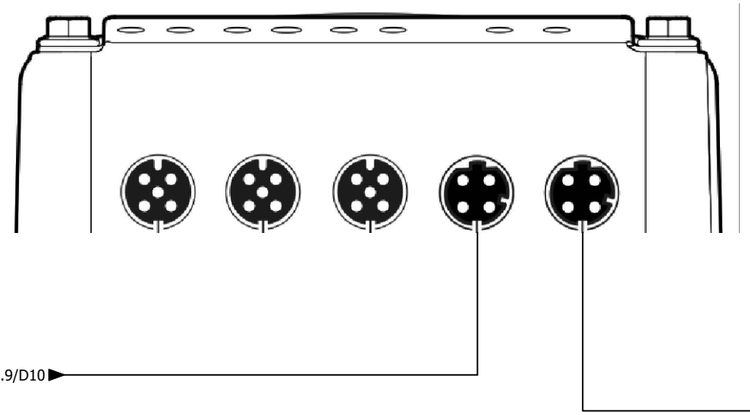
SHEET 20.9 OF 56

Assignment		
Contact	**Signal**	**Description**
1	TX+	Transmit line (+)
2	RX+	Receive line (+)
3	TX-	Transmit line (-)
4	RX-	Receive line (-)

20VFD4



X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK



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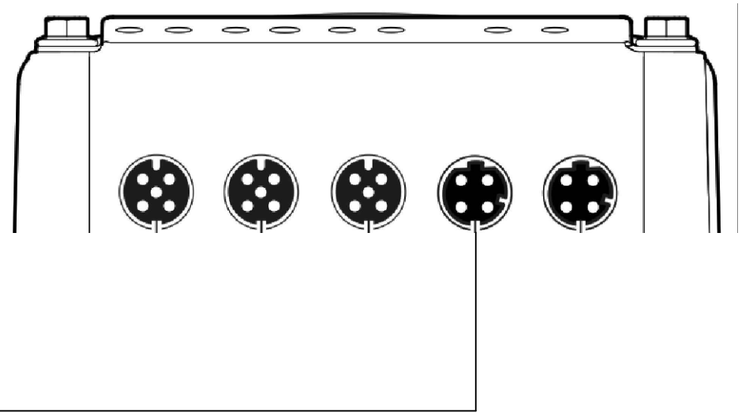

ROLLER
CONVEYOR 1

ROLLER
CONVEYOR 5

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE

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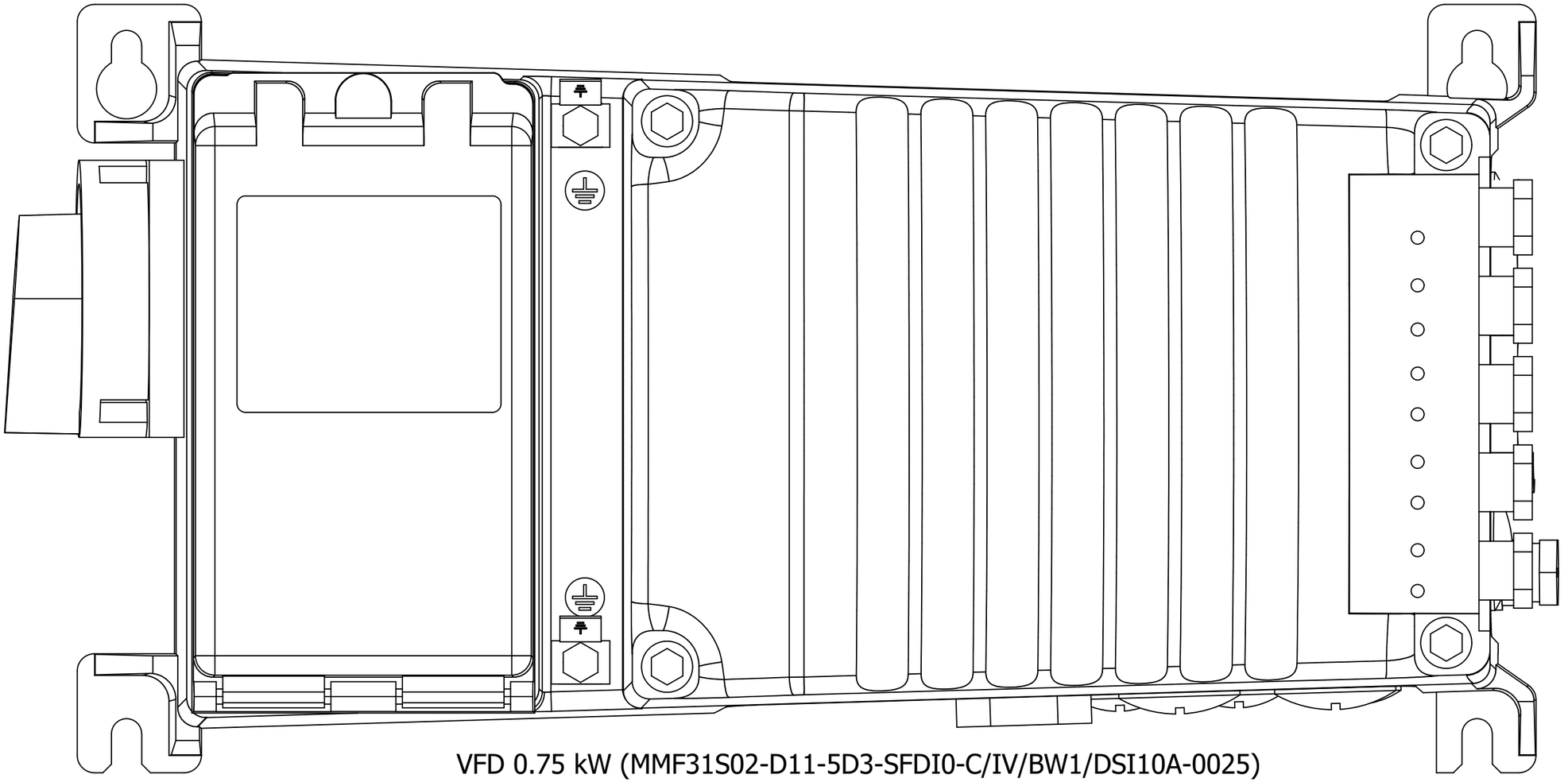
TOTAL PAGES: 171

SHEET 20.10 OF 56



VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

CAROUSEL CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 2 VFD's

21VFD1

CHAIN
CONVEYOR 1

21VFD2

Transfer
Conveyor 1

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 2

REV	DESCRIPTION	DATE

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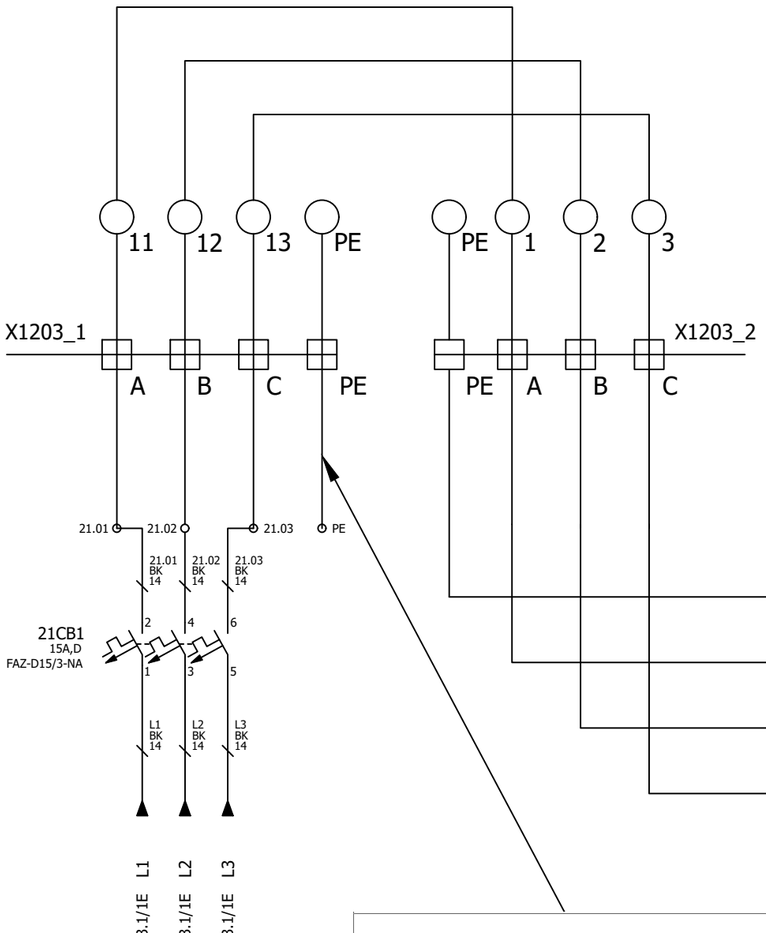
JOB # C40572

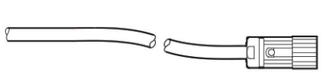
TOTAL PAGES: 171

SHEET 21.0 OF 56

CHAIN CONVEYOR 1

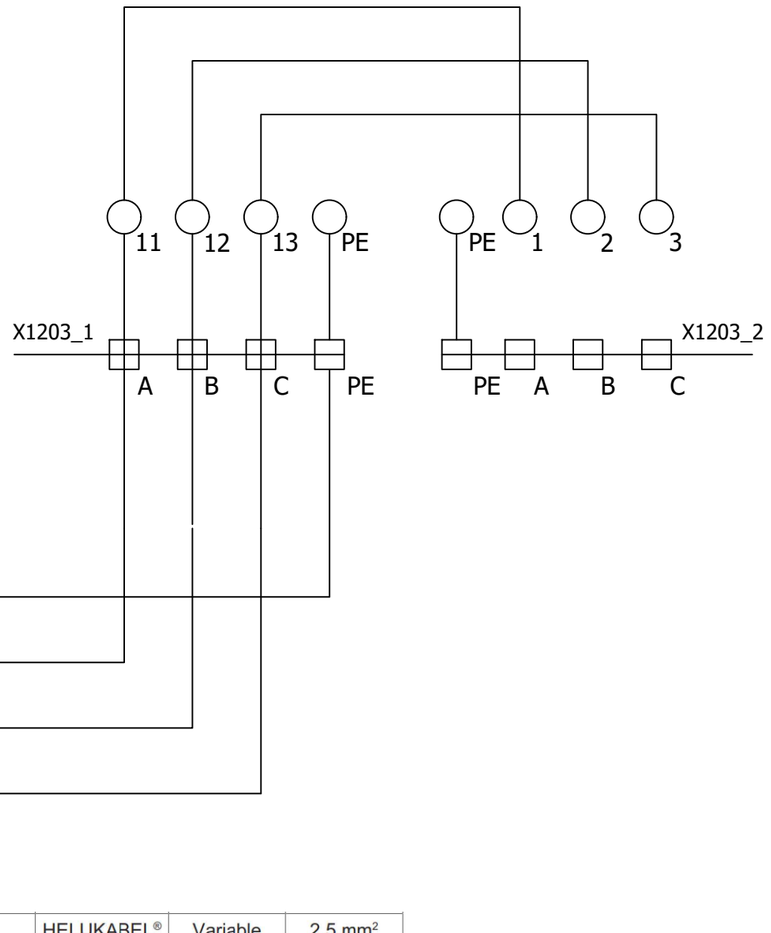
21VFD1



 Open — M23, coding ring: black, male	UL: 18153283	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

Transfer Conveyor 1

21VFD2



M23, coding ring: black, male — M23, coding ring: black, male	UL: 18153267	HELUKABEL® – JZ-602	Variable	2.5 mm² / AC 500 V

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 POWER

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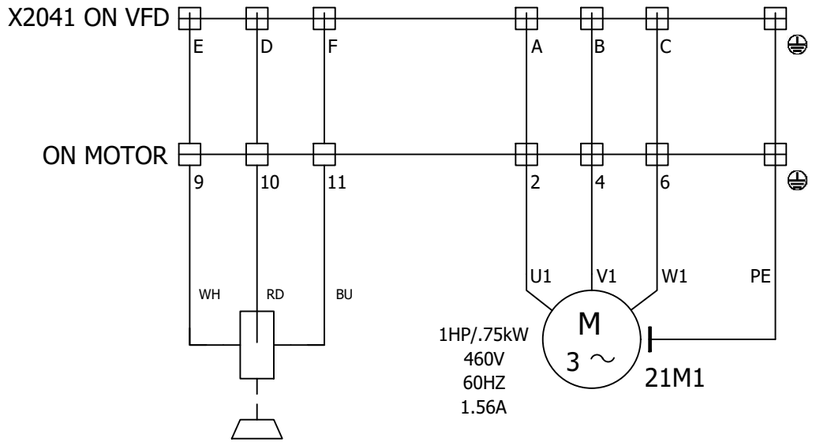
JOB # C40572

TOTAL PAGES: 171

SHEET 21.1 OF 56

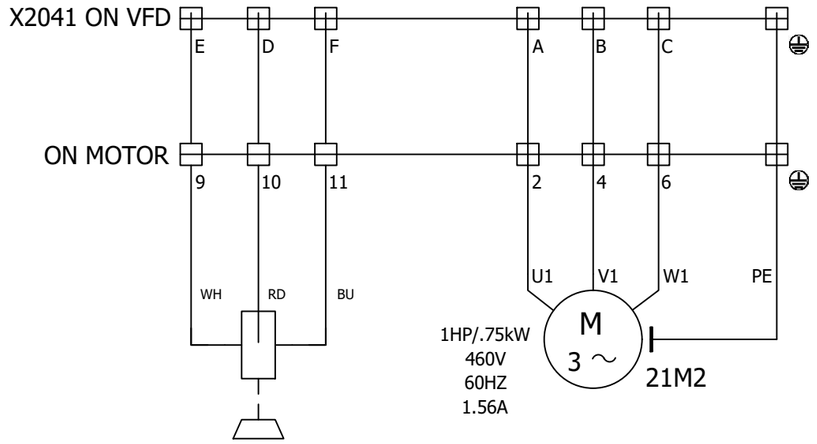
CHAIN CONVEYOR 1

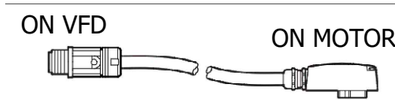
21VFD1



Transfer Conveyor 1

21VFD2





M23, coding ring: brown, male	IS2, female, ⅄	CE/UL: 28125940	LEONI LEHC® 000749	Variable	1.5 mm² / AC 500 V

Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 2 MOTOR POWER

REV	DESCRIPTION	DATE

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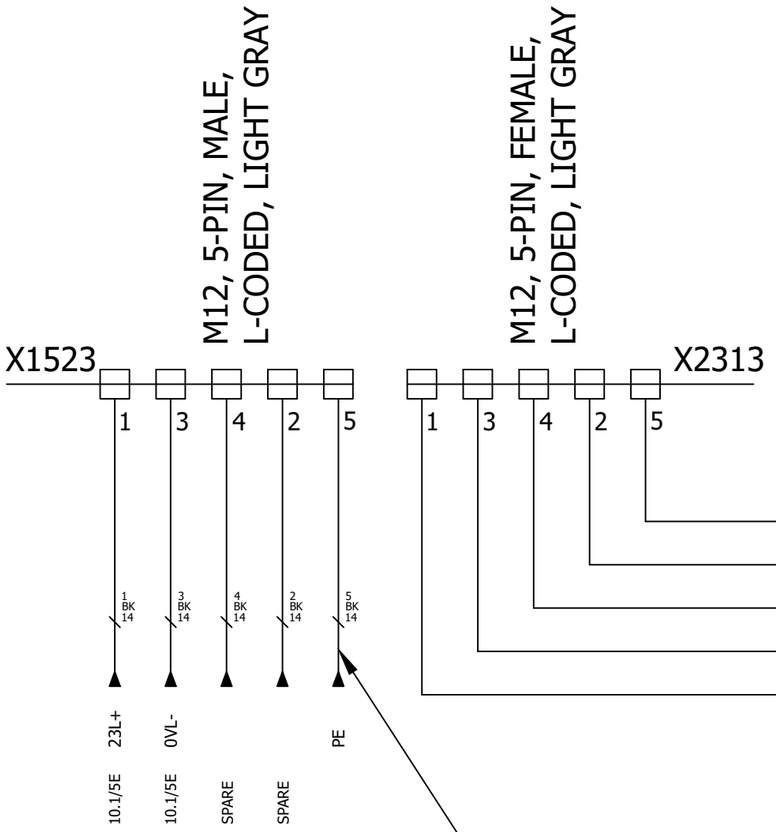
JOB # C40572

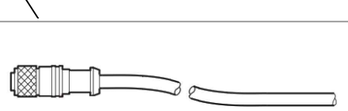
TOTAL PAGES: 171

SHEET 21.2 OF 56

CHAIN CONVEYOR 1

21VFD1



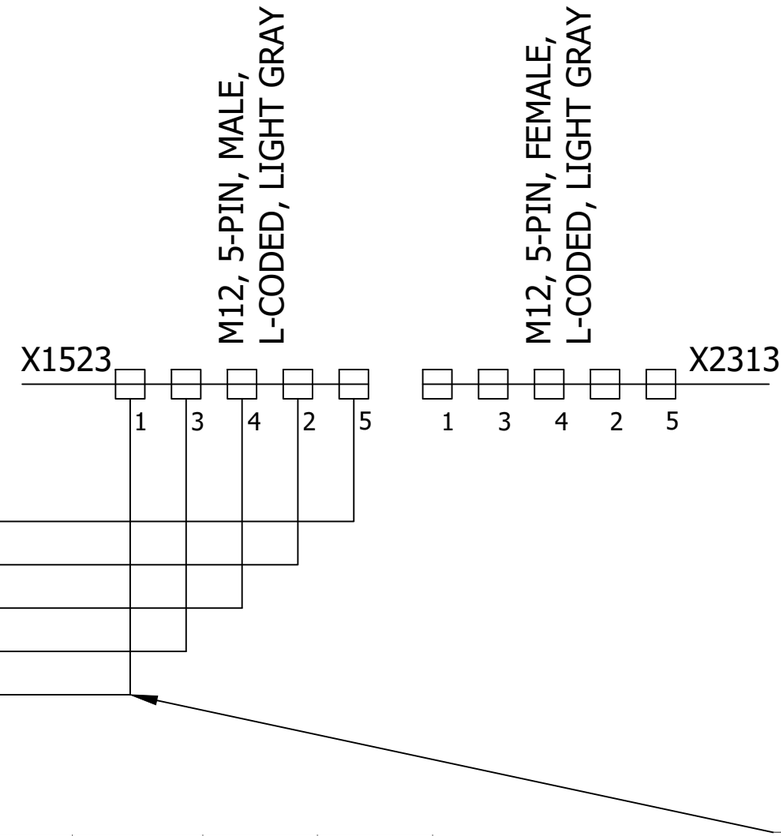


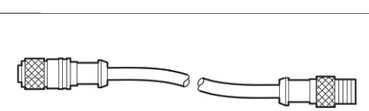
M12, 5-pin, L-coded, female — Open

CE/UL:	HELUKABEL® JZ-500	Variable	5 × 2.5 mm² / DC 60 V
28117786		✗	

Transfer Conveyor 1

21VFD2





M12, 5-pin, L-coded, female — M12, 5-pin, L-coded, male

CE/UL:	HELUKABEL® JZ-500	Variable	5 × 2.5 mm² / DC 60 V
28114345		✗	

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 2 VFD 24VDC POWER

REV	DESCRIPTION	DATE

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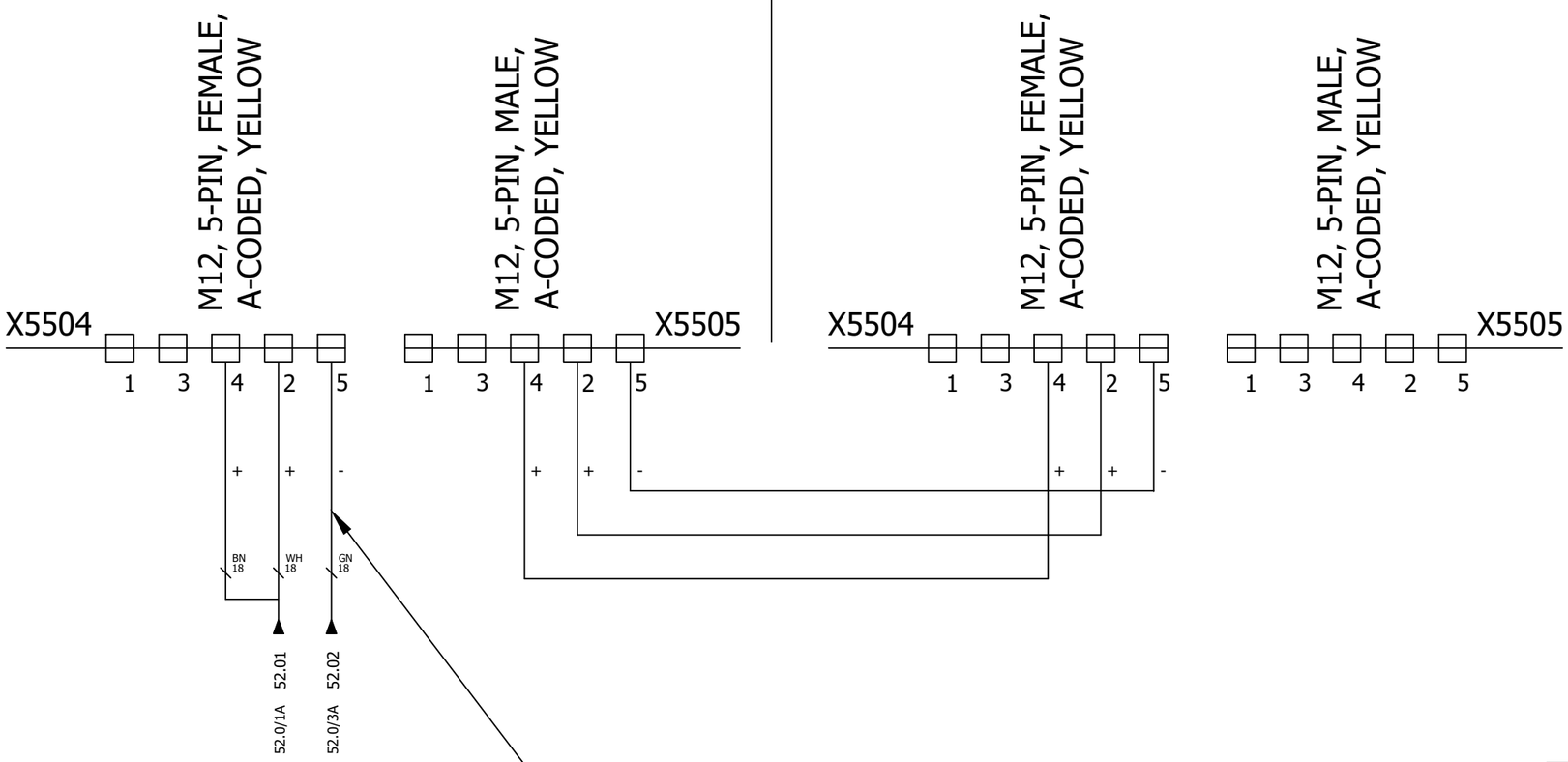
JOB # C40572

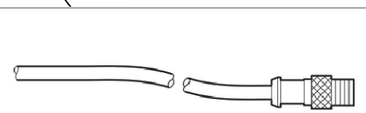
TOTAL PAGES: 171

SHEET 21.3 OF 56

CHAIN CONVEYOR 1

21VFD1

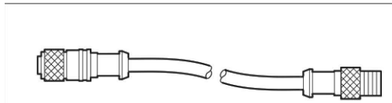




Open	M12, 5-pin, A-coded, male	CE/UL: 28110943	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V

Transfer Conveyor 1

21VFD2



M12, 5-pin, A-coded, female	M12, 5-pin, A-coded, male	CE/UL: 28110935	HELUKABEL® LiYCY	Variable	3 × 0.75 mm² / DC 60 V

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 2 VFD STO

REV	DESCRIPTION	DATE

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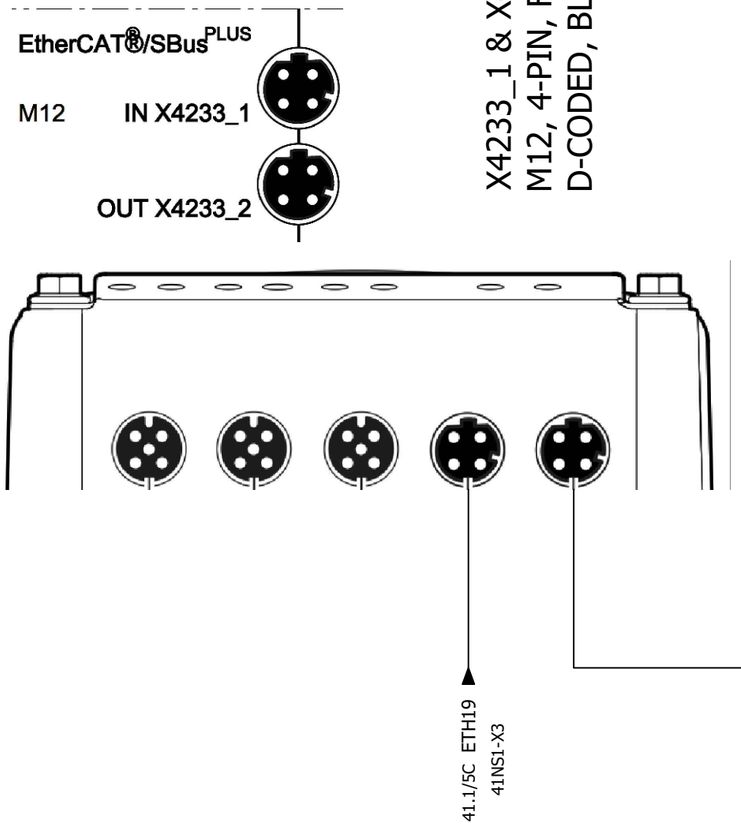
JOB # C40572

TOTAL PAGES: 171

SHEET 21.4 OF 56

Assignment		
Contact	**Signal**	**Description**
1	TX+	Transmit line (+)
2	RX+	Receive line (+)
3	TX-	Transmit line (-)
4	RX-	Receive line (-)

21VFD1



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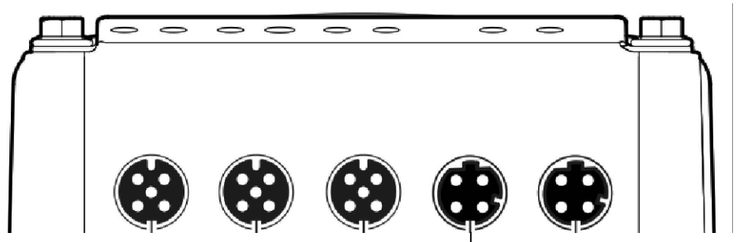

CHAIN CONVEYOR 1

Transfer Conveyor 1

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PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE

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PAGE DESCRIPTION: SPARE

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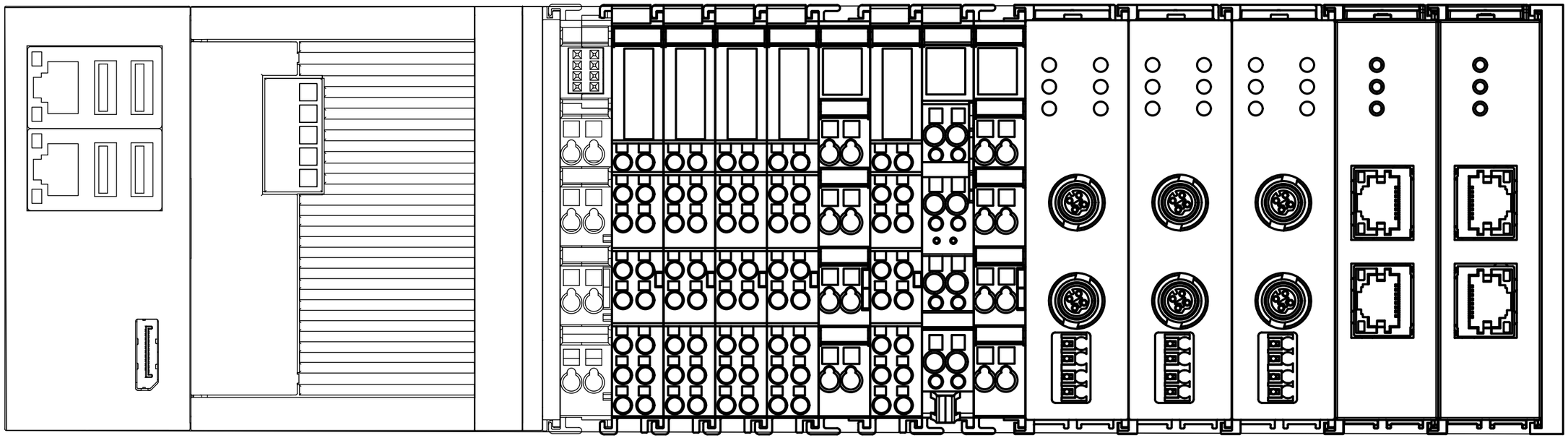
CX5330 - 0225

EL1809
EL1809
EL2809
EL2809
EL6910
EL1918
EL2912
EL9410
EL1322
EL1322
EL1322
EL6633
EK1122



23.0-1
23.0-2
23.0-3
23.0-4
23.0-5
23.0-6
23.0-7
23.0-8
23.0-9
23.0-10
23.0-11
23.0-12
23.0-13

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: PLC OVERVIEW RACK

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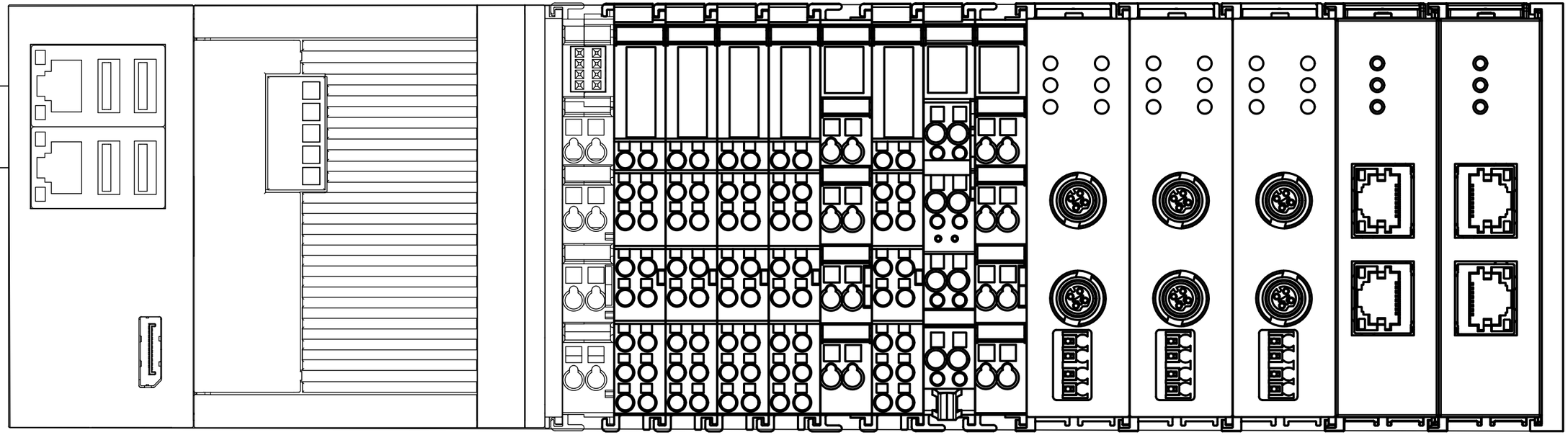
PANEL #7-10
FROM
23FMC1

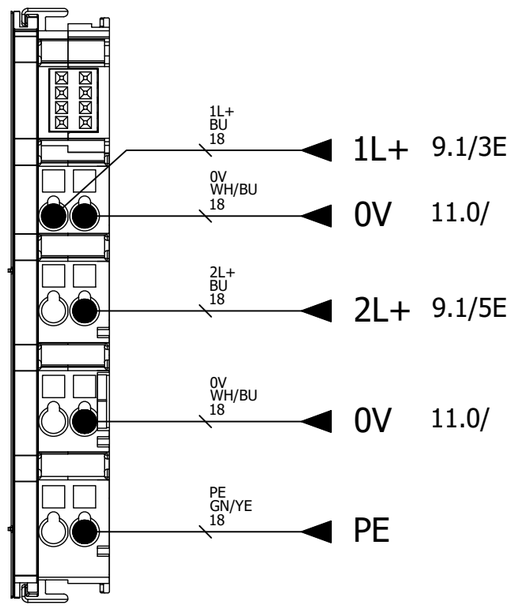
PANEL #6
FROM PLC
PANEL

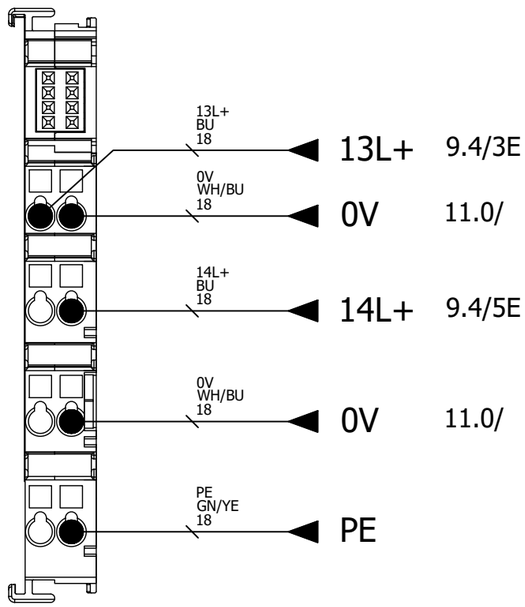
ETH1

39.1/3E
TO
39NS1-X1

ETH2







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LEDs for fieldbus diagnostics/power supply

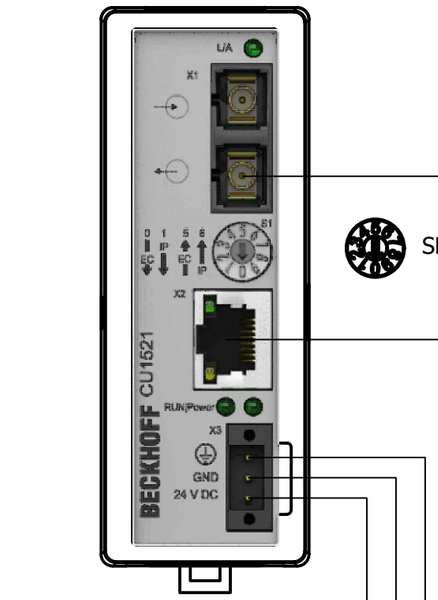
LED	Color	Display	State	Description
L/A (X1) green (x2)	green	off	-	no connection on the EtherCAT strand
		on	linked	EtherCAT device connected
		flashing	active	Communication with EtherCAT device
Power	green	off		No supply voltage
		on		24 V supply voltage present

LED diagnostics EtherCAT State Machine

LED	Color	Meaning	
RUN	green	This LED indicates the terminal's operating state:	
		off	State of the EtherCAT State Machine: **INIT** = initialization of the terminal
		single flash	State of the EtherCAT State Machine: **PREOP** = function for mailbox communication and different standard-settings set
		flashing	State of the EtherCAT State Machine: **SAFEOP** = verification of the sync manager channels and the distributed clocks. Outputs remain in safe state
		on	State of the EtherCAT State Machine: **OP** = normal operating state; mailbox and process data communication is possible
		flickering	State of the EtherCAT State Machine: **BOOTSTRAP** = function for terminal firmware updates
	orange	on	CU15x1 is in EtherCAT mode, rotary switch was moved during operation *Remedy*: Resetting or voltage reset
	red	on	CU15x1 is in Ethernet mode, rotary switch was moved during operation *Remedy*: Resetting or voltage reset
	red/ green	flashing	Invalid rotary switch position after power-on *Remedy*: Move rotary switch to valid position

IN PANEL
#7-10

23FMC1



23FE1 FROM PLC PANEL

SET TO 0

ETH1 TO PLC 23.1/2A

PE
GN/YE
20

5.0/3F 0V

0V
WH/BU
20

4.2/7E 9L+

9L+
BU
20

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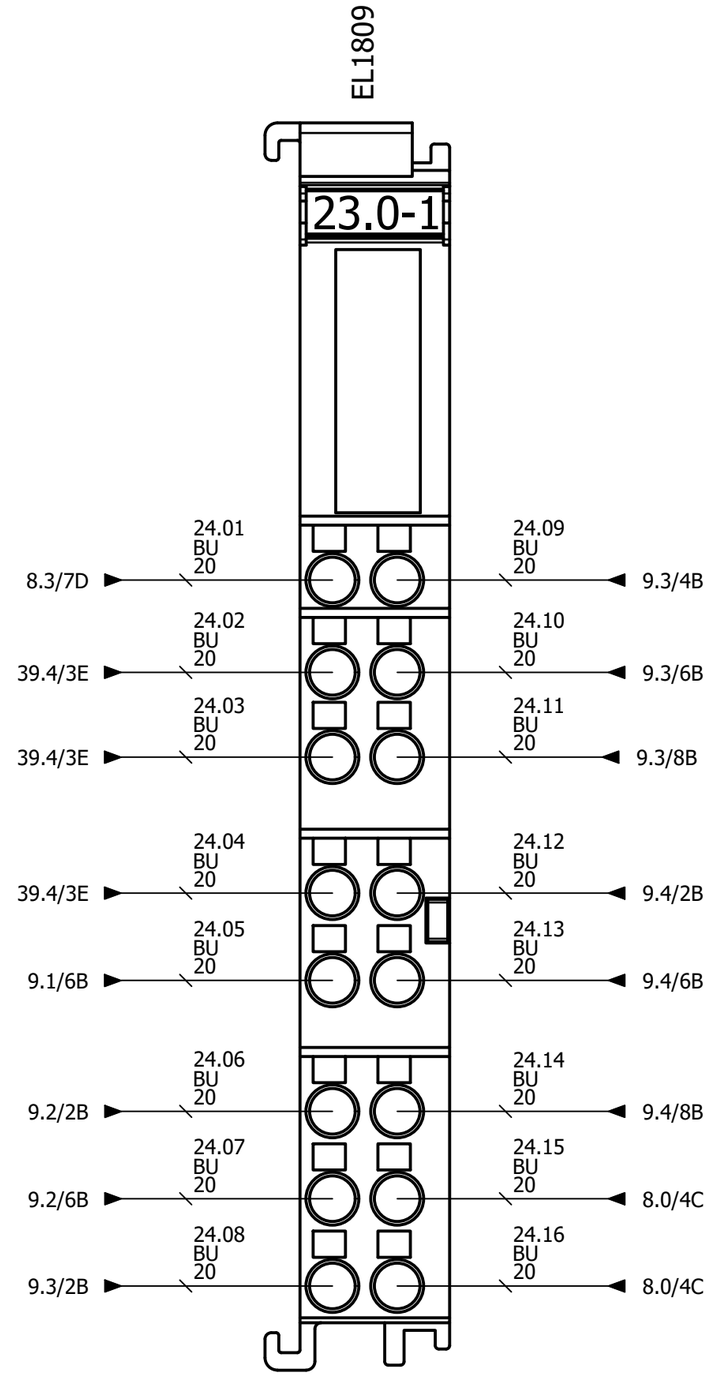

4.4/2E
11.0/

LM11
11L+ 0V

REDUNDANCY OK 8RS1
HANDHELD HMI KEY-SWITCH 1 NO
HANDHELD HMI KEY-SWITCH 2 NO
HANDHELD HMI CONNECTED
LM3-4 FAULT
LM5-6 FAULT
LM7-8 FAULT
LM9 FAULT

/9B /9B

8.3/7D
39.4/3E
39.4/3E
39.4/3E
9.1/6B
9.2/2B
9.2/6B
9.3/2B

9.3/4B
9.3/6B
9.3/8B
9.4/2B
9.4/6B
9.4/8B
8.0/4C
8.0/4C

/2F /2F

0V 11L+

LM10 FAULT
LM11 FAULT
LM12 FAULT
LM13-14 FAULT
LM15 FAULT
LM16 FAULT
8PS1 OUT 100%
8PS1 DC OK

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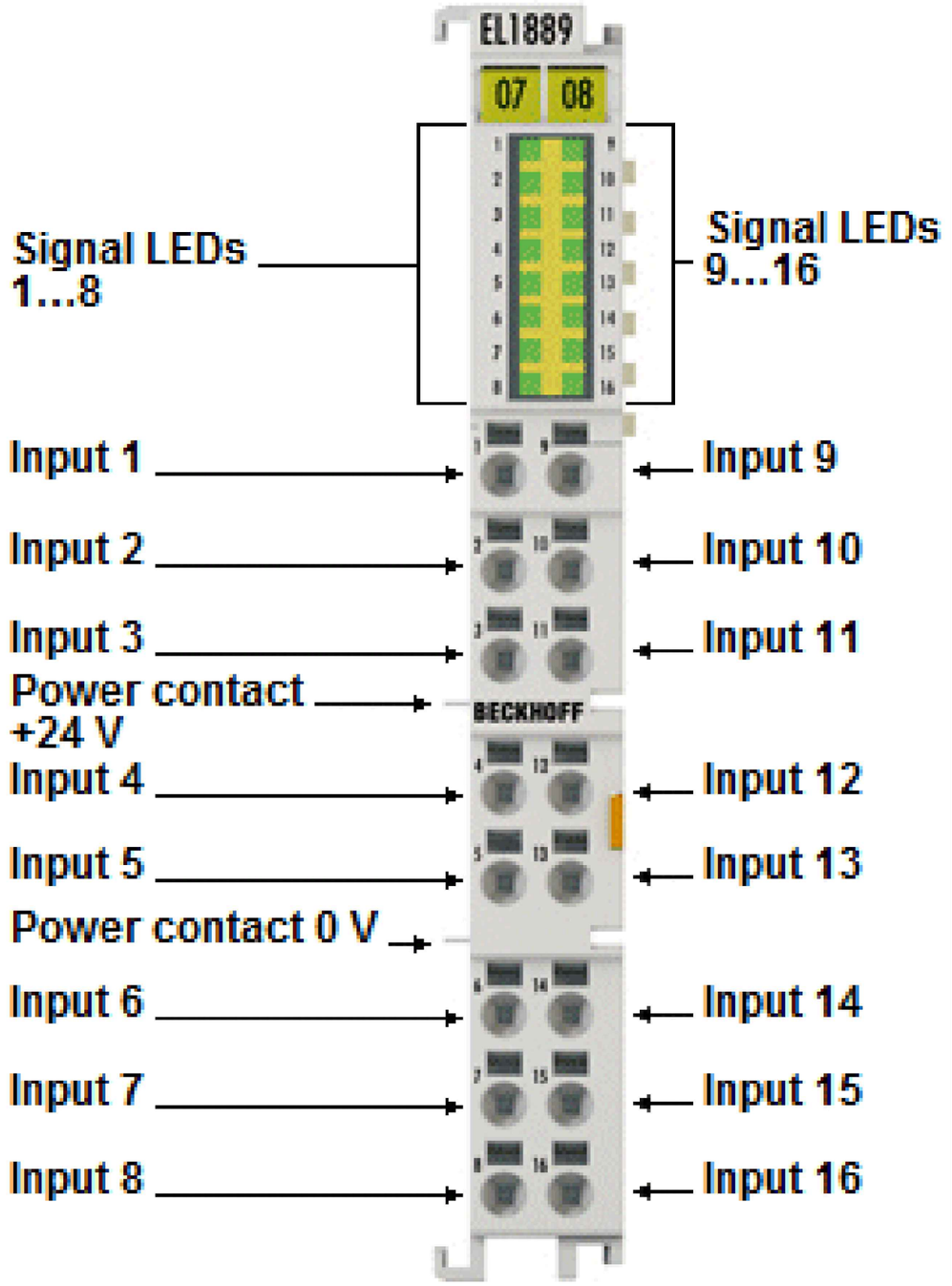

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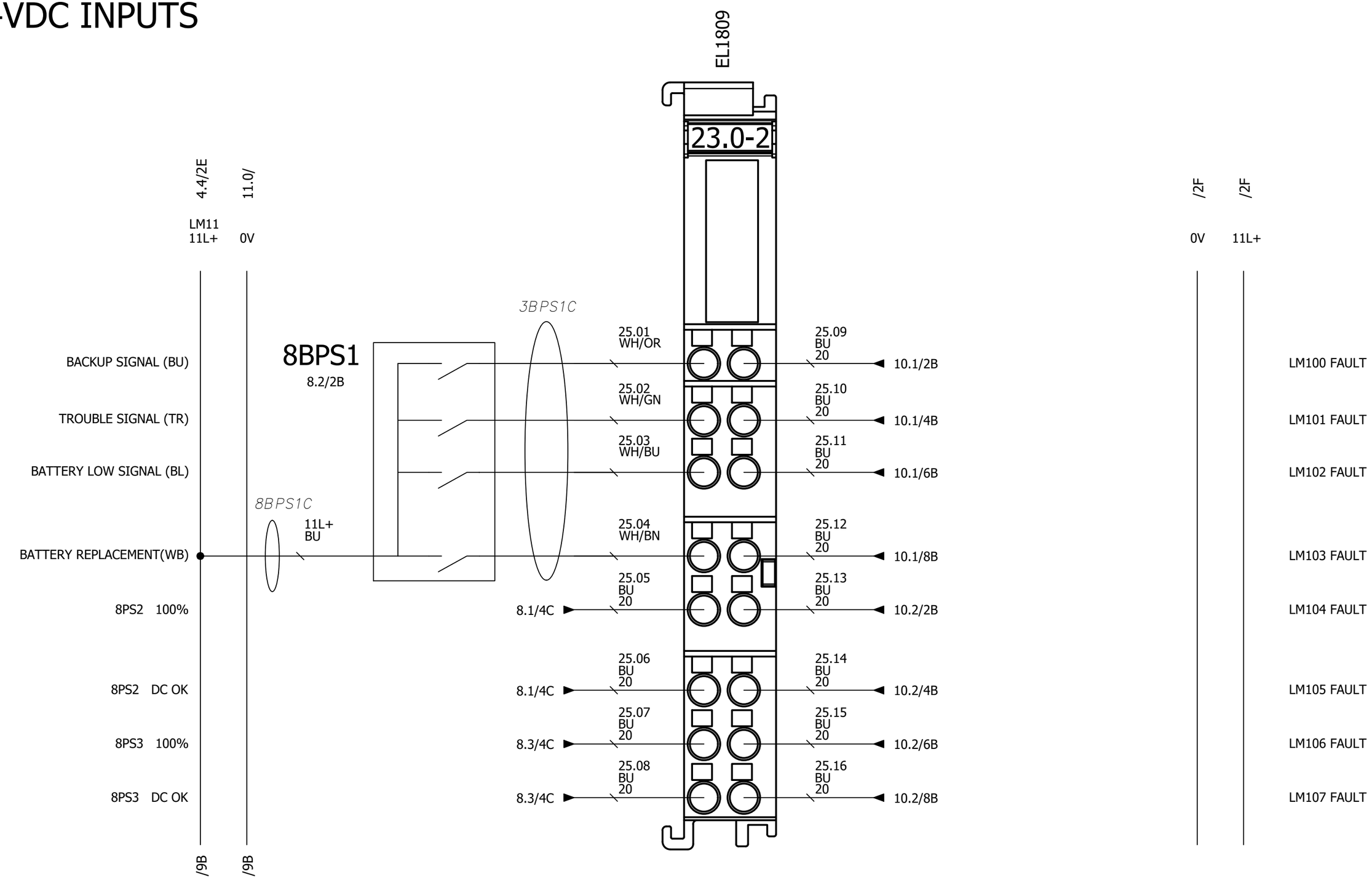

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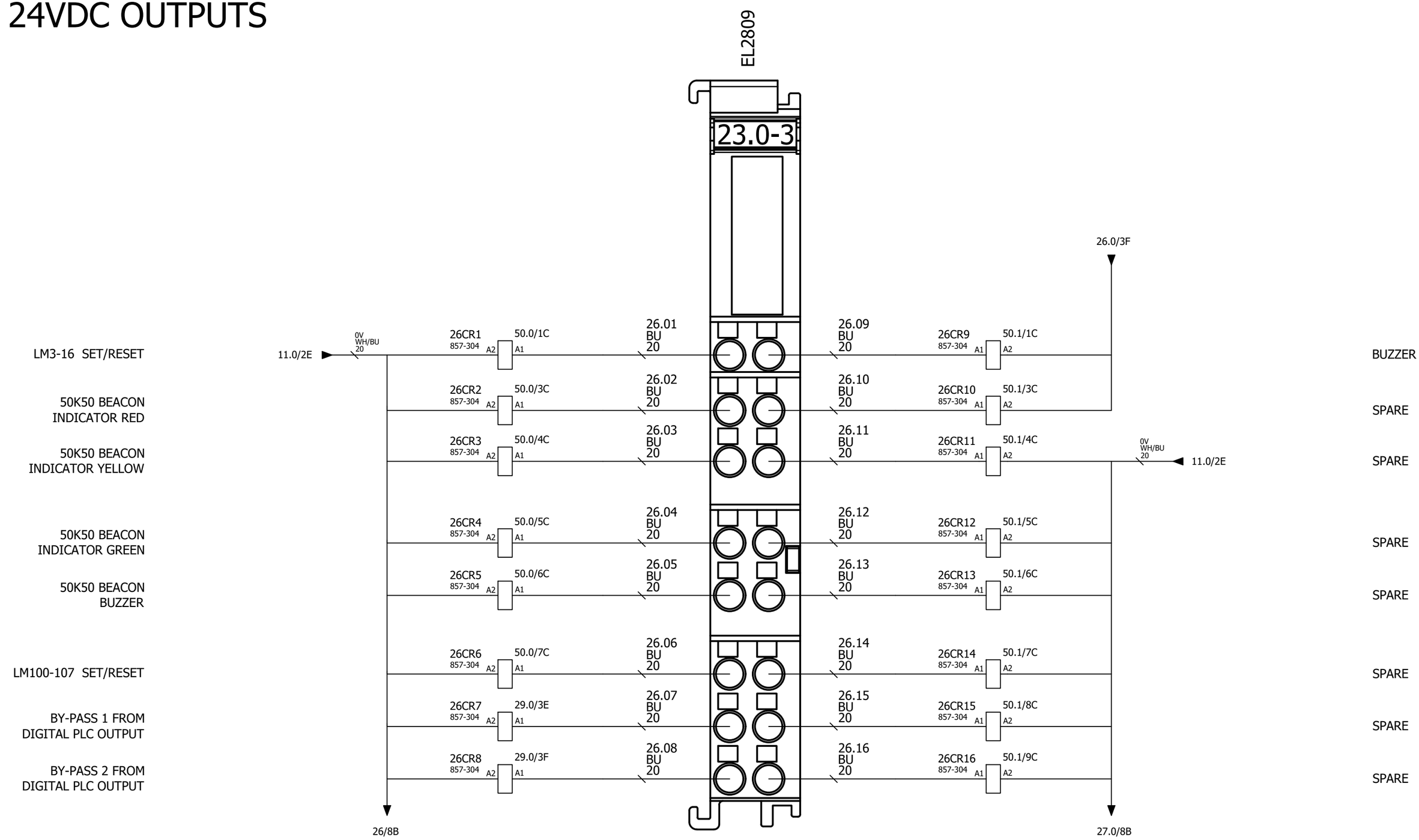

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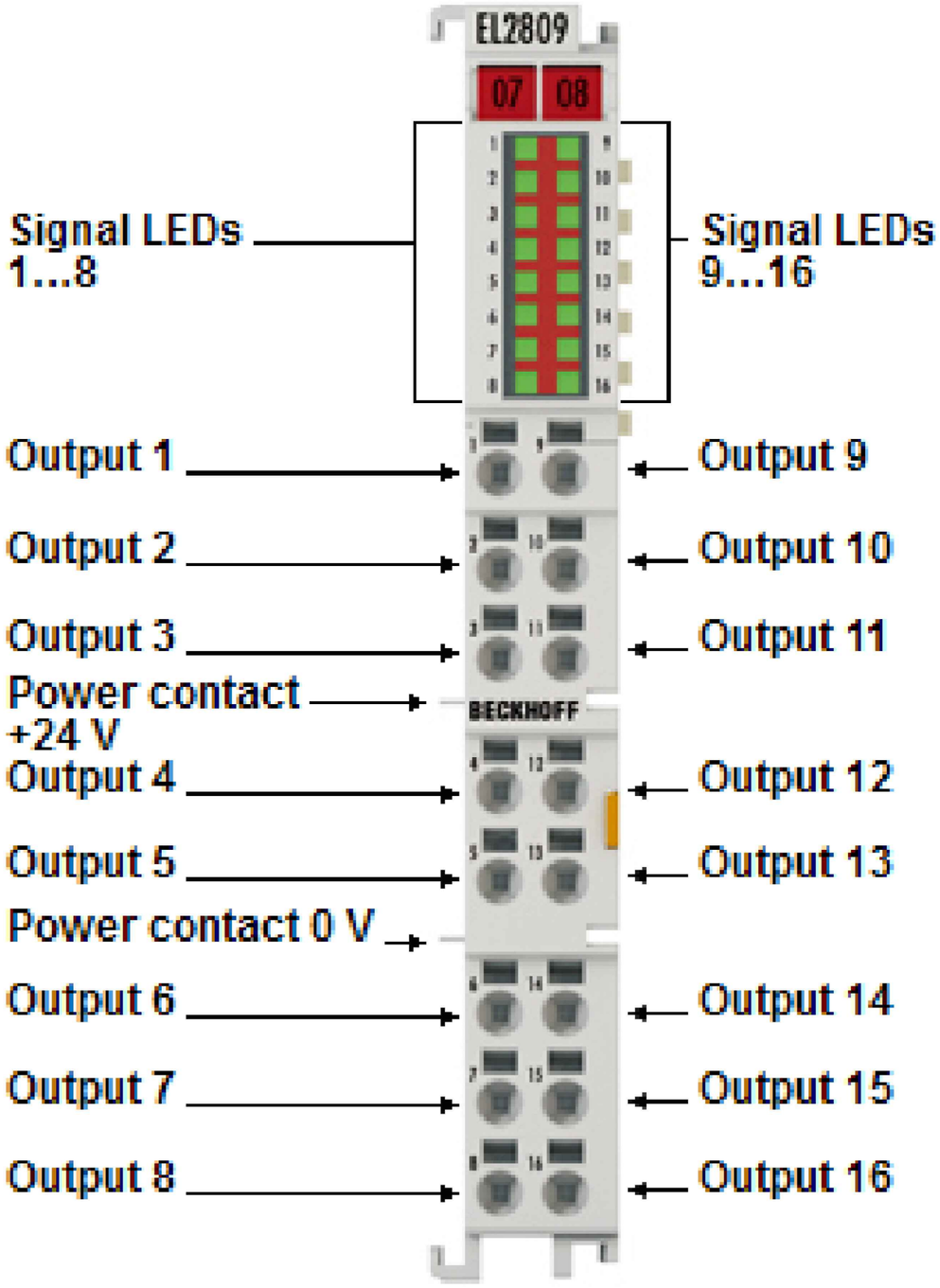

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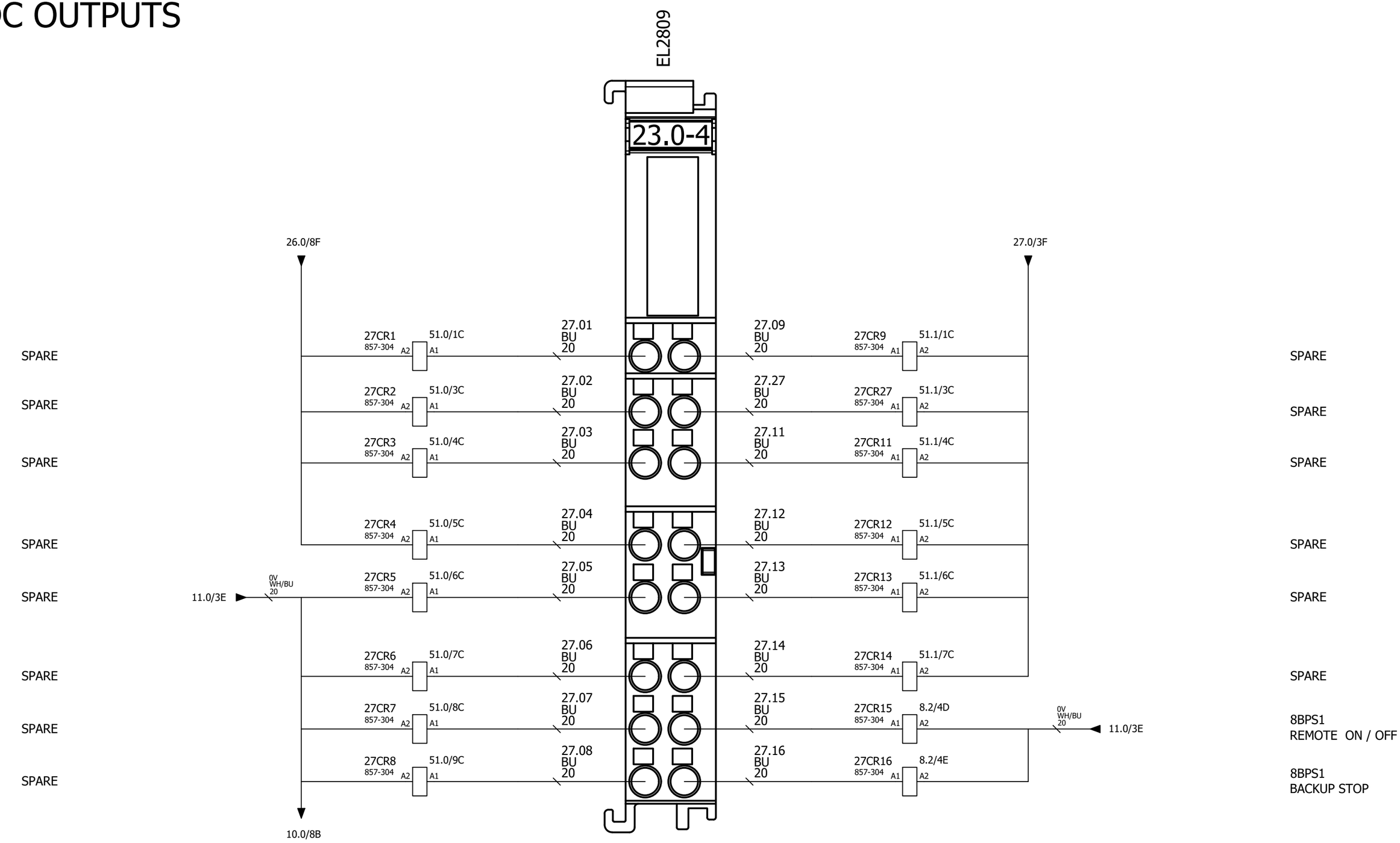

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PAGE DESCRIPTION: DRIVE LIFTING TRANSFER ROLLER CONVEYOR#3

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TOTAL PAGES: 171

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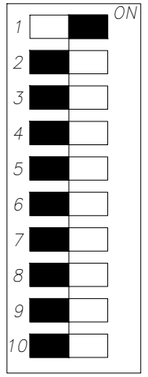
EL 6910 (SAFETY CPU CONTROLLER)

SET
DIPSWITCHES ON
THE MODULE



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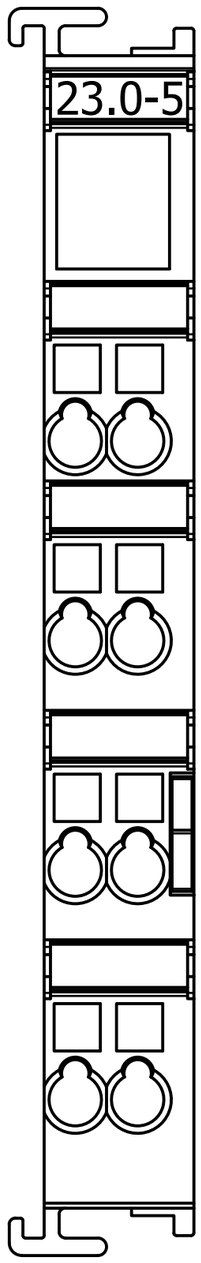

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PAGE DESCRIPTION: SAFETY CPU CONTROLLER 23.0-5

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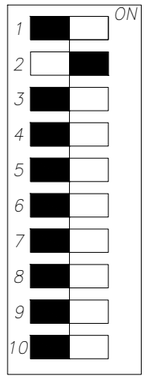
SAFETY INPUTS

SET
DIPSWITCHES ON
THE MODULE



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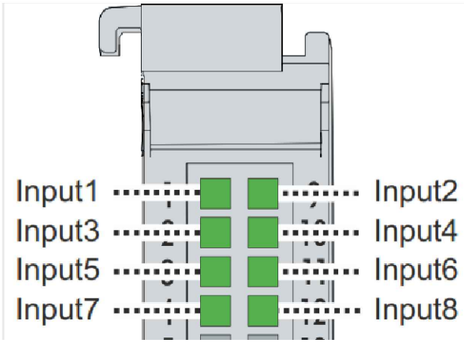

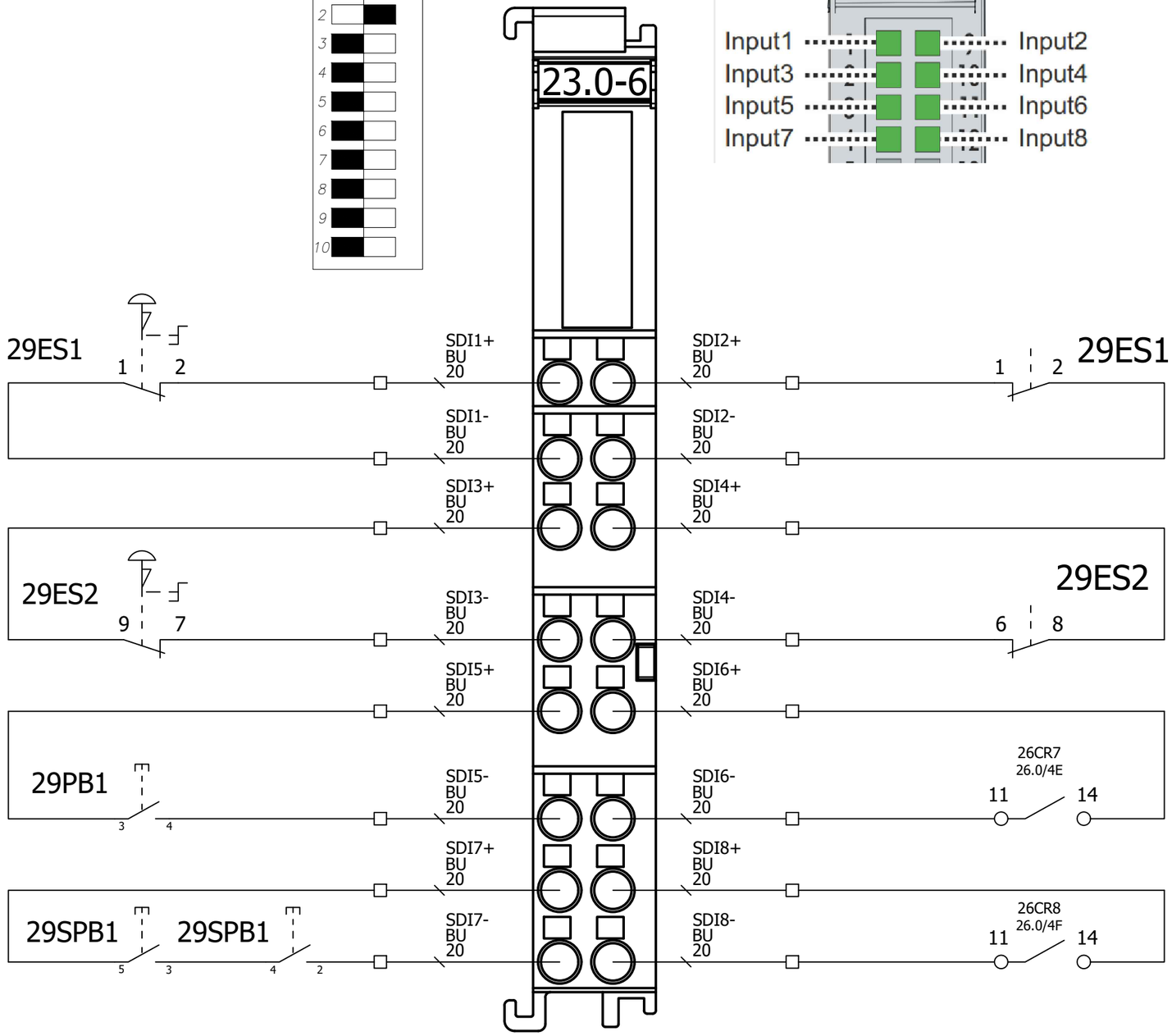

EMERGENCY STOP
ON DOOR

EMERGENCY STOP
ON DOOR

EMERGENCY STOP
ON HMI

EMERGENCY STOP
ON HMI

Emergency Stop
RESET ON DOOR

BY-PASS 1 FROM
DIGITAL PLC OUTPUT

Enabling switch
ON HMI

BY-PASS 2 FROM
DIGITAL PLC OUTPUT

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: SAFETY INPUTS 23.0-6

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TOTAL PAGES: 171

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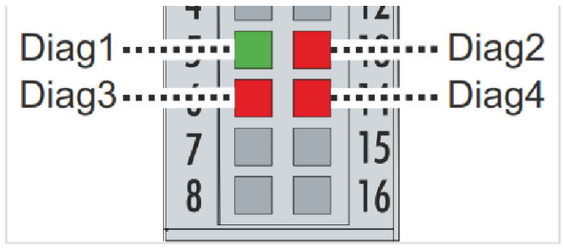
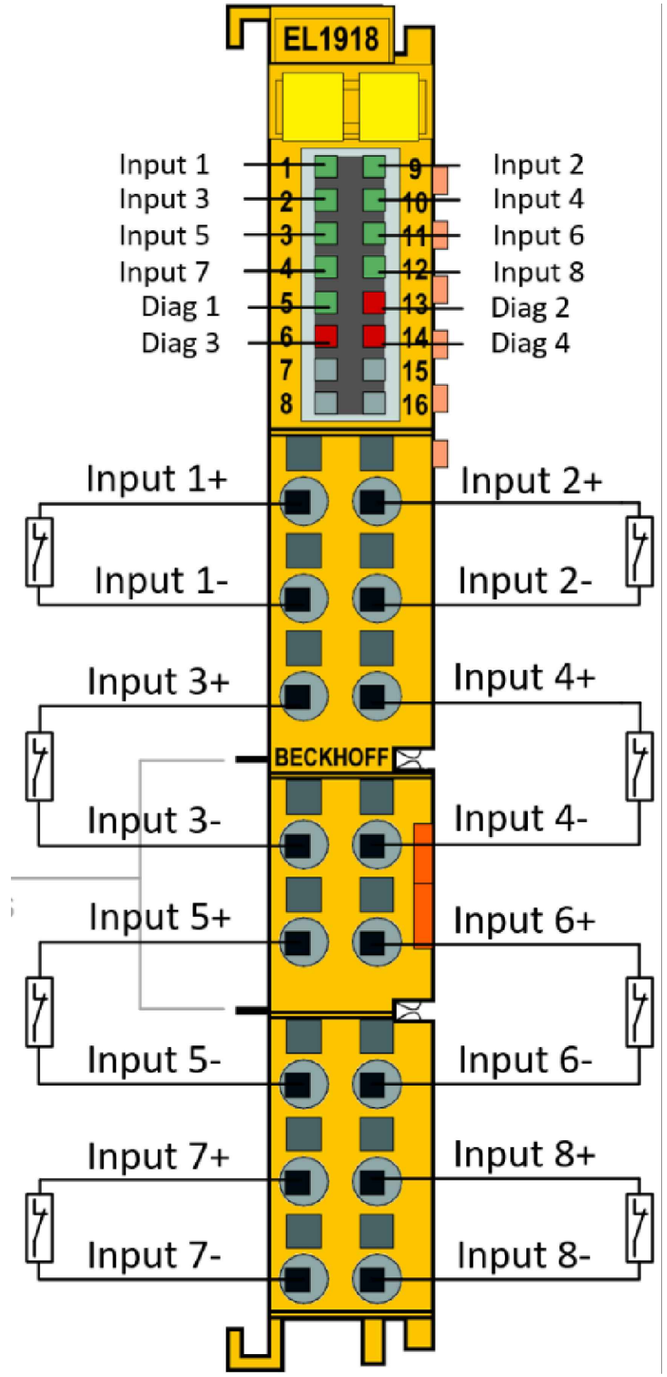
DIP switch										Address
1	2	3	4	5	6	7	8	9	10	
ON	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	1
OFF	**ON**	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	2
ON	**ON**	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	3
OFF	OFF	**ON**	OFF	OFF	OFF	OFF	OFF	OFF	OFF	4
ON	OFF	**ON**	OFF	OFF	OFF	OFF	OFF	OFF	OFF	5
OFF	**ON**	**ON**	OFF	OFF	OFF	OFF	OFF	OFF	OFF	6
ON	**ON**	**ON**	OFF	OFF	OFF	OFF	OFF	OFF	OFF	7
...	...	...	...	...	...	...	...	...	...	...
ON	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	**ON**	1023



Diag1	Diag2	Diag3	Diag4	Description
off	off	off	off	The device has no power supply.
lights up	off	-	-	The ambient conditions, operating voltage and internal tests are within the valid range.
lights up	flashes	-	-	Internal tests are outside the valid range. Diag2 outputs a logic error code. For more information, see the chapter Logic error codes [▶ 57].
off	flashes	-	-	Ambient conditions or electrical parameters are outside the valid range. Diag2 outputs an environment error code. For more information, see the chapter Environment error codes [▶ 57].
lights up	lights up	lights up	off	There is a global shutdown on µC1.
lights up	lights up	off	lights up	There is a global shutdown on µC2.
lights up	off	lights up	off	There is a global fault on µC1.
lights up	off	off	lights up	There is a global fault on µC2.
lights up	flickers	-	-	There is an error in a safe input or output module.



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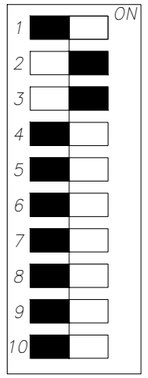
VFD's STO #1

SET
DIPSWITCHES ON
THE MODULE



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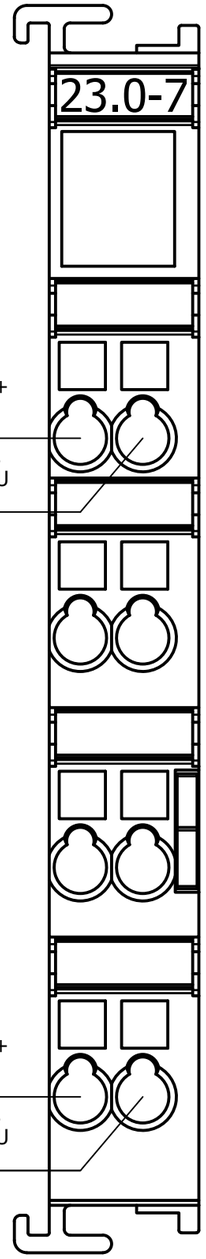
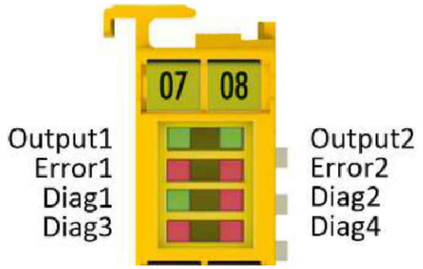
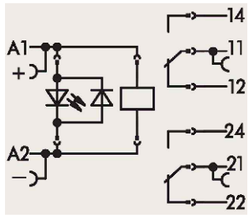




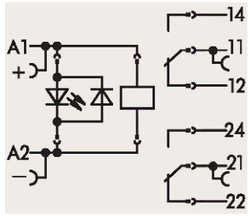
30SCR1
788-384
52.0/1C
A2
A1

SDO1+
BU
20

SDO1-
WH/BU
20



Safety relay SR2M DPDT (2 C/O), with force guided contacts and power indicator



Safety relay SR2M DPDT (2 C/O), with force guided contacts and power indicator

30SCR2
788-384
52.0/4C
A2
A1

SDO2+
BU
20

SDO2-
WH/BU
20

VFD's STO #1

CAROUSEL CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 5 VFD's

CAROUSEL CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

CAROUSEL CONVEYORS AREA VFD's
SECTION 3 TOTAL OF 4 VFD's

VFD's STO #2

VFD's STO #2

VTD STATIC AREA DRIVE

VTD MOBILE EXTRACTOR CONVEYOR AREA VFD

BUFFER CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 3 VFD's

BUFFER CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: SAFETY OUTPUTS 23.0-7

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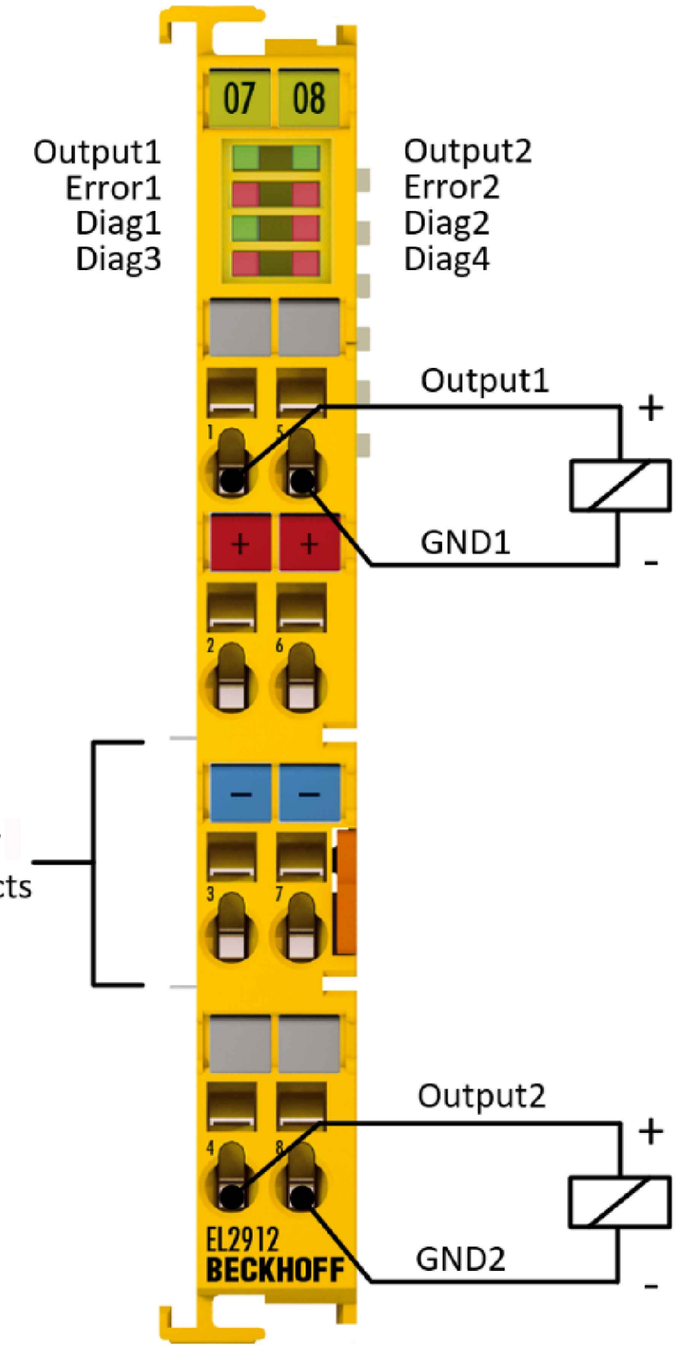
LED	lit	flashes	off
Diag 1 (green)	Environment variables, operating voltage and internal tests are in the valid range • If Diag 2 flashes, a logic error code applies	-	Environment variables, operating voltage and internal tests are outside the valid range • If Diag 2 flashes, an environment error code applies
Diag 2 (red)	Together with Diag 3 and 4: Global shutdown[1)] has occurred. (see diag history of the TwinSAFE components)	Logic or environment error code according to Diag1 and tables below is output	Together with Diag 3 and 4: Global fault[1)] has occurred. (see diag history of the TwinSAFE components)
Diag 3 (red)	Global fault or global shutdown on µC1[1)]	-	No global fault or global shutdown on µC1[1)]
Diag 4 (red)	Global fault or global shutdown on µC2[1)]	-	No global fault or global shutdown on µC2[1)]

LED	Color	Description
Output 1	green	Status and error display for the respective output
Error 1	red	LED is lit: output/error is set
Output 2	green	LED is not lit: output is not set or there is no error
Error 2	red	



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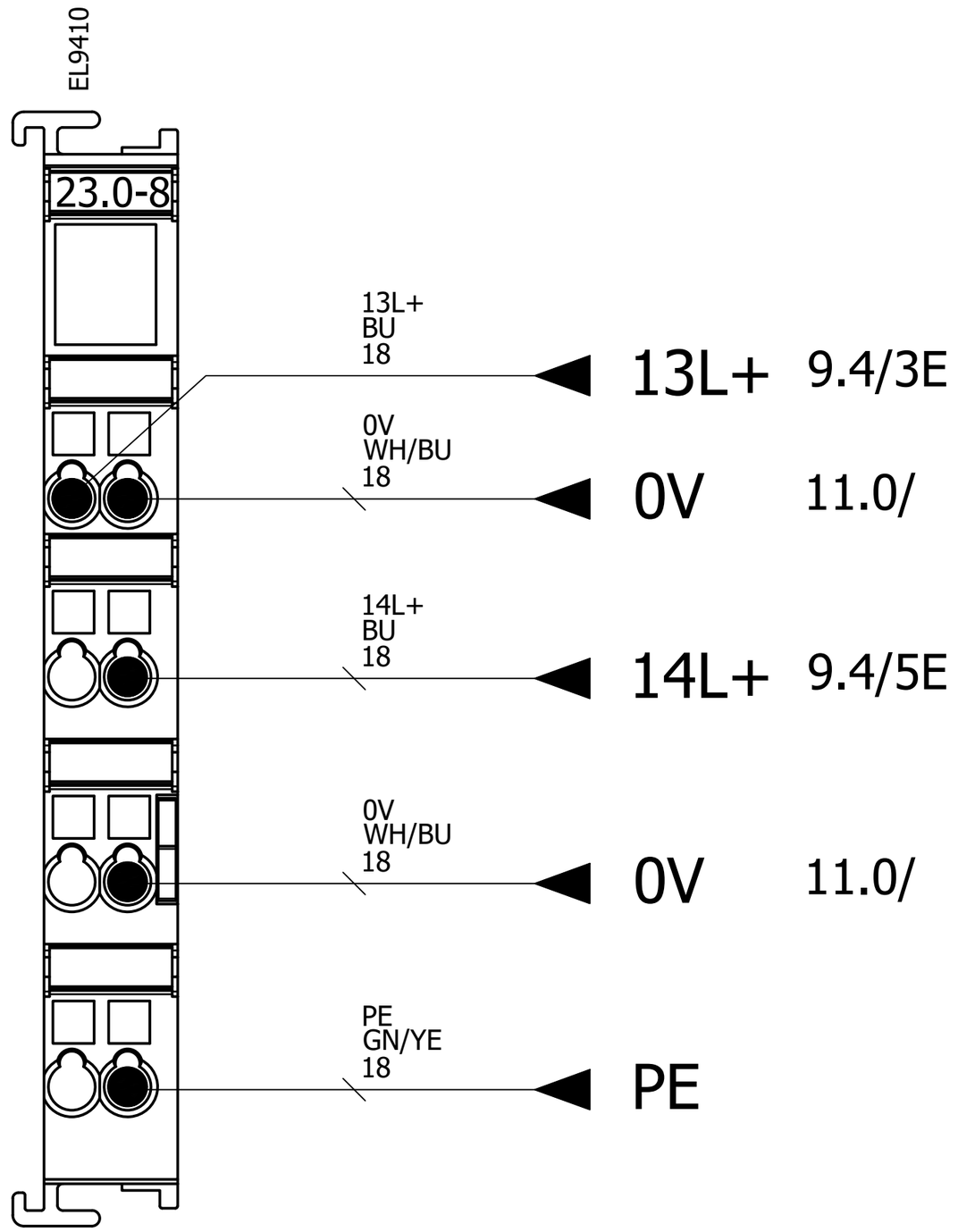

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: DC SUPPLY FOR NETWORK INTERFACE CARDS 23.0-8

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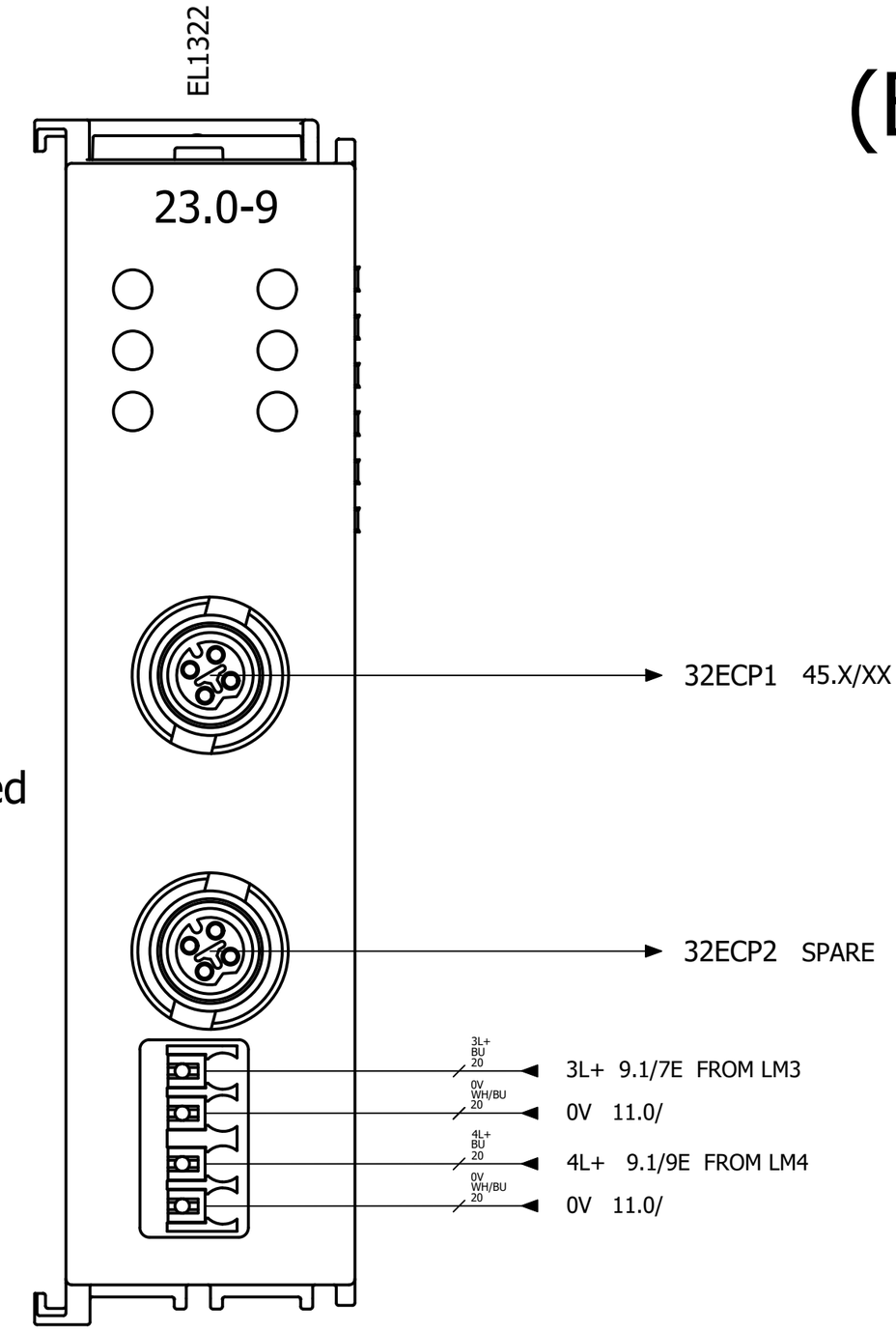
(EtherCAT P)

Terminal point		Description
Name	No.	
U_S 24 V	1	+24 V Feed-In U_S (System- and Sensor supply)
U_S 0 V	2	0 V of System- and Sensor supply U_S
U_P 24 V	3	+24 V Feed-In U_P (Peripheral voltage for Actors)
U_P 0 V	4	0 V of Peripheral voltage for Actors U_P

M8 socket, shielded, screw type, EtherCAT P-coded



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE CARD 23.0-9

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TOTAL PAGES: 171

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LED		Display	State	Description
Run	green	off	INIT	EtherCAT P module is in state Init
		flashing uniformly	PREOP	EtherCAT P module is in state Pre-Operational
		flashing slowly	SAFEOP	EtherCAT P module is in state Safe-Operational
		on	OP	EtherCAT P module is in state Operational

LEDs for fieldbus diagnostics

LED		Display	State	Description
X1 L/A	green	off	-	No connection on the previous EtherCAT P strand
		on	link	Previous EtherCAT P device connected
		flashing	active	Communication with previous EtherCAT P device
X2 L/A	green	off	-	No connection on the subsequent EtherCAT P strand
		on	link	Subsequent EtherCAT P device connected
		flashing	active	Communication with subsequent EtherCAT P device

LEDs power supply diagnostics

LED		Display	Description
U_S	green	off	System- and Sensor supply U_S not present
		on	System- and Sensor supply U_S present
	red	off	System- and Sensor supply U_S >= 19.4 V, no overload/ no case of short circuit
		on	System- and Sensor supply U_S < 19.4 V or overload/ case of short circuit (output current > 3 A)
U_P	green	off	Peripheral voltage for Actors U_P not present
		on	Peripheral voltage for Actors U_P present
	red	off	Peripheral voltage for Actors U_P >= 19.4 V, no overload/ no case of short circuit
		on	Peripheral voltage for Actors U_P < 19.4 V or overload/ case of short circuit (output current > 3 A)

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TOTAL PAGES: 171

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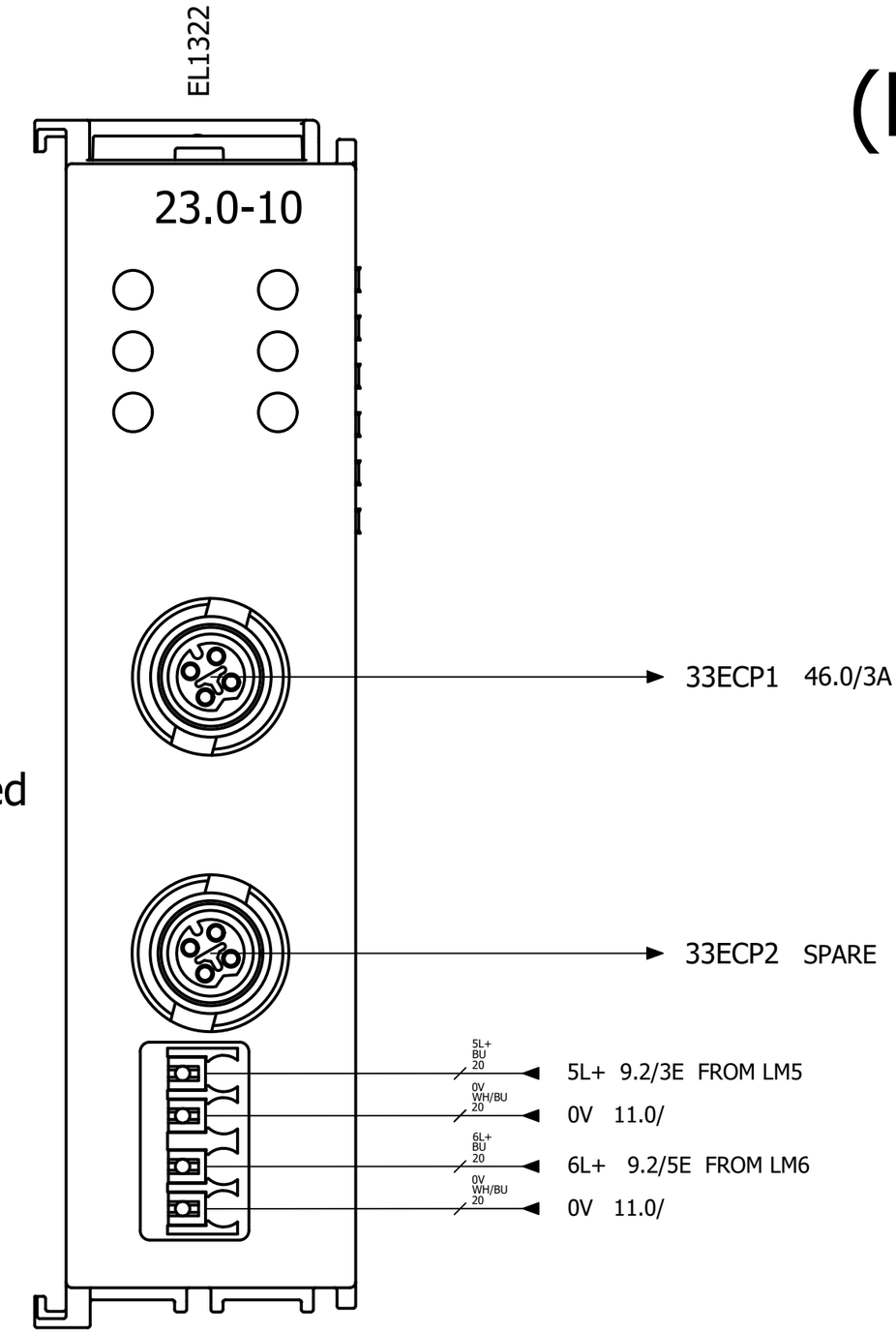
(EtherCAT P)

Terminal point		Description
Name	No.	
U_S 24 V	1	+24 V Feed-In U_S (System- and Sensor supply)
U_S 0 V	2	0 V of System- and Sensor supply U_S
U_P 24 V	3	+24 V Feed-In U_P (Peripheral voltage for Actors)
U_P 0 V	4	0 V of Peripheral voltage for Actors U_P

M8 socket, shielded, screw type, EtherCAT P-coded



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TOTAL PAGES: 171

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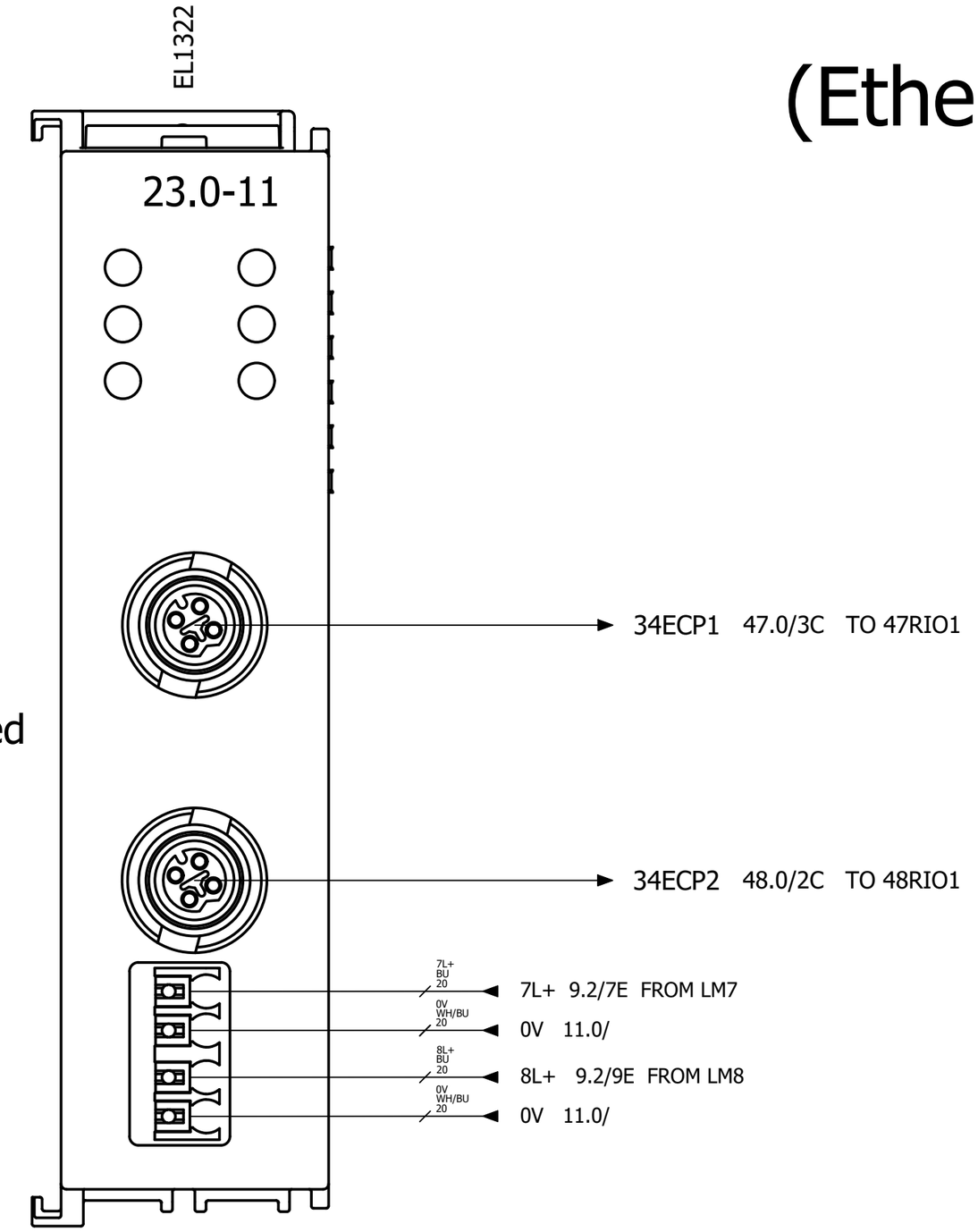
(EtherCAT P)

Terminal point		Description
Name	No.	
U_S 24 V	1	+24 V Feed-In U_S (System- and Sensor supply)
U_S 0 V	2	0 V of System- and Sensor supply U_S
U_P 24 V	3	+24 V Feed-In U_P (Peripheral voltage for Actors)
U_P 0 V	4	0 V of Peripheral voltage for Actors U_P

M8 socket, shielded, screw type, EtherCAT P-coded



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PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE CARD 23.0-11

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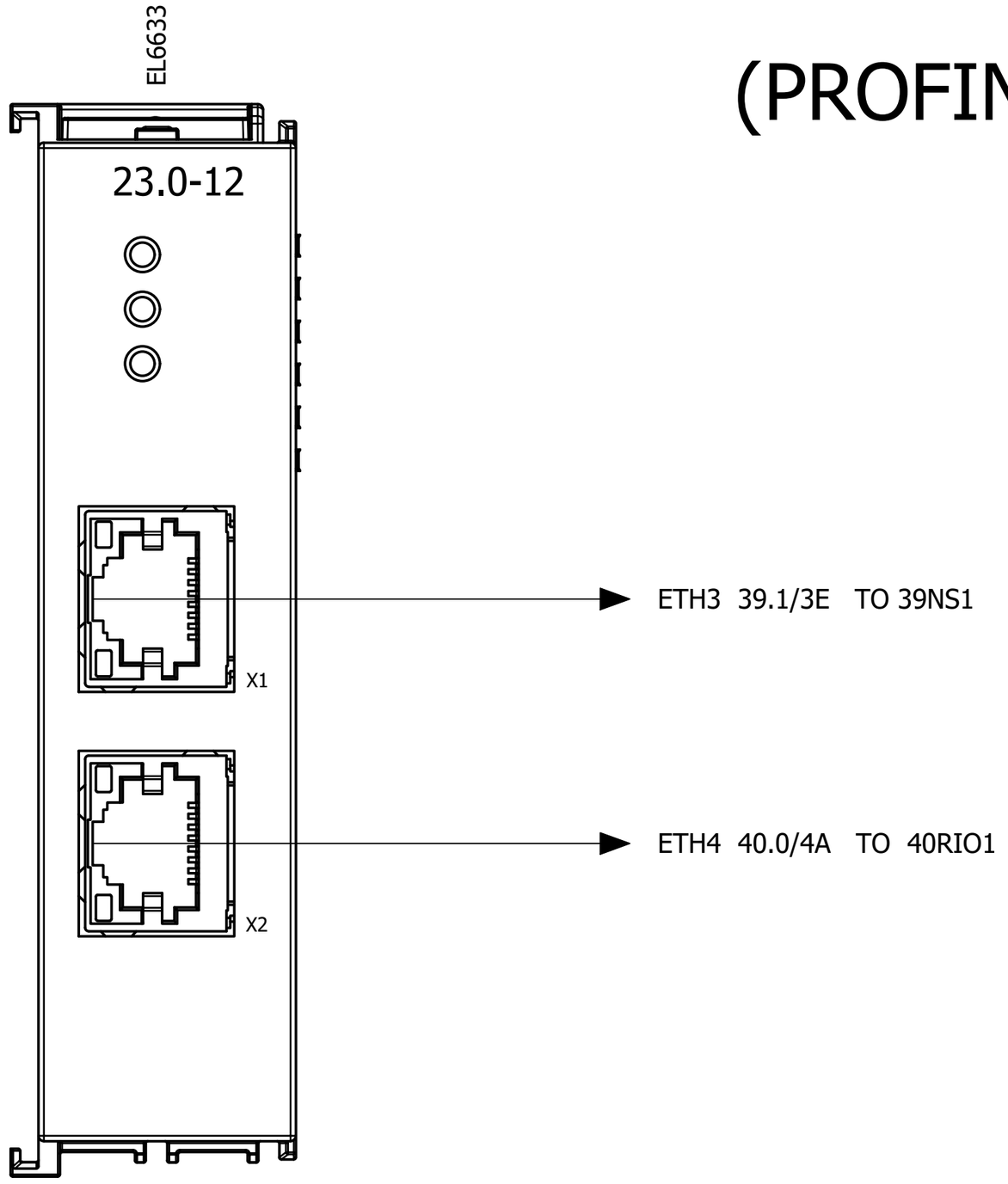
(PROFINET)



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LED		Display	Description
RUN	green	off	State of the EtherCAT State Machine: **INIT** = initialization of the terminal; **BOOTSTRAP** = function for firmware updates of the terminal
		flashing 200 ms	State of the EtherCAT State Machine: **PREOP** = function for mailbox communication and different standard-settings set
		off (1 s) on (200 ms)	State of the EtherCAT State Machine: **SAFEOP** = verification of the sync manager channels and the distributed clocks. Outputs remain in safe state
		on	State of the EtherCAT State Machine: **OP** = normal operating state; mailbox and process data communication is possible

Color green	Color red	Meaning
off	flashing 200 ms	Terminal starts
flashing 200 ms	off	No name
1 s off, 200 ms on	off	No IP address
on	off	EL terminal is parameterized

Colours green	Colours red	Meaning
off	flashing 500 ms	no AR established with any device
off	1 s off, 200 ms on	At least one device has not established an AR
1 s off, 200 ms on	off	At least one device has signaled an error, e.g. that there is a module difference or that the error bit for an IO-CR is set (Problem Indicator)
flashing 200 ms	off	At least one device is signaling that its status is 'Stop' (Provider State Stop)
on	off	All PROFINET devices are in data exchange mode

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: PROFINET NETWORK INTERFACE CARD 23.0-12

REV	DESCRIPTION	DATE

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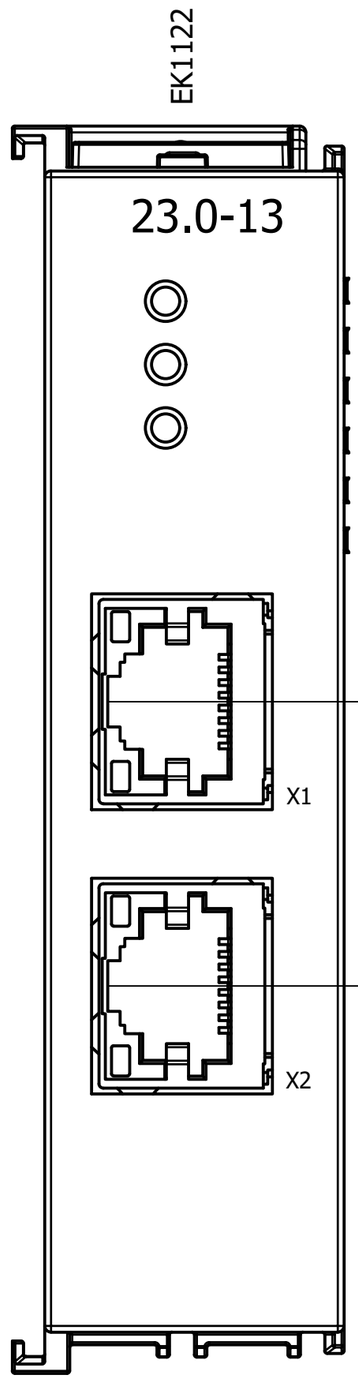
JOB # C40572

TOTAL PAGES: 171

SHEET 35.1 OF 56

(EtherCAT)



ETH5 41.1/7A 41NS1

ETH6 42.1/2E 42NS1

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PAGE DESCRIPTION: EtherCAT NETWORK INTERFACE CARD 23.0-13

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LEDs for fieldbus diagnostics

LED		Display	State	Description
LINK / ACT (X1/X2)	green	off	-	no connection on the EtherCAT strand
		on	linked	EtherCAT device connected
		flashing	active	Communication with EtherCAT device

LED diagnostics EtherCAT state machine

LED	Color	Meaning	
RUN	green	This LED indicates the terminal's operating state:	
		off	State of the EtherCAT State Machine: **INIT** = initialization of the terminal
		single flash	State of the EtherCAT State Machine: **PREOP** = function for mailbox communication and different standard-settings set
		flashing	State of the EtherCAT State Machine: **SAFEOP** = verification of the sync manager channels and the distributed clocks. Outputs remain in safe state
		on	State of the EtherCAT State Machine: **OP** = normal operating state; mailbox and process data communication is possible
		flickering	State of the EtherCAT State Machine: **BOOTSTRAP** = function for terminal firmware updates

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT NETWORK INTERFACE CARD 23.0-13

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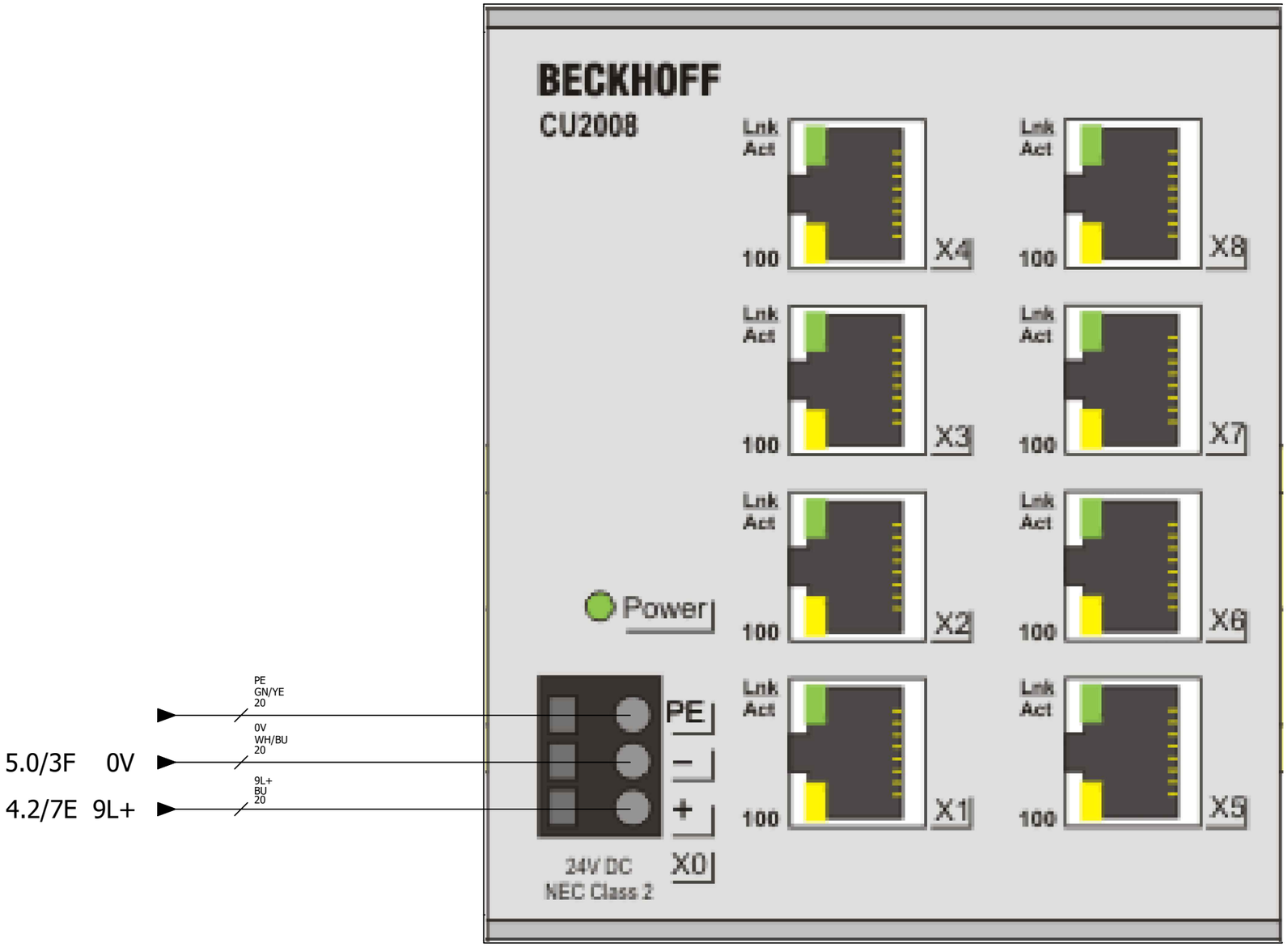
39NS1



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PAGE DESCRIPTION: NETWORK SWITCH 39NS1 POWER 24VDC

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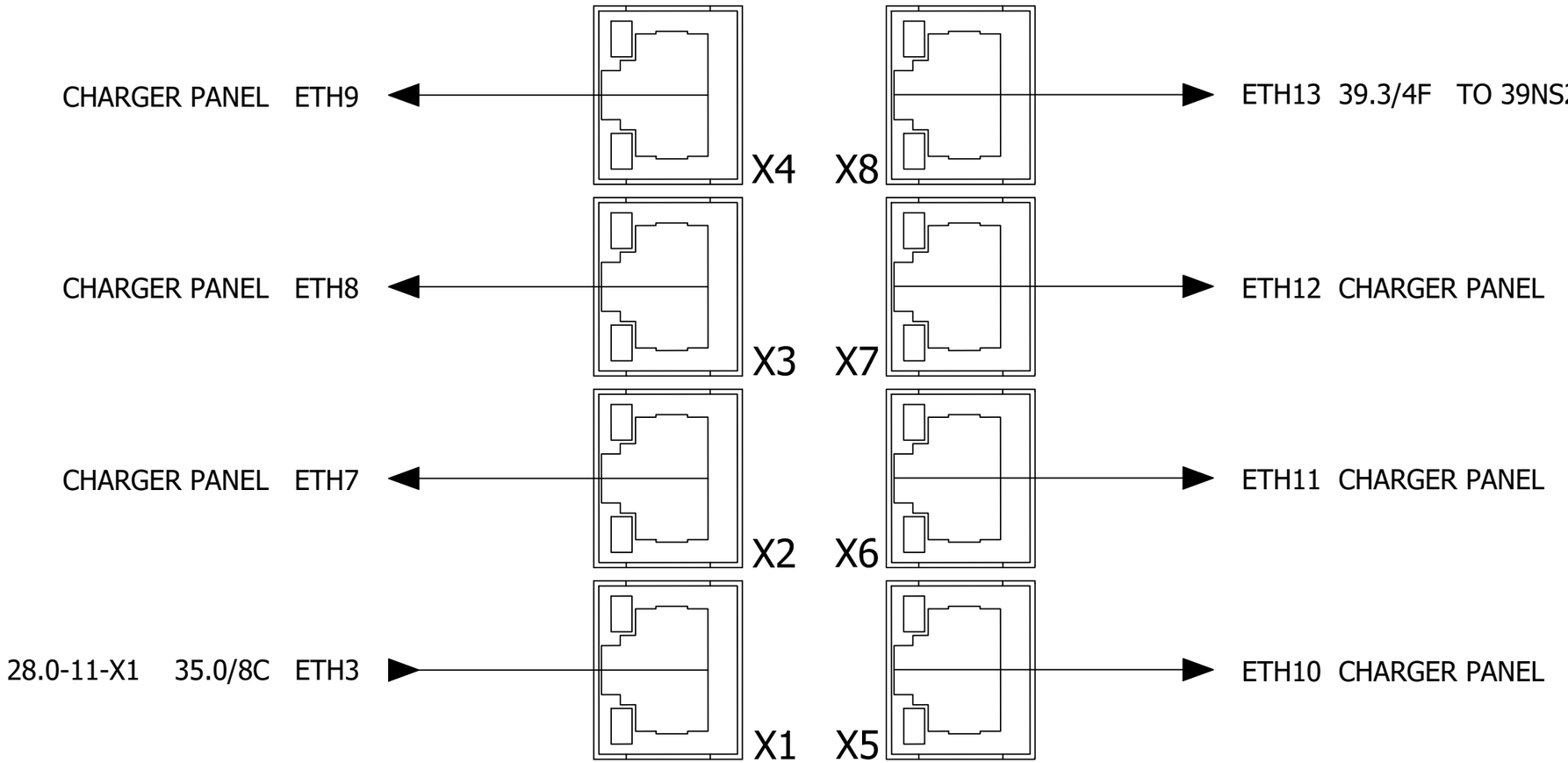
(PROFINET)

39NS1



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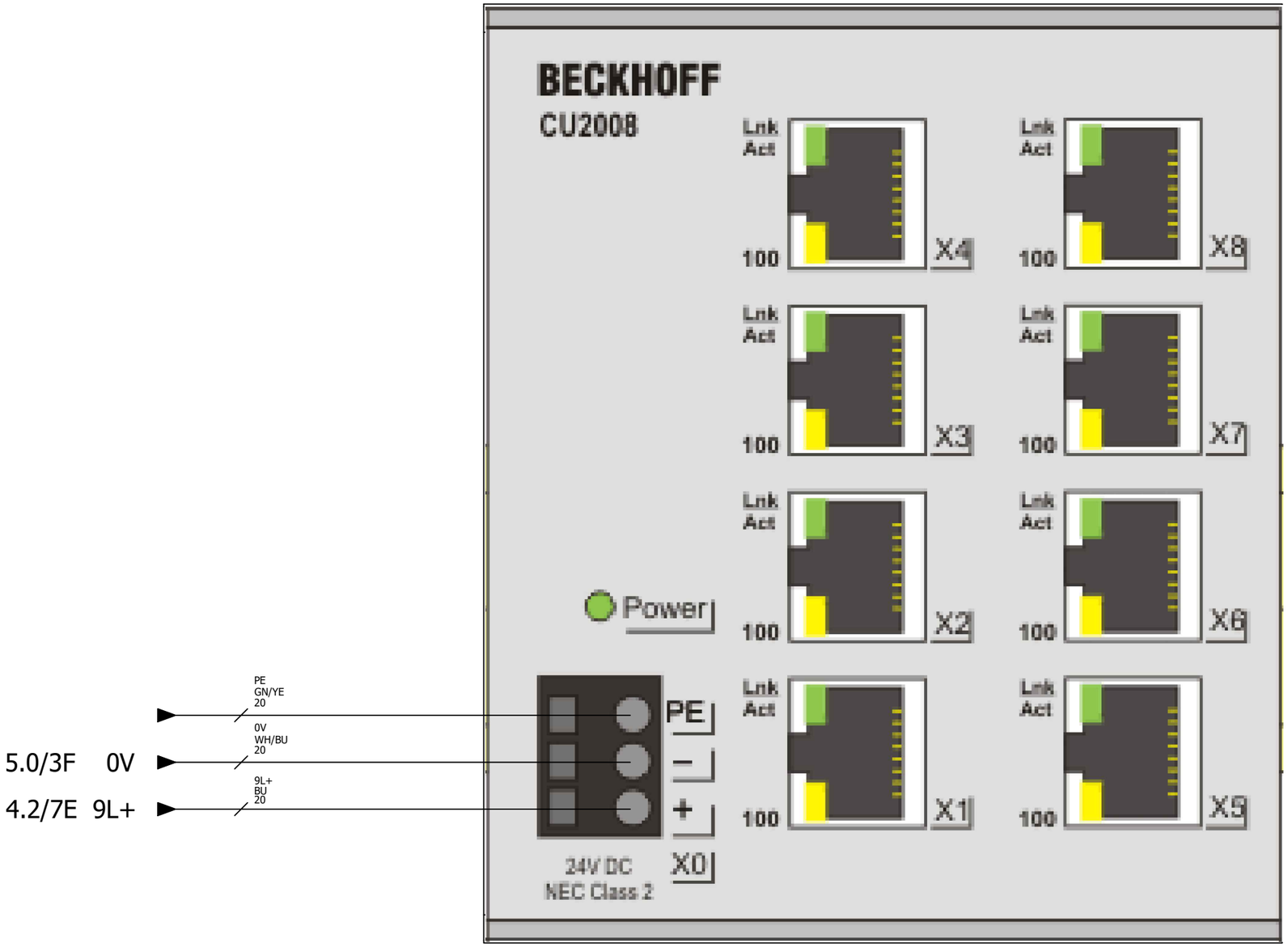
39NS2



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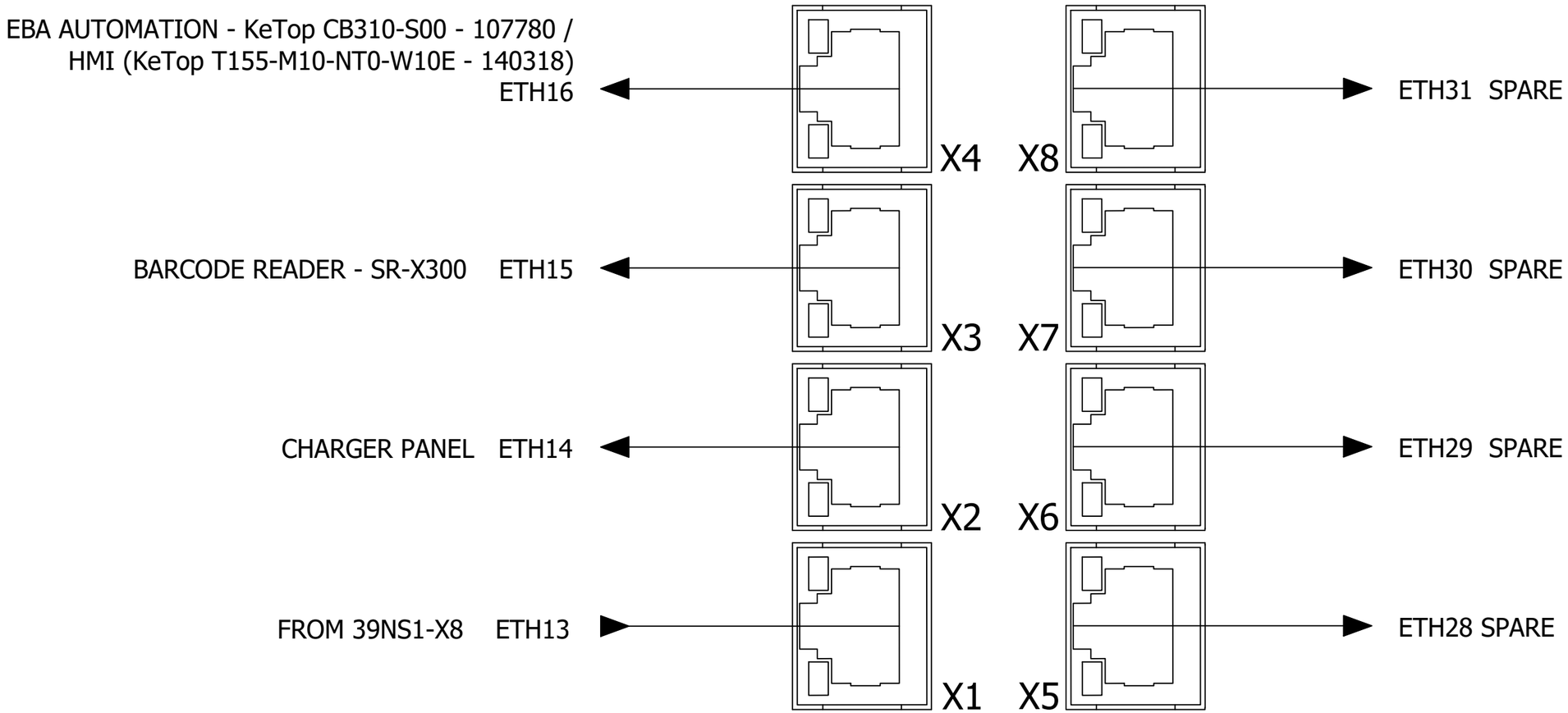
39NS2



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PAGE DESCRIPTION: NETWORK SWITCH 39NS2 PROFINET NETWORK INTERFACE

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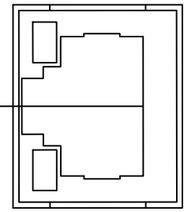
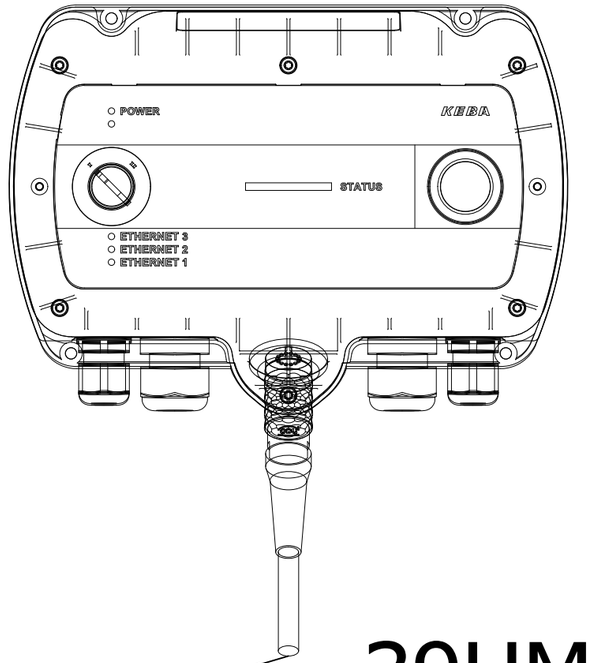
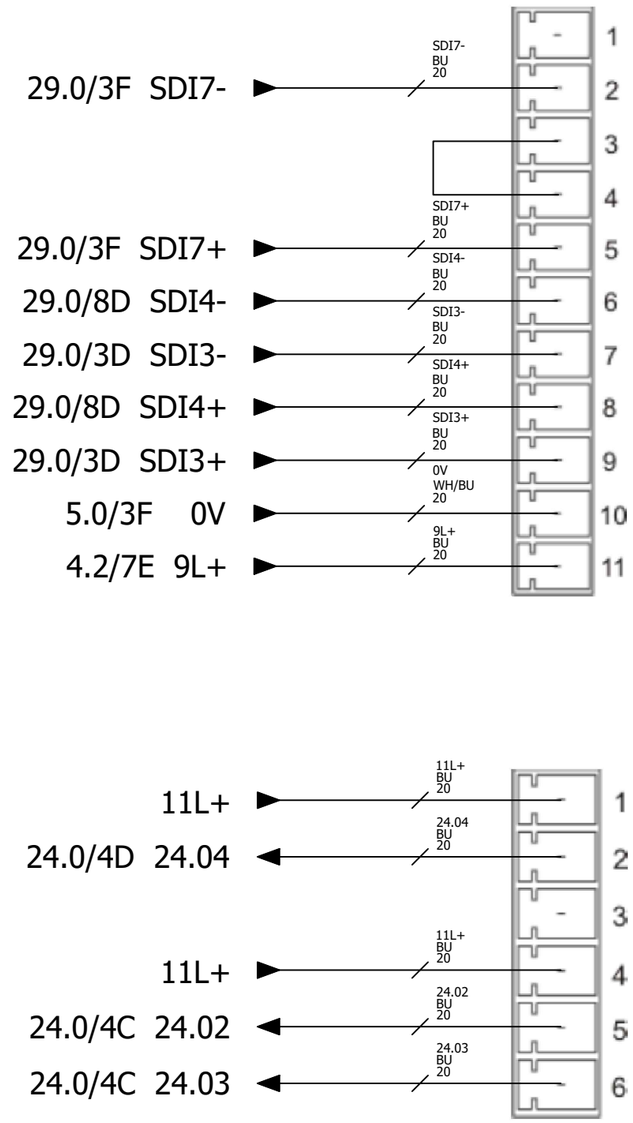
39CBOX

39NS2-X4 ETH16



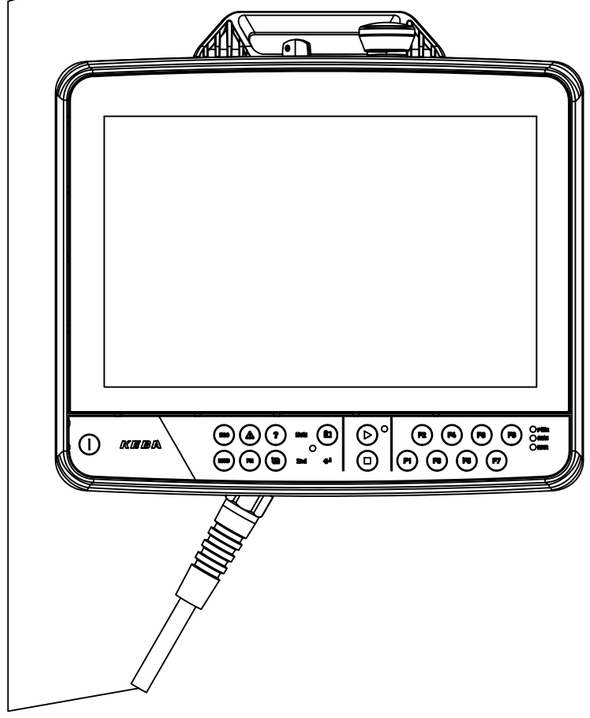




Pin no.	Signal designation	Description
1	N.C.	not connected
2	ZT2-	Approval control, circuit 2 negative
3	ZT1-	Approval control, circuit 1 negative
4	ZT2+	Approval control, circuit 2 positive
5	ZT1+	Approval control, circuit 1 positive
6	ES2	Emergency stop, circuit 2
7	ES1	Emergency stop, circuit 1
8	ES2	Emergency stop, circuit 2
9	ES1	Emergency stop, circuit 1
10	-	GND
11	+	24 V DC power supply

Pin no.	Signal designation	Description
1	DRY	Closed when handheld device is connected
2	DRY	
3	KEY3	Key-operated switch contact (3 pos): Position 3 (right switch position) Illuminated pushbutton: Supply (GND)
4	KEY2	Key-operated switch contact (3 pos): COMMON Key-operated switch contact (2 pos): COMMON Illuminated pushbutton: Supply (24 V)
5	KEY1	Key-operated switch contact (3 pos): Position 2 (middle switch position) Key-operated switch contact (2 pos): Position 2 (left switch position) Illuminated pushbutton: Circuit illuminated pushbutton
6	KEY0	Key-operated switch contact (3 pos): Position 1 (left switch position) Key-operated switch contact (2 pos): Position 1 (right switch position) Illuminated pushbutton: Circuit illuminated pushbutton

39HMI



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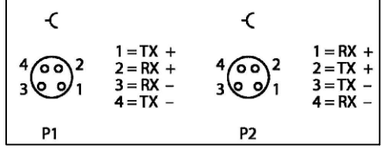
TOTAL PAGES: 171

SHEET 39.4 OF 56

P1 & P2 (PROFINET)

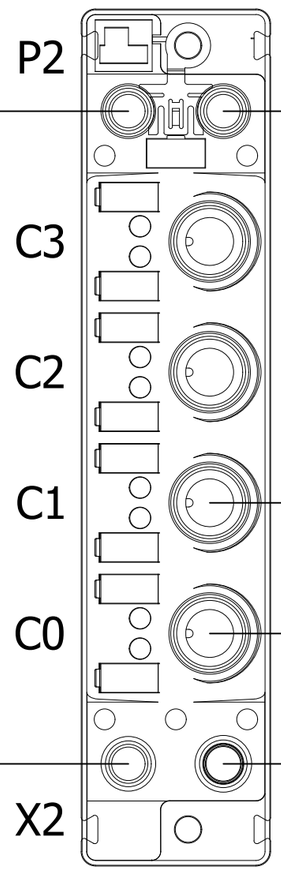
M8 x 1 Ethernet



23.0-12-X2 35.0/8C ETH4

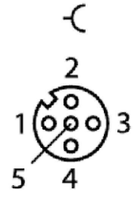
40BCS1 40.1/9C ETH4A

40RIO1



(IO LINK)

M12 x 1 I/O Port

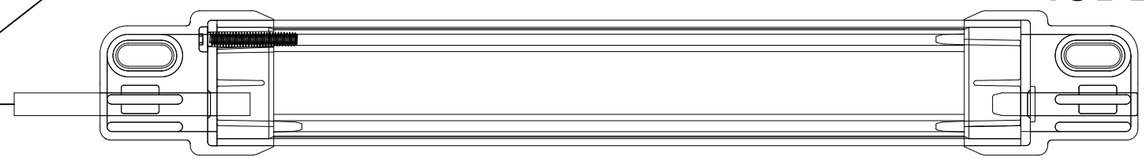


1 = V2 +
2 = DXP
3 = GND V2
4 = C/Q
5 = n.c.

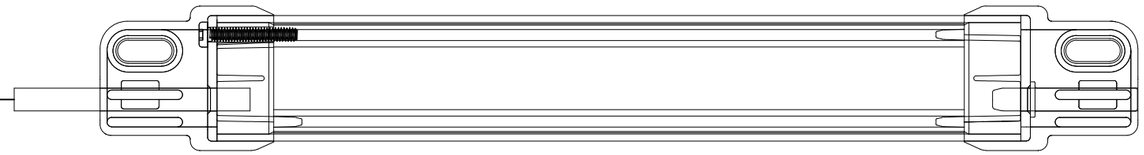
C0...C3

4-pin M12 Male	Pin	Wire Color	Description
	1	Brown	18 V DC to 30 V DC
	2	White	Not used
	3	Blue	DC common
	4	Black	IO-Link Communication

40BL2



40BL1

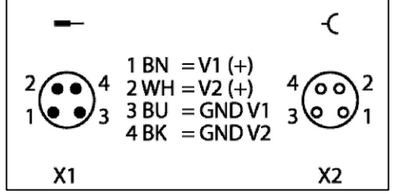


40.1/4E 40PC2

40PC1 XX.X/XX FROM 24L+ & 0VL-

VTD LIGHT
WLS15PX
BANNER IO LINK
LIGHTS

M8 x 1 Voltage Supply



TURCK TBEN-S2-4IOL

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 40RIO1 NETWORK + I/O + POWER

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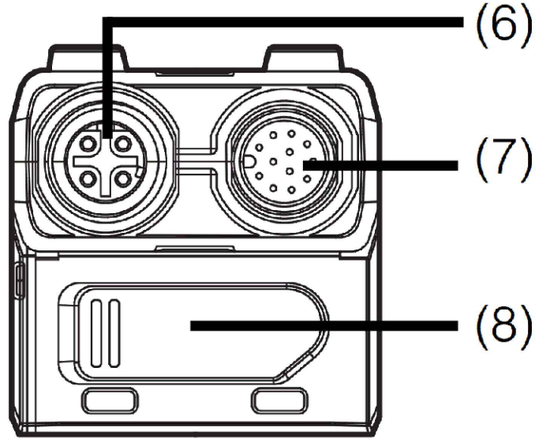
TOTAL PAGES: 171

SHEET 40.0 OF 56

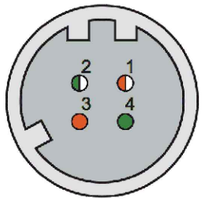
40BCS1

KEYENCE
SR-300X

Bottom



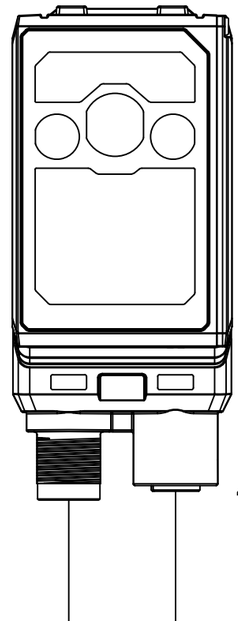
(6)	Ethernet port	Port for connecting the Ethernet cable: M12D 4pin (female)
(7)	Control port[*4]	Port for connecting the control cable: M12A 12pin (male)
(8)	USB port	Port for connecting USB memory or a USB cable[*3]: Type-C



M12D

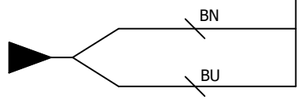
No.	Signal
1	TX+
2	RX+
3	TX-
4	RX-

Male connector, M12,
12-pin or 17-pin, A-coded



Female connector, M12,
4-pin, D-coded (Ethernet)

FROM 40RIO1 40.0/3B C40-2



(PROFINET)

FROM 40RIO1-P2 40.0/3B ETH4A

M12A

No.	Color
1	Brown
2	Blue
3	Purple
4	Yellow
5	Sky
6	Pink
7	Green
8	White/Blue
9	Black
10	White
11	Grey
12	Orange
SHELL	Braid

Wire color	Name	Description	Signal direction	Initial assignment
Brown	24 V	24 VDC	Input	-
Blue	0 V	Power GND	-	-
Orange	RXD (RD)	RS-232C Receive	Input	-
Yellow	TXD (SD)	RS-232C Send	Output	-
Purple	SGND	RS-232C GND	-	-
Green	IN1	Input signal 1	Input	Reading start
Gray	IN2	Input signal 2	Input	-
White and blue	IN COM	Input common	-	-
Pink	OUT1	Output signal 1	Output	Reading OK
Light blue	OUT2	Output signal 2	Output	Reading ERROR
White	OUT3	Output signal 3	Output	BUSY
Black	OUT COM	Output common	-	-
Black tube	FG	Frame ground	-	-

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 40BCS1 - 40BCC1 NETWORK + POWER

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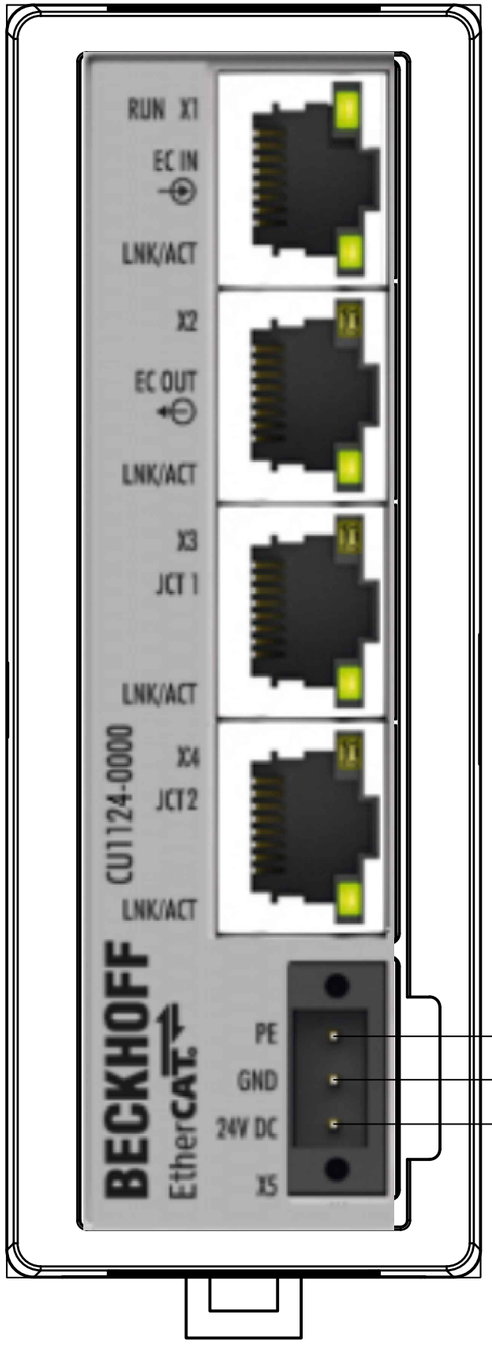
TOTAL PAGES: 171

SHEET 40.1 OF 56

(EtherCAT)



41NS1

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: NETWORK SWITCH 41NS1 POWER 24VDC

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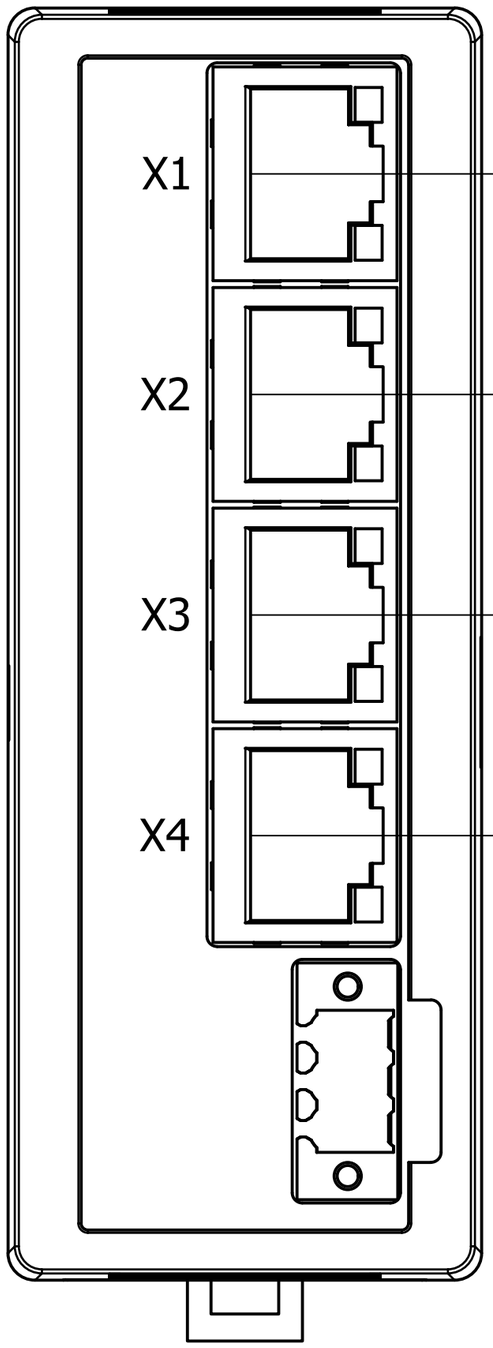
SHEET 41.0 OF 56

41NS1

(EtherCAT)



X1 → ETH5 FROM 5.0-13-X1 36.0/8C

X2 → ETH18 20VFD1 CAROUSEL CONVEYORS AREA SECTION #1 20.8/3E

X3 → ETH19 21VFD1 CAROUSEL CONVEYORS AREA SECTION #2 21.5/3E

X4 → ETH20 SPARE

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: NETWORK SWITCH 41NS1 POWER EtherCAT NETWORK INTERFACE

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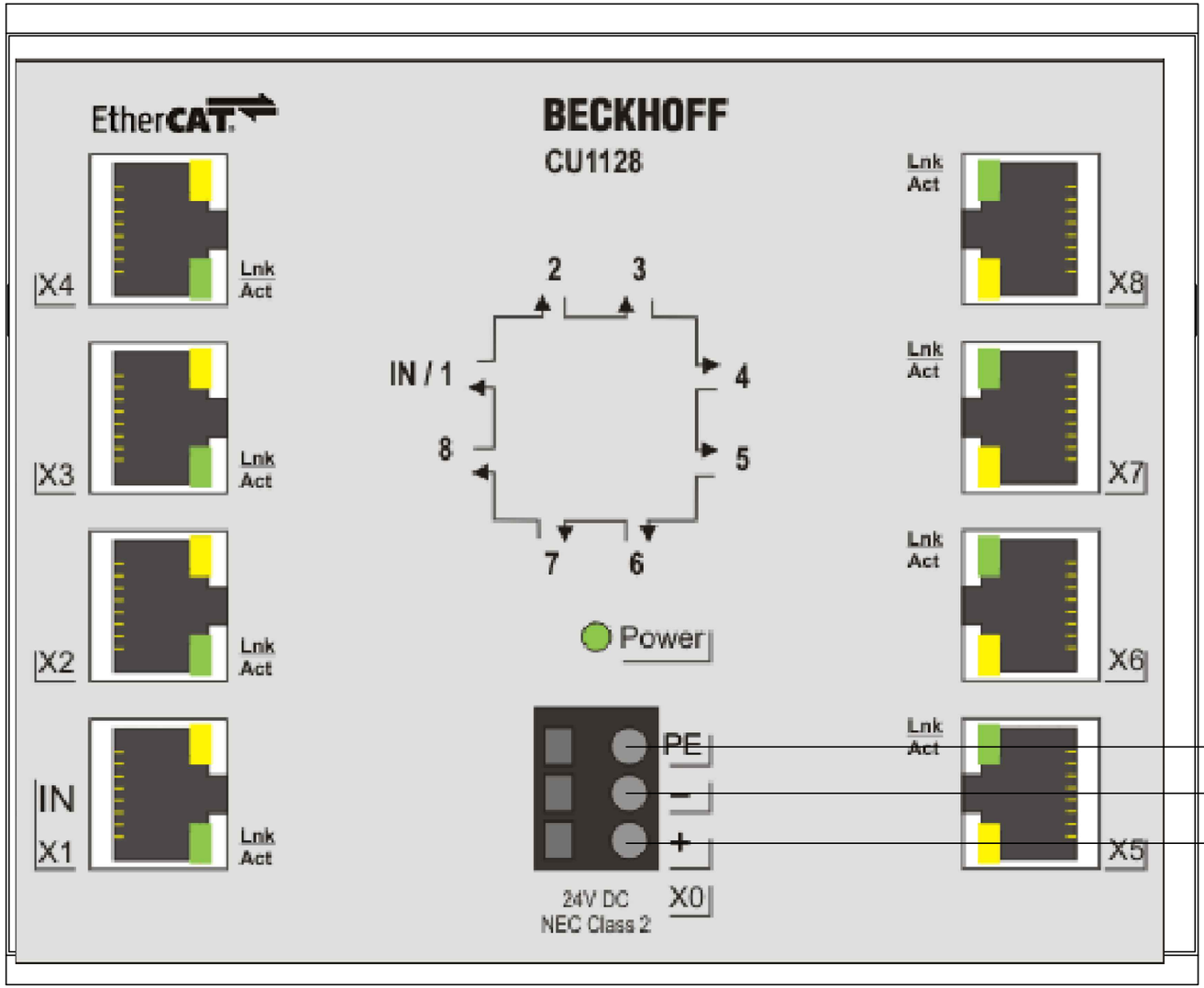
TOTAL PAGES: 171

SHEET 41.1 OF 56

42NS1

(EtherCAT)



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PAGE DESCRIPTION: NETWORK SWITCH 42NS1 POWER 24VDC

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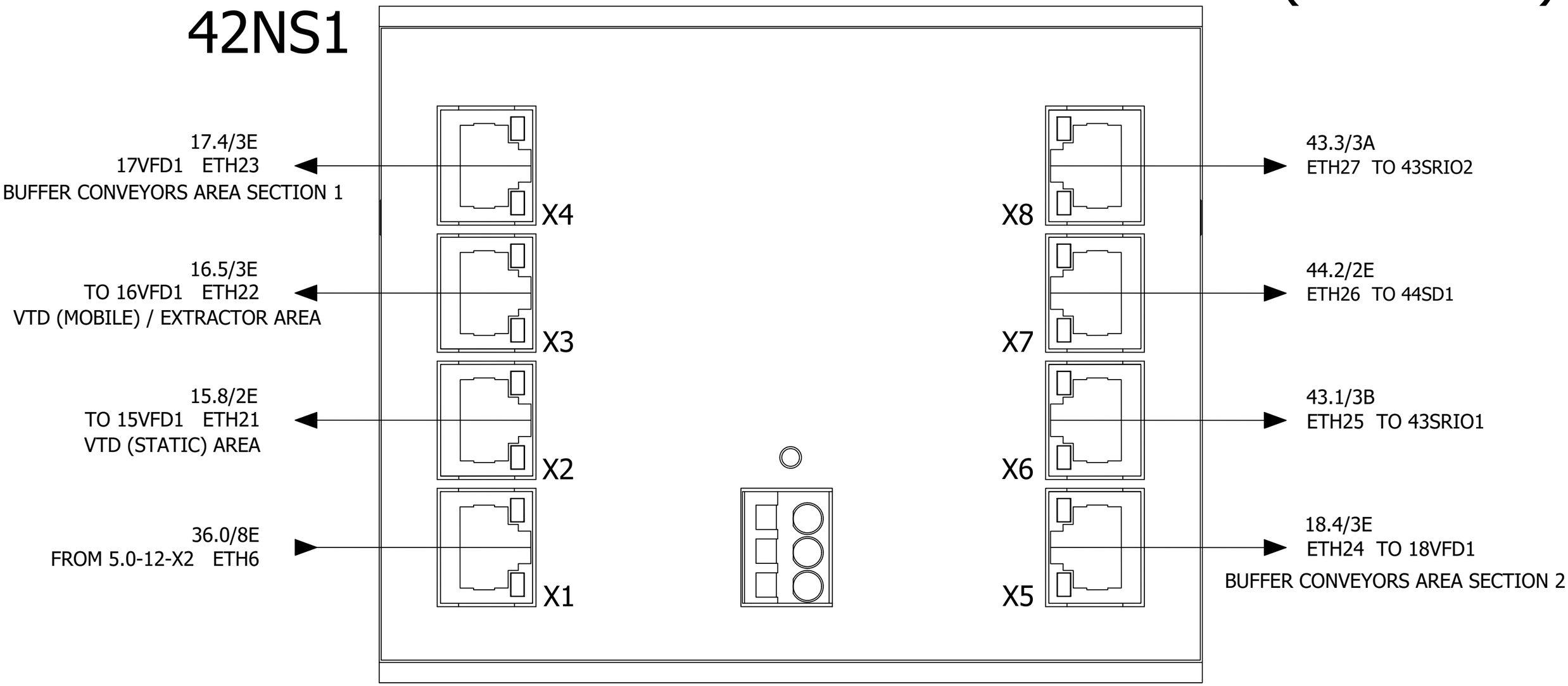
TOTAL PAGES: 171

SHEET 42.0 OF 56

(EtherCAT)

42NS1



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PAGE DESCRIPTION: RACK 01 PLC SAFETY DIGITAL IN MODULE 3

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TOTAL PAGES: 171

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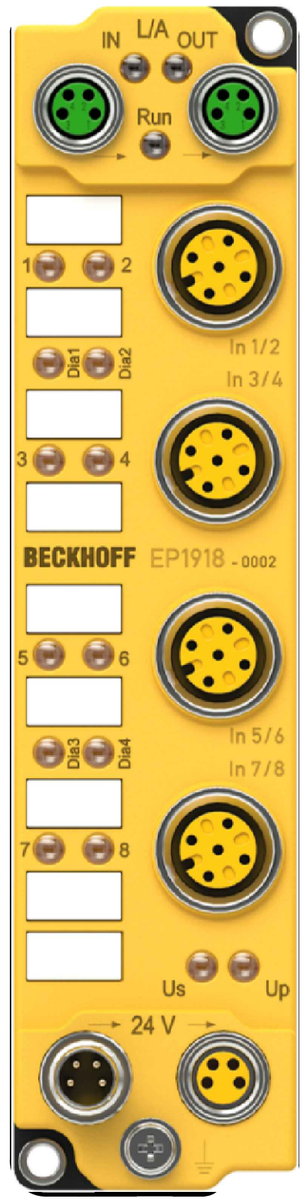
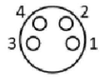
43SRIO1

LED displays

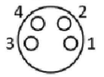
LED	Display	Meaning
IN Link/Act	off	no connection to the preceding EtherCAT module
	lit	LINK: connection to the preceding EtherCAT module
	flashes	ACT: communication with the preceding EtherCAT module
OUT Link/Act	off	no connection to the following EtherCAT module
	lit	LINK: connection to the following EtherCAT module
	flashes	ACT: Communication with the following EtherCAT module
Run	off	Status of the EtherCAT module is Init
	flashes quickly	Status of the EtherCAT module is pre-operational
	flashes slowly	Status of the EtherCAT module is safe-operational
	lit	Status of the EtherCAT module is operational



EtherCAT IN

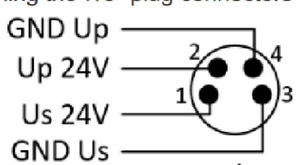


EtherCAT OUT

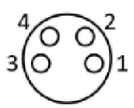


EtherCAT		Connector	Cable		Standard
Signal	**Description**	**M8**	**ZB9010, ZB9020, ZK1090-6292**	**ZB903x, ZK1090-31xx**	**TIA-568B**
Tx +	Transmit Data+	Pin 1	yellow[1]	orange/white[2]	white/orange
Tx -	Transmit Data-	Pin 4	orange[1]	orange[2]	orange
Rx +	Receive Data+	Pin 2	white[1]	blue/white[2]	white/green
Rx -	Receive Data-	Pin 3	blue[1]	blue[2]	green
Shield	Shield	Housing	Shield	Shield	Shield

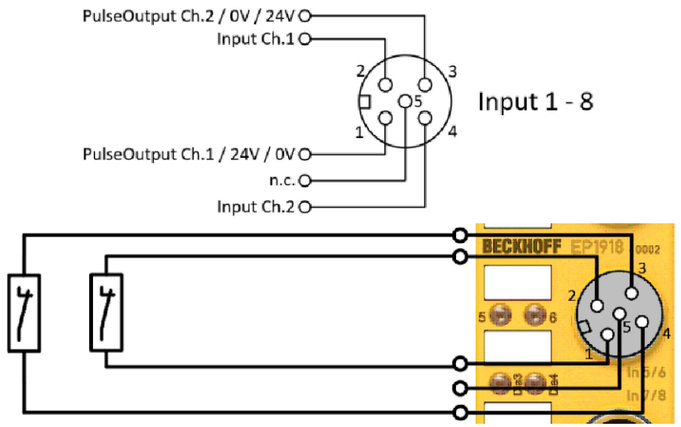
We recommend fastening the 7/8" plug connectors with a torque of **1.5 Nm**.



Us/Up IN



Us/Up OUT



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 43SRIO1 SAFETY DIGITAL IN MODULE

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TOTAL PAGES: 171

SHEET 43.0 OF 56

43SRIO1

ZK1090-3191-0100

FROM 42NS1-X6 ETH25
42.1/9D

43SC1 TO OVERTRAVEL UPPER-NC

43SC2 TO OVERTRAVEL LOWER-NC

ZK2020-3200-0100

FROM 24L+ & 0VL- 43PC1

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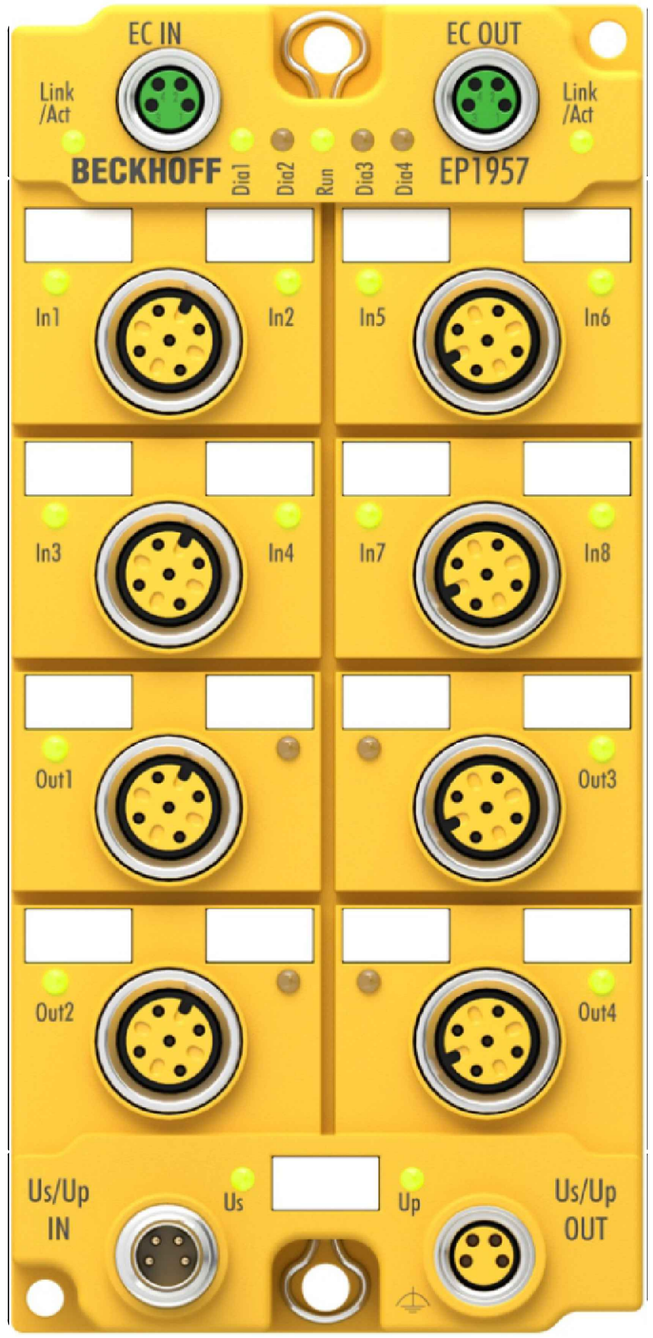
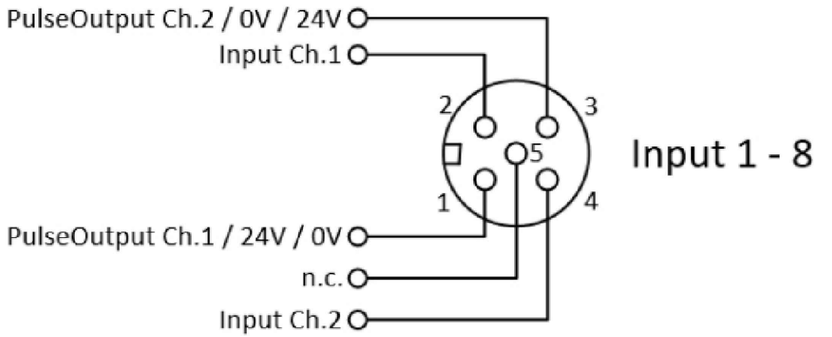
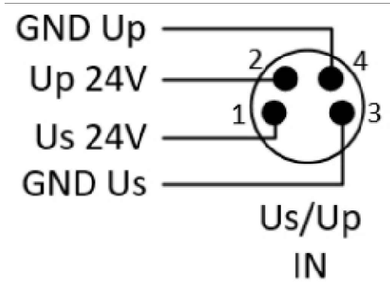
43SRIO2

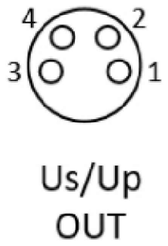
EtherCAT		Connector
Signal	**Description**	**M8**
Tx +	Transmit Data+	Pin 1
Tx -	Transmit Data-	Pin 4
Rx +	Receive Data+	Pin 2
Rx -	Receive Data-	Pin 3
Shield	Shield	Housing

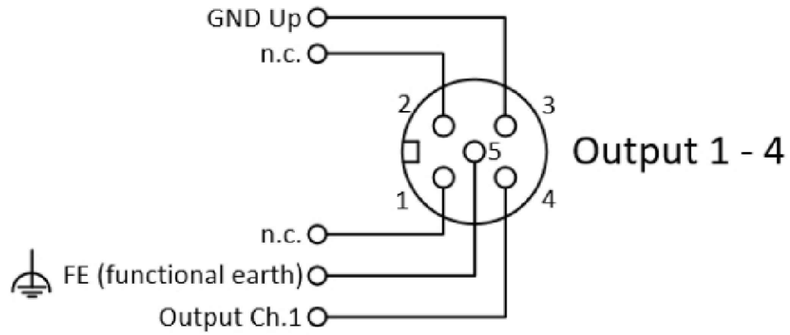
LED	Display	Meaning
IN Link/Act	off	no connection to the preceding EtherCAT module
	lit	LINK: connection to the preceding EtherCAT module
	flashes	ACT: communication with the preceding EtherCAT module
OUT Link/Act	off	no connection to the following EtherCAT module
	lit	LINK: connection to the following EtherCAT module
	flashes	ACT: Communication with the following EtherCAT module
Run	off	Status of the EtherCAT module is Init
	flashes quickly	Status of the EtherCAT module is pre-operational
	flashes slowly	Status of the EtherCAT module is safe-operational
	lit	Status of the EtherCAT module is operational











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PAGE DESCRIPTION: 43SRIO2 SAFETY DIGITAL IN/OUT MODULE

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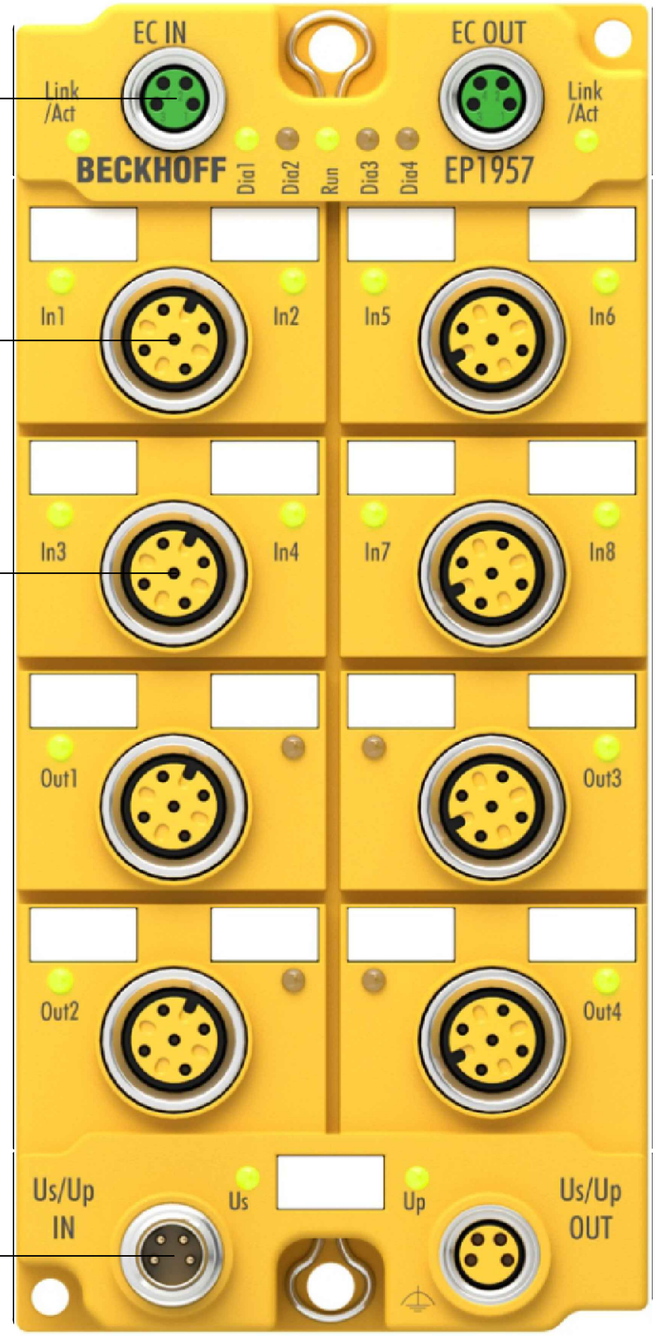
43SRIO2

42.1/9B FROM 42NS1-X8 ETH27

43.4/7C TO 43LC1 RECEIVER

43.4/5C TO 43LC2 SENDER

FROM 24L+ & 0VL- 12.2/



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PAGE DESCRIPTION: 43SRIO2 SAFETY DIGITAL IN/OUT MODULE

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System connection (M12, 5-pin)

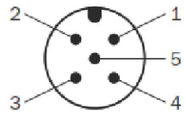


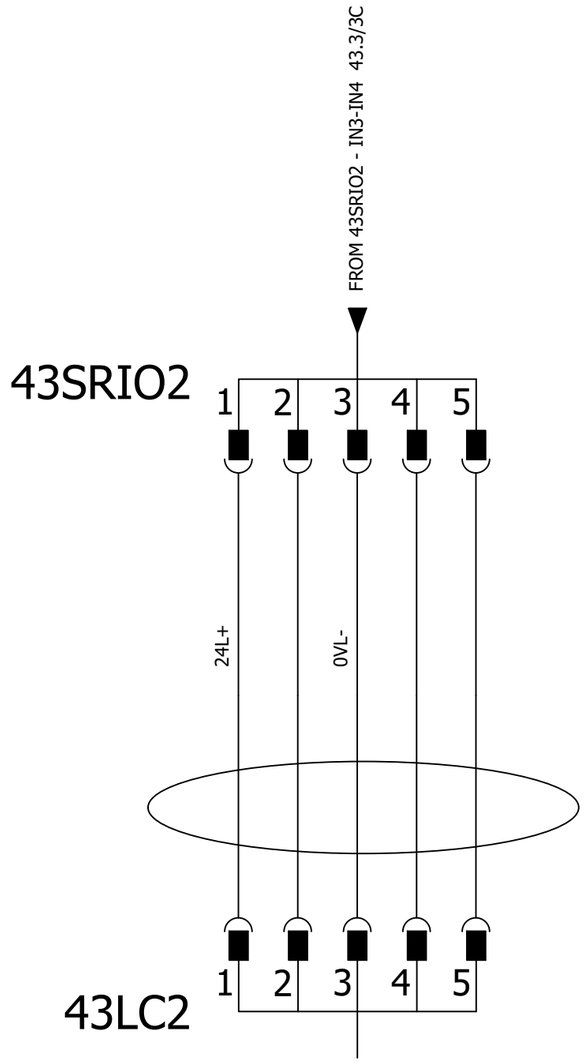
Figure 26: System connection (male connector, M12, 5-pin)

Pin	Wire color [1]	Sender	Receiver
1	Brown	+24 V DC (voltage supply input)	+24 V DC (voltage supply input)
2	White	Reserved	OSSD1 (switching output 1)
3	Blue	0 V DC (voltage supply input)	0 V DC (voltage supply input)
4	Black	Reserved	OSSD2 (switching output 2)
5	Gray	Not assigned	Not assigned

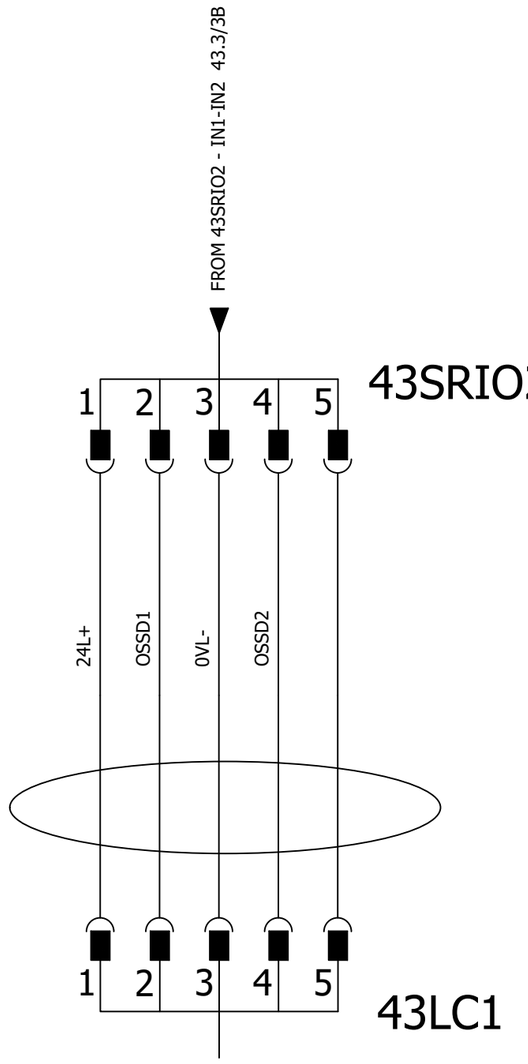
LIGHT CURTAIN - CAROUSEL CONVEYORS

SICK - RECEIVER - C4C-EV13530B10000 PART # 1107661

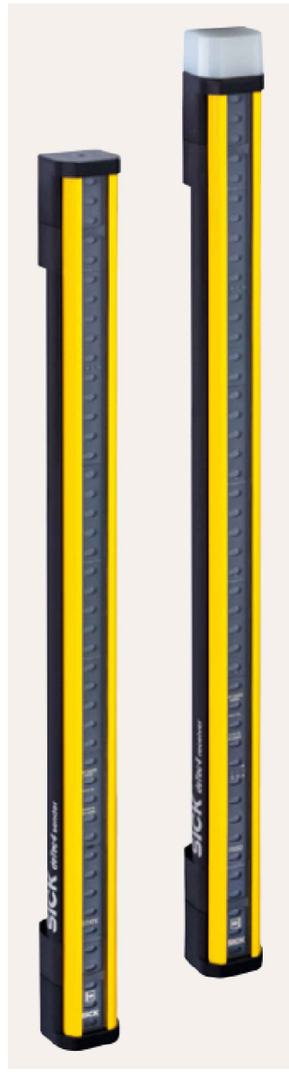
SICK - SENDER - C4C-SV13530B10000 PART # 1107660



SENDER



RECEIVER



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: LIGHT CURTAIN - CAROUSEL CONVEYORS

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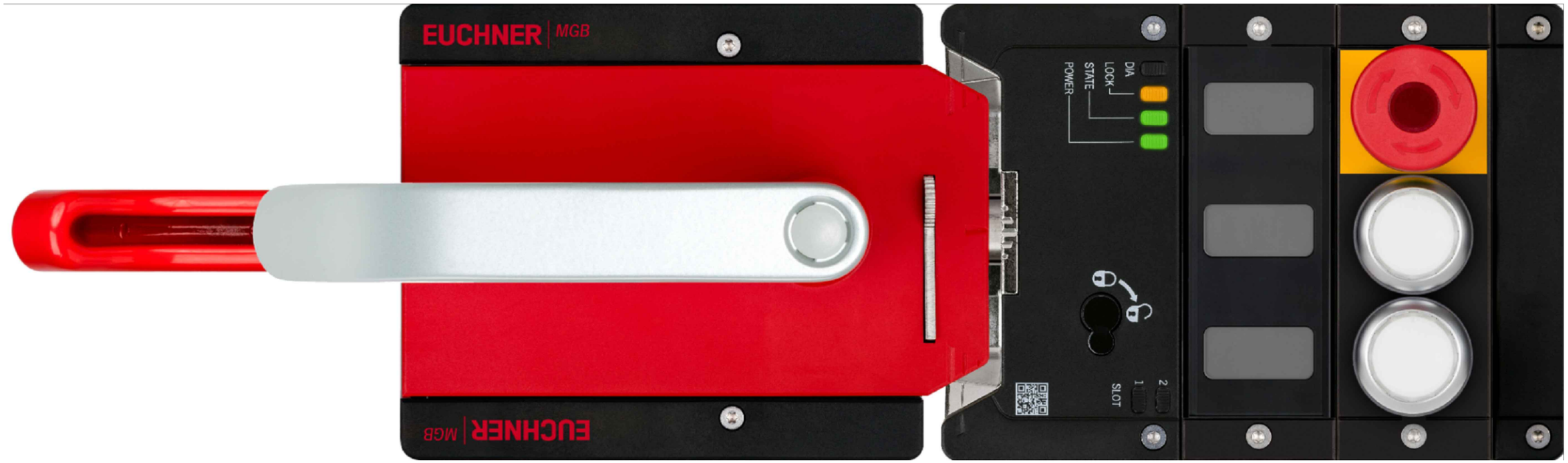
JOB # C40572

TOTAL PAGES: 171

SHEET 43.4 OF 56

44SD1



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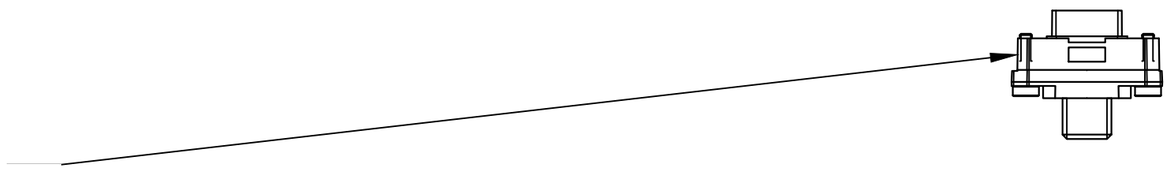

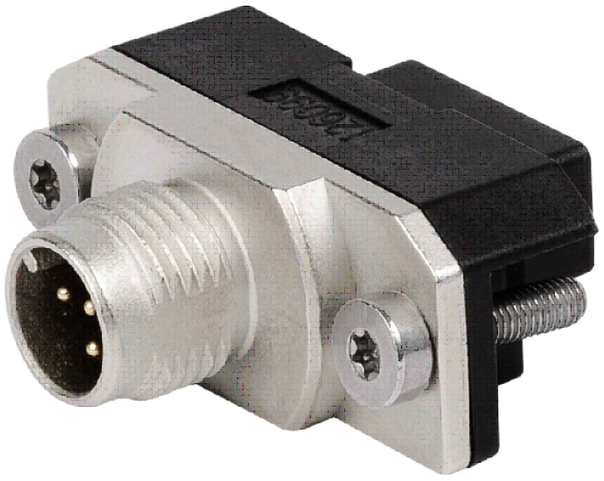

direct plug

AC-MC-SB-M-A-157024

MBM-EC-S1-MLI-3B-173401

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: Guard lock with guard lock monitoring

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TOTAL PAGES: 171

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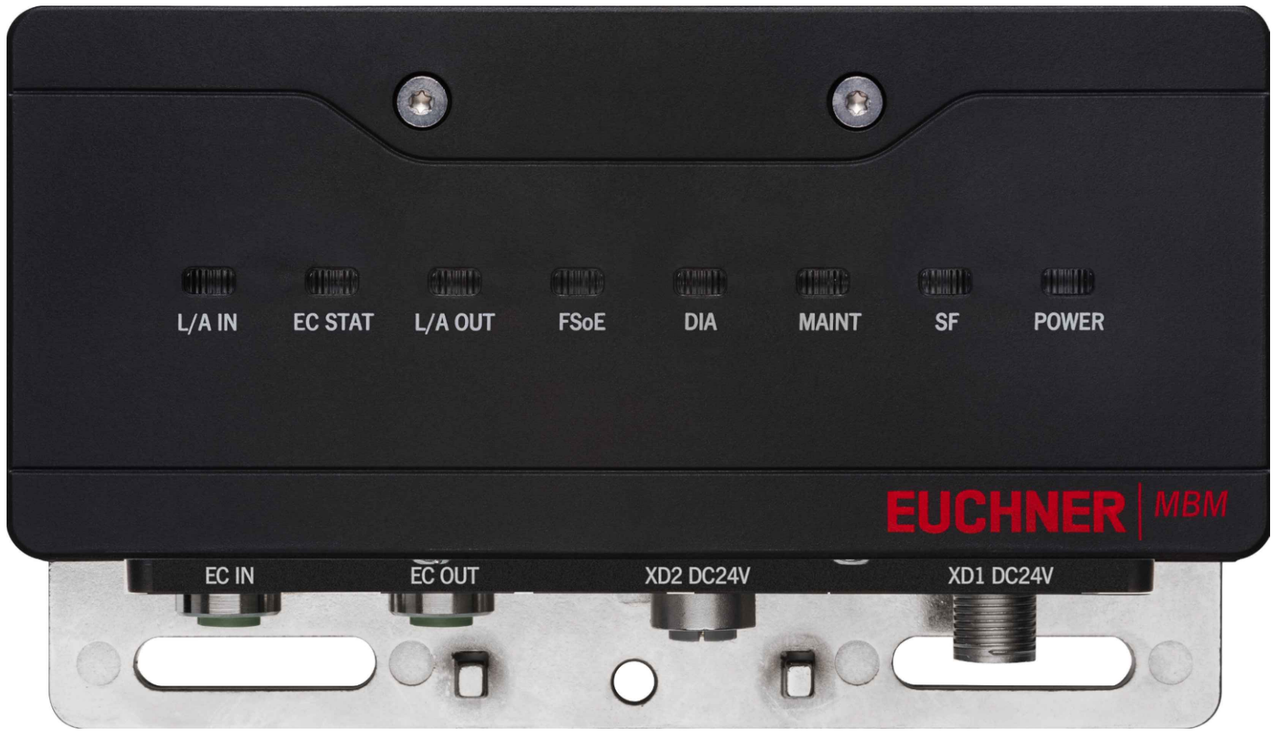
LED	Description
Bus module MBM	
L/A IN	Connection correct: statically on Data transfer: flashing at 10 Hz Color: green
EC STAT	Normal state: statically on Flashing codes according to EtherCAT specification Color: green/red
L/A OUT	Connection correct: statically on Data transfer: flashing at 10 Hz Color: green
FSoE	Normal state: statically on Color: green FSoE inactive/passivated: statically on Color: yellow FSoE Reset/Init State: flashing Color: yellow FSoE error: statically on Color: red FSoE – no connection to control system: off
DIA	Indicates errors Color: red
MAINT	Indicates the error flashing code Color: red/green/yellow
SF	System fault: statically on (see chapter *12. Diagnostics, troubleshooting and aids*) Color: red
Power	Power Up: flashing at 5 Hz Normal state: statically on Color: green



44SD1

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: Guard lock with guard lock monitoring

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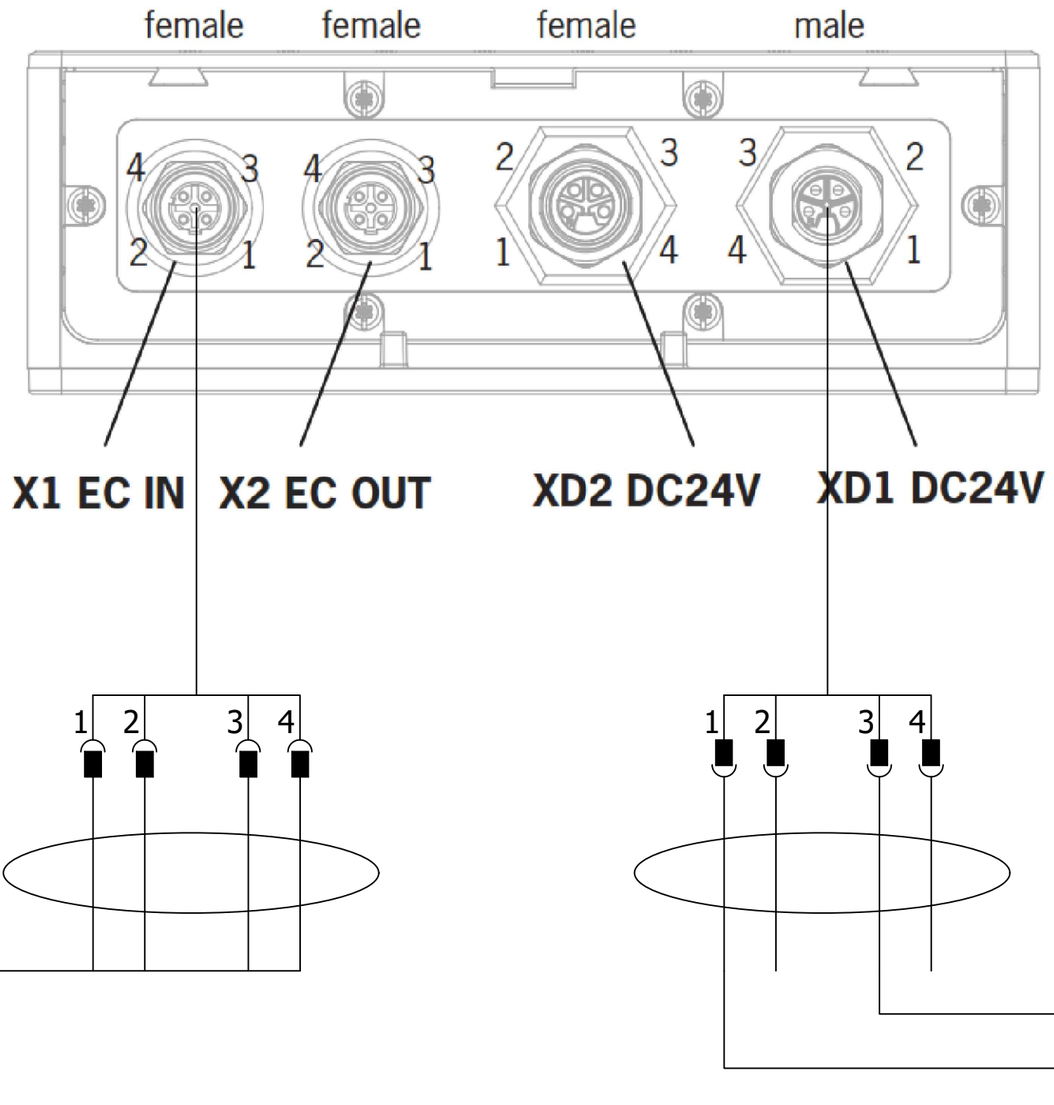
TOTAL PAGES: 171

SHEET 44.1 OF 56

Terminal assignment for version with Power M12 plug (L-coded), 4-pin, and bus M12 plug (D-coded)

Pin	Description
X1.1	Transmit Data TX+
X1.2	Receive Data RX+
X1.3	Transmit Data TX–
X1.4	Receive Data RX–
Functional earth on plug housing	

Pin	Description
X2.1	Transmit Data TX+
X2.2	Receive Data RX+
X2.3	Transmit Data TX–
X2.4	Receive Data RX–
Functional earth on plug housing	



FROM 42NS2-X7
42.1/9C ETH26

Pin	Description
XD1.1	L1 operating voltage US DC 24 V
XD1.2	N2 auxiliary voltage UP 1) 0 V
XD1.3	N1 operating voltage US 0 V
XD1.4	L2 auxiliary voltage UP 1) DC 24 V

Pin	Description
XD2.1	L1 operating voltage US DC 24 V
XD2.2	N2 auxiliary voltage UP 1) 0 V
XD2.3	N1 operating voltage US 0 V
XD2.4	L2 auxiliary voltage UP 1) DC 24 V

1) The auxiliary voltage is not required for the MGB2 system.

0VL- 12.0/

24L+ 12.2/

44SD1

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: Guard lock with guard lock monitoring

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TOTAL PAGES: 171

SHEET 44.2 OF 56

CAROUSEL CONVEYORS AREA SECTION 3

FROM 23.0-9-X1 32.0/7C 32ECP1

45RIO1

45ECP1

45RIO2

45ECP2

45RIO3

45ECP3

45RIO4

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

REV	DESCRIPTION	DATE

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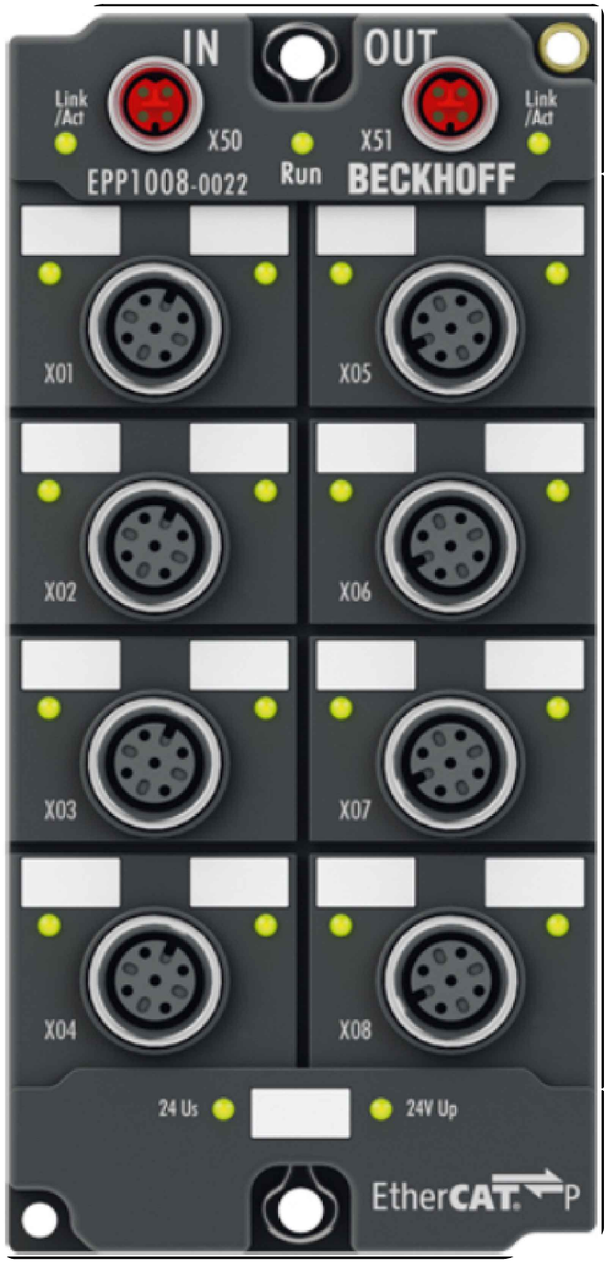
JOB # C40572

TOTAL PAGES: 171

SHEET 45.0 OF 56

EPP1008-0022



EtherCAT P
M8 socket, 4-pin, P-coded, shielded

1 | Tx+ / GND_S
2 | Rx+ / GND_P
3 | Rx- / U_P
4 | Tx- / U_S

DIGITAL INPUTS
M12 socket, 5-pin, A-coded

1 | +24 V U_S
2 | n.c.
3 | GND_S
4 | Input
5 | n.c.

CAROUSEL
CONVEYOR
POPUP 1-2

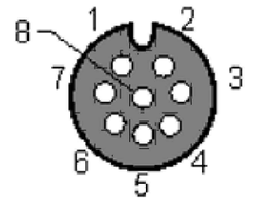
CAROUSEL CONVEYOR
TURNTABLE+ROLLER

EPP5001-0002

EtherCAT P
M8 socket, 4-pin, P-coded, shielded

1 | Tx+ / GND_S
2 | Rx+ / GND_P
3 | Rx- / U_P
4 | Tx- / U_S

SSI encoder input
M12 socket, 8-pin



1 | GND_S
2 | Sensor Supply
3 | Clock +
4 | Clock -
5 | Data +
6 | Data -
7 | Set
8 | Direction

CAROUSEL
CONVEYOR
TURNTABLE
EPP5001-0002
SSI ENCODER

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL
PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

REV	DESCRIPTION	DATE

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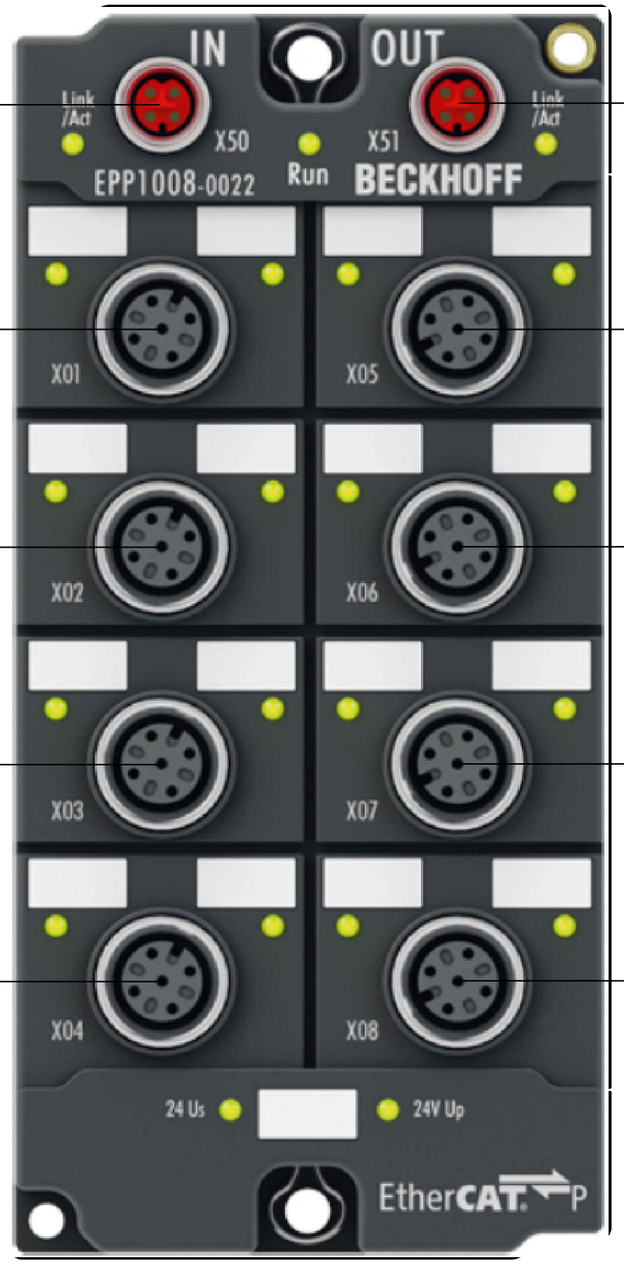
JOB # C40572
TOTAL PAGES: 171
SHEET 45.1 OF 56

EPP1008-0022

45RIO1

FROM 23.0-9-X1 32.0/7C 32ECP1 → 32ECP1

45ECP1 → 45ECP1 45.3/3B TO 45RIO2-X50



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X05 → ?????????

????????? → X02

X06 → ?????????

????????? → X03

X07 → ?????????

????????? → X04

X08 → ?????????

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 45RIO1 REMOTE I/O INPUT LAYOUT

REV	DESCRIPTION	DATE

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JOB # C40572

TOTAL PAGES: 171

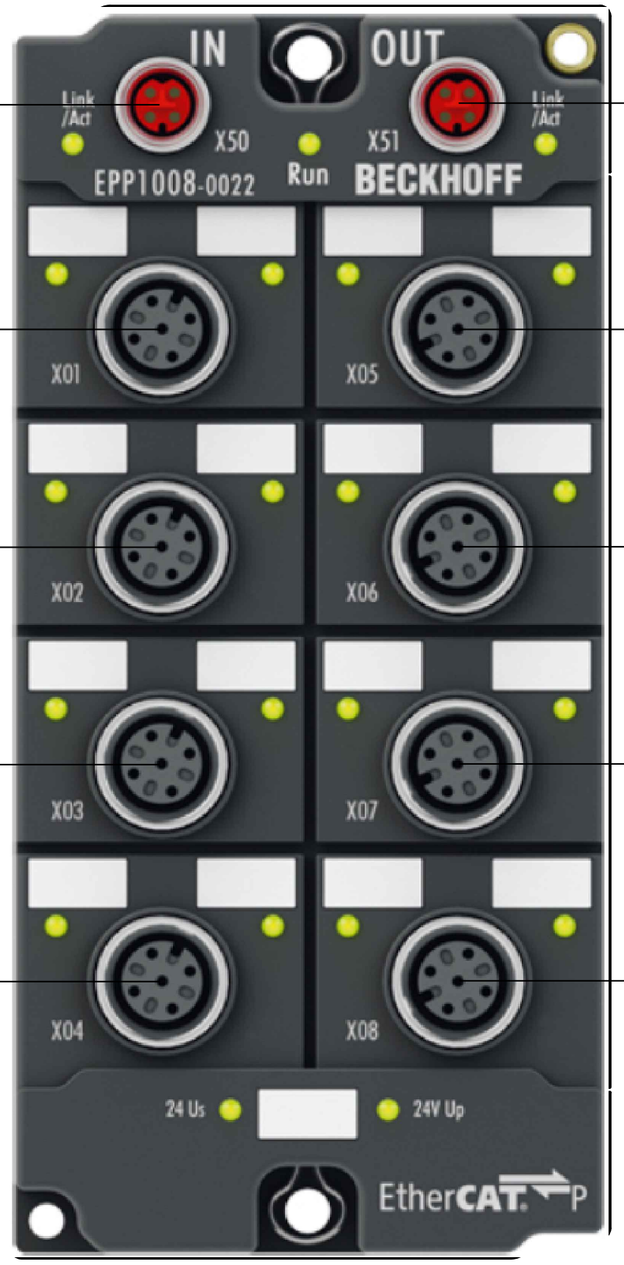
SHEET 45.2 OF 56

EPP1008-0022

45RIO2

FROM 45RIO1-X51 45.1/8B 45RIO1 → 45ECP1

45ECP2 → 45ECP2 45.4/3B TO 45RIO3-X50



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 45RIO2 REMOTE I/O INPUT LAYOUT

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TOTAL PAGES: 171

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EPP1008-0022

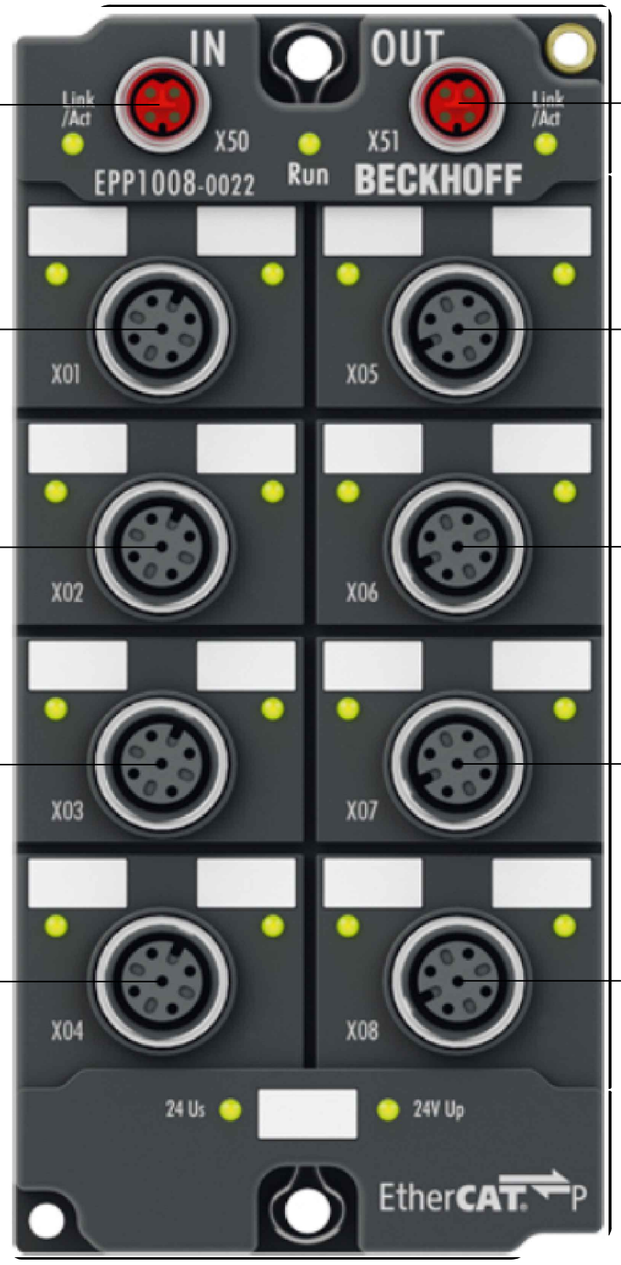
45RIO3

FROM 45RIO2-X51 45.2/8B 45ECP2

45ECP2

45ECP3

45ECP3 45.5/3B TO 45RIO4-X50



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 45RIO3 REMOTE I/O INPUT LAYOUT

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SHEET 45.4 OF 56

EPP5001-0002

45RIO4

FROM 45RIO3-X51 45.4/8B 45ECP3 → 45ECP3

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 45RIO4 REMOTE I/O INPUT LAYOUT

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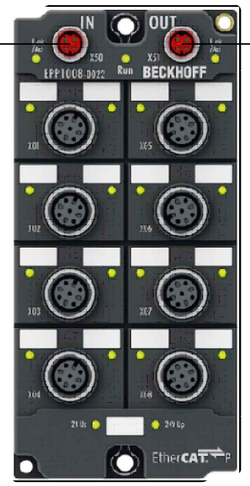
TOTAL PAGES: 171

SHEET 45.5 OF 56

CAROUSEL CONVEYORS AREA SECTION 1 & 2

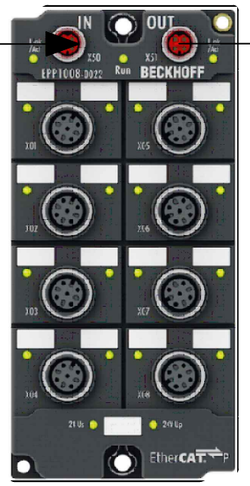
FROM 23.0-10-X1 33.0/7C 33ECP1

46RIO1



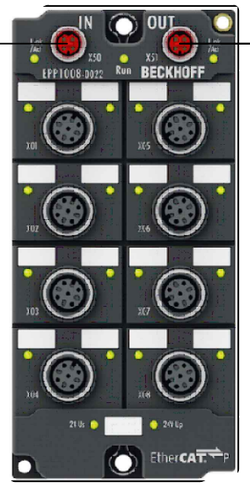
46ECP1

46RIO2



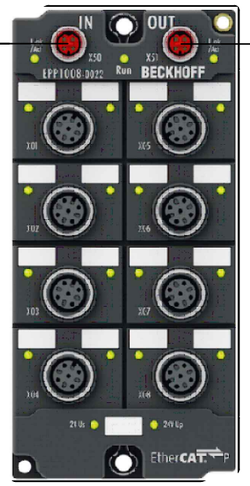
46ECP2

46RIO3



46ECP3

46RIO4



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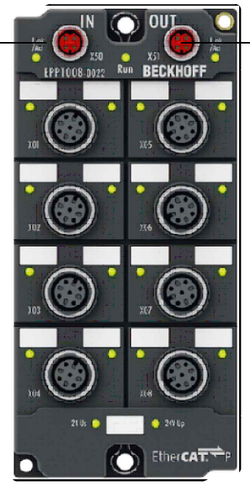
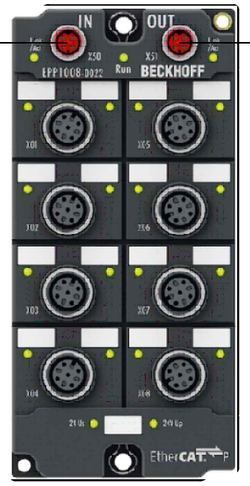


46ECP5

46RIO6



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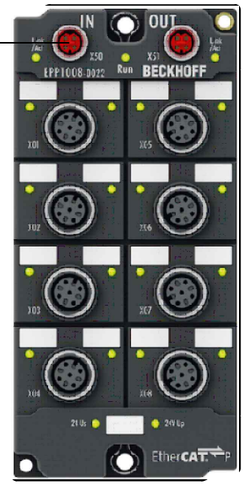
46RIO7



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

REV	DESCRIPTION	DATE

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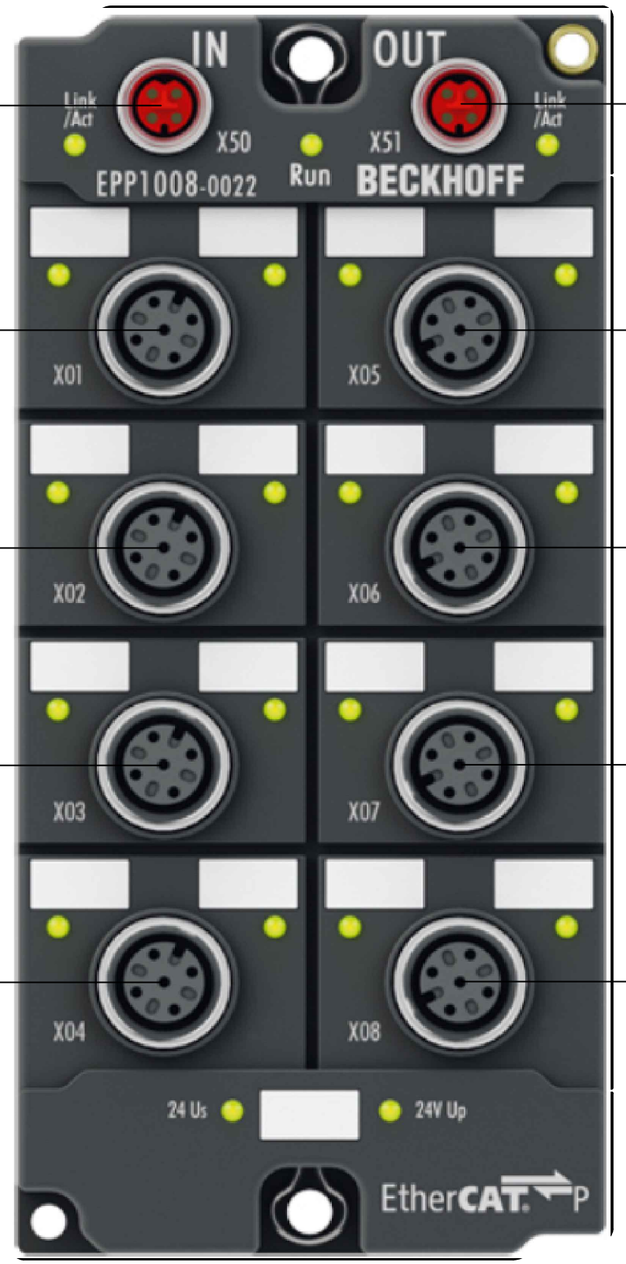
JOB # C40572

TOTAL PAGES: 171

SHEET 46.0 OF 56

46RIO1

FROM 23.0-10-X1 33.0/7C 33ECP1 → 33ECP1



46ECP1 → 46ECP1 46.2/3B TO 46RIO2-X50

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PAGE DESCRIPTION: 46RIO1 REMOTE I/O INPUT LAYOUT

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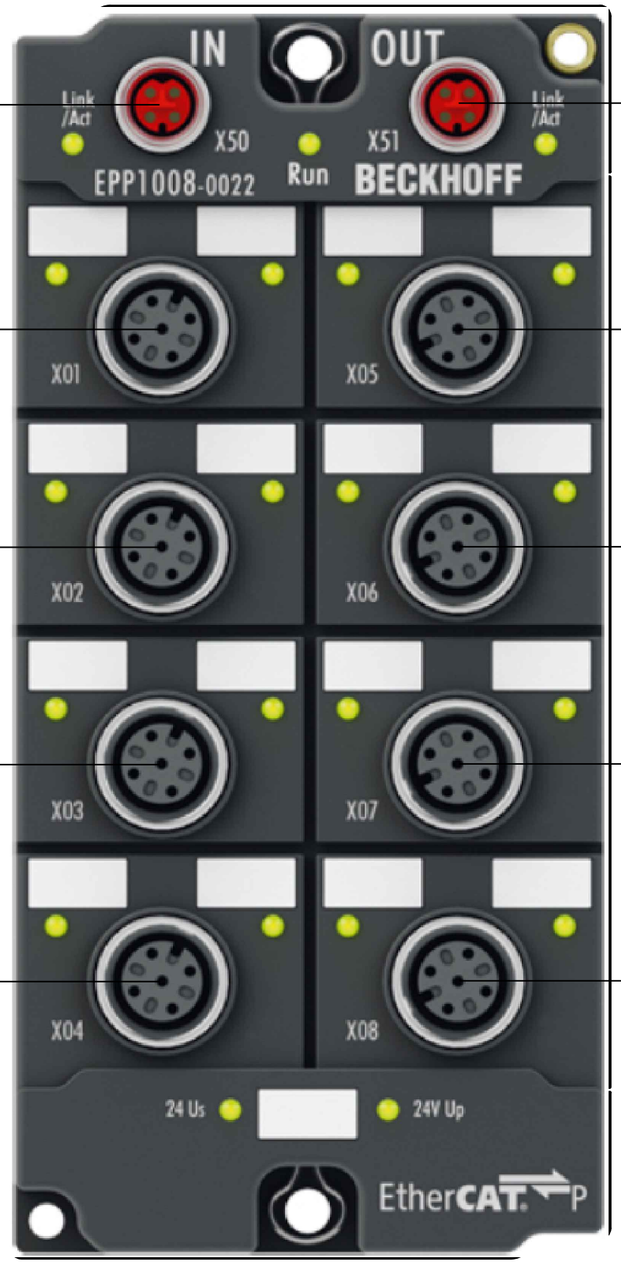
EPP1008-0022

46RIO2

FROM 46RIO1-X51 46.1/8B 46ECP1 → 46ECP1

46ECP2 → 46ECP2 46.3/3B TO 46RIO3-X50



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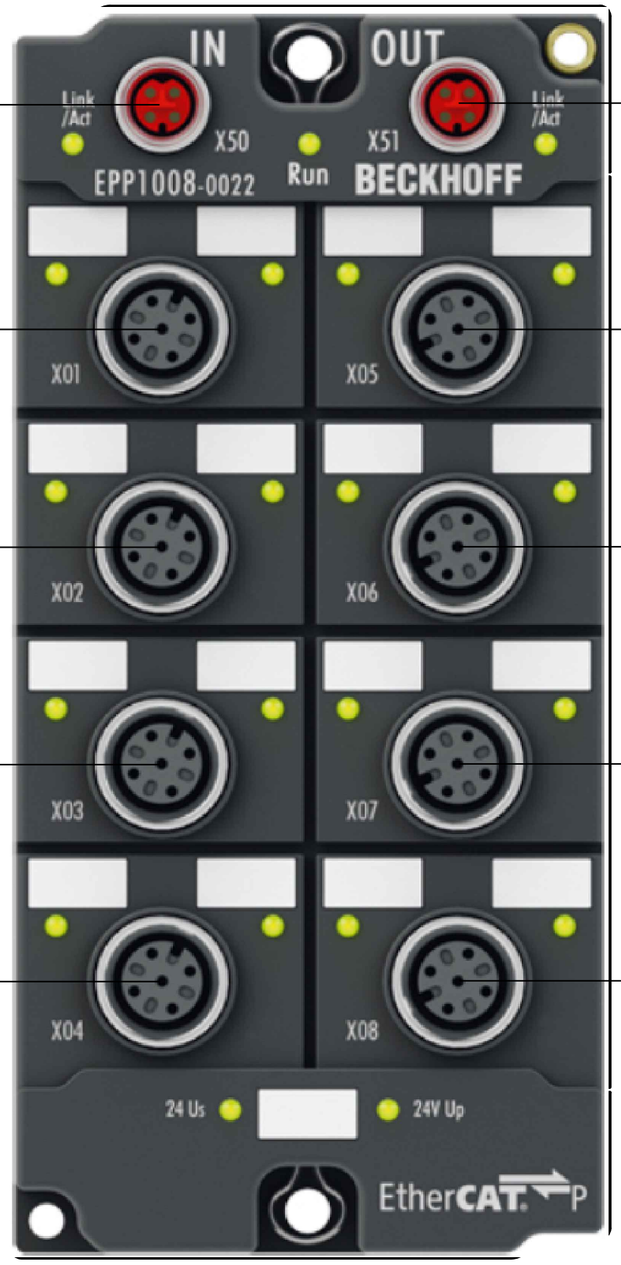
46RIO3

FROM 46RIO2-X51 46.2/8B 46ECP2

46ECP2

46ECP3

46ECP3 46.4/3B TO 46RIO4-X50



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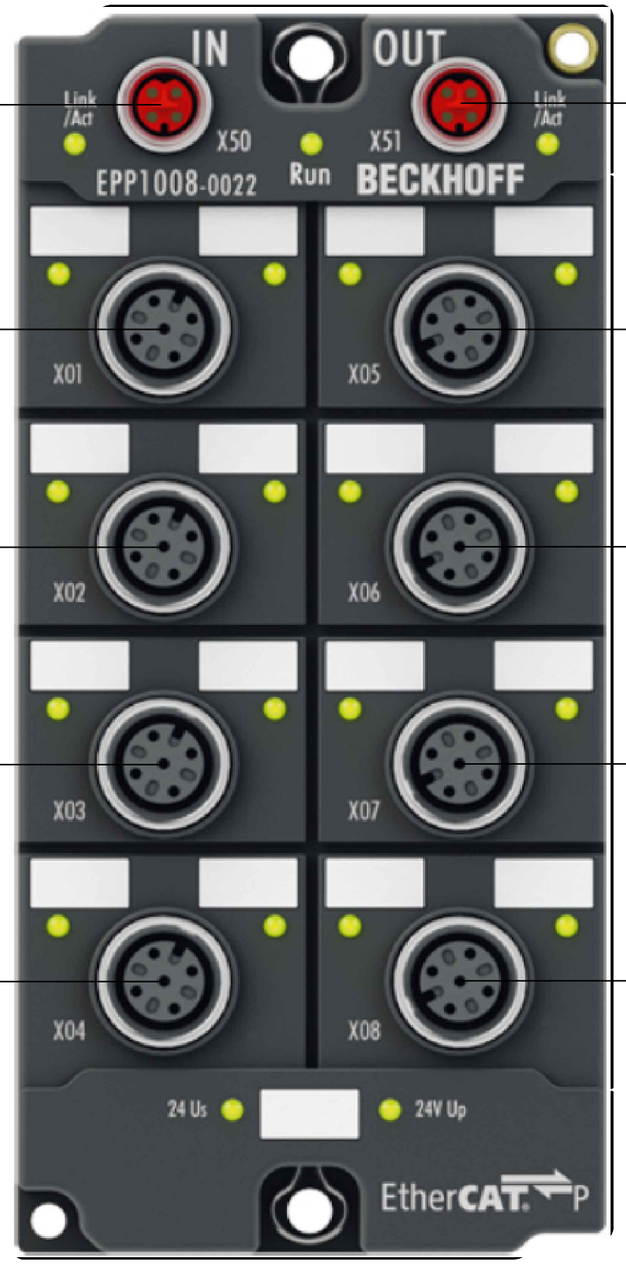
46RIO4

FROM 46RIO3-X51 46.3/8B 46ECP3

46ECP3



46ECP4

46ECP4 46.5/3B TO 46RIO5-X50

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PAGE DESCRIPTION: 46RIO4 REMOTE I/O INPUT LAYOUT

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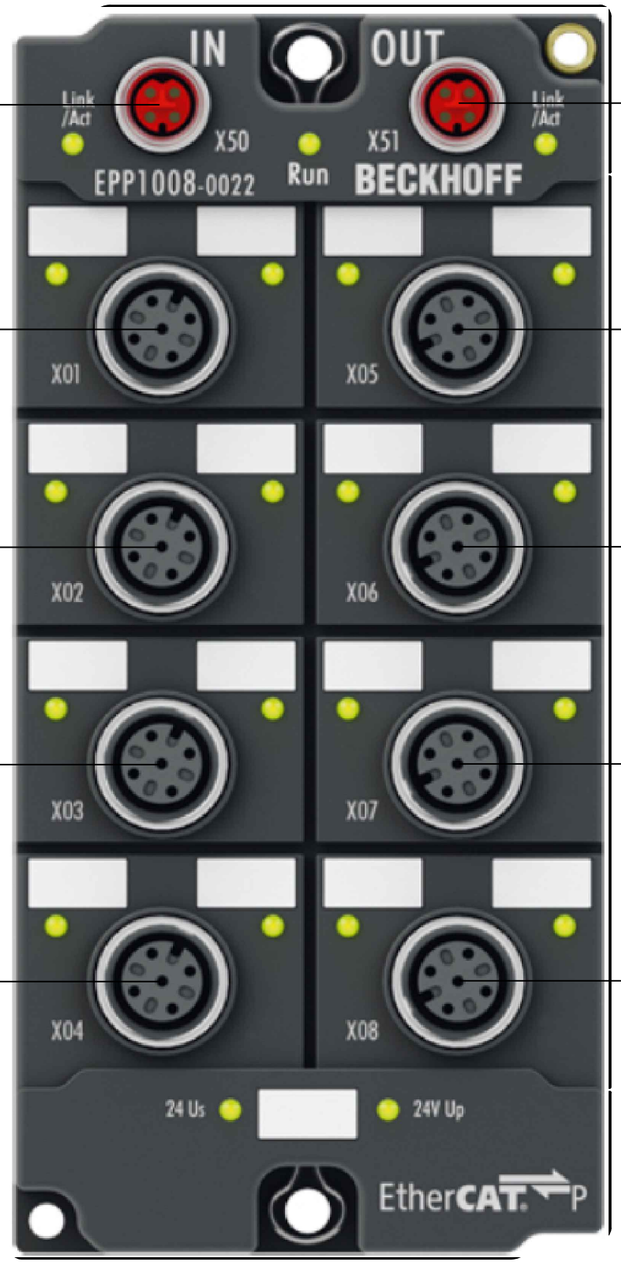
46RIO5

FROM 46RIO4-X51 46.4/8B 46ECP4 → 46ECP4

46ECP5 → 46ECP5 46.6/3B TO 46RIO6-X50



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X05 → ?????????

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X06 → ?????????

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X07 → ?????????

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X08 → ?????????

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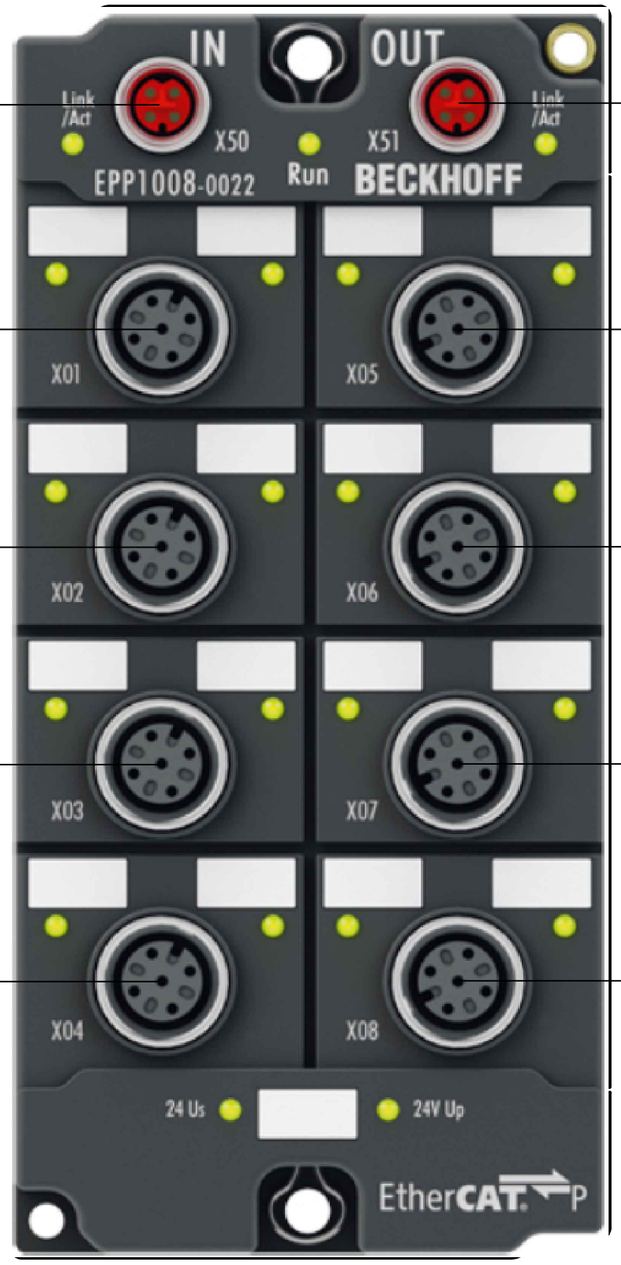
46RIO6

FROM 46RIO5-X51 46.5/8B 46ECP5 → 46ECP5

46ECP6 → 46ECP6 46.6/3B TO 46RIO7-X50



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PAGE DESCRIPTION: 46RIO6 REMOTE I/O INPUT LAYOUT

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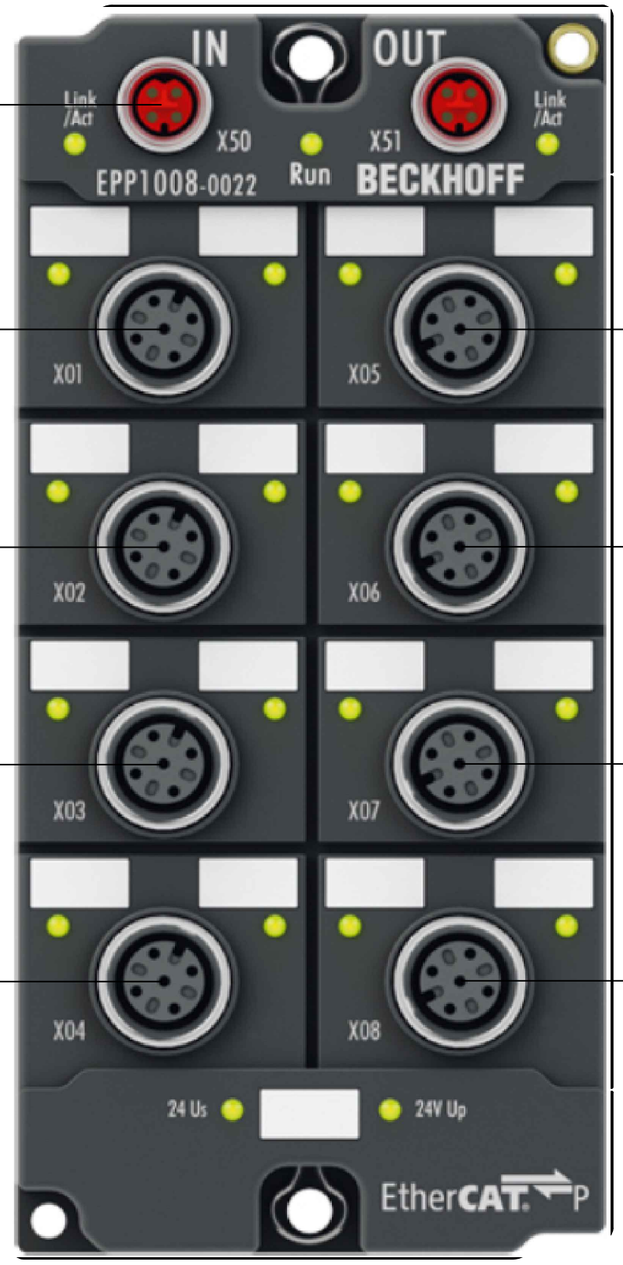
EPP1008-0022

46RIO7

FROM 46RIO6-X51 46.6/8B 46ECP6

46ECP6



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PAGE DESCRIPTION: 46RIO7 REMOTE I/O INPUT LAYOUT

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VTD (MOBILE) / EXTRACTOR AREA

FROM 23.0-11-X1 34.0/7C 34ECP1

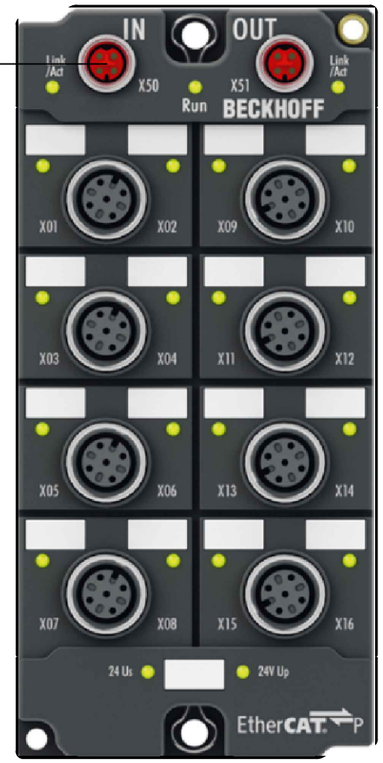
47ECP1

47RIO1

EPP2328-0002

EXTRACTOR CONVEYOR LOCKING PIN

47RIO2



EPP1809-0022

EXTRACTOR CONVEYOR

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

REV	DESCRIPTION	DATE

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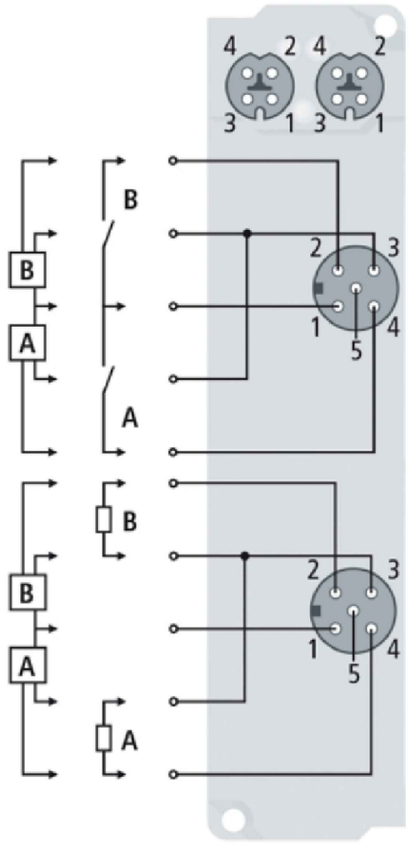
47RIO1
EPP2328-0002

Contact	Signal	Voltage	Core color [1]
1	Tx +	GND_S	yellow
2	Rx +	GND_P	white
3	Rx -	U_P: peripheral voltage, +24 V_{DC}	blue
4	Tx -	U_S: control voltage, +24 V_{DC}	orange
Housing	Shield	Shield	Shield

2 x M8 socket, 4-pin, P-coded, shielded

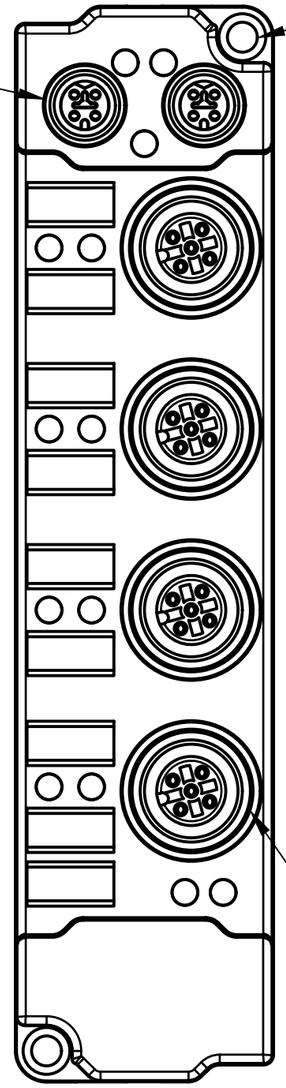


1 | Tx+ / GND_S
2 | Rx+ / GND_P
3 | Rx- / U_P
4 | Tx- / U_S

1 | +24 V U_S
2 | Input B
3 | GND_S
4 | Input A
5 | n.c.

1 | +24 V U_P
2 | Output B
3 | GND_P
4 | Output A
5 | n.c.

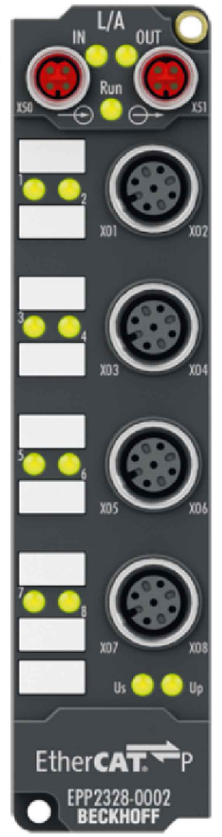
Connector diameter	Tightening torque
M8	0.4 Nm
M12	0.6 Nm



The upper mounting holes also serve as a connection for functional earth (FE).
Make sure that the box is grounded to low impedance via the functional earth (FE) connection. You can achieve this, for example, by mounting the box on a grounded machine bed.

EtherCAT P input

EtherCAT P downstream connection



Digital inputs

Digital outputs

4 x M12 socket, 5-pin, A-coded

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 47RIO1 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

REV	DESCRIPTION	DATE

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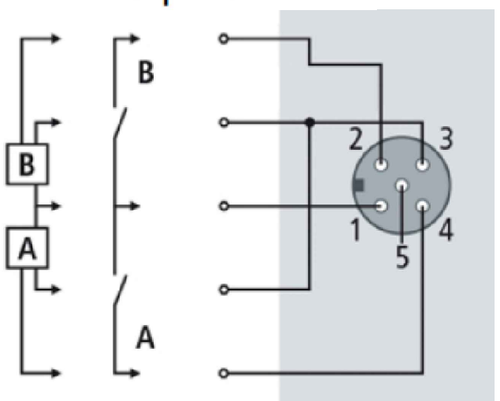
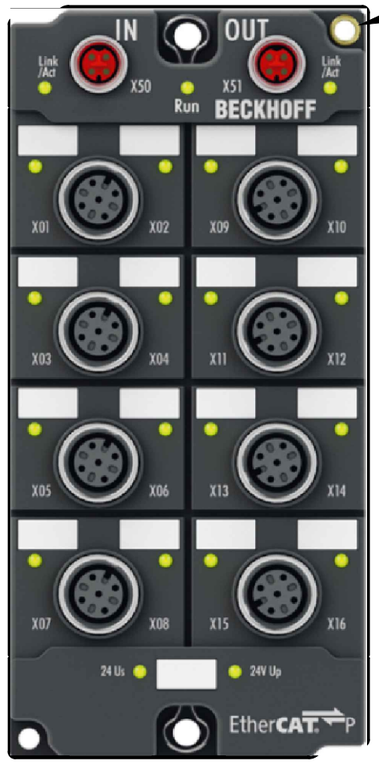
1 | Tx+ / GND_S
2 | Rx+ / GND_P
3 | Rx- / U_P
4 | Tx- / U_S

1 | +24 V U_S
2 | Input B
3 | GND_S
4 | Input A
5 | n.c.



47RIO2



EPP1809-0022

The upper mounting holes also serve as a connection for functional earth (FE).
Make sure that the box is grounded to low impedance via the functional earth (FE) connection. You can achieve this, for example, by mounting the box on a grounded machine bed.

Pin assignment

Pin	Function	Wire color [1)]
1	U_S	brown
2	Input B	white
3	GND_S	blue
4	Input A	black
5	-	gray

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL
PAGE DESCRIPTION: 47RIO2 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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PLEASE REFERENCE DOCKING PINS J-BOX ELECTRICAL DRAWING

47RIO1
EPP2328-0002

FROM 23.0-11-X1 34.0/7C 34ECP1 → 34ECP1

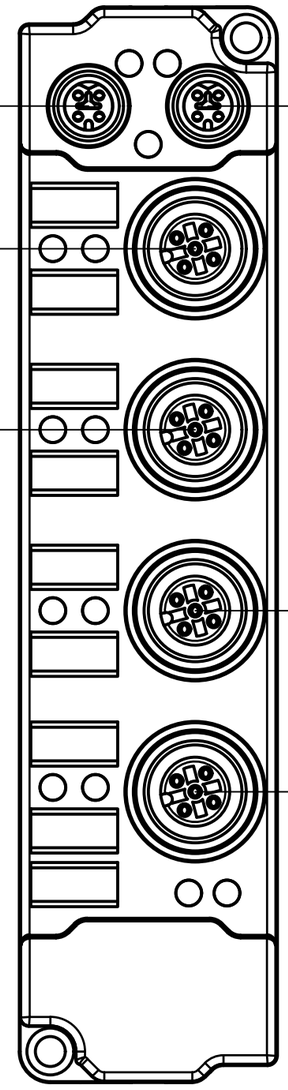
47ECP1 → 47ECP1 47.4/3B TO 47RIO2-X50

FROM LEFT LOCKING PINS EXTENDED PROXIMITY SENSORS — II

FROM RIGHT LOCKING PINS EXTENDED PROXIMITY SENSORS — III

FROM LEFT LOCKING PINS RETRACTED PROXIMITY SENSORS — II

FROM RIGHT LOCKING PINS RETRACTED PROXIMITY SENSORS — III



II — TO 3KA1 IN JUNCTION BOX.

III — TO 3KA2 IN JUNCTION BOX.

II — TO 3KA3 IN JUNCTION BOX.

III — TO 3KA4 IN JUNCTION BOX.

X4

6ES71946KA000XA0

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 47RIO1 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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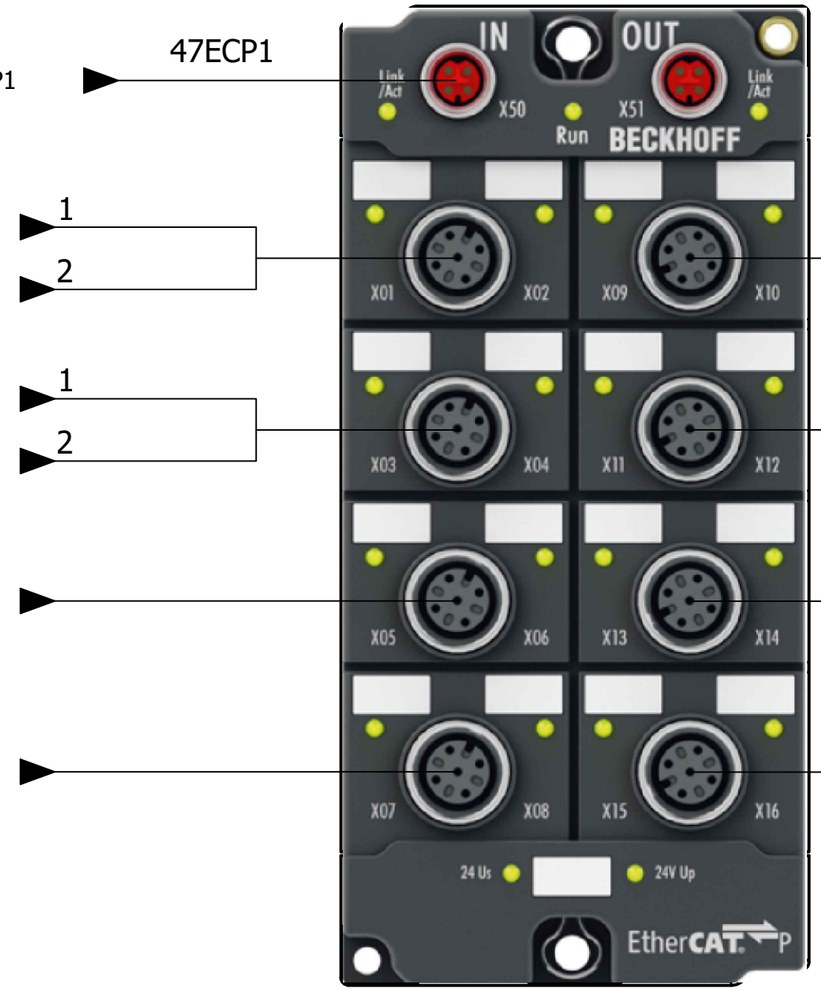
47RIO2

FROM 47RIO1-X51 47.3/7B 34ECP1

47ECP1



EPP1809-0022

LEVEL DETECT 1 SENSOR 1

LEVEL DETECT 2 SENSOR 2

CREEP SENSOR 1

SLOWDOWN SENSOR 2

PALLET DETECTION #1

PALLET DETECTION #2

PALLET DETECTION #3

PALLET DETECTION #4

PALLET DETECTION #5

PALLET DETECTION #6

SHUTTLE DETECTION #1

SHUTTLE DETECTION #2

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 47RIO2 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

REV	DESCRIPTION	DATE

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JOB # C40572

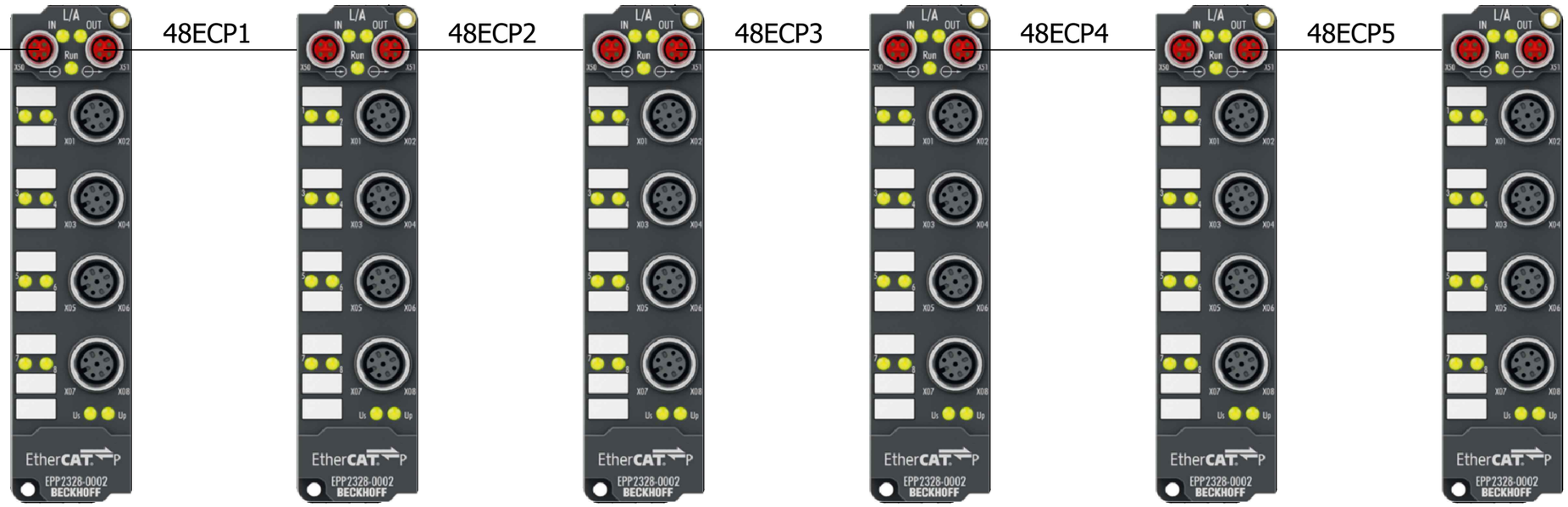
TOTAL PAGES: 171

SHEET 47.4 OF 56

BUFFER CONVEYORS AREA SECTION 1 & 2

FROM 23.0-11-X2 34.0/7E 34ECP2

48RIO1 — 48ECP1 — 48RIO2 — 48ECP2 — 48RIO3 — 48ECP3 — 48RIO4 — 48ECP4 — 48RIO5 — 48ECP5 — 48RIO6



EPP2328-0002 — CONVEYOR WITH STOPPER LEVEL 1

EPP2328-0002 — CONVEYOR WITH STOPPER LEVEL 2

EPP2328-0002 — CONVEYOR WITH STOPPER LEVEL 3

EPP2328-0002 — CONVEYOR WITH STOPPER LEVEL 4

EPP2328-0002 — CONVEYOR WITH STOPPER LEVEL 5

EPP2328-0002 — CONVEYOR WITH STOPPER LEVEL 6

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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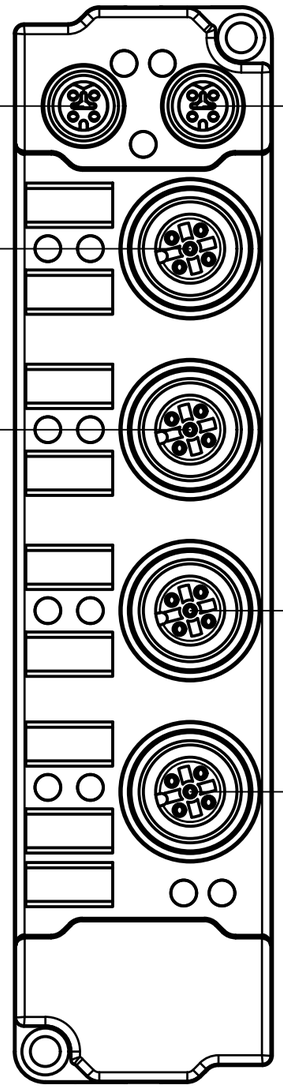
TOTAL PAGES: 171

SHEET 48.0 OF 56

CONVEYOR
WITH
STOPPER
LEVEL 1

48RIO1
EPP2328-0002



FROM 23.0-11-X2 34.0/7E 34ECP2 → 34ECP2

34ECP2 → 48ECP1

48ECP1 48.2/4B TO 48RIO2-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2

FOR WIRING METHOD, PLEASE REFER TO PAGE 47.1

TO 53RR1-(3) PG 53.0/5A

TO 53RR2-(3) PG 53.0/7E

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X2

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: 48RIO1 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE

REV	DESCRIPTION	DATE
	LAYOUT	

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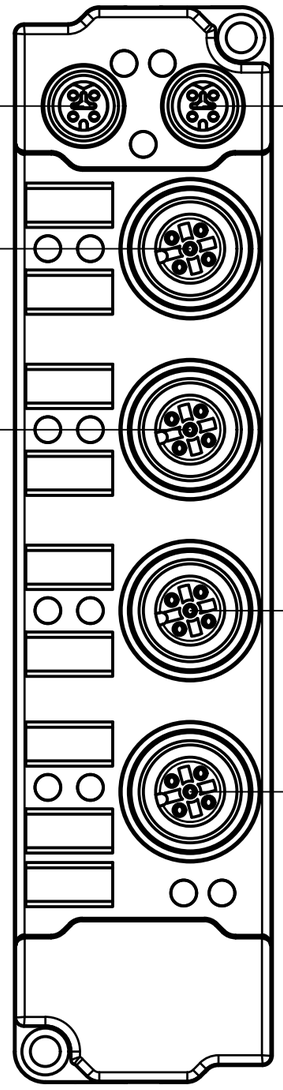
TOTAL PAGES: 171

CONVEYOR
WITH
STOPPER
LEVEL 2

48RIO2
EPP2328-0002



FROM 48RIO1-X51 48.1/7B 48ECP1 → 48ECP1

48ECP2 → 48ECP2 48.3/4B TO 48RIO3-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2

FOR WIRING METHOD, PLEASE REFER TO PAGE 47.1

TO 53RR3-(3) PG 53.1/5A

TO 53RR4-(3) PG 53.1/7E

ZS2000-4722

X2

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PAGE DESCRIPTION: 48RIO2 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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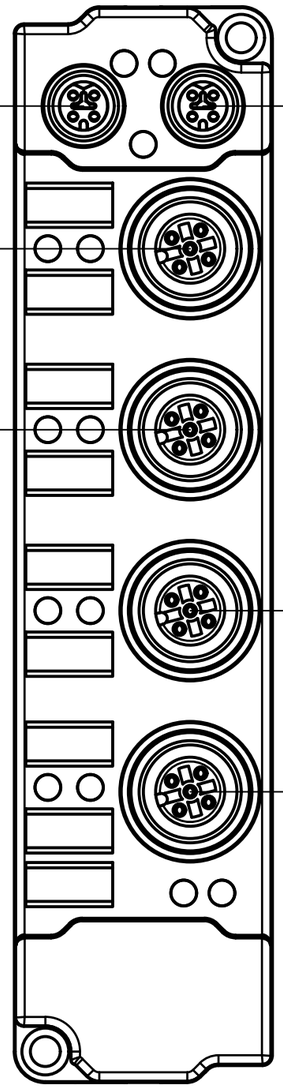
TOTAL PAGES: 171

CONVEYOR
WITH
STOPPER
LEVEL 3

48RIO3
EPP2328-0002



FROM 48RIO2-X51 48.2/7B 48ECP2 → 48ECP2

48ECP3 → 48ECP3 48.4/4B TO 48RIO4-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2

TO 53RR5-(3) PG 53.2/5A

TO 53RR6-(3) PG 53.2/7E

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X4

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PAGE DESCRIPTION: 48RIO3 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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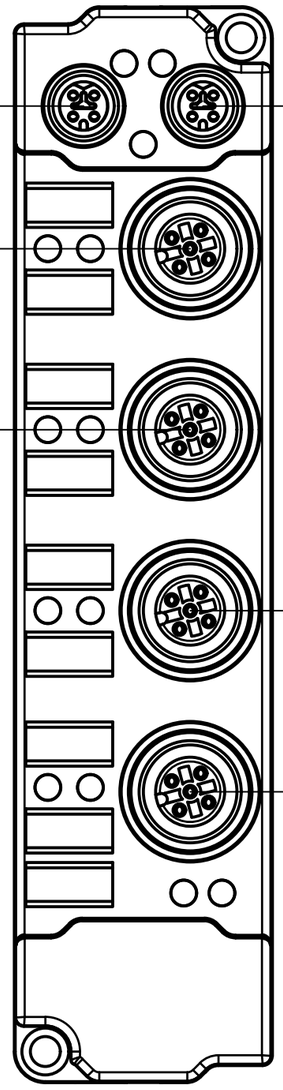
TOTAL PAGES: 171

CONVEYOR
WITH
STOPPER
LEVEL 4

48RIO4
EPP2328-0002



FROM 48RIO3-X51 48.3/7B 48ECP3 → 48ECP3

48ECP4 → 48ECP4 48.5/4B TO 48RIO5-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2

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X4

TO 53RR7-(3) PG 53.3/5A

TO 53RR8-(3) PG 53.3/7E

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PAGE DESCRIPTION: 48RIO4 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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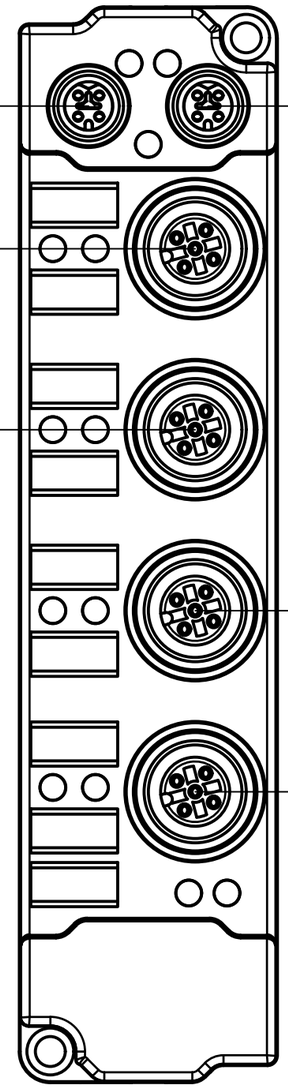
TOTAL PAGES: 171

CONVEYOR
WITH
STOPPER
LEVEL 5

48RIO5
EPP2328-0002



FROM 48RIO4-X51 48.4/7B 48ECP4 → 48ECP4

48ECP5 → 48ECP5 48.6/4B TO 48RIO6-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2

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X4

TO 53RR9-(3) PG 53.4/5A

TO 53RR10-(3) PG 53.4/7E

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PAGE DESCRIPTION: 48RIO5 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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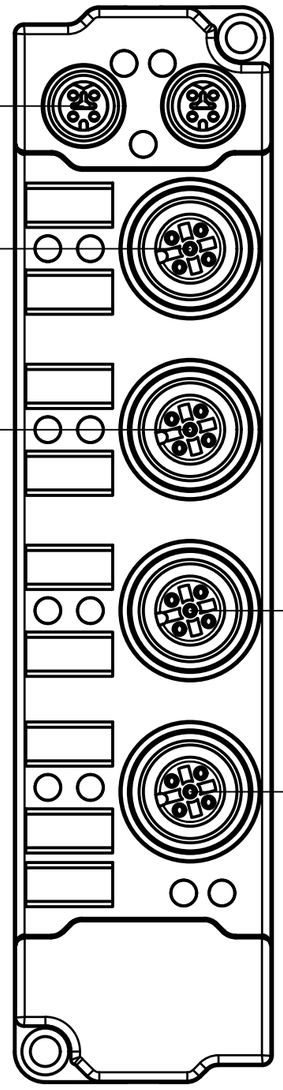
TOTAL PAGES: 171

CONVEYOR
WITH
STOPPER
LEVEL 6

48RIO6
EPP2328-0002



FROM 48RIO5-X51 48.5/7B 48ECP5 ▶ 48ECP5

SAFETY STOPPER UPPER SENSOR ▶ 1

SAFETY STOPPER LOWER SENSOR ▶ 2

BUFFER CONVEYOR SENSOR 1 ▶ 1

BUFFER CONVEYOR SENSOR 2 ▶ 2

TO 53RR11-(3) PG 53.5/5A ◀

TO 53RR12-(3) PG 53.5/7E ◀

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X4

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PAGE DESCRIPTION: 48RIO6 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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PAGE DESCRIPTION: SPARE

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JOB # C40572

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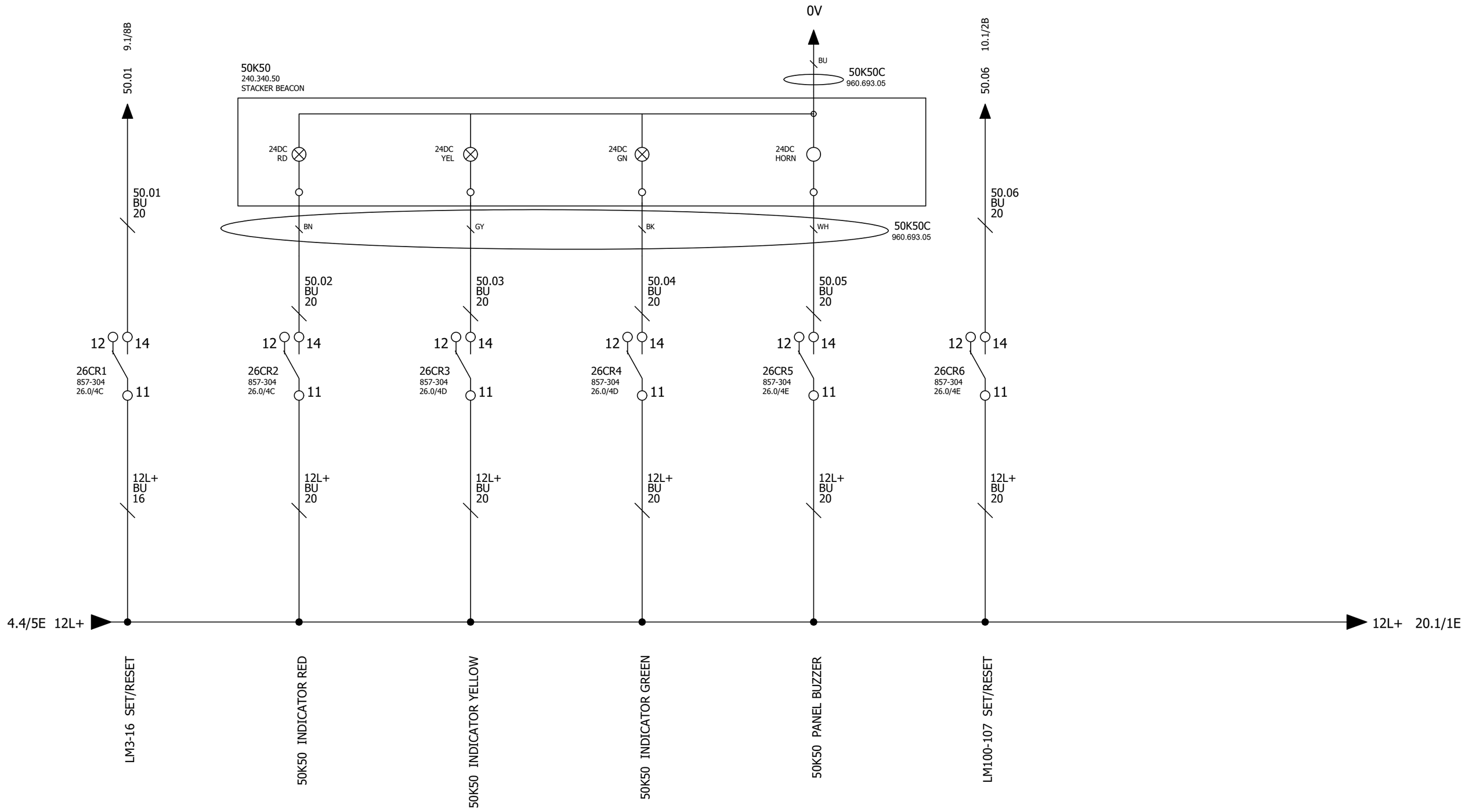
SHEET 49.0 OF 56



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PAGE DESCRIPTION: RELAYS

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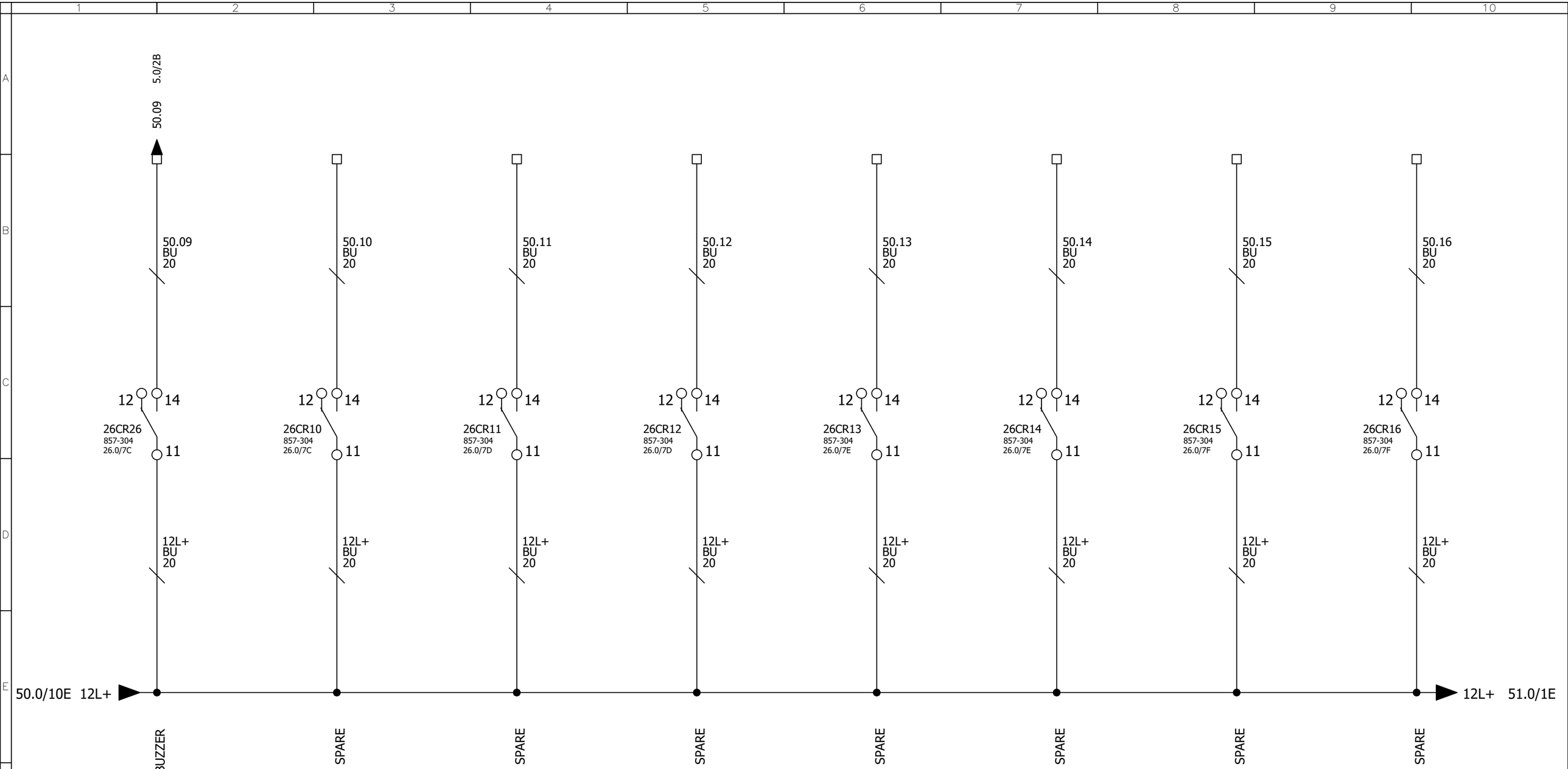
TOTAL PAGES: 171

SHEET 50.0 OF 56



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: RELAYS

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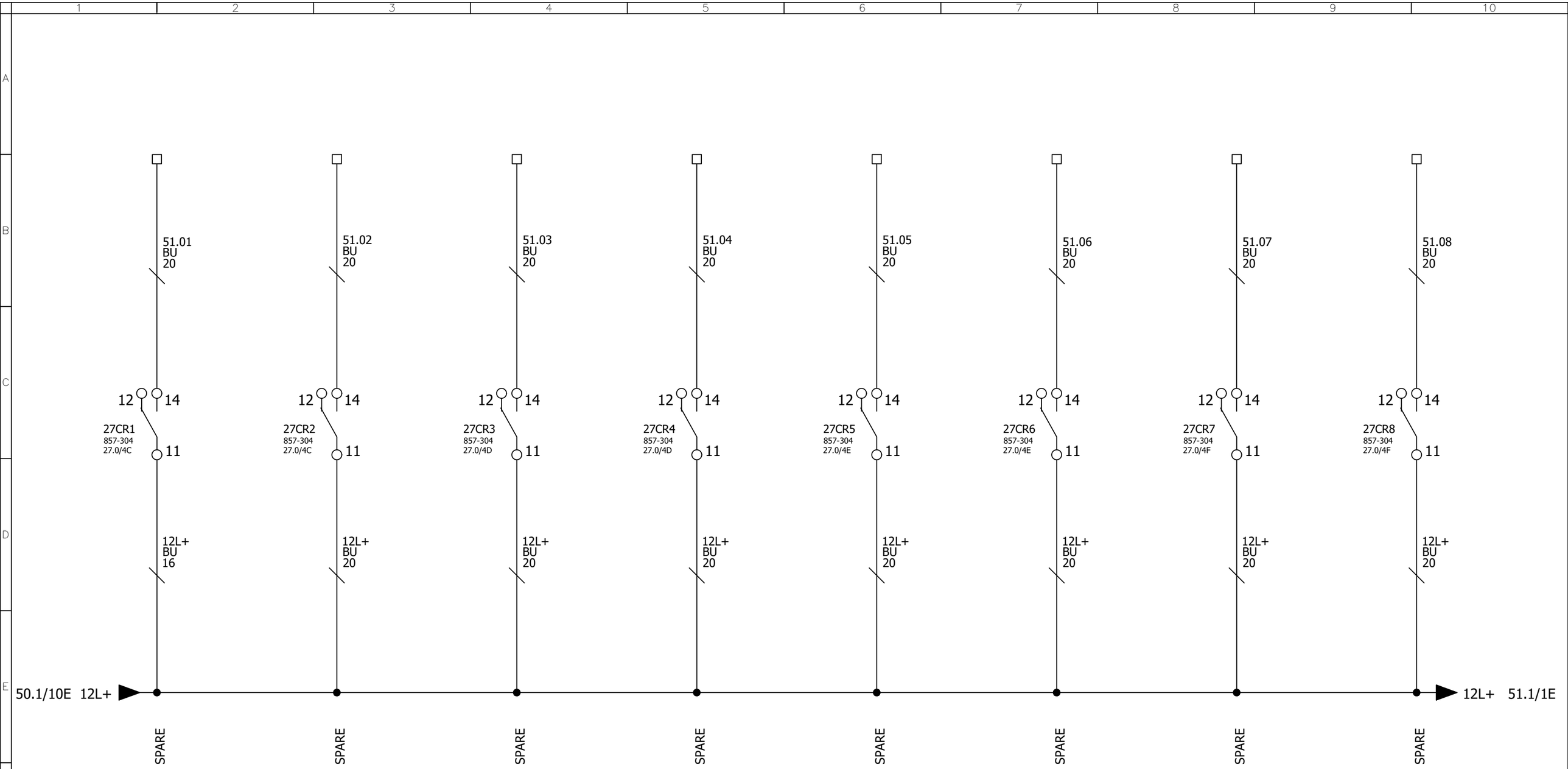
TOTAL PAGES: 171

SHEET 50.1 OF 56



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PAGE DESCRIPTION: RELAYS

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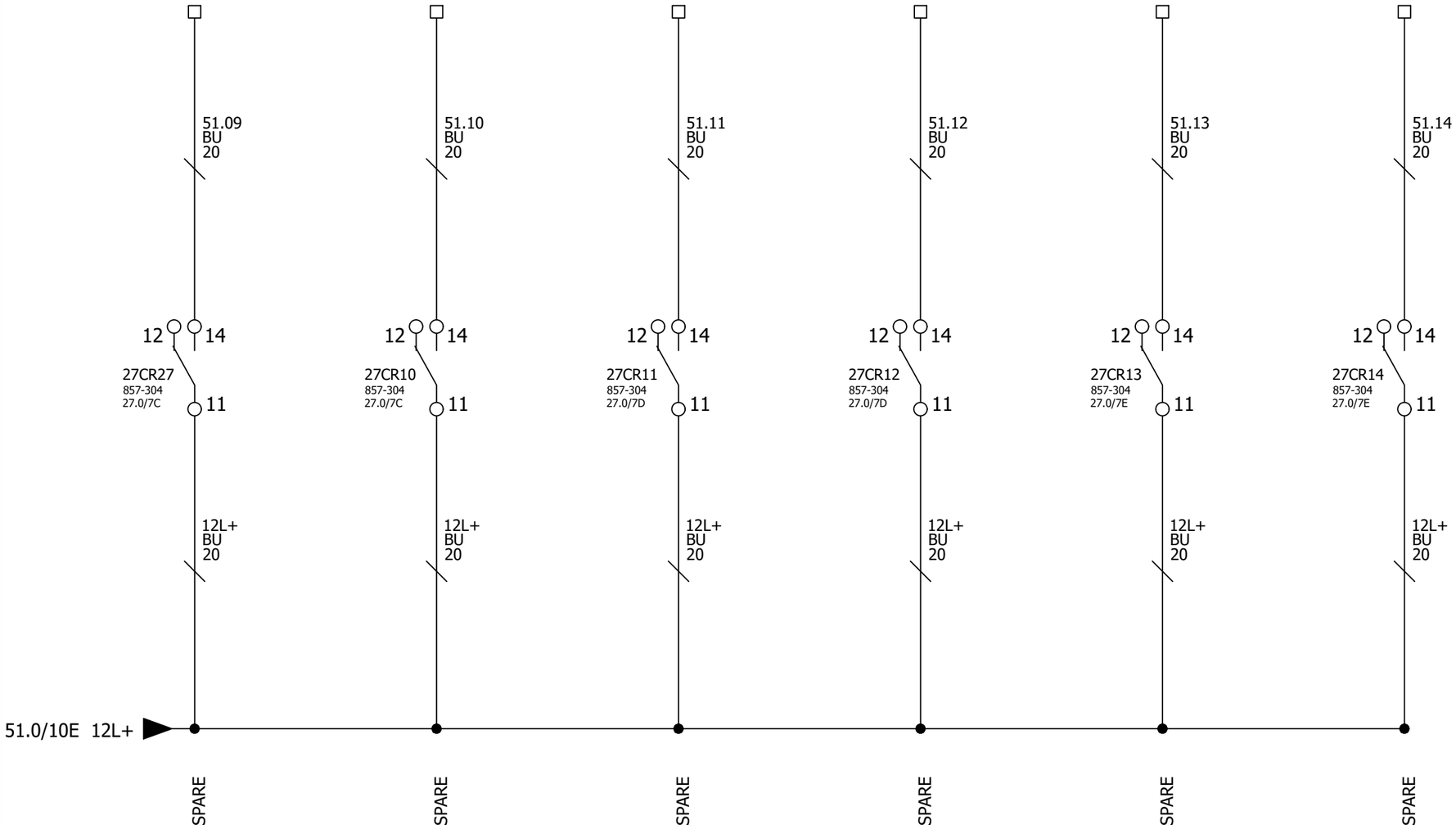
JOB # C40572

TOTAL PAGES: 171

SHEET 51.0 OF 56



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: RELAYS

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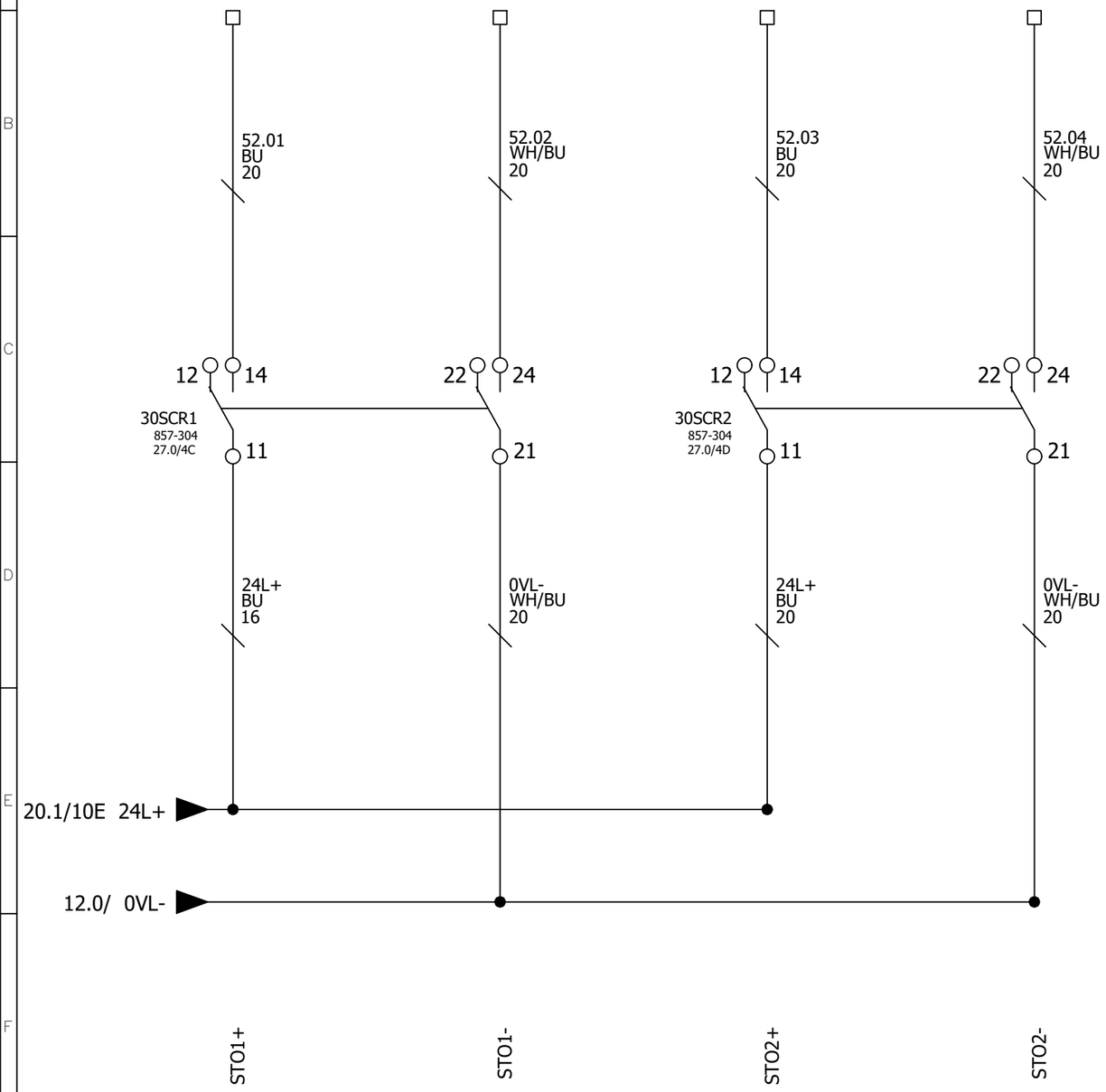
TOTAL PAGES: 171

SHEET 51.1 OF 56



VFD's STO #1

CAROUSEL CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 5 VFD's

CAROUSEL CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

VFD's STO #2

VTD STATIC AREA DRIVE

VTD MOBILE EXTRACTOR CONVEYOR AREA VFD

BUFFER CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 3 VFD's

BUFFER CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: SAFETY RELAYS

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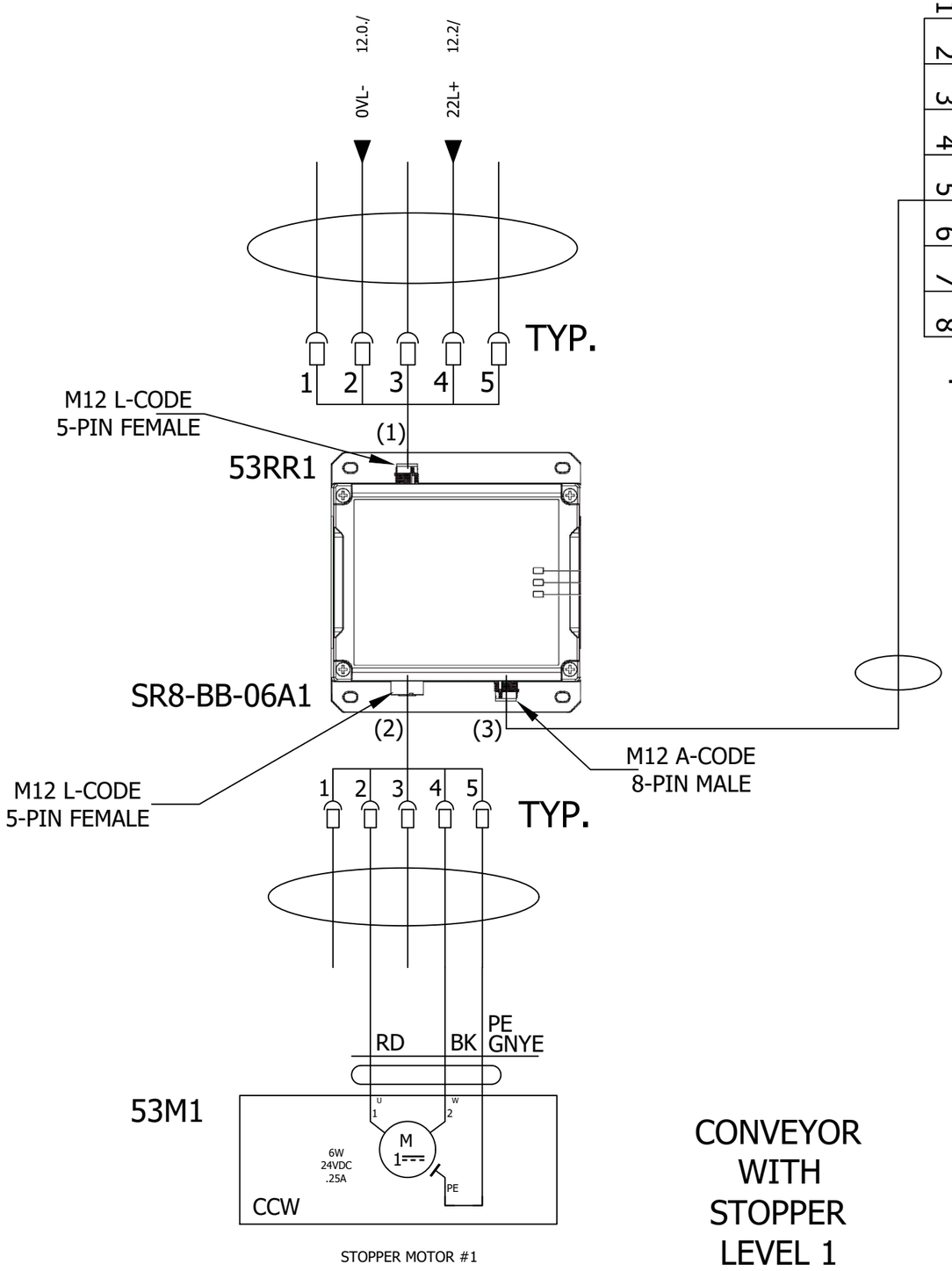
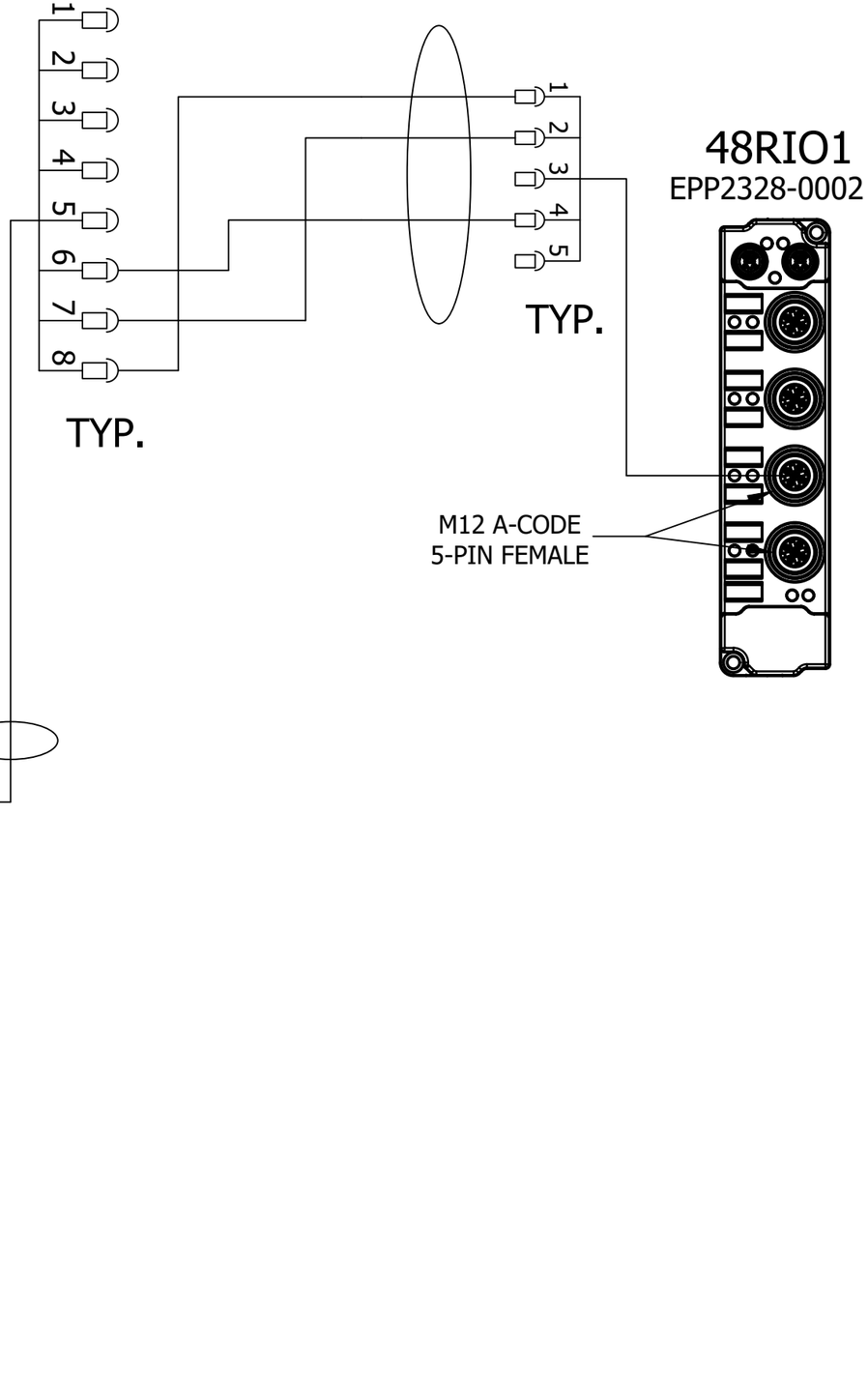
TOTAL PAGES: 171

SHEET 52.0 OF 56

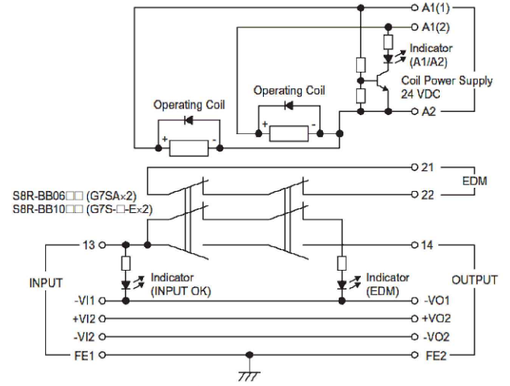


STOPPER MOTOR #1

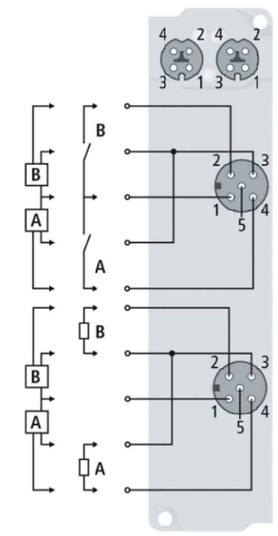


CONVEYOR WITH STOPPER LEVEL 1

SR8-BB-06A1



No.	Connector Name	Shape	Pin Number/Name
(1)	Input Terminal M12-L (Plug)		FE: FE1 (Functional Earth ⏚) 1: +VI2 2: -VI1 3: -VI2 4: 13
(2)	Output Terminal M12-L (Socket)		FE: FE2 1: +VO2 2: -VO1 3: -VO2 4: 14
(3)	EDM/Coil Terminal M12-A (Plug)		1: 21 2: 22 3: N/A 4: N/A 5: N/A 6: A1 (1) 7: A1 (2) 8: A2



1 | Tx+ / GND_S
2 | Rx+ / GND_P
3 | Rx- / U_P
4 | Tx- / U_S

1 | +24 V U_S
2 | Input B
3 | GND_S
4 | Input A
5 | n.c.

1 | +24 V U_P
2 | Output B
3 | GND_P
4 | Output A
5 | n.c.

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: LEVEL 1 STOPPER MOTORS

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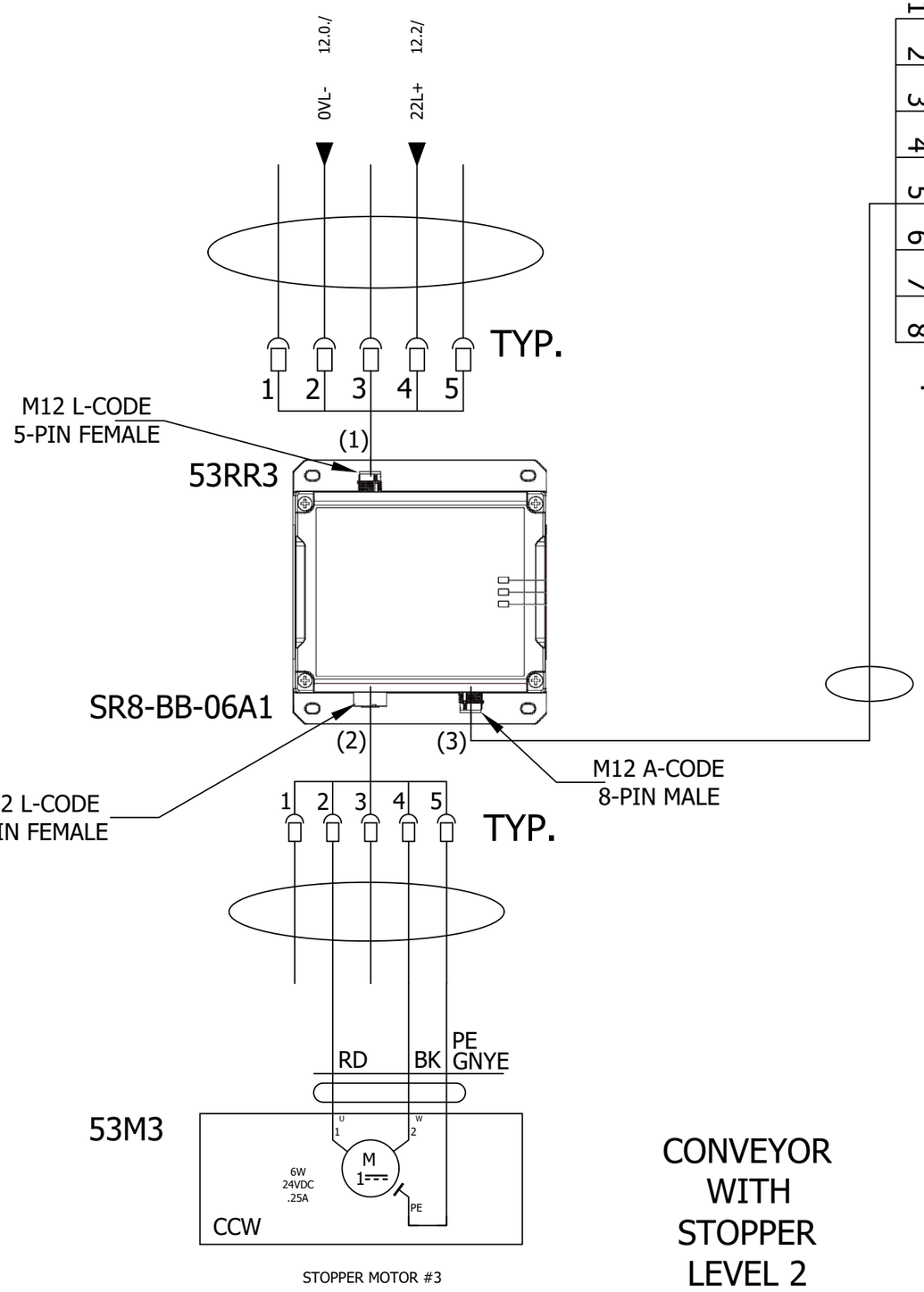

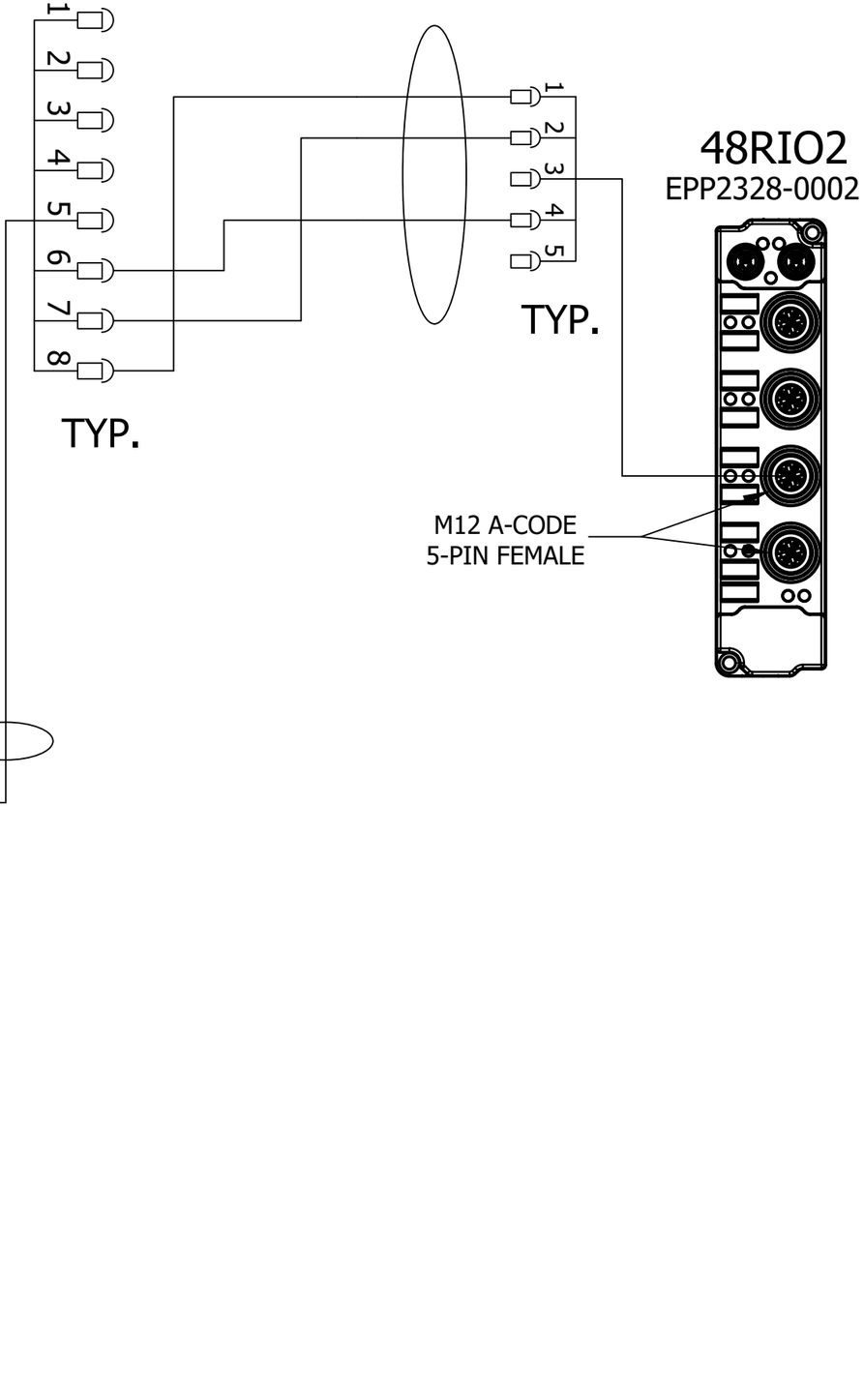

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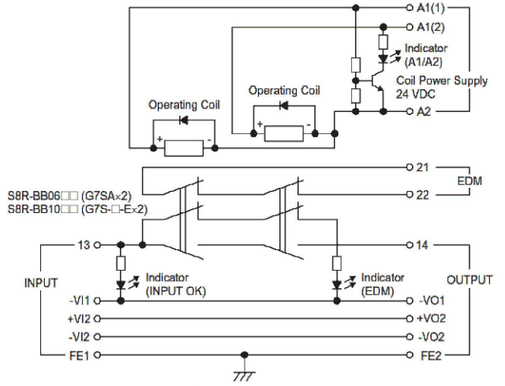
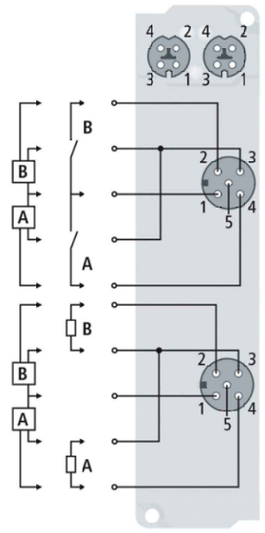


No.	Connector Name	Shape	Pin Number/Name
(1)	Input Terminal M12-L (Plug)		FE: FE1 (Functional Earth ⏚) 1: +VI2 2: -VI1 3: -VI2 4: 13
(2)	Output Terminal M12-L (Socket)		FE: FE2 1: +VO2 2: -VO1 3: -VO2 4: 14
(3)	EDM/Coil Terminal M12-A (Plug)		1: 21 2: 22 3: N/A 4: N/A 5: N/A 6: A1 (1) 7: A1 (2) 8: A2



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CONVEYOR WITH STOPPER LEVEL 2

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PAGE DESCRIPTION: LEVEL 2 STOPPER MOTORS

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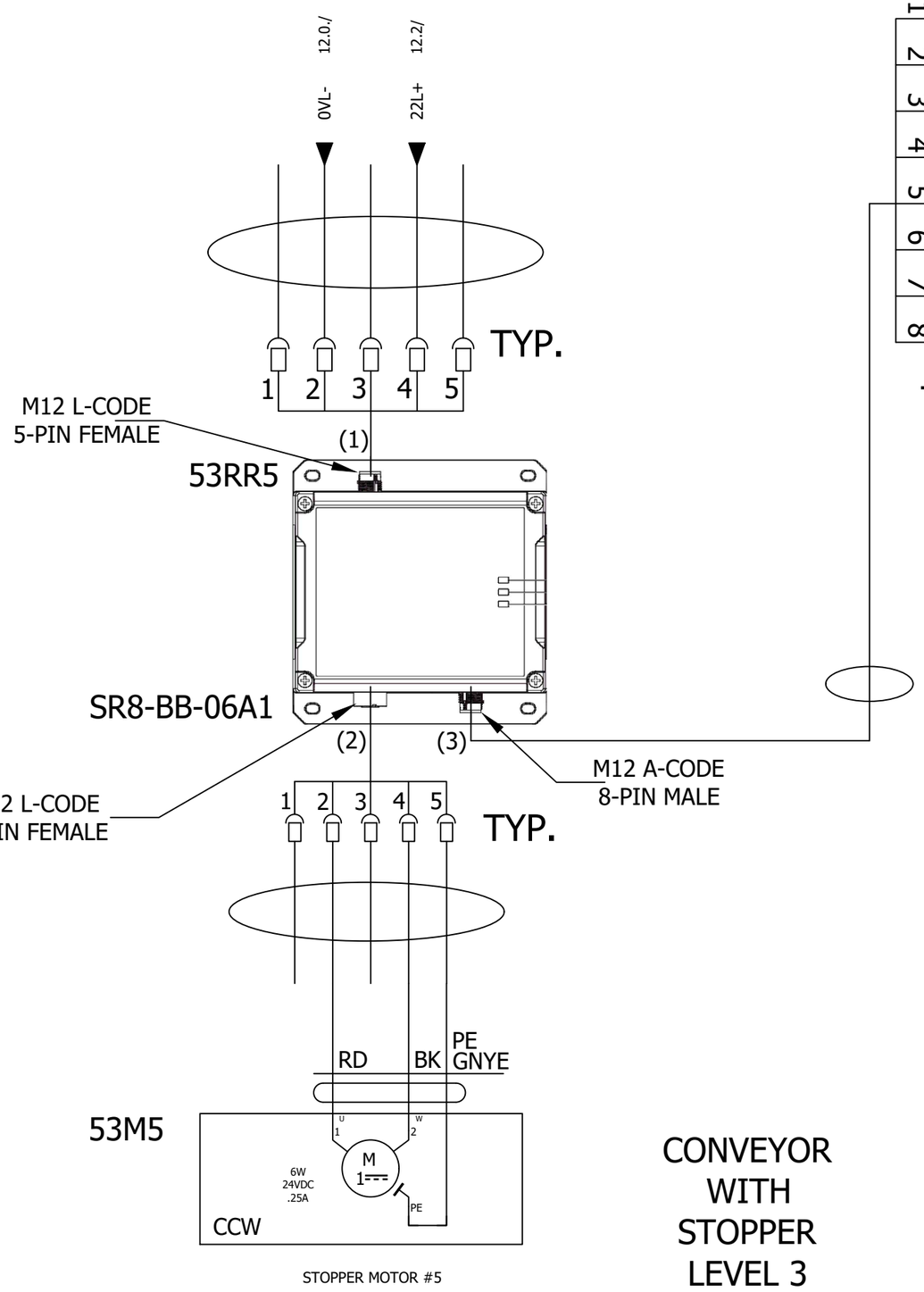
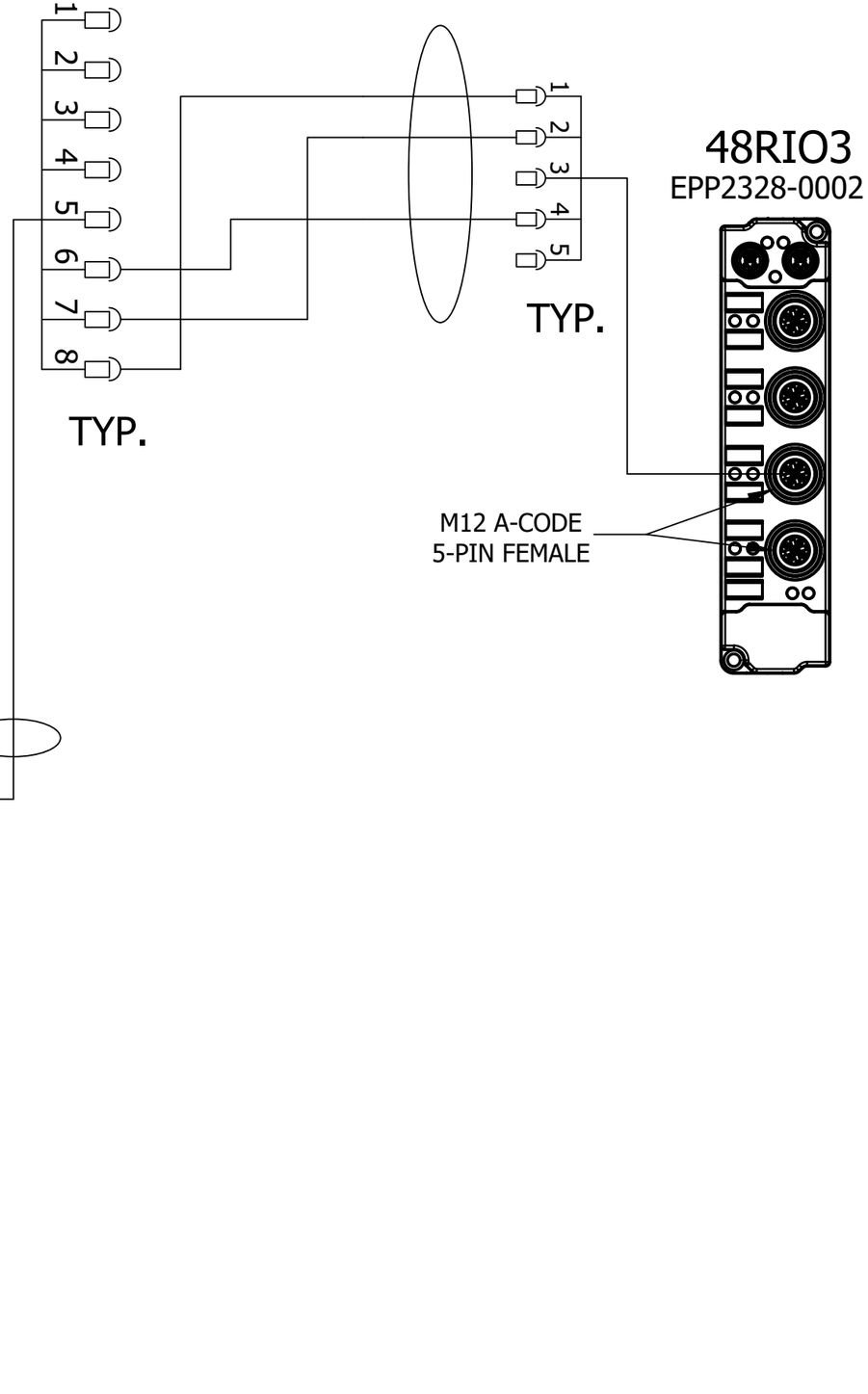
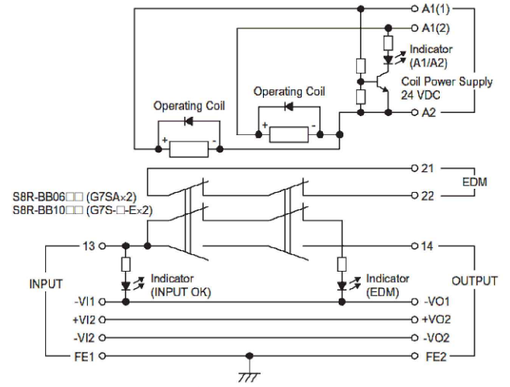


STOPPER MOTOR #5

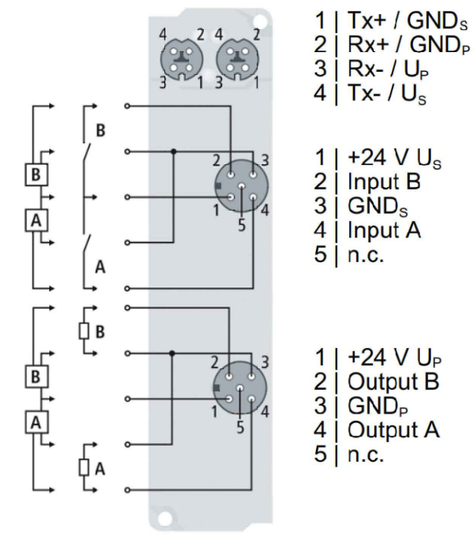


CONVEYOR WITH STOPPER LEVEL 3

SR8-BB-06A1



No.	Connector Name	Shape	Pin Number/Name
(1)	Input Terminal M12-L (Plug)		FE: FE1 (Functional Earth ⏚) 1: +VI2 2: -VI1 3: -VI2 4: 13
(2)	Output Terminal M12-L (Socket)		FE: FE2 1: +VO2 2: -VO1 3: -VO2 4: 14
(3)	EDM/Coil Terminal M12-A (Plug)		1: 21 2: 22 3: N/A 4: N/A 5: N/A 6: A1 (1) 7: A1 (2) 8: A2



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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: LEVEL 3 STOPPER MOTORS

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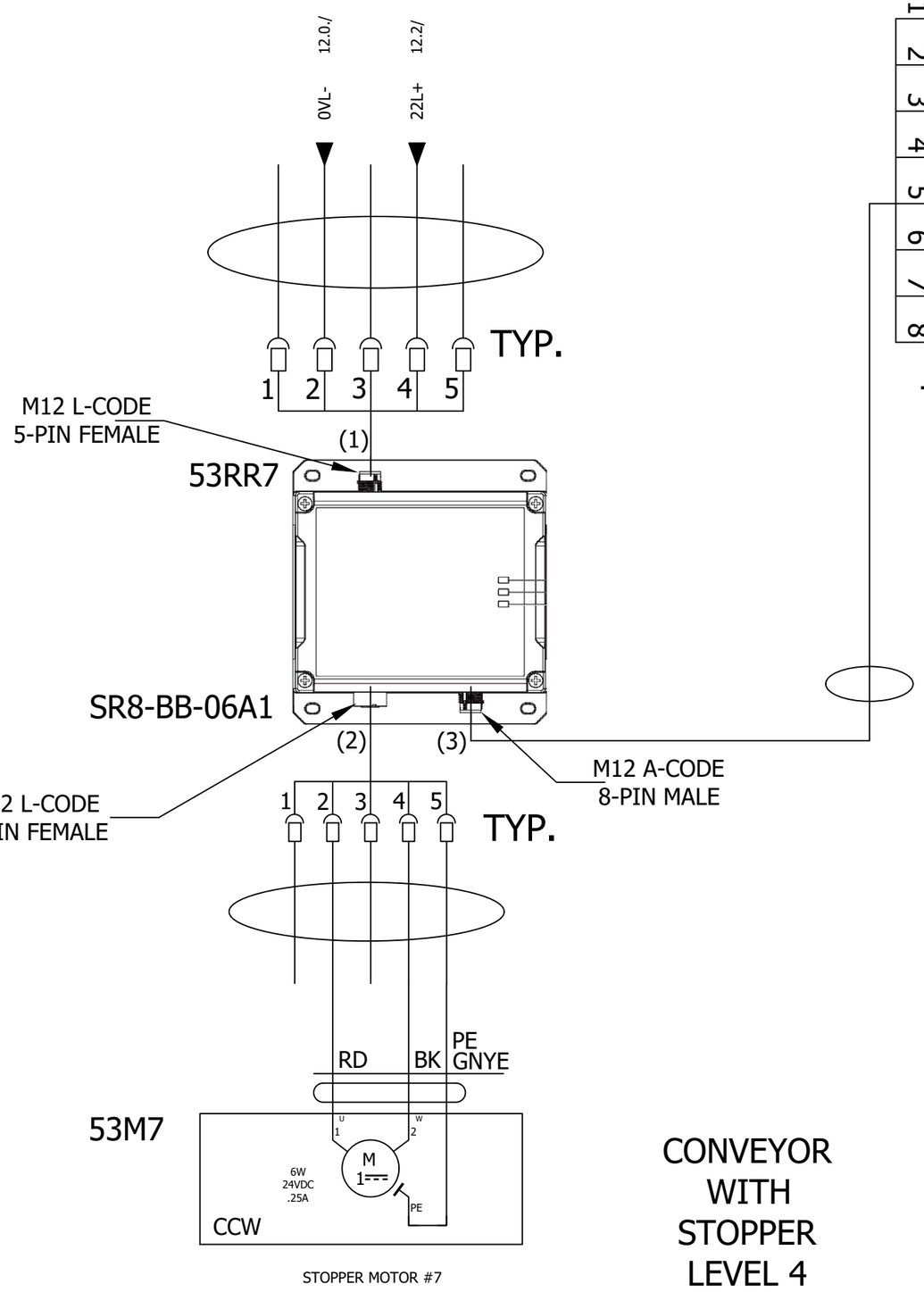
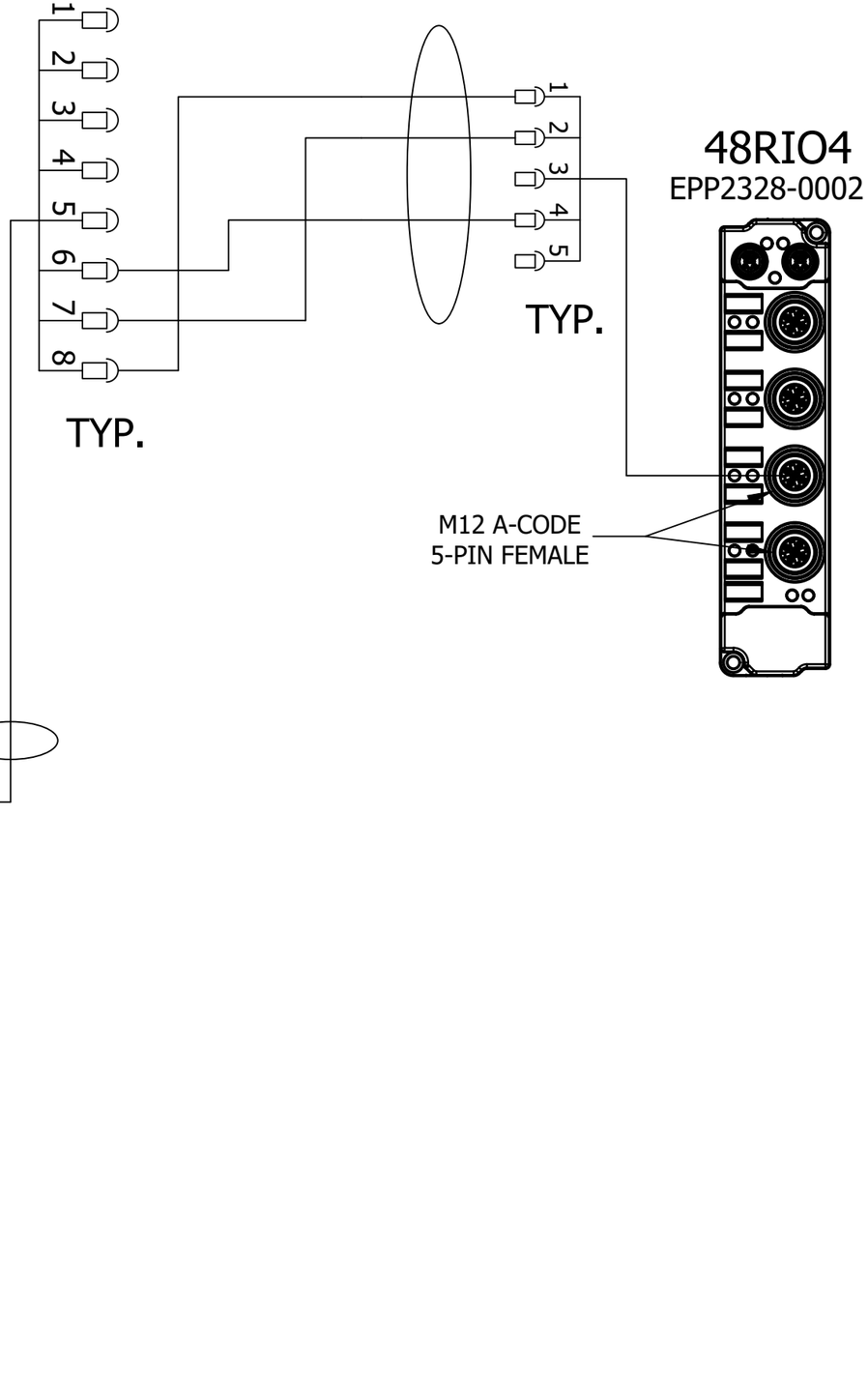
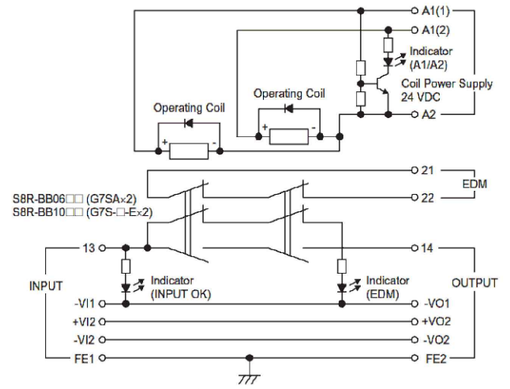


STOPPER MOTOR #7

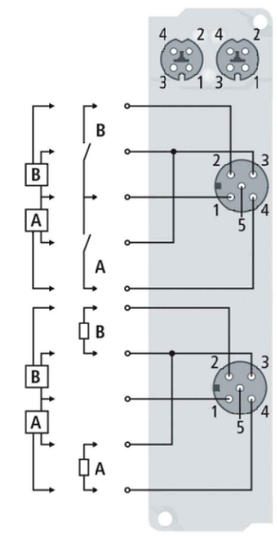


CONVEYOR WITH STOPPER LEVEL 4

SR8-BB-06A1



No.	Connector Name	Shape	Pin Number/Name
(1)	Input Terminal M12-L (Plug)		FE: FE1 (Functional Earth ⏚) 1: +VI2 2: -VI1 3: -VI2 4: 13
(2)	Output Terminal M12-L (Socket)		FE: FE2 1: +VO2 2: -VO1 3: -VO2 4: 14
(3)	EDM/Coil Terminal M12-A (Plug)		1: 21 2: 22 3: N/A 4: N/A 5: N/A 6: A1 (1) 7: A1 (2) 8: A2



1 | Tx+ / GND_S
2 | Rx+ / GND_P
3 | Rx- / U_P
4 | Tx- / U_S

1 | +24 V U_S
2 | Input B
3 | GND_S
4 | Input A
5 | n.c.

1 | +24 V U_P
2 | Output B
3 | GND_P
4 | Output A
5 | n.c.

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: LEVEL 4 STOPPER MOTORS

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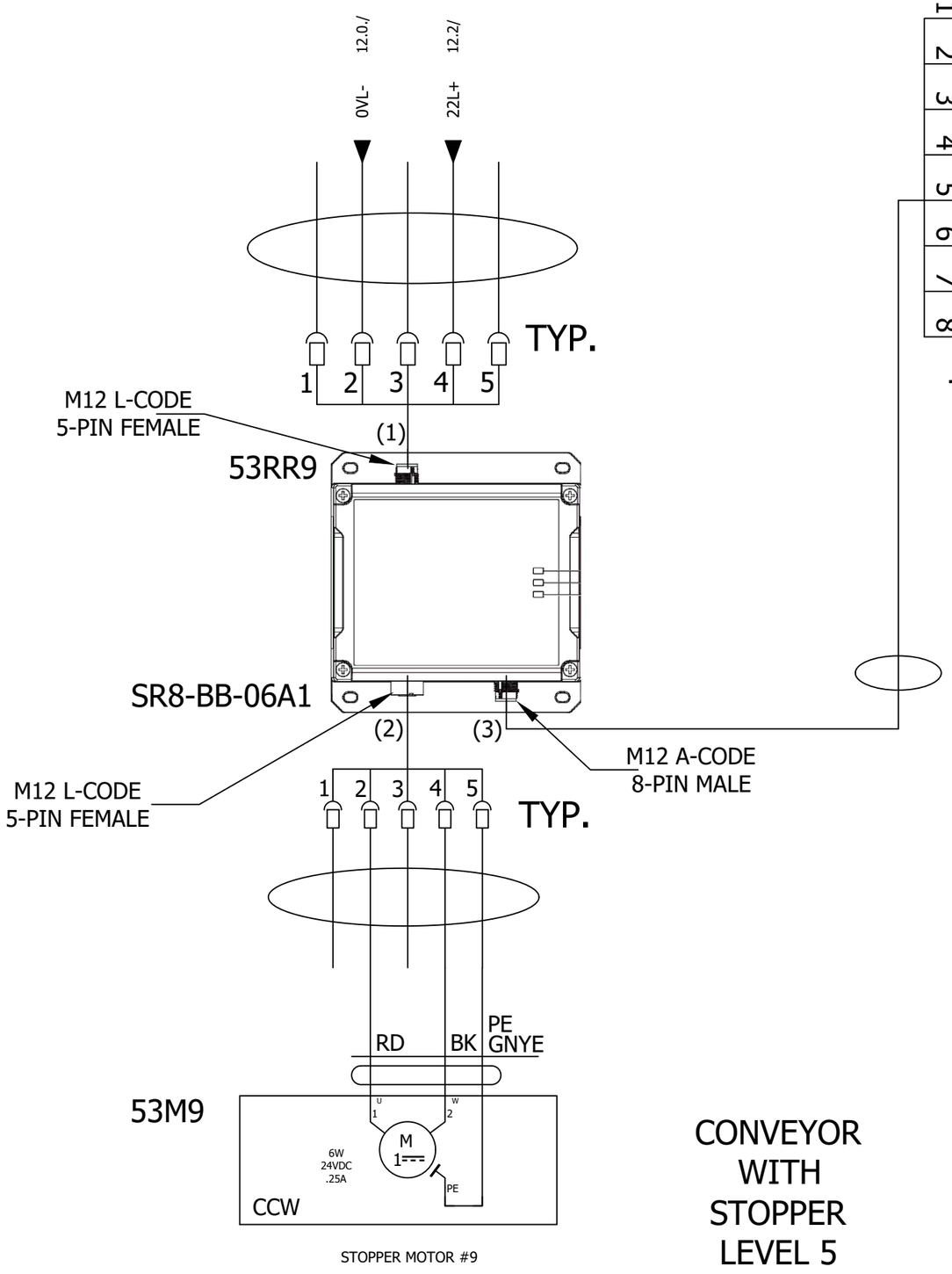
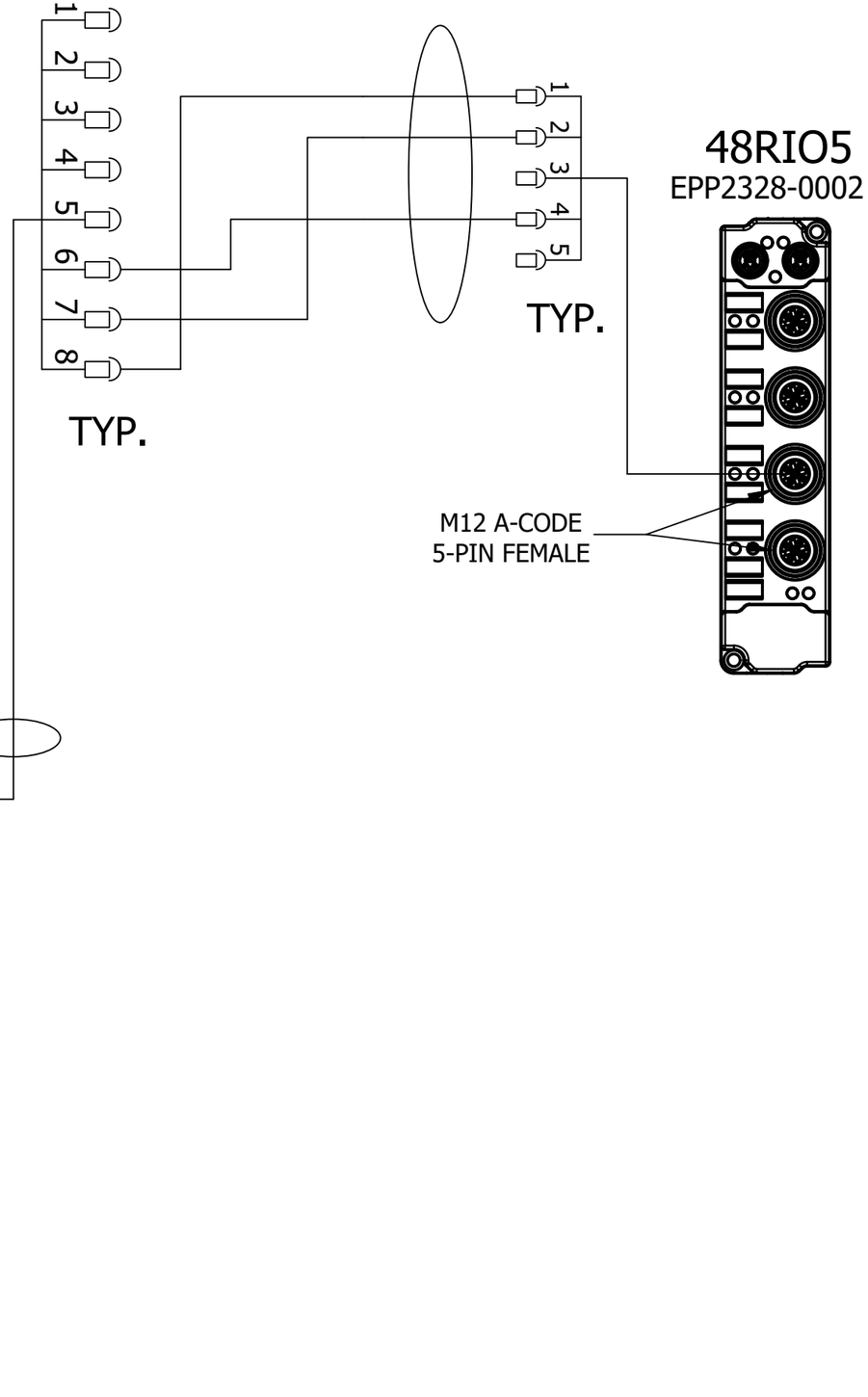
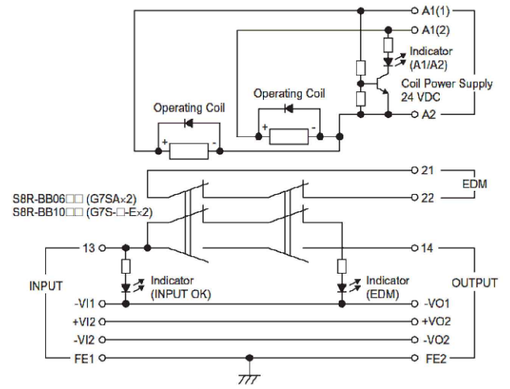


STOPPER MOTOR #9

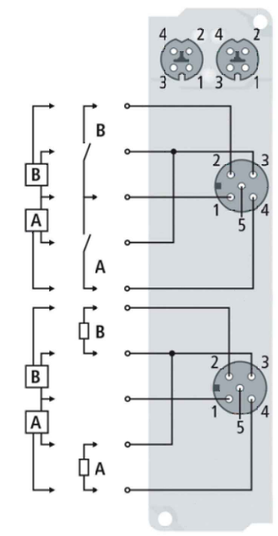


CONVEYOR WITH STOPPER LEVEL 5

SR8-BB-06A1



No.	Connector Name	Shape	Pin Number/Name
(1)	Input Terminal M12-L (Plug)		FE: FE1 (Functional Earth ⏚) 1: +VI2 2: -VI1 3: -VI2 4: 13
(2)	Output Terminal M12-L (Socket)		FE: FE2 1: +VO2 2: -VO1 3: -VO2 4: 14
(3)	EDM/Coil Terminal M12-A (Plug)		1: 21 2: 22 3: N/A 4: N/A 5: N/A 6: A1 (1) 7: A1 (2) 8: A2



1 | Tx+ / GND_S
2 | Rx+ / GND_P
3 | Rx- / U_P
4 | Tx- / U_S

1 | +24 V U_S
2 | Input B
3 | GND_S
4 | Input A
5 | n.c.

1 | +24 V U_P
2 | Output B
3 | GND_P
4 | Output A
5 | n.c.

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PAGE DESCRIPTION: LEVEL 5 STOPPER MOTORS

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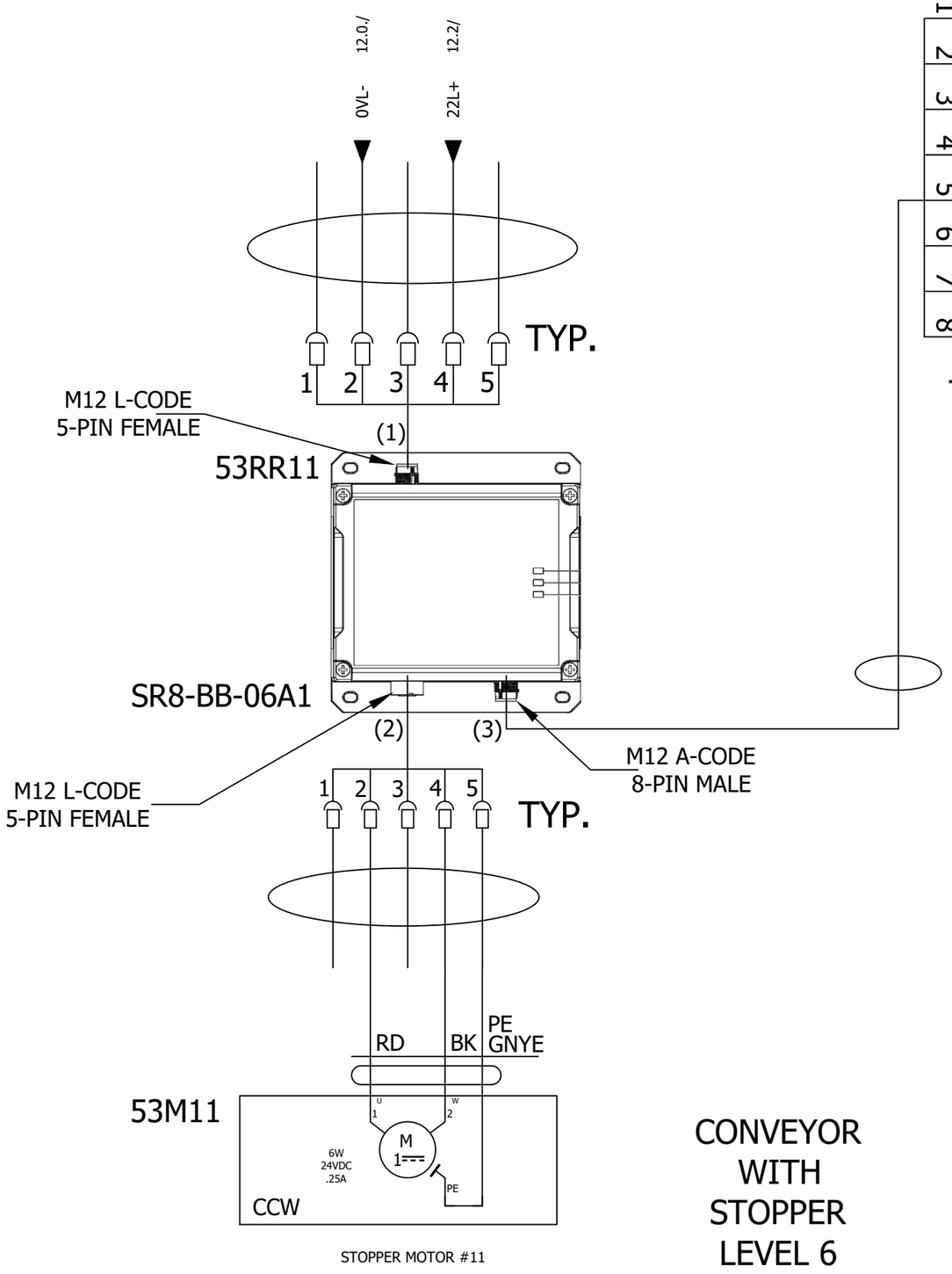

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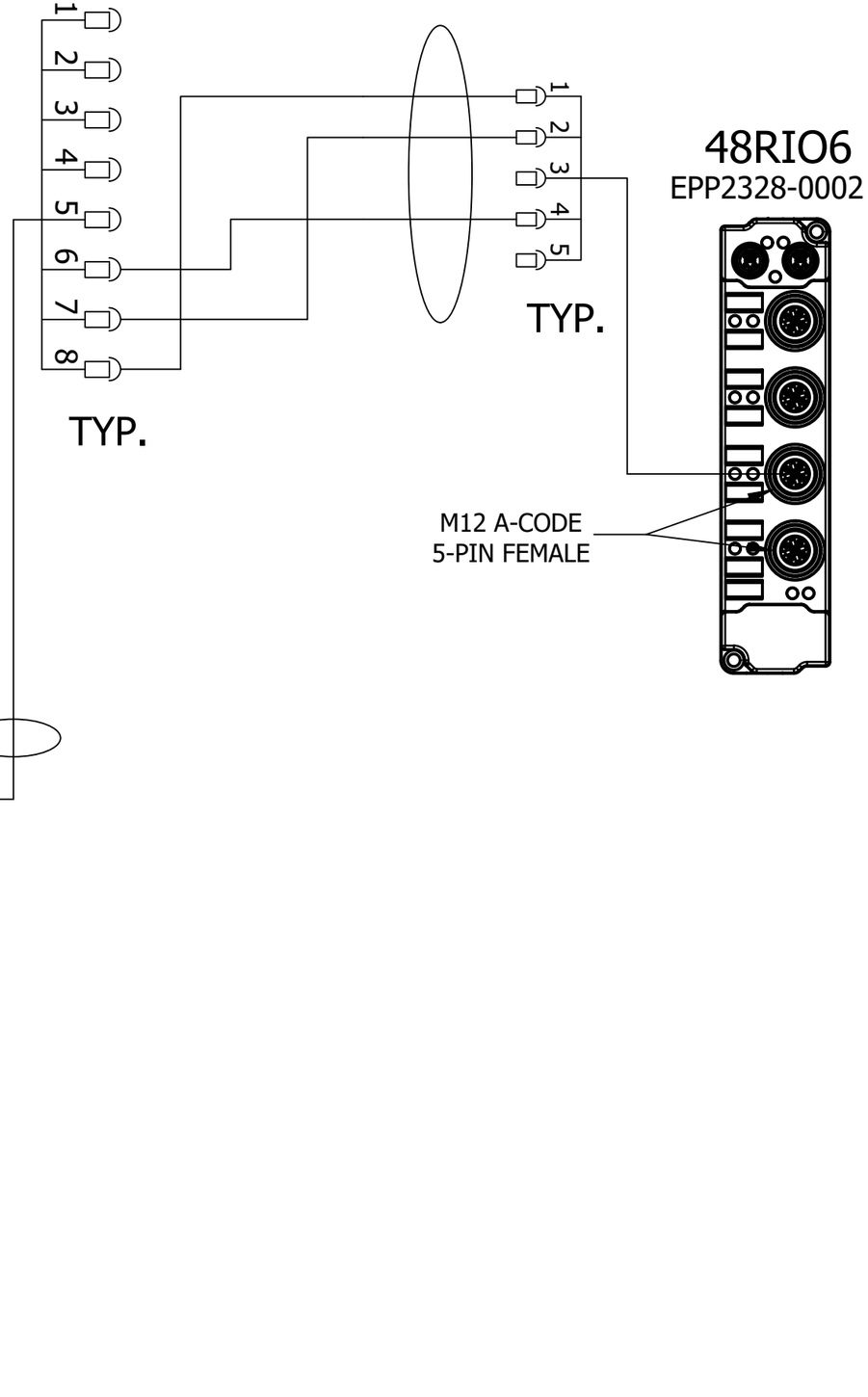
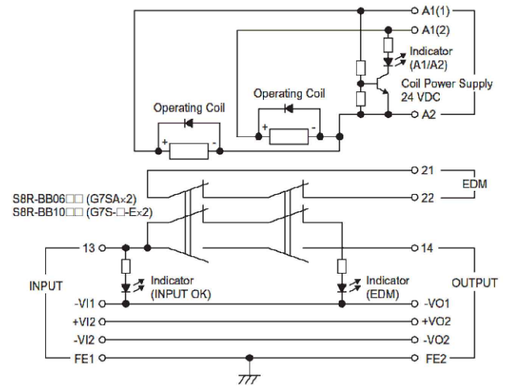


CONVEYOR WITH STOPPER LEVEL 6

SR8-BB-06A1



No.	Connector Name	Shape	Pin Number/Name
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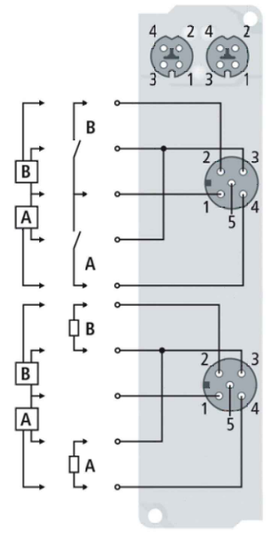

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PAGE DESCRIPTION: LEVEL 6 STOPPER MOTORS

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PROJECT DESCRIPTION: SpaceX Project OUTBOUND VTD PANEL

PAGE DESCRIPTION: SPARE

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PAUL	Carlos A	C40572
DATE	DATE	CURRENT REVISION:
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07663

JOB # C40572

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Component Tag(s)	Manufacturer / Vendor	Part Number	Description	Qty / Panel	Qty / 5 Panels	Drawing Sheet(s)
7THM1	FINDER	7T.81.0.000.2301	Compact, DIN rail-mounted panel thermostat, Temperature Setting Range: -20°C to +40°C (-4°F to +104°F), Contact Configuration: SPST-NO (Normally Open), Contact Rating: 10 A at 250 V AC	1	5	7.0
7FAN1	Hawa	3156-0291-40-71-00	Filter Fan, 460 V, UL, cUL, CE, CUT-OUT (mm) 291 x 291, NEMA Type 12, 427 CFM, RAL 7035 (Lt. Grey)	2	10	7.0
7C1	EATON	XTMC9A10TD	mini contactor, 8.80A, 24 Vdc, 1NO, 9A	1	5	7.0
7CB1	EATON	FAZ-D1/3-NA	miniature circuit breaker, current-limiting, 2A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC	1	5	7.0
8BPS1	Omron	S8BA-24D24D960SBF	Main body Uninterruptible Power Supply (UPS) / Control Unit Part 960w	1	5	8.0, 8.2
8PS1, 8PS2, 8PS3	Omron	S8VK-WB48024	DIN Rail Power Supplies 3Phs 480V Pwr Sup 20A 24VDC	2	10	8.0, 8.1, 8.3
8PS1BAT	OMRON	S8BA-S960L	Battery unit 7800 mAh	1	5	8.0
8CB1, 8CB3, 8CB5	EATON	FAZ-D1.5/3-NA	miniature circuit breaker, current-limiting, 1.5A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC	1	5	8.0, 8.1, 8.3
8CB2, 8CB4, 8CB6, 8CB7	EATON	FAZ-C20/1-NA-DC-SP	489 Industrial miniature circuit breaker-supplementary protector,Single package,Medium levels of inrush current are expected,20A,10 kAIC,Single-pole,125 Vdc per pole,5-10X	1	5	8.0, 8.1, 8.3
LM1-16; LM100-107	LUTZE	716401	Electronic load monitoring up to DC 10 A Single-channel design, Adjustable current range: DC 1 A – 10 A Adjustable characteristics, fast, medium, slow 1, -2, -3	24	120	9.0, 10.0
15CB1	EATON	FAZ-D30/3-NA	miniature circuit breaker, current-limiting, 30A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC	1	5	15.1
17CB1, 20CB1, 21CB1	EATON	FAZ-D15/3-NA	miniature circuit breaker, current-limiting, 15A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC	3	15	17.1, 20.1, 21.1
16CB1	EATON	FAZ-D5/3-NA	miniature circuit breaker, current-limiting, 5A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC	1	5	16.2
18CB1	EATON	FAZ-D7/3-NA	miniature circuit breaker, current-limiting, 7A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC	2	10	18.1
26CR1-26CR16; 27CR1-27CR16	WAGO	857-304	Relay module; Nominal input voltage: 24 VDC; 1 changeover contact; Limiting continuous current: 6 A; Yellow status indicator	32	160	26.0, 27.0, 50.0, 50.1, 51.0, 51.1, 52.0
50K50C	WERMA	240.340.50	Multi-Color (RGB) with Buzzer, UL	1	5	50.0
50K50C	Werma	960.693.05	5 Pole, Female Right Angle, 2M, 4A, MDC (A-coded)	1	5	50.0
4PI1	Panduit	VS2-AVT-3PB-16	VeriSafe™ Testers VeriSafe 2.0 Absence of Voltage Tester. Includes Three-Phase Battery-powered Indicator Module, 16ft (4.8m) System Cable, and 10ft (3.0m) Sensor Leads	1	5	4.0
5PI2; 5PI3	EATON	M22-T-W-W	Press-to-Test unit, 22.5 mm, Momentary, Illuminated, Bezel: Silver, IP67, IP69K, NEMA 4X, 13, 12-30 Vac/dc, White button	2	10	5.0

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8RS1	Omron	S8VK-R20	Redundancy Unit	1	5	8.3
4PI1	PANDUIT	VS2-AVT-3P-ISO		1	5	4.0
8RS1	Omron	S8VK-R20	Redundancy unit	1	5	8.3
15VFD1	SEW-EURODRIVE	MPX23A-DSI10A-MPI-0460-5E3-CA-C	15 kW VFD	1	5	15.0
	SEW-EURODRIVE	DR2L132L4/BE20HR/TF/AK8W	11 kW motor; drawing also shows FA77B gear unit	1	5	15.2
15DE1	SICK	DL50-P2228	Distance sensor / SSI connection	1	5	15.4
16VFD1; 17VFD1-17VFD4; 18VFD1-18VFD3; 20VFD1-20VFD5; 21VFD1-21VFD2	SEW-EURODRIVE	MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025	0.75 kW VFD	15	75	16.0, 17.0, 18.0, 20.0, 21.0
	Siemens	6ES71946KA000XA0	ET200 Y cable	8	40	16.7, 16.8, 17.11, 17.12, 18.6, 18.7, 47.3
16PE1-5; 17PE1-23	SICK	GL6-P1112	Sensor	25	125	16.8, 17.12, 18.7
17M1-17M4; 18M1-18M3; 20M1-20M5; 21M1-21M2	SEW-EURODRIVE	DRN80M4/BE1HR/ISU/TF	0.75 kW motor	14	70	17.3, 17.4, 18.2, 20.3, 20.4, 21.2
	Beckhoff	CX5330-0225	PLC/controller	1	5	23.0
23.0-13	Beckhoff	EK1122	EtherCAT module	2	10	23.0, 36.0
23.0-1, 23.0-2	Beckhoff	EL1809	24 VDC input terminal	2	10	23.0, 24.0, 25.0
23.0-3, 23.0-4	Beckhoff	EL2809	24 VDC output terminal	2	10	23.0, 26.0, 27.0
23.0-8	Beckhoff	EL9410	DC supply terminal for network interface cards	1	5	23.0, 31.0
23.0-12	Beckhoff	EL6633	PROFINET interface terminal	1	5	23.0, 35.0
23.0-7	Beckhoff	EL2912	Safety output terminal	1	5	23.0, 30.0
23.0-5	Beckhoff	EL6910	Safety CPU controller	1	5	23.0, 28.0
23.0-6	Beckhoff	EL1918	Safety input terminal	1	5	23.0, 29.0
	Beckhoff	EL1322	PLC terminal	1	5	23.0, 32.0, 33.0, 34.0
	EBA Automation	CB310-S00	KeTop component - 107780	1	5	39.3
	EBA Automation	T155-M10-NT0-W10E	KeTop HMI - 140318	1	5	39.3
		SR-X300	Barcode reader	1	5	39.3
40RIO1	TURCK	TBEN-S2-4IOL	I/O-Link master / remote I/O device	1	5	40.0
	KEYENCE	SR-300X	Device shown on 40.1	1	5	40.1
	Beckhoff	ZK1090-3191-0100	Cable / connection item	1	5	43.1
	Beckhoff	ZK2020-3200-0100	Cable / connection item	1	5	43.1
43LC1	SICK	C4C-EV13530B10000	Light curtain receiver; drawing Part # 1107661	1	5	43.4
43LC2	SICK	C4C-SV13530B10000	Light curtain sender; drawing Part # 1107660	1	5	43.4

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Component Tag(s)	Manufacturer / Vendor	Part Number	Description	Qty / Panel	Qty / 5 Panels	Drawing Sheet(s)
44SD1		MGB2-L1HE-MLI-U-DA-R-165567	Guard lock device	1	5	44.0
		AC-MC-SB-M-A-157024	Guard lock associated component	1	5	44.0
		MBM-EC-S1-MLI-3B-173401	Guard lock associated component	1	5	44.0
47RIO1; 48RIO1-48RIO6	Beckhoff	EPP2328-0002	EtherCAT P remote I/O module	7	35	47.0, 47.1, 47.3, 48.0, 48.1, 48.2, 48.3, 48.4, 48.5, 48.6, 53.0, 53.1, 53.2, 53.3, 53.4, 53.5
45RIO1-45RIO3; 46RIO2-46RIO7	Beckhoff	EPP1008-0022	EtherCAT P remote I/O module	9	45	45.1, 45.2, 45.3, 45.4, 46.2, 46.3, 46.4, 46.5, 46.6, 46.7
45RIO4	Beckhoff	EPP5001-0002	EtherCAT P remote I/O module	1	5	45.1, 45.5
47RIO2	Beckhoff	EPP1809-0022	EtherCAT P remote I/O module	1	5	47.0, 47.2, 47.4
	Beckhoff	ZS2000-4722	Connection/accessory shown with remote I/O	7	35	47.4, 48.1, 48.2, 48.3, 48.4, 48.5, 48.6
53RR1, 53RR3, 53RR5, 53RR7, 53RR9, 53RR11	Omron	SR8-BB-06A1	Stopper motor interface/device	6	30	53.0, 53.1, 53.2, 53.3, 53.4, 53.5

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